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FILE: public\js\appointments-queue.js

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/**

* public/js/appointments-queue.js

* Safe Appointments & Patient Queue Module for Multi-Facility Platform.

* Handles visit types, queue statuses, routing guards, and priority triage badges.

*/

window.APPOINTMENT_TYPES = {

OPD_APPOINTMENT: { id: 'opd_app', name_en: 'OPD Appointment', name_ar: '???? ????????? ?????????' },

WALK_IN: { id: 'walk_in', name_en: 'Walk-In', name_ar: '???? ???? ????' },

ER_VISIT: { id: 'er_visit', name_en: 'ER Visit', name_ar: '????? ?????????' },

FOLLOW_UP: { id: 'follow_up', name_en: 'Clinical Follow-up', name_ar: '??????? ???????' },

INPATIENT_ADMISSION_REQUEST: { id: 'inpatient_adm', name_en: 'Inpatient Admission Request', name_ar: '??? ??
??? ?????' },

INTERNAL_REFERRAL: { id: 'internal_ref', name_en: 'Internal Department Referral', name_ar: '????? ??????? ??
????????' }
};

window.QUEUE_STATUSES = {

SCHEDULED_PLACEHOLDER: { id: 'scheduled', name_en: 'Scheduled', name_ar: '?????' },

ARRIVED_PLACEHOLDER: { id: 'arrived', name_en: 'Arrived', name_ar: '??? ?????????' },

WAITING_TRIAGE: { id: 'waiting_triage', name_en: 'Waiting Triage', name_ar: '???????? ??????' },

WAITING_CLINIC: { id: 'waiting_clinic', name_en: 'Waiting Clinic', name_ar: '???????? ?????????' },

IN_ENCOUNTER: { id: 'in_encounter', name_en: 'In Encounter', name_ar: '???? ?????????' },

REFERRED_PLACEHOLDER: { id: 'referred', name_en: 'Referred', name_ar: '??? ?????????' },

ADMISSION_REQUESTED_PLACEHOLDER: { id: 'adm_requested', name_en: 'Admission Requested', name_ar: '????? ????
???' }
};

window.MOCK_QUEUE_DATA = [

{ queueNo: 'OPD-001', type: 'opd_app', status: 'waiting_triage', priority: 'normal', waitTimeMin: 15, deptPage: 14 },

{ queueNo: 'OPD-002', type: 'walk_in', status: 'waiting_clinic', priority: 'normal', waitTimeMin: 25, deptPage: 14 },

{ queueNo: 'ER-911', type: 'er_visit', status: 'waiting_triage', priority: 'high', waitTimeMin: 5, deptPage: 21 }
];

window.getAppointmentsForFacility = function(facilityType) {

if (facilityType === 'health_unit') {

return [window.APPOINTMENT_TYPES.OPD_APPOINTMENT, window.APPOINTMENT_TYPES.WALK_IN];

}

if (facilityType === 'phc' || facilityType === 'polyclinic') {

return [window.APPOINTMENT_TYPES.OPD_APPOINTMENT, window.APPOINTMENT_TYPES.WALK_IN, window.APPOINTMENT_TYPES.FOLLOW_UP, window.APPOINTMENT_TYPES.INTERNAL_REFERRAL];

}

return Object.values(window.APPOINTMENT_TYPES);

};

```

window.canBookAppointmentPlaceholder = function(facilityType, deptPage, appointmentTypeId) {
  const allowedDepts = window.getDepartmentsByFacilityType(facilityType);
  if (allowedDepts !== null && !allowedDepts.includes(deptPage)) {
    return false;
  }

  // Prevent Inpatient Admission bookings in primary care / clinic facilities
  if (appointmentTypeId === 'inpatient_adm' && ['health_unit', 'phc', 'polyclinic'].includes(facilityType)) {
    return false;
  }

  return true;
};

window.getTriagePriorityBadges = function(priorityCode) {
  if (priorityCode === 'high') {
    return { class: 'danger', label_en: 'Urgent (Level 2)', label_ar: '???? (??????? 2)' };
  }
  return { class: 'info', label_en: 'Routine (Level 4)', label_ar: '???? (??????? 4)' };
};

```

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=====
FILE: public\js\cardiology-station.js
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```

```

/**
 * cardiology-station.js
 * Cardiology Specialist Station - Stitch Google Design
 * Implements the "Command Center" for Cardiologists
 */

const CardiologyStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Cardiology Command Center', '???? ????
? ????? ?????')}</h1>
          <div class="flex gap-2">
            <button onclick="CardiologyStation.openECGUpload()" class="stitch-btn-primary px-4 py-
2 bg-blue-600 text-white rounded-lg shadow-sm hover:bg-blue-700 transition-all">
              ${tr('Upload ECG', '??? ????? ?????')}
            </button>
            <button onclick="CardiologyStation.openCathLab()" class="stitch-btn-secondary px-4 py-
2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('Cath Lab Report', '????? ????????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">

```

```
<!-- Left: Patient Context -->
<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
    <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Patient Vitals', '??????? ????')}</h3>
    <div id="cardiology-vitals-grid" class="grid grid-cols-2 gap-4">
      <!-- Dynamic Vitals Gauges -->
      <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
        <span class="block text-xs text-slate-500">${tr('BP', '???')}</span>
        <span class="text-lg font-bold text-slate-800">120/80</span>
      </div>
      <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
        <span class="block text-xs text-slate-500">${tr('HR', '???')}</span>
        <span class="text-lg font-bold text-slate-800">72 bpm</span>
      </div>
    </div>
  </div>

  <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
    <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('HF Risk Score', '??? ????')}</h3>
    <div class="flex flex-col items-center">
      <div class="stitch-radial-gauge w-32 h-32 relative">
        <svg class="w-full h-full" viewBox="0 0 100 100">
          <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
          <circle cx="50" cy="50" r="40" stroke="#ef4444" stroke-width="8" fill=
"none"
          stroke-dasharray="251.2" stroke-dashoffset="60" stroke-linecap="ro
und" />
        </svg>
        <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">75%</span>
      </div>
      <p class="mt-4 text-xs text-red-500 font-medium">${tr('High Risk - Immediate R
eview', '??? - ????')}</p>
    </div>
  </div>
</div>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-blue-600 text-blue-600 font-medium"
>${tr('ECG Analysis', '????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Cath Lab',
'????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Nuclear',
'????')}</button>
    </div>
    <div id="cardiology-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
```



```

window.BILLING_STATUSES = {
  CHARGE_PREVIEW: { id: 'charge_prev', name_en: 'Charge Preview', name_ar: '?????? ??????' },
  DOCUMENTATION_MISSING: { id: 'doc_missing', name_en: 'Documentation Missing', name_ar: '???????? ????' },
  ELIGIBILITY_PREVIEW: { id: 'elig_prev', name_en: 'Eligibility Preview', name_ar: '?????? ????????' },
  CLAIM_DRAFT: { id: 'claim_draft', name_en: 'Claim Draft', name_ar: '????? ?????????' },
  REVIEW_REQUIRED: { id: 'review_req', name_en: 'Review Required', name_ar: '????? ????????' },
  SUBMISSION_DISABLED: { id: 'sub_disabled', name_en: 'Submission Disabled', name_ar: '???????? ????' }
};

window.MOCK_PAYERS = [
  { id: 'Tawuniya', name_en: 'Tawuniya Insurance', name_ar: '????????? ????????' },
  { id: 'Bupa', name_en: 'Bupa Arabia', name_ar: '???? ????????' },
  { id: 'Medgulf', name_en: 'Medgulf', name_ar: '???????' }
];

window.MOCK_CHARGE_ITEMS = {
  consultation: { code: 'CONS_01', name_en: 'Consultation Fee', name_ar: '???? ?????????', price: 150 },
  cbc: { code: 'CBC', name_en: 'Complete Blood Count', name_ar: '???? ?? ?????', price: 100 },
  cxr: { code: 'CXR', name_en: 'Chest X-Ray', name_ar: '???? ????? ?????', price: 120 }
};

window.getChargeCapturePreview = function(encounterTypeId, itemsList = []) {
  const charges = [window.MOCK_CHARGE_ITEMS.consultation];
  itemsList.forEach(itemCode => {
    const item = Object.values(window.MOCK_CHARGE_ITEMS).find(x => x.code === itemCode);
    if (item) charges.push(item);
  });
  return charges;
};

window.getEligibilityPreview = function(payerId) {
  return {
    payerId: payerId,
    status: 'active',
    copayPercent: 10,
    maxLimit: 5000,
    warnings: []
  };
};

window.getClaimDraftPreview = function(encounterTypeId, charges = []) {
  const total = charges.reduce((sum, item) => sum + item.price, 0);
  return {
    status: 'Claim Draft Created',
    totalAmount: total,
    nphiesReady: false,
    warnings: ['Validation Warning: Simulated environment. Submission is disabled.']
  };
};

window.getMissingDocumentationWarnings = function(encounterTypeId, noteWritten) {
  const warnings = [];
  if (!noteWritten) {

```

```

        warnings.push({
            type: 'missing_soap',
            text_en: 'Missing Documentation: SOAP clinical note is required before billing.',
            text_ar: '??????? ?????: ??? ????? ????????? ?????? SOAP ??? ????????'
        });
    }
    return warnings;
};

```

```

window.getClaimValidationWarnings = function(charges) {
    const warnings = [];
    if (charges.length === 0) {
        warnings.push({
            type: 'no_charges',
            text_en: 'No billable charges captured for this encounter.',
            text_ar: '?? ????? ????? ?????? ??????? ????? ????????'
        });
    }
    return warnings;
};

```

```

window.canCreateFinalInvoice = function(facilityType, deptPage) {
    return false;
};

```

```

window.canPostFinancialEntry = function(facilityType, deptPage) {
    return false;
};

```

```

window.canSubmitNphiesClaim = function(facilityType, deptPage) {
    return false;
};

```

```

=====
FILE: public\js\clinical-orders.js
=====

```

```

/**
 * public/js/clinical-orders.js
 * Safe Clinical Orders Module for Multi-Facility Platform.
 * Manages clinical order types, catalog lookups, and safety warnings.
 */

```

```

window.ORDER_TYPES = {
    LAB: { id: 'lab', name_en: 'Laboratory Order', name_ar: '??? ?????' },
    RADIOLOGY: { id: 'radiology', name_en: 'Radiology Order', name_ar: '??? ?????' },
    MEDICATION: { id: 'medication', name_en: 'Medication Order', name_ar: '???? ?????' },
    PROCEDURE: { id: 'procedure', name_en: 'Procedure Order', name_ar: '??? ?????? ???' },
    CONSULTATION: { id: 'consultation', name_en: 'Consultation Order', name_ar: '??? ??????? ?????' }
};

```

```

window.MOCK_ORDER_CATALOG = {

```

```

lab: [
  { code: 'CBC', name_en: 'Complete Blood Count', name_ar: '???? ?? ?????', risk: 'low' },
  { code: 'LFT', name_en: 'Liver Function Test', name_ar: '????? ?????', risk: 'low' },
  { code: 'BG', name_en: 'Blood Grouping', name_ar: '????? ????? ?????', risk: 'low' }
],
radiology: [
  { code: 'CXR', name_en: 'Chest X-Ray', name_ar: '???? ????? ?????', risk: 'low' },
  { code: 'MRI_BRAIN', name_en: 'Brain MRI', name_ar: '???? ????????? ?????', risk: 'medium' },
  { code: 'CT_ABD', name_en: 'CT Abdomen', name_ar: '???? ??????? ?????', risk: 'medium' }
],
medication: [
  { code: 'PARA500', name_en: 'Paracetamol 500mg', name_ar: '???????????? 500 ???', risk: 'low' },
  { code: 'AM0500', name_en: 'Amoxicillin 500mg', name_ar: '???????????? 500 ???', risk: 'medium' }
],
procedure: [
  { code: 'SUT1', name_en: 'Laceration Suture', name_ar: '????? ???', risk: 'medium' },
  { code: 'LP', name_en: 'Lumbar Puncture', name_ar: '??? ?????', risk: 'high' }
],
consultation: [
  { code: 'CARD_CONS', name_en: 'Cardiology Consultation', name_ar: '??????? ????? ???', risk: 'low' },
  { code: 'NEUR_CONS', name_en: 'Neurology Consultation', name_ar: '??????? ????? ?? ??????', risk: 'low' }
]
};

```

```

window.getOrderTypesForEncounter = function(facilityType, encounterTypeId) {
  if (facilityType === 'health_unit') {
    return [window.ORDER_TYPES.LAB, window.ORDER_TYPES.CONULTATION];
  }
  if (facilityType === 'phc') {
    return [window.ORDER_TYPES.LAB, window.ORDER_TYPES.RADIOLOGY, window.ORDER_TYPES.CONULTATION];
  }
  // General Hospital and Medical City allow all
  return Object.values(window.ORDER_TYPES);
};

```

```

window.canCreateOrderDraft = function(facilityType, deptPage, orderTypeId) {
  // Check if department is allowed
  const allowedDepts = window.getDepartmentsByFacilityType(facilityType);
  if (allowedDepts !== null && !allowedDepts.includes(deptPage)) {
    return false;
  }

  // Health Unit restricts advanced orders
  if (facilityType === 'health_unit' && ['radiology', 'procedure', 'medication'].includes(orderTypeId)) {
    return false;
  }

  // PHC restricts advanced procedures
  if (facilityType === 'phc' && orderTypeId === 'procedure') {
    return false;
  }

  return true;
};

```

```

window.getOrderSafetyWarnings = function(orderTypeCode, itemCode) {
  const warnings = [];
  if (orderTypeCode === 'medication') {
    warnings.push({
      type: 'allergy',
      text_en: 'Simulated Allergy Check: Please verify patient allergies before prescribing.',
      text_ar: '??? ?????? ??????: ??? ???? ?? ?????? ?????? ?? ?? ??????'
    });
  }

  // High risk procedures require consent
  const catalog = window.MOCK_ORDER_CATALOG[orderTypeCode] || [];
  const item = catalog.find(x => x.code === itemCode);
  if (item && item.risk === 'high') {
    warnings.push({
      type: 'consent',
      text_en: 'High-Risk Procedure: Signed Patient Consent Form is required.',
      text_ar: '???? ? ? ???? ?????: ????? ?????? ?????? ?????? ??????'
    });
  }

  return warnings;
};

=====
FILE: public\js\clinical-pharmacy.js
=====

/**
 * public/js/clinical-pharmacy.js
 * Safe Clinical Pharmacy & FEFO Inventory Module for Multi-Facility Platform.
 * Handles medication safety review, batch expiry preview, and controlled substance alerts.
 */

window.PHARMACY_REVIEW_STATUSES = {
  DRAFT_ORDER: { id: 'draft', name_en: 'Draft Order', name_ar: '??? ?????' },
  NEEDS_PHARMACIST_REVIEW: { id: 'review', name_en: 'Needs Pharmacist Review', name_ar: '???? ??????? ???????' },
  SAFETY_WARNING: { id: 'warning', name_en: 'Safety Warning Triggered', name_ar: '???? ?????? ????' },
  READY_FOR_PREVIEW: { id: 'preview', name_en: 'Ready for Preview', name_ar: '???? ????????' },
  DISPENSE_DISABLED: { id: 'disabled', name_en: 'Dispense Blocked', name_ar: '???? ?????' }
};

window.MOCK_MEDICATION_BATCHES = {
  'PARA500': [
    { batchCode: 'B-PR-982', expiryDate: '2026-08-30', quantity: 120, fefoRank: 1 },
    { batchCode: 'B-PR-441', expiryDate: '2027-02-15', quantity: 350, fefoRank: 2 }
  ],
  'AM0500': [
    { batchCode: 'B-AM-103', expiryDate: '2026-07-15', quantity: 80, fefoRank: 1 },
    { batchCode: 'B-AM-552', expiryDate: '2026-12-01', quantity: 200, fefoRank: 2 }
  ]
};

```



```

    ]
};

window.getFefoBatchPreview = function(medicationCode) {
    const batches = window.MOCK_MEDICATION_BATCHES[medicationCode] || [];
    // Sort by expiry date (FEFO: First Expired, First Out)
    return [...batches].sort((a, b) => new Date(a.expiryDate) - new Date(b.expiryDate));
};

window.getMedicationSafetyWarnings = function(medicationCode) {
    const warnings = [];
    if (medicationCode === 'AM0500') {
        warnings.push({
            type: 'allergy',
            text_en: 'Allergy Warning: Penicillin class drug. Cross-sensitivity risk.',
            text_ar: '????? ??????: ??? ? ? ? ? ??????. ?? ?????? ??????????'
        });
    }
    return warnings;
};

window.getControlledMedicationWarnings = function(medicationCode) {
    const warnings = [];
    if (medicationCode === 'MORPHINE' || medicationCode === 'FENTANYL') {
        warnings.push({
            type: 'controlled',
            text_en: 'Controlled Substance: Requires double-signature and narcotic prescription registration.',
            text_ar: '???? ????? ??????: ????? ?????? ?????? ?????? ? ? ? ? ?????? ??????'
        });
    }
    return warnings;
};

window.canPreviewDispense = function(facilityType, deptPage, medicationCode) {
    const allowedDepts = window.getDepartmentsByFacilityType(facilityType);
    if (allowedDepts !== null && !allowedDepts.includes(deptPage)) {
        return false;
    }
    return true;
};

window.canFinalizeDispense = function(facilityType, deptPage, medicationCode) {
    // Always blocked in this simulation phase
    return false;
};

```

```

=====
FILE: public\js\clinical-results.js
=====

```

```

/**
 * public/js/clinical-results.js

```

```
* Safe Clinical Results & Documentation Module for Multi-Facility Platform.  
* Manages clinical results, diagnostic reports, and SOAP notes.  
*/
```

```
window.RESULT_TYPES = {  
  LAB_RESULT: { id: 'lab', name_en: 'Laboratory Result', name_ar: '????? ??????' },  
  RADIOLOGY_RESULT: { id: 'radiology', name_en: 'Radiology Report', name_ar: '?????? ??????' },  
  PROCEDURE_NOTE: { id: 'procedure', name_en: 'Procedure Note', name_ar: '?????? ??????? ??????' },  
  CONSULTATION_RESPONSE: { id: 'consultation', name_en: 'Consultation Response', name_ar: '?? ?????????? ??????' },  
  CLINICAL_NOTE: { id: 'note', name_en: 'Clinical Note', name_ar: '????????? ??????????' }  
};
```

```
window.MOCK_RESULTS = {  
  lab: [  
    { code: 'CBC', name_en: 'Complete Blood Count', value: 'Hb: 14.2 g/dL (Normal)', status: 'Reviewed', abnormal: false },  
    { code: 'LFT', name_en: 'Liver Function Test', value: 'ALT: 85 U/L (High)', status: 'Needs Review', abnormal: true }  
  ],  
  radiology: [  
    { code: 'CXR', name_en: 'Chest X-Ray', value: 'Clear lung fields. No cardiomegaly.', status: 'Needs Review', abnormal: false },  
    { code: 'CT_ABD', name_en: 'CT Abdomen', value: 'Mild splenomegaly noted.', status: 'Needs Review', abnormal: true }  
  ],  
  procedure: [  
    { code: 'SUT1', name_en: 'Laceration Suture', value: '3 sutures applied to left forearm. Clean wound.', status: 'Draft', abnormal: false }  
  ],  
  consultation: [  
    { code: 'CARD_CONS', name_en: 'Cardiology Consultation', value: 'Recommend ambulatory blood pressure monitoring.', status: 'Needs Review', abnormal: false }  
  ]  
};
```

```
window.getResultsForEncounter = function(facilityType, encounterTypeId) {  
  // Returns allowed result types for the facility  
  if (facilityType === 'health_unit') {  
    return [window.RESULT_TYPES.LAB_RESULT, window.RESULT_TYPES.CLINICAL_NOTE];  
  }  
  if (facilityType === 'phc' || facilityType === 'polyclinic') {  
    return [window.RESULT_TYPES.LAB_RESULT, window.RESULT_TYPES.RADIOLOGY_RESULT, window.RESULT_TYPES.CLINICAL_NOTE];  
  }  
  return Object.values(window.RESULT_TYPES);  
};
```

```
window.getAbnormalResultWarnings = function(resultTypeCode, itemCode) {  
  const warnings = [];  
  const items = window.MOCK_RESULTS[resultTypeCode] || [];  
  const result = items.find(x => x.code === itemCode);  
  
  if (result && result.abnormal) {
```

```
warnings.push({
  type: 'abnormal',
  text_en: 'Abnormal Finding: Immediate physician review and clinical correlation required.',
  text_ar: '????? ??? ??????: ????? ?????? ????? ?? ?????? ??????? ?????? ?????? ??????.'
});
}
return warnings;
};
```

```
window.canAcknowledgeResult = function(facilityType, deptPage, resultTypeCode) {
  // Check if department has access
  const allowedDepts = window.getDepartmentsByFacilityType(facilityType);
  if (allowedDepts !== null && !allowedDepts.includes(deptPage)) {
    return false;
  }
  return true;
};
```

```
=====
FILE: public\js\critical-station.js
=====
```

```
/**
 * critical-station.js
 * Critical Care & Emergency Specialist Station - Stitch Google Design
 */

const CriticalStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-900 min-h-screen text-white">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-white">${tr('Critical Care Command Center', '???? ?????
??????? ??????? ???????')}</h1>
          <div class="flex gap-2">
            <button onclick="CriticalStation.openTriage()" class="stitch-btn-primary px-4 py-2 bg-
red-600 text-white rounded-lg shadow-sm hover:bg-red-700 transition-all">
              ${tr('Quick Triage', '??? ???')}
            </button>
            <button onclick="CriticalStation.openAnesthesiaLog()" class="stitch-btn-secondary px-4
py-2 bg-slate-800 border border-slate-700 text-slate-300 rounded-lg shadow-sm hover:bg-slate-700 transition-a
ll">
              ${tr('Anesthesia Log', '??? ?????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <!-- Left: High-Acuity Vitals -->
          <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-slate-800 rounded-xl shadow-sm border border-slate-700"
```

```
>
        <h3 class="text-sm font-semibold text-slate-400 mb-4 uppercase tracking-wider">${t
r('Real-time Vitals', '???????? ?????? ???????')}</h3>
        <div id="critical-vitals-grid" class="grid grid-cols-2 gap-4">
            <div class="stitch-gauge-item p-3 bg-slate-700 rounded-lg text-center">
                <span class="block text-xs text-slate-400">${tr('MAP', '????? ?????')}</sp
an>

                <span class="text-lg font-bold text-red-500">58 mmHg</span>
            </div>
            <div class="stitch-gauge-item p-3 bg-slate-700 rounded-lg text-center">
                <span class="block text-xs text-slate-400">${tr('SpO2', '???? ?????????')}<
/span>

                <span class="text-lg font-bold text-emerald-400">92%</span>
            </div>
        </div>
    </div>

    <div class="stitch-card p-4 bg-slate-800 rounded-xl shadow-sm border border-slate-700"
>
        <h3 class="text-sm font-semibold text-slate-400 mb-4 uppercase tracking-wider">${t
r('Crash Risk', '??? ?????????')}</h3>
        <div class="flex flex-col items-center">
            <div class="stitch-radial-gauge w-32 h-32 relative">
                <svg class="w-full h-full" viewBox="0 0 100 100">
                    <circle cx="50" cy="50" r="40" stroke="#334155" stroke-width="8" fill=
"none" />

                    <circle cx="50" cy="50" r="40" stroke="#ef4444" stroke-width="8" fill=
"none"

                    stroke-dasharray="251.2" stroke-dashoffset="40" stroke-linecap="ro
und" />
                </svg>
                <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-white">High</span>
            </div>
            <p class="mt-4 text-xs text-red-500 font-bold animate-pulse">${tr('Sepsis Bund
le Required', '????? ?????? ??? ????')}</p>
        </div>
    </div>

    <!-- Center: Active Workspace -->
    <div class="col-span-6 space-y-6">
        <div class="stitch-card p-6 bg-slate-800 rounded-xl shadow-sm border border-slate-700
min-h-[600px]">
            <div class="flex border-b border-slate-700 mb-6">
                <button class="px-4 py-2 border-b-2 border-red-600 text-red-600 font-medium">${
tr('Triage Log', '??? ?????')}</button>

                <button class="px-4 py-2 text-slate-500 hover:text-slate-300">${tr('ICU Flowsh
eet', '???? ???????? ?????')}</button>

                <button class="px-4 py-2 text-slate-500 hover:text-slate-300">${tr('AI Predict
ions', '??????? AI')}</button>
            </div>
            <div id="critical-workspace-content" class="space-y-4">
                <div class="p-8 text-center text-slate-500 border-2 border-dashed border-slate
```

```

                                ${tr('No active critical session. Please start a triage or ICU log.', '??
???? ???? ???? ?????. ????. ??? ?? ?? ?? ?? ?? ???? ?????.')}}
                                </div>
                            </div>
                        </div>
                    </div>

                <!-- Right: AI Insights -->
                <div class="col-span-3 space-y-6">
                    <div class="stitch-card p-4 bg-red-900/20 rounded-xl shadow-sm border border-red-900/3
0">

                        <div class="flex items-center gap-2 mb-4">
                            <span class="text-xl">?</span>
                            <h3 class="text-sm font-bold text-red-400 uppercase tracking-wider">${tr('Crit
ical AI', '???? ?????? ??????')}</h3>
                        </div>
                        <div id="ai-insights-panel" class="space-y-4">
                            <div class="p-3 bg-slate-800 rounded-lg border border-red-900/50 shadow-sm">
                                <p class="text-xs text-red-400 font-semibold mb-1">${tr('Deterioration Ale
rt', '???? ???? ?????')}</p>
                                <p class="text-sm text-slate-300 italic">"Hemodynamic instability detected
. Pattern matches Septic Shock. Suggest starting Norepinephrine."</p>
                                <button class="mt-2 text-xs text-red-400 underline font-medium">${tr('View
Evidence', '?? ????')}</button>
                            </div>
                        </div>
                    </div>
                </div>
            </div>
        `;
    },

    openTriage: () => { alert('Triage Modal Triggered'); },
    openAnesthesiaLog: () => { alert('Anesthesia Log Modal Triggered'); }
};
```

```
/**
 * dermat-station.js
 * Dermatology Specialist Station - Stitch Google Design
 */
```

```
const DermStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
```

```

<header class="flex justify-between items-center mb-6">
  <h1 class="text-2xl font-bold text-slate-800">${tr('Dermatology Command Center', '???? ??
?? ?????? ??????')}</h1>
  <div class="flex gap-2">
    <button onclick="DermStation.openLesionLog()" class="stitch-btn-primary px-4 py-2 bg-p
ink-600 text-white rounded-lg shadow-sm hover:bg-pink-700 transition-all">
      ${tr('Log Lesion', '???? ?? ?????')}
    </button>
    <button onclick="DermStation.openCosmeticLog()" class="stitch-btn-secondary px-4 py-2
bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
      ${tr('Cosmetic Procedure', '???? ?????')}
    </button>
  </div>
</header>

<div class="grid grid-cols-12 gap-6">
  <div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
      <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Skin Profile', '??? ?????')}</h3>
      <div id="derm-vitals-grid" class="grid grid-cols-2 gap-4">
        <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
          <span class="block text-xs text-slate-500">${tr('Skin Type', '??? ?????')}
        </span>

          <span class="text-lg font-bold text-slate-800">Fitzpatrick III</span>
        </div>
        <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
          <span class="block text-xs text-slate-500">${tr('Allergies', '????????')}<
        </span>

          <span class="text-lg font-bold text-red-600">Lidocaine</span>
        </div>
      </div>
    </div>
    <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
      <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Biopsy Status', '???? ?????')}</h3>
      <div class="flex flex-col items-center">
        <div class="stitch-radial-gauge w-32 h-32 relative">
          <svg class="w-full h-full" viewBox="0 0 100 100">
            <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

            <circle cx="50" cy="50" r="40" stroke="#ec4899" stroke-width="8" fill=
"none"

              stroke-dasharray="251.2" stroke-dashoffset="100" stroke-linecap="r
ound" />
          </svg>
          <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">Pending</span>
        </div>
        <p class="mt-4 text-xs text-pink-600 font-medium">${tr('Awaiting Pathology', '
?? ?????? ?????????')}</p>
      </div>
    </div>
  </div>
</div>

```

```

        <div class="col-span-6 space-y-6">
            <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
                <div class="flex border-b border-slate-200 mb-6">
                    <button class="px-4 py-2 border-b-2 border-pink-600 text-pink-600 font-medium"
>${tr('Lesion Map', '???? ????')}</button>
                    <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Cosmetic L
og', '?? ????')}</button>
                    <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('AI Vision'
, '??? ????? ????')}</button>
                </div>
                <div id="derm-workspace-content" class="space-y-4">
                    <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
                        ${tr('No active lesion. Please map a new lesion to begin.', '?? ??? ? ?
???. ??? ? ? ? ? ? ? ?')}
                    </div>
                </div>
            </div>
        </div>

        <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-pink-50 rounded-xl shadow-sm border border-pink-100">
                <div class="flex items-center gap-2 mb-4">
                    <span class="text-xl">?</span>
                    <h3 class="text-sm font-bold text-pink-900 uppercase tracking-wider">${tr('AI
Insights', '??? ????? ????')}</h3>
                </div>
                <div id="ai-insights-panel" class="space-y-4">
                    <div class="p-3 bg-white rounded-lg border border-pink-200 shadow-sm">
                        <p class="text-xs text-pink-700 font-semibold mb-1">${tr('Visual Match', '
???? ????')}</p>
                        <p class="text-sm text-slate-600 italic">"Lesion morphology is 88% similar
to Basal Cell Carcinoma. Recommend biopsy."</p>
                        <button class="mt-2 text-xs text-pink-600 underline font-medium">${tr('Vie
w Evidence', '??? ?????')}</button>
                    </div>
                </div>
            </div>
        </div>
    </div>
    `;
},

openLesionLog: () => { alert('Lesion Log Modal Triggered'); },
openCosmeticLog: () => { alert('Cosmetic Log Modal Triggered'); }
};

```

```

=====
FILE: public\js\diagnostics-station.js

```

=====

```
/**
 * diagnostics-station.js
 * Advanced Diagnostics Specialist Station - Stitch Google Design
 */

const DiagnosticsStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Advanced Diagnostics Command Center',
'???? ????? ????????? ?????????')}</h1>
          <div class="flex gap-2">
            <button onclick="DiagnosticsStation.openScanUpload()" class="stitch-btn-primary px-4 p
y-2 bg-indigo-600 text-white rounded-lg shadow-sm hover:bg-indigo-700 transition-all">
              ${tr('Upload Scan', '??? ???')}
            </button>
            <button onclick="DiagnosticsStation.openLabResult()" class="stitch-btn-secondary px-4
py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('Lab Results', '???? ?????????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Diagnostic Status', '???? ?????????')}</h3>
              <div class="space-y-3">
                <div class="flex justify-between text-sm">
                  <span>${tr('Radiology Report', '???? ?????????')}</span>
                  <span class="text-emerald-600 font-bold">Final</span>
                </div>
                <div class="flex justify-between text-sm">
                  <span>${tr('Lab Verification', '???? ?????????')}</span>
                  <span class="text-amber-500 font-bold">Pending</span>
                </div>
              </div>
            </div>
            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Criticality Index', '???? ?????????')}</h3>
              <div class="flex flex-col items-center">
                <div class="stitch-radial-gauge w-32 h-32 relative">
                  <svg class="w-full h-full" viewBox="0 0 100 100">
                    <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
                    <circle cx="50" cy="50" r="40" stroke="#ef4444" stroke-width="8" fill=
"none"
                    stroke-dasharray="251.2" stroke-dashoffset="60" stroke-linecap="ro
```



```
und" />

        </svg>
        <span class="absolute inset-0 flex items-center justify-center text-xl font-
t-bold text-slate-800">High</span>
    </div>
    <p class="mt-4 text-xs text-red-600 font-medium">${tr('Critical Value Detected
', '?? ?????? ??? ????')}</p>
    </div>
  </div>
</div>

<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-indigo-600 text-indigo-600 font-med
ium">${tr('Radiology', '????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Laboratory
', '????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Functional
', '????')}</button>
    </div>
    <div id="diag-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
        ${tr('No active study. Please select a scan or lab result to begin.', '??
???? ?????? ????. ??? ???? ???? ?? ???? ?????? ????.')}}
      </div>
    </div>
  </div>
</div>

<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-indigo-50 rounded-xl shadow-sm border border-indigo-100
">
    <div class="flex items-center gap-2 mb-4">
      <span class="text-xl">?</span>
      <h3 class="text-sm font-bold text-indigo-900 uppercase tracking-wider">${tr('A
I Insights', '?? ?????? ??????')}</h3>
    </div>
    <div id="ai-insights-panel" class="space-y-4">
      <div class="p-3 bg-white rounded-lg border border-indigo-200 shadow-sm">
        <p class="text-xs text-indigo-700 font-semibold mb-1">${tr('Suggested Find
ing', '????? ??????')}</p>
        <p class="text-sm text-slate-600 italic">"Symmetry analysis suggests early
-stage focal lesion in the left lobe. Similar to Case # la- la- la."</p>
        <button class="mt-2 text-xs text-indigo-600 underline font-medium">${tr('V
iew Evidence', '?? ?????')}</button>
      </div>
    </div>
  </div>
</div>
</div>
</div>
```

```

    `;
},

openScanUpload: () => { alert('Scan Upload Modal Triggered'); },
openLabResult: () => { alert('Lab Result Modal Triggered'); }
};

```

```

=====
FILE: public\js\doctor-station.js
=====

```

```

/**
 * =====
 * DOCTOR STATION ? World-Class Clinical Workstation (v3)
 * ???? ????? ? ????? ????? (Epic / Cerner / Oracle Health)
 * =====
 * Three-Panel Layout:
 *   LEFT  ? ????? ?????? ?????? (Wait Queue) ?? Triage Levels
 *   CENTER ? ????? ??????? (11 ?????: Summary,History,Problems,
 *       Medications,Allergies,Vitals,Notes,Orders,Results,History+)
 *   RIGHT ? ???? CPOE ????????? (Dx,Rx,Lab,Rad,Nursing,Diet,IV,Referral,Discharge)
 * =====
 * v3 New: Orders Board, Lab Results Viewer, Extended History,
 *       BMI/BP in header, Triage badge, Insurance info,
 *       Diet/IV/Nursing orders, Rx templates, Refill support
 * =====
 */

```

```

'use strict';

```

```

/* ---- Globals scoped to this module ---- */
window._DS = window._DS || {
  selectedPatientId: null,
  selectedPatientData: null,
  activeTab: 'summary',
  activeOrderSection: 'diagnosis',
  rxItems: [],           // ???? ?????????? ??? ???????
  activeEncounterId: null,
  waitTimer: null,       // setInterval for live wait-time refresh
};

```

```

/* ===== */
/* MAIN ENTRY POINT ? replaces the old renderDoctor function */
/* ===== */
async function renderDoctor(el) {
  // Kill any previous wait-list refresh timer
  if (window._DS.waitTimer) { clearInterval(window._DS.waitTimer); window._DS.waitTimer = null; }
  window._DS.selectedPatientId = null;
  window._DS.rxItems = [];

  // Skeleton while loading
  el.innerHTML = `

```

```

<div class="page-title">???? ${tr('Doctor Station', '???? ?????')}
  <span style="font-size:12px;font-weight:400;color:var(--text-dim);margin-right:8px">
    ${tr('Epic-class Clinical Workstation', '???? ?????? ?????? ?????')}
  </span>
</div>
<div class="ds-layout" id="dsLayout">
  <div class="ds-left-panel">
    <div class="ds-panel-header"><span>? ${tr('Wait Queue', '????? ?????????')}</span>
      <div class="skeleton-bar" style="width:30px;height:18px;border-radius:10px"></div>
    </div>
    ${[1,2,3,4].map(() => `
      <div class="ds-wait-item">
        <div class="ds-wait-avatar" style="background:#e5e7eb"></div>
        <div class="ds-wait-info">
          <div class="skeleton-bar mb-4" style="height:13px;width:80%"></div>
          <div class="skeleton-bar" style="height:10px;width:60%"></div>
        </div>
      </div>
    `).join('')}
  </div>
  <div class="ds-center-panel" style="align-items:center;justify-content:center">
    <div style="text-align:center;color:var(--text-dim);padding:40px">
      <div style="font-size:64px;margin-bottom:16px;opacity:0.4">????</div>
      <p>${tr('Loading...', '???? ?????????...')}</p>
    </div>
  </div>
  <div class="ds-right-panel">
    <div class="ds-panel-header"><span>? ${tr('Orders', '????????')}</span></div>
    <div style="padding:16px">
      <div class="skeleton-bar mb-12" style="height:40px;border-radius:10px"></div>
      <div class="skeleton-bar mb-12" style="height:40px;border-radius:10px"></div>
      <div class="skeleton-bar" style="height:40px;border-radius:10px"></div>
    </div>
  </div>
</div>
`;

// Load wait queue + current user in parallel
let waitQueue = [], drugs = [], currentUserData = {}, allServices = [];
try {
  [waitQueue, drugs, currentUserData, allServices] = await Promise.all([
    API.get('/api/doctor/wait-queue').catch(() => API.get('/api/patients').then(p => p.filter(x => x.status
=== 'Waiting' || x.status === 'With Doctor'))),
    API.get('/api/pharmacy/drugs').catch(() => []),
    API.get('/api/auth/me').catch(() => ({})),
    API.get('/api/medical/services').catch(() => []),
  ]);
} catch (e) {
  el.innerHTML = `
    <div class="page-title">???? ${tr('Doctor Station', '???? ?????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load Doctor Station', '??? ?????? ????? ??????')}</h3>
      <p>${escapeHTML(e.message || String(e))}</p>
  `;
}

```

```

        <button class="btn btn-primary" onclick="navigateTo(3)">? ${tr('Retry', '????? ???????')}</button>
    </div>`;
    return;
}

window._DS.drugs = drugs;
window._DS.allServices = allServices;
window._DS.currentUser = currentUserData.user || currentUserData;

// Render the full three-panel layout
el.innerHTML = `
    <div class="page-title" style="margin-bottom:12px">
        ??? ${tr('Doctor Station', '???? ?????')}
        <span style="font-size:12px;font-weight:500;color:var(--text-dim);margin-inline-start:12px">
            ${tr('Physician:', '?????:')} <strong>${escapeHTML(window._DS.currentUser?.name || window._DS.current
User?.username || 'Doctor')}</strong>
        </span>
        <span style="font-size:12px;font-weight:500;color:var(--text-dim);margin-inline-start:12px">
            ? ${new Date().toLocaleDateString(isArabic ? 'ar-SA' : 'en-US', { weekday:'long', year:'numeric', mont
h:'long', day:'numeric' })}
        </span>
    </div>
    <div class="ds-layout" id="dsLayout">

        <!-- ===== LEFT PANEL: Wait Queue ===== -->
        <div class="ds-left-panel" id="dsWaitPanel">
            <div class="ds-panel-header">
                <span>? ${tr('Wait Queue', '????? ???????')}</span>
                <div class="ds-badge-count" id="dsWaitCount">${waitQueue.length}</div>
            </div>
            <div id="dsWaitList" style="flex:1;overflow-y:auto"></div>
            <div style="padding:10px;border-top:1px solid var(--border)">
                <button class="btn btn-sm w-full" onclick="window.dsRefreshWaitQueue()"
                    style="background:var(--primary-glow);color:var(--primary);border:1px solid var(--primary);font-si
ze:11px">
                    ? ${tr('Refresh', '????')}
                </button>
            </div>
        </div>

        <!-- ===== CENTER PANEL: Patient Chart ===== -->
        <div class="ds-center-panel" id="dsCenterPanel">
            <div id="dsCenterContent" style="flex:1;display:flex;align-items:center;justify-content:center;flex-di
rection:column;padding:40px;color:var(--text-dim);text-align:center">
                <div style="font-size:72px;margin-bottom:20px;opacity:0.3">?</div>
                <h3 style="font-size:18px;font-weight:700;margin-bottom:8px;color:var(--text-dim)">
                    ${tr('Select a patient from the wait queue', '???? ?????? ?? ????? ???????')}
                </h3>
                <p style="font-size:13px;opacity:0.7">
                    ${tr('The patient chart will appear here', '????? ?????? ??????? ?????? ???')}
                </p>
            </div>
        </div>
    </div>

```

```

<!-- ===== RIGHT PANEL: Orders ===== -->
<div class="ds-right-panel" id="dsOrdersPanel">
  <div class="ds-panel-header"><span>? ${tr('Orders', '??????')}</span></div>
  <div id="dsOrdersContent" style="flex:1;padding:16px;display:flex;align-items:center;justify-content:center;flex-direction:column;color:var(--text-dim)">
    <div style="font-size:48px;opacity:0.25;margin-bottom:12px">?</div>
    <p style="font-size:12px">${tr('Select a patient to write orders', '???? ?????? ?????? ??????')}</p>
  >
  </div>
</div>

`;

// Render wait list
window.dsRenderWaitList(waitQueue);

// Live timer refresh every 60s
window._DS.waitTimer = setInterval(window.dsRefreshWaitQueue, 60000);
}

/* ===== */
/* WAIT QUEUE RENDERING */
/* ===== */
window.dsRenderWaitList = function(patients) {
  const list = document.getElementById('dsWaitList');
  if (!list) return;
  // Queue rows carry the queue-row id in `id` and the real patient id in `patient_id`.
  // Resolve to the patient id and drop duplicate queue rows for the same patient.
  if (!Array.isArray(patients)) patients = [];
  const seenPids = new Set();
  patients = patients.filter(p => {
    const pid = p.patient_id || p.id;
    if (!pid || seenPids.has(pid)) return false;
    seenPids.add(pid);
    return true;
  });
  const countEl = document.getElementById('dsWaitCount');
  if (countEl) countEl.textContent = patients.length;

  if (!patients.length) {
    list.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)">
      <div style="font-size:36px;margin-bottom:8px">?</div>
      <p style="font-size:12px">${tr('No patients waiting', '?? ??? ???? ?? ???????')}</p>
    </div>`;
    return;
  }

  list.innerHTML = patients.map(p => {
    const pid = p.patient_id || p.id;
    const name = isArabic ? (p.name_ar || p.name_en || '-') : (p.name_en || p.name_ar || '-');
    const initial = (name || '?').charAt(0).toUpperCase();
    const waitMin = Math.round(parseFloat(p.wait_minutes || 0));
    const timerClass = waitMin > 60 ? 'critical' : waitMin > 30 ? 'warning' : 'normal';

```

```

const timerText = waitMin > 0 ? (waitMin + ' ' + tr('min', '?')) : tr('Now', '????');
const isActive = window._DS.selectedPatientId === pid;
const isWithDoc = p.status === 'With Doctor';
return `
  <div class="ds-wait-item ${isActive ? 'active' : ''} ${isWithDoc ? 'with-doctor' : ''}"
    onclick="window.dsSelectPatient(${safeId(pid)})"
    data-pid="${safeId(pid)}">
    <div class="ds-wait-avatar">${escapeHTML(initial)}</div>
    <div class="ds-wait-info">
      <div class="ds-wait-name">${escapeHTML(name)}</div>
      <div class="ds-wait-meta">
        ${p.file_number ? '?? ' + escapeHTML(String(p.file_number)) + ' . ' : ''}
        ${escapeHTML(p.department || p.chief_complaint || tr('General', '???'))}
        ${isWithDoc ? ' · <span style="color:#16a34a;font-weight:700">???? ' + tr('With Doctor', '?? ?????
?) + '</span>' : ''}
      </div>
    </div>
    <div class="ds-wait-timer ${timerClass}">${escapeHTML(timerText)}</div>
  </div>`;
}).join('');
};

window.dsRefreshWaitQueue = async function() {
  try {
    const q = await API.get('/api/doctor/wait-queue').catch(() =>
      API.get('/api/patients').then(p => p.filter(x => x.status === 'Waiting' || x.status === 'With Doctor'))
    );
    window.dsRenderWaitList(q);
  } catch (e) { /* silent */ }
};

/* ===== */
/* SELECT PATIENT ? Load Chart */
/* ===== */

window.dsSelectPatient = async function(patientId) {
  if (!patientId) return;
  window._DS.selectedPatientId = patientId;
  window._DS.rxItems = [];
  window._DS.activeTab = 'summary';

  // Highlight in wait list
  document.querySelectorAll('.ds-wait-item').forEach(el => {
    el.classList.toggle('active', parseInt(el.dataset.pid) === patientId);
  });

  // Show loading in center & right
  const center = document.getElementById('dsCenterContent') || document.getElementById('dsCenterPanel');
  const ordersContent = document.getElementById('dsOrdersContent');
  if (center) center.innerHTML = `<div style="padding:40px;text-align:center;color:var(--text-dim)"><div style
="font-size:32px;margin-bottom:12px">?</div><p>${tr('Loading chart...', '???? ????? ??????...')}</p></div>`;
  if (ordersContent) ordersContent.innerHTML = `<div style="padding:40px;text-align:center;color:var(--text-di
m)"><div style="font-size:24px;margin-bottom:12px">?</div></div>`;

  // Fetch all chart data in parallel

```

```

let chart = {}, vitals = [], problems = [], allergies = [], medications = [];
try {
  [chart, vitals, problems, allergies, medications] = await Promise.all([
    API.get(`/api/patients/${patientId}/chart`).catch(() => ({})),
    API.get(`/api/patients/${patientId}/vitals`).catch(() => []),
    API.get(`/api/patients/${patientId}/problems`).catch(() => []),
    API.get(`/api/patients/${patientId}/allergies`).catch(() => []),
    API.get(`/api/patients/${patientId}/medications`).catch(() => []),
  ]);
} catch (e) { console.error('Chart load error:', e); }

// API.get resolves with the error JSON (not a throw) on 4xx/5xx ? normalize shapes
const asArray = v => (Array.isArray(v) ? v : []);
vitals = asArray(vitals);
problems = asArray(problems);
allergies = asArray(allergies);
medications = asArray(medications);
if (!chart || typeof chart !== 'object' || chart.error) chart = {};

const patient = chart.patient || {};
window._DS.selectedPatientData = { patient, chart, vitals, problems, allergies, medications };

// Render Patient Header + Chart Tabs
window.dsRenderPatientChart(patient, chart, vitals, problems, allergies, medications);
// Render Orders Panel
window.dsRenderOrdersPanel(patient, chart.records || []);
};

/* ===== */
/* CENTER PANEL: Patient Chart */
/* ===== */
window.dsRenderPatientChart = function(patient, chart, vitals, problems, allergies, medications) {
  const centerPanel = document.getElementById('dsCenterPanel');
  if (!centerPanel) return;

  const records = chart.records || [];
  const name = isArabic ? (patient.name_ar || patient.name_en || '-') : (patient.name_en || patient.name_ar || '-');
  const initial = (name || '?').charAt(0).toUpperCase();
  const age = patient.age || patient.dob ? (patient.age || tr('N/A', '??? ???')) : '?';
  const gender = patient.gender || '?';
  const fileNo = patient.file_number || '?';

  // Build alert chips
  const allergyChips = allergies.length
    ? `>? ${allergies.length} ${tr('Allergy', '???????')}</span>`
    : '';
  const pendingOrders = (chart.orders || []).filter(o => o.status === 'Pending' || o.status === 'pending');
  const pendingChip = pendingOrders.length
    ? `>? ${pendingOrders.length} ${tr('Pending', '????')}</span>`
    : '';
  const activeProblems = problems.filter(p => p.status === 'active');
  const problemChip = activeProblems.length

```

```

    ? `<span class="ds-alert-chip critical">?? ${activeProblems.length} ${tr('Active Problem', '????? ?????')}<
/span>`
    : '';

// --- ???? BMI ???? BP ???? ???? ---
const _lv = {};
vitals.forEach(v => { if (!_lv[v.score_type]) _lv[v.score_type] = v.score_value; });
const _w = parseFloat(_lv['weight']||0), _h = parseFloat(_lv['height']||0);
const bmiVal = (_w>0&&_h>0) ? (_w/((_h/100)**2)).toFixed(1) : null;
const bmiLabel = bmiVal ? (parseFloat(bmiVal)<18.5?'? ??:parseFloat(bmiVal)<25?'? ??:parseFloat(bmiVal)
l)<30?'? ??:'? ??:'? ??:') : null;
const bpRaw = _lv['blood_pressure']||'';
const bpSys = parseInt((bpRaw.split('/')[0])||0);
const bpColor = bpSys>=160?'#dc2626':bpSys>=140?'#ea580c':bpSys>0?'#16a34a':'';
const triageLevel = chart.queueInfo?.triage_level || null;
const triageColors = {1:'#dc2626',2:'#ea580c',3:'#ca8a04',4:'#16a34a',5:'#6b7280'};
const triageLabels = {1:tr('Resuscitation','?????'),2:tr('Emergent','?????'),3:tr('Urgent','?????'),4:tr('Le
ss Urgent','??? ??????'),5:tr('Non-Urgent','?????')};

centerPanel.innerHTML = `
<!-- Patient Header Banner v3 -->
<div class="ds-patient-header">
  <div class="ds-patient-avatar">${escapeHTML(initial)}</div>
  <div class="ds-patient-info" style="flex:1">
    <div style="display:flex;align-items:center;gap:8px;flex-wrap:wrap">
      <div class="ds-patient-name">${escapeHTML(name)}</div>
      ${triageLevel ? `<span style="background:${triageColors[triageLevel]}|#6b7280';color:#fff;padding:
2px 8px;border-radius:20px;font-size:10px;font-weight:700">T${triageLevel} ${escapeHTML(triageLabels[triageLev
el]}|'|')</span>` : ''}
      ${patient.blood_type ? `<span style="background:#7c3aed22;color:#7c3aed;padding:2px 8px;border-radiu
s:20px;font-size:10px;font-weight:700">? ${escapeHTML(patient.blood_type)}</span>` : ''}
    </div>
    <div class="ds-patient-meta" style="flex-wrap:wrap">
      <span class="ds-patient-meta-item">?? ${escapeHTML(String(fileNo))}</span>
      <span class="ds-patient-meta-item">? ${tr('Age','????')}: ${escapeHTML(String(age))}</span>
      <span class="ds-patient-meta-item">${gender==='??'||gender==='Male'?':gender==='????'||gender==='
Female'?':'?'} ${escapeHTML(gender)}</span>
      ${patient.national_id ? `<span class="ds-patient-meta-item">? ${escapeHTML(patient.national_id)}</sp
an>` : ''}
      ${bmiVal ? `<span class="ds-patient-meta-item" title="BMI">? BMI: <strong>${bmiVal}</strong> ${bmiL
abel}</span>` : ''}
      ${bpRaw ? `<span class="ds-patient-meta-item" style="color:${bpColor}||'inherit';font-weight:700">?
BP: ${escapeHTML(bpRaw)}</span>` : ''}
      ${patient.insurance_company ? `<span class="ds-patient-meta-item">? ${escapeHTML(patient.insurance_c
ompany)}</span>` : ''}
      ${patient.insurance_number ? `<span class="ds-patient-meta-item">? ${escapeHTML(patient.insurance_nu
mber)}</span>` : ''}
      ${chart.queueInfo?.exam_room_name ? `<span class="ds-patient-meta-item">? ${escapeHTML(chart.queueIn
fo.exam_room_name)}</span>` : ''}
    </div>
    <div class="ds-patient-alerts">
      ${allergyChips}${problemChip}${pendingChip}
      ${patient.status === 'With Doctor' ? `<span class="ds-alert-chip info">???? ${tr('With Doctor','? ?
????')}</span>` : ''}

```



```

    </div>
</div>
<div class="ds-patient-actions">
    <button class="btn btn-sm btn-primary" onclick="window.dsMarkWithDoctor(${safeId(patient.id)})"
        style="font-size:11px;padding:6px 12px" id="btnMarkDoctor">
        ??? ${tr('Start Visit','?? ?????')}
    </button>
    <button class="btn btn-sm" onclick="window.dsSignEncounter()"
        style="font-size:11px;padding:6px 12px;background:#7c3aed;color:#fff;border-color:#7c3aed">
        ?? ${tr('Sign & Close','???? ?????')}
    </button>
</div>
</div>

<!-- Chart Tabs (v3: 10 tabs) -->
<div class="ds-chart-tabs" id="dsChartTabs">
    $[
        { id: 'summary',      icon: '?', en: 'Summary',      ar: '?????' },
        { id: 'history',      icon: '?', en: 'History',      ar: '?????', count: records.length },
        { id: 'problems',     icon: '??', en: 'Problems',     ar: '?????', count: problems.length },
        { id: 'medications',  icon: '?', en: 'Medications',  ar: '?????', count: medications.length },
        { id: 'allergies',    icon: '?', en: 'Allergies',    ar: '?????', count: allergies.length },
        { id: 'vitals',       icon: '??', en: 'Vitals',       ar: '?????', count: vitals.length },
        { id: 'notes',        icon: '?', en: 'Notes',        ar: '?????' },
        { id: 'orders',       icon: '?', en: 'Orders',       ar: '?????' },
        { id: 'consents',     icon: '?', en: 'Consents',     ar: '?????' },
        { id: 'results',      icon: '?', en: 'Results',      ar: '?????' },
        { id: 'history_ext',  icon: '????', en: 'History+',   ar: '?????+' },
        { id: 'pulmonology',  icon: '?', en: 'Pulmonology', ar: '????? ??????' },
        { id: 'gastro',       icon: '?', en: 'Gastro',       ar: '????? ??????' },
        { id: 'surgery',      icon: '?', en: 'Surgery',      ar: '????? ??????' },
        { id: 'obgyn',       icon: '?', en: 'OBGYN',       ar: '????? ??????' },
    ].map(t => `
        <div class="ds-tab ${t.id === window._DS.activeTab ? 'active' : ''}"
            onclick="window.dsSwitchTab('${t.id}')" data-tab="${t.id}">
            ${t.icon} ${tr(t.en, t.ar)}
            ${t.count ? `<span class="ds-tab-badge">${t.count}</span>` : ''}
        </div>
    `).join('')}
</div>

<!-- Tab Content -->
<div class="ds-tab-content" id="dsTabContent"></div>
`;

// Render active tab
window.dsSwitchTab(window._DS.activeTab);
};

/* ===== */
/* TAB SWITCHING */
/* ===== */
window.dsSwitchTab = function(tabId) {
    window._DS.activeTab = tabId;

```

```

// Update active tab styling
document.querySelectorAll('.ds-tab').forEach(el => {
  el.classList.toggle('active', el.dataset.tab === tabId);
});
const content = document.getElementById('dsTabContent');
if (!content) return;
const d = window._DS.selectedPatientData || {};

switch (tabId) {
  case 'summary':      content.innerHTML = window.dsTabSummary(d); break;
  case 'history':      content.innerHTML = window.dsTabHistory(d); break;
  case 'problems':     content.innerHTML = window.dsTabProblems(d); break;
  case 'medications':  content.innerHTML = window.dsTabMedications(d); break;
  case 'allergies':    content.innerHTML = window.dsTabAllergies(d); break;
  case 'vitals':       content.innerHTML = window.dsTabVitals(d); break;
  case 'notes':
    content.innerHTML = window.dsTabNotes(d);
    window.dsInitSoapNotes();
    break;
  case 'orders':       window.dsTabOrders(content); break;
  case 'consents':     window.dsTabConsents(content); break;
  case 'results':      window.dsTabResults(content); break;
  case 'history_ext':  window.dsTabHistoryExt(content); break;
  case 'pulmonology':  window.dsTabPulmonology(d); break;
  case 'gastro':       window.dsTabGastroenterology(d); break;
  case 'surgery':      window.dsTabSurgery(d); break;
  case 'obgyn':        window.dsTabObgyn(d); break;
  default: content.innerHTML = `<p>${tr('Coming soon', '?????')}</p>`;
}
};

/* ===== */
/*  TAB: SUMMARY                                */
/* ===== */

window.dsTabPulmonology = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const pulmRecords = records.filter(r => r.department === 'pulmonology');
  const lastPulm = pulmRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- PFT Analysis Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Pulmonary Function Test (PFT)', '????? ????? ????')}</strong>
        </div>
        <button class="btn btn-sm btn-primary" onclick="window.dsOpenPftModal()">+ ${tr('New PFT', '????? ?
        ???')}</button>
      </div>
      <div id="pulmPftChart" style="height:200px;background:#f9fafb;border-radius:8px;margin-bottom:12px;dis
        play:flex;align-items:center;justify-content:center;color:var(--text-dim);border:1px dashed var(--border)">
        <div style="text-align:center">
          <div style="font-size:24px">?</div>
        </div>
      </div>
    </div>
  `;
};

```

```

        <p style="font-size:11px">${tr('Flow-Volume Loop Visualization', '????? ?????? ?????? ??????')}</p>
    </div>
</div>
<div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
    <div class="ds-vital-card">
        <div class="ds-vital-label">FEV1/FVC</div>
        <div class="ds-vital-value">${escapeHTML(lastPulm.fev1_fvc_ratio || '?')}</div>
    </div>
    <div class="ds-vital-card">
        <div class="ds-vital-label">GOLD Stage</div>
        <div class="ds-vital-value" style="color:var(--primary);font-weight:700">${escapeHTML(lastPulm.gol
d_stage || '?')}</div>
    </div>
    <div class="ds-vital-card">
        <div class="ds-vital-label">DLCO</div>
        <div class="ds-vital-value">${escapeHTML(lastPulm.dlco_actual || '?')}</div>
    </div>
</div>

<!-- Sleep Study Section -->
<div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
        <strong style="font-size:13px">? ${tr('Sleep Study (Polysomnography)', '????? ??????')}</strong>
        <button class="btn btn-sm btn-primary" onclick="window.dsOpenSleepModal()">+ ${tr('New Study', '????
? ??????')}</button>
    </div>
    <div class="ds-problem-row" style="margin-bottom:8px">
        <div class="ds-med-icon">?</div>
        <div class="ds-med-info">
            <div class="ds-med-name">${tr('AHI Index:', '???? AHI:')} ${escapeHTML(lastPulm.ahi_index || '?')}
        </div>
            <div class="ds-med-detail">${tr('Severity:', '?????:')} ${escapeHTML(lastPulm.sleep_severity || '?
')}</div>
        </div>
        <div class="ds-med-status active">${escapeHTML(lastPulm.sleep_diagnosis || '?')}</div>
    </div>
</div>

<!-- Cor Pulmonale Bridge (S-MODE) -->
<div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
    <div class="ds-cds-alert-icon">?</div>
    <div class="ds-cds-alert-body">
        <div class="ds-cds-alert-title">${tr('Cardio-Pulmonary Bridge', '??? ?????? ??????')}</div>
        <div class="ds-cds-alert-desc">${tr('Check for Cor Pulmonale risk based on current FEV1 and Right Ve
ntricle Pressure', '???? ?? ?????? ??? ?????? ?????? ?????? ??? FEV1 ????? ?????? ??????')}</div>
        <button class=" la btn btn-sm" onclick="window.dsCheckCorPulmonale()" style="margin-top:8px;backgrou
nd:#ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
            ? ${tr('Run Risk Analysis', '????? ??????')}</button>
    </div>
</div>
</div>
';

```

```

};

window.dsOpenPftModal = async function() {
  const pid = window._DS.selectedPatientId;
  const encounterId = window._DS.activeEncounterId;
  if (!pid) return;

  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = `
    <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:450px;box-shadow:0 10px 25px rgba(0,0,0,0.2)">
      <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('PFT Results', '???? ???? ???? ????')}</h3>
      <div class="ds-soap-grid" style="grid-template-columns:1fr 1fr;gap:12px">
        <div class="form-group">
          <label style="font-size:11px;font-weight:700">FEV1 Actual</label>
          <input class="form-input" id="pft_fev1_act" type="number" step="0.01">
        </div>
        <div class="form-group">
          <label style="font-size:11px;font-weight:700">FEV1 Predicted</label>
          <input class="form-input" id="pft_fev1_pred" type="number" step="0.01">
        </div>
        <div class="form-group">
          <label style="font-size:11px;font-weight:700">FVC Actual</label>
          <input class="form-input" id="pft_fvc_act" type="number" step="0.01">
        </div>
        <div class="form-group">
          <label style="font-size:11px;font-weight:700">FVC Predicted</label>
          <input class="form-input" id="pft_fvc_pred" type="number" step="0.01">
        </div>
      </div>
      <div class="form-group mt-12">
        <label style="font-size:11px;font-weight:700">${tr('Interpretation', '??????')}</label>
        <textarea class="form-input" id="pft_interp" rows="2"></textarea>
      </div>
      <div style="display:flex;gap:10px;margin-top:20px">
        <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '????')}</button>
        <button class="btn btn-primary" style="flex:1" onclick="window.dsSavePft()">? ${tr('Save & Analyze', '??? ?????')}</button>
      </div>
    </div>
  `;
  modal.classList.add('fixed');
  document.body.appendChild(modal);
};

window.dsSavePft = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    encounter_id: window._DS.activeEncounterId,
    fev1_actual: document.getElementById('pft_fev1_act').value,
    fev1_predicted: document.getElementById('pft_fev1_pred').value,
  }

```

```

    fvc_actual: document.getElementById('pft_fvc_act').value,
    fvc_predicted: document.getElementById('pft_fvc_pred').value,
    interpretation: document.getElementById('pft_interp').value,
  };
  try {
    const res = await API.post('/api/pulmonology/pft', data);
    showToast(tr('PFT Saved & Analyzed!', '?? ??? ?????? PFT!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

window.dsCheckCorPulmonale = async function() {
  const pid = window._DS.selectedPatientId;
  const mockData = {
    patient_id: pid,
    fev1: 25,
    oxygenSat: 88,
    rightVentriclePressure: 30,
    ecgFindings: 'Right Axis Deviation'
  };
  try {
    const res = await API.post('/api/pulmonology/cor-pulmonale-check', mockData);
    const risk = res.riskAssessment;
    const color = risk.risk === 'HIGH' ? '#dc2626' : '#16a34a';
    showToast(`Risk: ${risk.risk} - ${risk.finding}`, risk.risk === 'HIGH' ? 'error' : 'success');
  } catch (e) { showToast(e.message, 'error'); }
};

/* ===== */
/* TAB: GASTROENTEROLOGY (S-MODE Gold Standard) */
/* ===== */
window.dsTabGastroenterology = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const gastroRecords = records.filter(r => r.department === 'gastroenterology');
  const lastGastro = gastroRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- Endoscopy Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Endoscopy Report', '????? ???????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenGastroModal()">+ ${tr('New Procedure',
'????? ?????')}</button>
        </div>
        <div class="ds-problem-row" style="margin-bottom:8px">
          <div class="ds-med-icon">?</div>
          <div class="ds-med-info">
            <div class="ds-med-name">${escapeHTML(lastGastro.procedure_type || tr('No procedure recorded', '??
???? ?????? ?????'))}</div>
            <div class="ds-med-detail">${tr('Findings:', '???????')} ${escapeHTML(lastGastro.findings || '?')}
          </div>
        </div>
      </div>
    </div>
  `
};

```

```

        <div class="ds-med-status active">${escapeHTML(lastGastro.recommendation || '?')}</div>
    </div>
</div>

<!-- Hepatic Markers Section -->
<div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
        <strong style="font-size:13px">? ${tr('Hepatic Markers', '????? ?????')}</strong>
        <button class="btn btn-sm btn-primary" onclick="window.dsOpenHepaticModal()">+ ${tr('Add Markers', '
????? ??????')}</button>
    </div>
    <div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
        <div class="ds-vital-card">
            <div class="ds-vital-label">Bilirubin</div>
            <div class="ds-vital-value">${escapeHTML(lastGastro.bilirubin_total || '?')}</div>
        </div>
        <div class="ds-vital-card">
            <div class="ds-vital-label">ALT/AST</div>
            <div class="ds-vital-value">${escapeHTML(lastGastro.alt_level || '?')} / ${escapeHTML(lastGastro.a
st_level || '?')}</div>
        </div>
        <div class="ds-vital-card">
            <div class="ds-vital-label">Albumin</div>
            <div class="ds-vital-value">${escapeHTML(lastGastro.albumin || '?')}</div>
        </div>
    </div>
</div>

<!-- Child-Pugh Calculator (S-MODE) -->
<div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
    <div class="ds-cds-alert-icon">??</div>
    <div class="ds-cds-alert-body">
        <div class="ds-cds-alert-title">${tr('Child-Pugh Scoring', '????? Child-Pugh')}</div>
        <div class="ds-cds-alert-desc">${tr('Assess liver cirrhosis severity and prognosis', '????? ??? ????
????? ??????? ??????')}</div>
        <button class="btn btn-sm" onclick="window.dsCalculateChildPugh()" style="margin-top:8px;background:
#ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
            ? ${tr('Open Calculator', '??? ??????')}
        </button>
    </div>
</div>
</div>
`;
};

```

```

window.dsOpenGastroModal = async function() {
    const pid = window._DS.selectedPatientId;
    if (!pid) return;

    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
    modal.innerHTML = `
        <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:450px;box-shadow:0 10px 25px r

```

```

gba(0,0,0,0.2)">
  <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('Endoscopy Report', '???? ?')}</h3>
  <div class="form-group mb-8">
    <label style="font-size:11px;font-weight:700">Procedure Type</label>
    <select class="form-input" id="gastro_proc_type">
      <option value="Gastroscopy">Gastroscopy</option>
      <option value="Colonoscopy">Colonoscopy</option>
      <option value="ERCP">ERCP</option>
    </select>
  </div>
  <div class="form-group mb-8">
    <label style="font-size:11px;font-weight:700">Findings</label>
    <textarea class="form-input" id="gastro_findings" rows="3"></textarea>
  </div>
  <div style="display:flex;gap:10px;margin-top:20px">
    <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '????')}</b
utton>
    <button class="btn btn-primary" style="flex:1" onclick="window.dsSaveGastro()">? ${tr('Save Report', '
??? ?????')}</button>
  </div>
</div>
`;
modal.classList.add('fixed');
document.body.appendChild(modal);
};

window.dsSaveGastro = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    procedure_type: document.getElementById('gastro_proc_type').value,
    findings: document.getElementById('gastro_findings').value,
  };
  try {
    await API.post('/api/gastro/endoscopy', data);
    showToast(tr('Report Saved!', '?? ??? ??????!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

window.dsCalculateChildPugh = async function() {
  const bilirubin = prompt('Enter Bilirubin (mg/dL):');
  const albumin = prompt('Enter Albumin (g/dL):');
  const ascites = prompt('Ascites (none/mild/severe):');
  const encephalopathy = prompt('Encephalopathy (none/grade1-2/severe):P');
  const inr = prompt('INR:');

  try {
    const res = await API.post('/api/gastro/child-pugh', { bilirubin, albumin, ascites, encephalopathy, inr })
    ;
    const score = res.childPugh;
    showToast(`Child-Pugh Class: ${score.class} (${score.description}) - Score: ${score.score}`, 'success');
  } catch (e) { showToast(e.message, 'error'); }
};

```

```

return `
  <!-- Quick Vitals Summary -->
  ${Object.keys(latestVitals).length ? `
    <div class="ds-vital-grid" style="margin-bottom:16px">
      ${['blood_pressure', 'temperature', 'pulse', 'spo2', 'weight', 'glucose'].filter(k => latestVitals[k]).map(k
=> {
        const v = latestVitals[k];
        const icons = { blood_pressure: '?', temperature: '??', pulse: '?', spo2: '?', weight: '??', glucose:
'? ' };
        const units = { blood_pressure: 'mmHg', temperature: '°C', pulse: 'bpm', spo2: '%', weight: 'kg', gluc
ose: 'mg/dL' };
        const labels = { blood_pressure: tr('BP', '??? ???'), temperature: tr('Temp', '???????'), pulse: tr('Pu
lse', '?????'), spo2: tr('SpO2', '?????'), weight: tr('Weight', '?????'), glucose: tr('Glucose', '????????') };
        const val = v.score_value || '?';
        const abnormal = (k==='temperature' && parseFloat(val) > 38.5) ||
          (k==='pulse' && (parseFloat(val) < 60 || parseFloat(val) > 100)) ||
          (k==='spo2' && parseFloat(val) < 95) ||
          (k==='glucose' && parseFloat(val) > 200);
        return `<div class="ds-vital-card ${abnormal ? 'abnormal' : ''}">
          <div style="font-size:18px">${icons[k]||'?'}</div>
          <div class="ds-vital-value">${escapeHTML(String(val))}</div>
          <div class="ds-vital-unit">${units[k]||''}</div>
          <div class="ds-vital-label">${labels[k]||k}</div>
        </div>`;
      }).join('')}
    </div>
  ` : ''}

  <!-- Last Visit -->
  ${lastRecord ? `
    <div class="card mb-12" style="border-radius:12px;padding:14px 16px;border:1px solid var(--border)">
      <div style="font-weight:700;font-size:13px;color:var(--primary);margin-bottom:8px">? ${tr('Last Encounte
r', '??? ?????')}
      <span style="font-size:11px;font-weight:400;color:var(--text-dim);margin-inline-start:8px">${escapeHTM
L(lastRecord.created_at?.split('T')[0] || '')}</span>
    </div>
    <div style="font-size:13px;color:var(--on-surface)"><strong>${tr('Diagnosis:', '???????')}</strong> ${e
scapeHTML(lastRecord.diagnosis || lastRecord.treatment || '-')}</div>
    ${lastRecord.notes ? `<div style="font-size:12px;color:var(--text-dim);margin-top:4px">${escapeHTML(last
Record.notes)}</div>` : ''}
  </div>
  ` : `<div class="card mb-12" style="padding:14px;border-radius:12px;text-align:center;color:var(--text-dim
)"><span style="font-size:24px">?</span><p style="margin-top:8px;font-size:12px">${tr('No previous encounters'
, '?? ??? ???? ????')}</p></div>`}

  <!-- Active Problems -->
  <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:uppercas
e;letter-spacing:0.5px">?? ${tr('Active Problems', '???????')}</div>
  ${activeProblems.length ? activeProblems.map(p => `
    <div class="ds-problem-row">
      <div class="ds-problem-dot active"></div>

```



```

    <div class="ds-problem-name">${escapeHTML(p.problem_name || '-')}</div>
    <div class="ds-problem-icd">${escapeHTML(p.icd_code || '')}</div>
  </div>
  `).join('') : `<div style="color:var(--text-dim);font-size:12px;padding:8px;margin-bottom:12px">? ${tr('None recorded', '?? ???')}</div>`

  <!-- Current Medications -->
  <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-top:16px;margin-bottom:8px;text-transform:uppercase;letter-spacing:0.5px">? ${tr('Current Medications', '?????? ?????')}</div>
  ${activeMeds.length ? activeMeds.map(m => `
    <div class="ds-med-row">
      <div class="ds-med-icon">?</div>
      <div class="ds-med-info">
        <div class="ds-med-name">${escapeHTML(m.drug_name || m.description || '-')}</div>
        <div class="ds-med-detail">${escapeHTML(m.quantity || '')} ${escapeHTML(m.notes || '')}</div>
      </div>
      <div class="ds-med-status active">${tr('Active', '????')}</div>
    </div>
  `).join('') : `<div style="color:var(--text-dim);font-size:12px;padding:8px">? ${tr('None recorded', '?? ? ???')}</div>`}

  <!-- Allergies Warning -->
  ${allergies.length ? `
    <div style="margin-top:16px">
      <div style="font-weight:700;font-size:12px;color:#b91c1c;margin-bottom:8px;text-transform:uppercase;letter-spacing:0.5px">? ${tr('Allergies', '????????')}</div>
      ${allergies.slice(0,3).map(a => `
        <div class="ds-cds-alert warning">
          <div class="ds-cds-alert-icon">?</div>
          <div class="ds-cds-alert-body">
            <div class="ds-cds-alert-title">${escapeHTML(a.allergen || '-')}</div>
            <div class="ds-cds-alert-desc">${escapeHTML(a.reaction || '')} ${a.severity ? '. ' + escapeHTML(a.severity) : ''}</div>
          </div>
        </div>
      `).join('')}
    </div>` : ''}
  `;
} // End of dsTabAllergies

/* ===== */
/* TAB: PULMONOLOGY (S-MODE Gold Standard) */
/* ===== */
window.dsTabPulmonology = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const pulmRecords = records.filter(r => r.department === 'pulmonology');
  const lastPulm = pulmRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- PFT Analysis Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Pulmonary Function Test (PFT)', '????? ????? ?????')}</strong>

```

```

g>
    <button class="btn btn-sm btn-primary" onclick="window.dsOpenPftModal()">+ ${tr('New PFT', '????? ?
???')}</button>
  </div>
  <div id="pulmPftChart" style="height:200px;background:#f9fafb;border-radius:8px;margin-bottom:12px;dis
play:flex;align-items:center;justify-content:center;color:var(--text-dim);border:1px dashed var(--border)">
    <div style="text-align:center">
      <div style="font-size:24px">?</div>
      <p style="font-size:11px">${tr('Flow-Volume Loop Visualization', '???? ???? ?????? ??????')}</p>
    </div>
  </div>
  <div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
    <div class="ds-vital-card">
      <div class="ds-vital-label">FEV1/FVC</div>
      <div class="ds-vital-value">${escapeHTML(lastPulm.fev1_fvc_ratio || '?')}</div>
    </div>
    <div class="ds-vital-card">
      <div class="ds-vital-label">GOLD Stage</div>
      <div class="ds-vital-value" style="color:var(--primary);font-weight:700">${escapeHTML(lastPulm.gol
d_stage || '?')}</div>
    </div>
    <div class="ds-vital-card">
      <div class="ds-vital-label">DLCO</div>
      <div class="ds-vital-value">${escapeHTML(lastPulm.dlco_actual || '?')}</div>
    </div>
  </div>

  <!-- Sleep Study Section -->
  <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
      <strong style="font-size:13px">? ${tr('Sleep Study (Polysomnography)', '???? ?????')}</strong>
      <button class="btn btn-sm btn-primary" onclick="window.dsOpenSleepModal()">+ ${tr('New Study', '????
? ?????')}</button>
    </div>
    <div class="ds-problem-row" style="margin-bottom:8px">
      <div class="ds-med-icon">?</div>
      <div class="ds-med-info">
        <div class="ds-med-name">${tr('AHI Index:', '???? AHI:')} ${escapeHTML(lastPulm.ahi_index || '?')}
      </div>
        <div class="ds-med-detail">${tr('Severity:', '?????:')} ${escapeHTML(lastPulm.sleep_severity || '?
')}</div>
      </div>
      <div class="ds-med-status active">${escapeHTML(lastPulm.sleep_diagnosis || '?')}</div>
    </div>
  </div>

  <!-- Cor Pulmonale Bridge (S-MODE) -->
  <div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
    <div class="ds-cds-alert-icon">?</div>
    <div class="ds-cds-alert-body">
      <div class="ds-cds-alert-title">${tr('Cardio-Pulmonary Bridge', '??? ????? ?????')}</div>
      <div class="ds-cds-alert-desc">${tr('Check for Cor Pulmonale risk based on current FEV1 and Right Ve
ntricle Pressure', '???? ?? ????? ?? ????? ?????? ?????? ??? FEV1 ??? ?????? ?????')}</div>
    </div>
  </div>

```

```

        <button class=" la btn btn-sm" onclick="window.dsCheckCorPulmonale()" style="margin-top:8px;backgroundrou
nd:#ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
            ? ${tr('Run Risk Analysis', '????? ??????')}
        </button>
    </div>
</div>
</div>
`;
};

window.dsOpenPftModal = async function() {
    const pid = window._DS.selectedPatientId;
    const encounterId = window._DS.activeEncounterId;
    if (!pid) return;

    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
    modal.innerHTML = `
        <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:450px;box-shadow:0 10px 25px r
gba(0,0,0,0.2)">
            <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('PFT Results', '????? ?????? ?????? ??????')}</h3>
            <div class="ds-soap-grid" style="grid-template-columns:1fr 1fr;gap:12px">
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FEV1 Actual</label>
                    <input class="form-input" id="pft_fev1_act" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FEV1 Predicted</label>
                    <input class="form-input" id="pft_fev1_pred" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FVC Actual</label>
                    <input class="form-input" id="pft_fvc_act" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FVC Predicted</label>
                    <input class="form-input" id="pft_fvc_pred" type="number" step="0.01">
                </div>
            </div>
            <div class="form-group mt-12">
                <label style="font-size:11px;font-weight:700">${tr('Interpretation', '???????')}</label>
                <textarea class="form-input" id="pft_interp" rows="2"></textarea>
            </div>
            <div style="display:flex;gap:10px;margin-top:20px">
                <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '?????')}</b
utton>
                <button class="btn btn-primary" style="flex:1" onclick="window.dsSavePft()">? ${tr('Save & Analyze', '
??? ??????')}</button>
            </div>
        </div>
    `;
    modal.classList.add('fixed');
    document.body.appendChild(modal);

```

```

};

window.dsSavePft = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    encounter_id: window._DS.activeEncounterId,
    fev1_actual: document.getElementById('pft_fev1_act').value,
    fev1_predicted: document.getElementById('pft_fev1_pred').value,
    fvc_actual: document.getElementById('pft_fvc_act').value,
    fvc_predicted: document.getElementById('pft_fvc_pred').value,
    interpretation: document.getElementById('pft_interp').value,
  };
  try {
    const res = await API.post('/api/pulmonology/pft', data);
    showToast(tr('PFT Saved & Analyzed!', '?? ?? ???? PFT!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

window.dsCheckCorPulmonale = async function() {
  const pid = window._DS.selectedPatientId;
  const mockData = {
    patient_id: pid,
    fev1: 25,
    oxygenSat: 88,
    rightVentriclePressure: 30,
    ecgFindings: 'Right Axis Deviation'
  };
  try {
    const res = await API.post('/api/pulmonology/cor-pulmonale-check', mockData);
    const risk = res.riskAssessment;
    const color = risk.risk === 'HIGH' ? '#dc2626' : '#16a34a';
    showToast(`Risk: ${risk.risk} - ${risk.finding}`, risk.risk === 'HIGH' ? 'error' : 'success');
  } catch (e) { showToast(e.message, 'error'); }
};

/* ===== */
/* TAB: GASTROENTEROLOGY (S-MODE Gold Standard) */
/* ===== */
window.dsTabGastroenterology = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const gastroRecords = records.filter(r => r.department === 'gastroenterology');
  const lastGastro = gastroRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- Endoscopy Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Endoscopy Report', '???? ????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenGastroModal()">+ ${tr('New Procedure',
'???? ????')}</button>
        </div>

```

```

<div class="ds-problem-row" style="margin-bottom:8px">
  <div class="ds-med-icon">?</div>
  <div class="ds-med-info">
    <div class="ds-med-name">${escapeHTML(lastGastro.procedure_type || tr('No procedure recorded', '??
???? ????? ?????'))}</div>
    <div class="ds-med-detail">${tr('Findings:', '???????')} ${escapeHTML(lastGastro.findings || '?')}
  </div>
  <div class="ds-med-status active">${escapeHTML(lastGastro.recommendation || '?')}</div>
</div>

<!-- Hepatic Markers Section -->
<div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
  <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
    <strong style="font-size:13px">? ${tr('Hepatic Markers', '?????? ?????')}</strong>
    <button class="btn btn-sm btn-primary" onclick="window.dsOpenHepaticModal()">+ ${tr('Add Markers', '
????? ??????')}</button>
  </div>
  <div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
    <div class="ds-vital-card">
      <div class="ds-vital-label">Bilirubin</div>
      <div class="ds-vital-value">${escapeHTML(lastGastro.bilirubin_total || '?')}</div>
    </div>
    <div class="ds-vital-card">
      <div class="ds-vital-label">ALT/AST</div>
      <div class="ds-vital-value">${escapeHTML(lastGastro.alt_level || '?')} / ${escapeHTML(lastGastro.a
st_level || '?')}</div>
    </div>
    <div class="ds-vital-card">
      <div class="ds-vital-label">Albumin</div>
      <div class="ds-vital-value">${escapeHTML(lastGastro.albumin || '?')}</div>
    </div>
  </div>
</div>

<!-- Child-Pugh Calculator (S-MODE) -->
<div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
  <div class="ds-cds-alert-icon">??</div>
  <div class="ds-cds-alert-body">
    <div class="ds-cds-alert-title">${tr('Child-Pugh Scoring', '????? Child-Pugh')}</div>
    <div class="ds-cds-alert-desc">${tr('Assess liver cirrhosis severity and prognosis', '????? ??? ???
????? ??????? ??????')}</div>
    <button class="btn btn-sm" onclick="window.dsCalculateChildPugh()" style="margin-top:8px;background:
#ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
      ? ${tr('Open Calculator', '??? ??????')}
    </button>
  </div>
</div>
</div>
';
};

```

```

window.dsOpenGastroModal = async function() {

```

```

const pid = window._DS.selectedPatientId;
if (!pid) return;

const modal = document.createElement('div');
modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
modal.innerHTML = `
  <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:450px;box-shadow:0 10px 25px rgba(0,0,0,0.2)">
    <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('Endoscopy Report', '???? ?????')}</h3>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">Procedure Type</label>
      <select class="form-input" id="gastro_proc_type">
        <option value="Gastroscopy">Gastroscopy</option>
        <option value="Colonoscopy">Colonoscopy</option>
        <option value="ERCP">ERCP</option>
      </select>
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">Findings</label>
      <textarea class="form-input" id="gastro_findings" rows="3"></textarea>
    </div>
    <div style="display:flex;gap:10px;margin-top:20px">
      <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '????')}</button>
      <button class="btn btn-primary" style="flex:1" onclick="window.dsSaveGastro()">? ${tr('Save Report', '???? ?????')}</button>
    </div>
  </div>
`;
modal.classList.add('fixed');
document.body.appendChild(modal);
};

```

```

window.dsSaveGastro = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    procedure_type: document.getElementById('gastro_proc_type').value,
    findings: document.getElementById('gastro_findings').value,
  };
  try {
    await API.post('/api/gastro/endoscopy', data);
    showToast(tr('Report Saved!', '?? ??? ??????!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

```

```

window.dsCalculateChildPugh = async function() {
  const bilirubin = prompt('Enter Bilirubin (mg/dL):');
  const albumin = prompt('Enter Albumin (g/dL):');
  const ascites = prompt('Ascites (none/mild/severe):');
  const encephalopathy = prompt('Encephalopathy (none/grade1-2/severe):');
  const inr = prompt('INR:');

```

```

try {
  const res = await API.post('/api/gastro/child-pugh', { bilirubin, albumin, ascites, encephalopathy, inr })
;
  const score = res.childPugh;
  showToast(`Child-Pugh Class: ${score.class} (${score.description}) - Score: ${score.score}`, 'success');
} catch (e) { showToast(e.message, 'error'); }
};

/* ===== */
/* TAB: SURGERY (S-MODE Gold Standard) */
/* ===== */
window.dsTabSurgery = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const surgRecords = records.filter(r => r.department === 'surgery');
  const lastSurg = surgRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- Surgical Encounter Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Surgical Encounter', '?????? ???????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenSurgeryModal()">+ ${tr('New Procedure', '????? ?????')}</button>
        </div>
        <div class="ds-problem-row" style="margin-bottom:8px">
          <div class="ds-med-icon">?</div>
          <div class="ds-med-info">
            <div class="ds-med-name">${escapeHTML(lastSurg.surgical_approach || tr('No procedure recorded', '?
? ???? ?????? ?????'))}</div>
            <div class="ds-med-detail">${tr('Diagnosis:', '???????')} ${escapeHTML(lastSurg.post_op_diagnosis
|| '?')}</div>
          </div>
          <div class="ds-med-status active">${escapeHTML(lastSurg.status || '?')}</div>
        </div>
      </div>

      <!-- Wound Tracking Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Wound & Drain Tracking', '???? ?????? ???????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenWoundModal()">+ ${tr('Log Check', '????
? ???')}</button>
        </div>
        <div class="ds-vital-grid" style="grid-template-columns:repeat(2,1fr)">
          <div class="ds-vital-card">
            <div class="ds-vital-label">Wound Status</div>
            <div class="ds-vital-value">${escapeHTML(lastSurg.wound_status || '?')}</div>
          </div>
          <div class="ds-vital-card">
            <div class="ds-vital-label">Drainage (ml)</div>
            <div class="ds-vital-value">${escapeHTML(lastSurg.drainage_amount_ml || '?')}</div>
          </div>
        </div>
      </div>
    </div>
  `
}

```

```

    </div>
</div>

<!-- Surgical Risk Assessment (S-MODE) -->
<div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
    <div class="ds-cds-alert-icon">??</div>
    <div class="ds-cds-alert-body">
        <div class="ds-cds-alert-title">${tr('Surgical Risk Assessment', '????? ??????? ?????????')}</div>
        <div class="ds-cds-alert-desc">${tr('Calculate ASA score and perioperative risk', '???? ???? ASA ???
??? ?? ??? ???????')}</div>
        <button class="btn btn-sm" onclick="window.dsCalculateSurgRisk()" style="margin-top:8px;background:#
ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
            ? ${tr('Run Risk Analysis', '????? ???????')}
        </button>
    </div>
</div>
</div>
`;
};

window.dsOpenSurgeryModal = async function() {
    const pid = window._DS.selectedPatientId;
    if (!pid) return;

    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
    modal.innerHTML = `
        <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:500px;box-shadow:0 10px 25px r
gba(0,0,0,0.2)">
            <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('Surgical Procedure', '??????? ?????????')}</h3>
            <div class="ds-soap-grid" style="grid-template-columns:1fr 1fr;gap:12px">
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">Surgical Approach</label>
                    <select class="form-input" id="surg_approach">
                        <option value="Open">Open</option>
                        <option value="Laparoscopic">Laparoscopic</option>
                        <option value="Robotic">Robotic</option>
                        <option value="Combined">Combined</option>
                    </select>
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">Anesthesia Type</label>
                    <input class="form-input" id="surg_anesthesia" placeholder="General, Spinal, etc.">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">Op Time (min)</label>
                    <input class="form-input" id="surg_time" type="number">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">Blood Loss (ml)</label>
                    <input class="form-input" id="surg_blood" type="number">
                </div>
            </div>
        </div>
    `;
};

```



```

<div class="form-group mt-12">
  <label style="font-size:11px;font-weight:700">${tr('Post-Op Diagnosis', '????? ?? ??? ???????')}</label>
  <textarea class="form-input" id="surg_post_diag" rows="2"></textarea>
</div>
<div style="display:flex;gap:10px;margin-top:20px">
  <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '????')}</button>
  <button class="btn btn-primary" style="flex:1" onclick="window.dsSaveSurgery()">? ${tr('Save Procedure', '??? ???????')}</button>
</div>
</div>
`;
modal.classList.add('fixed');
document.body.appendChild(modal);
};

window.dsSaveSurgery = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    patient_id: pid,
    surgeon_id: window._DS.currentUser?.id,
    surgical_approach: document.getElementById('surg_approach').value,
    anesthesia_type: document.getElementById('surg_anesthesia').value,
    operation_time_min: document.getElementById('surg_time').value,
    blood_loss_ml: document.getElementById('surg_blood').value,
    post_op_diagnosis: document.getElementById('surg_post_diag').value,
  };
  try {
    await API.post('/api/surgery/encounter', data);
    showToast(tr('Surgical Record Saved!', '?? ??? ????? ??????!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

window.dsCalculateSurgRisk = async function() {
  const asa = prompt('Enter ASA Score (1-5):');
  const age = prompt('Enter Patient Age:');
  const comorbidities = prompt('Number of Comorbidities:');
  const emergency = prompt('Is it an Emergency? (Yes/No):');

  try {
    const res = await API.post('/api/surgery/risk-assessment', {
      asaScore: parseInt(asa),
      age: parseInt(age),
      comorbiditiesCount: parseInt(comorbidities),
      emergencyStatus: emergency === 'Yes' ? 'Emergency' : 'Elective'
    });
    const risk = res.riskAssessment;
    showToast(`Surgical Risk: ${risk.category} - ${risk.recommendation}`, 'info');
  } catch (e) { showToast(e.message, 'error'); }
};

```

```

/* ===== */
/* TAB: HISTORY (Timeline) */
/* ===== */
window.dsTabHistory = function({ chart = {} }) {

  const records = chart.records || [];
  const orders = chart.orders || [];
  const invoices = chart.invoices || [];

  // Merge events
  const events = [
    ...records.map(r => ({ date: r.created_at, type: 'visit', title: r.diagnosis || r.treatment || tr('Encounter', '????'), detail: r.notes || '' })),
    ...orders.filter(o => o.type && o.type.includes('lab')).map(o => ({ date: o.created_at, type: 'lab', title: o.description || tr('Lab Order', '?? ????'), detail: o.status || '' })),
    ...orders.filter(o => o.type && (o.type.includes('rad') || o.type.includes('xray'))).map(o => ({ date: o.created_at, type: 'radiology', title: o.description || tr('Radiology', '????'), detail: o.status || '' })),
    ...orders.filter(o => o.type && (o.type.includes('med') || o.type.includes('rx'))).map(o => ({ date: o.created_at, type: 'pharmacy', title: o.description || tr('Medication', '????'), detail: o.status || '' })),
  ].sort((a, b) => new Date(b.date) - new Date(a.date));

  if (!events.length) return `<div style="text-align:center;padding:32px;color:var(--text-dim)">
    <div style="font-size:48px;margin-bottom:12px;opacity:0.3">?</div>
    <p>${tr('No history found', '?? ???? ???? ????')}</p>
  </div>`;

  return `<div class="ds-timeline">
    ${events.map(ev => `
      <div class="ds-timeline-item">
        <div class="ds-timeline-dot ${ev.type}"></div>
        <div class="ds-timeline-card">
          <div class="ds-timeline-date">${escapeHTML(ev.date?.split('T')[0] || '?')}</div>
          <div class="ds-timeline-title">${escapeHTML(ev.title)}</div>
          ${ev.detail ? `<div class="ds-timeline-detail">${escapeHTML(ev.detail)}</div>` : ''}
        </div>
      </div>
    `).join('')}
  </div>`;
};

/* ===== */
/* TAB: PROBLEMS */
/* ===== */
window.dsTabProblems = function({ problems = [] }) {

  if (!problems.length) return `<div style="text-align:center;padding:32px;color:var(--text-dim)">
    <div style="font-size:48px;margin-bottom:12px;opacity:0.3">??</div>
    <p>${tr('No problems recorded', '?? ???? ???? ????')}</p>
    <button class="btn btn-sm btn-primary" style="margin-top:12px" onclick="window.dsAddProblem()">
      + ${tr('Add Problem', '???? ????')}
    </button>
  </div>`;

  const statusDot = { active: 'active', controlled: 'controlled', resolved: 'resolved' };
  const statusLabel = { active: { en: 'Active', ar: '???' }, controlled: { en: 'Controlled', ar: '???? ??????' } };

```

```

' }, resolved: { en: 'Resolved', ar: '?????' } }];

return `
<div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
  <strong style="font-size:13px">${tr('Problem List', '????? ????????')} (${problems.length})</strong>
  <button class="btn btn-sm btn-primary" onclick="window.dsAddProblem()">+ ${tr('Add', '?????')}</button>
</div>
${problems.map(p => {
  const dot = statusDot[p.status] || 'active';
  const lbl = statusLabel[p.status] || { en: p.status, ar: p.status };
  return `<div class="ds-problem-row">
    <div class="ds-problem-dot ${dot}" title="${tr(lbl.en, lbl.ar)}"></div>
    <div>
      <div class="ds-problem-name">${escapeHTML(p.problem_name || '-')}</div>
      <div style="font-size:11px;color:var(--text-dim)">${tr('Onset:', '???????')} ${escapeHTML(p.onset_date || '?')}</div>
    </div>
    <div class="ds-problem-icd">${escapeHTML(p.icd_code || '')}</div>
    <span class="badge badge-${dot === 'active' ? 'danger' : dot === 'controlled' ? 'warning' : 'success'}" style="font-size:10px">${tr(lbl.en, lbl.ar)}</span>
  </div>`;
}).join('')}
`;
};

/* ===== */
/* TAB: MEDICATIONS */
/* ===== */
window.dsTabMedications = function({ medications = [] }) {
  if (!medications.length) return `<div style="text-align:center;padding:32px;color:var(--text-dim)">
    <div style="font-size:48px;margin-bottom:12px;opacity:0.3">?</div>
    <p>${tr('No medications recorded', '?? ???? ????? ?????')}</p>
  </div>`;

  return `
    <div style="font-weight:700;font-size:13px;margin-bottom:12px">? ${tr('Medication List', '????? ????????')} (${medications.length})</div>
    ${medications.map(m => `
      <div class="ds-med-row">
        <div class="ds-med-icon">?</div>
        <div class="ds-med-info">
          <div class="ds-med-name">${escapeHTML(m.drug_name || m.description || '-')}</div>
          <div class="ds-med-detail">${escapeHTML(m.quantity || '')} · ${escapeHTML(m.notes || '')} · ${escapeHTML(m.created_at?.split('T')[0] || '')}</div>
        </div>
        <div class="ds-med-status ${m.status === 'Completed' ? 'stopped' : m.status === 'Pending' ? 'pending' : 'active'}">
          ${escapeHTML(isArabic ? (m.status === 'Completed' ? '?????' : m.status === 'Pending' ? '????' : '???') : (m.status || 'Active'))}
        </div>
      </div>
    `).join('')}
  `;
};

```

```

/* ===== */
/* TAB: ALLERGIES */
/* ===== */
window.dsTabAllergies = function({ allergies = [] }) {
  if (!allergies.length) return `<div style="text-align:center;padding:32px;color:var(--text-dim)">
    <div style="font-size:48px;margin-bottom:12px;opacity:0.3">?</div>
    <p style="color:#16a34a;font-weight:700">${tr('No Known Drug Allergies (NKDA)', '?? ?????? ?????? ??????')}
  </p>
  </div>`;

  const sevColor = { severe: '#b91c1c', moderate: '#c2410c', mild: '#92400e', unknown: '#64748b' };
  return `
    <div style="margin-bottom:12px">
      <div class="ds-cds-alert critical" style="margin-bottom:12px">
        <div class="ds-cds-alert-icon">?</div>
        <div class="ds-cds-alert-body">
          <div class="ds-cds-alert-title">${tr('Allergy Alert', '???? ?????')} ? ${allergies.length} ${tr(
'recorded', '????')}</div>
          <div class="ds-cds-alert-desc">${tr('Check before prescribing any new medication', '???? ??? ?? ??
???? ?????')}</div>
        </div>
      </div>
      ${allergies.map(a => `
        <div class="ds-problem-row" style="border-right:4px solid ${sevColor[a.severity]} || '#64748b'">
          <div class="ds-med-icon">?</div>
          <div class="ds-med-info">
            <div class="ds-med-name" style="color:#b91c1c">${escapeHTML(a.allergen || '-')}</div>
            <div class="ds-med-detail">${tr('Reaction:', '??????:')} ${escapeHTML(a.reaction || '?')} · ${tr(
'Type:', '?????:')} ${escapeHTML(a.allergen_type || '?')}</div>
          </div>
          <div class="ds-med-status stopped">${escapeHTML(a.severity || tr('Unknown', '?? ?????'))}</div>
        </div>
      `).join('')}
    </div>
  `;
};

/* ===== */
/* TAB: VITALS */
/* ===== */
window.dsTabVitals = function({ vitals = [] }) {
  if (!vitals.length) return `<div style="text-align:center;padding:32px;color:var(--text-dim)">
    <div style="font-size:48px;margin-bottom:12px;opacity:0.3">??</div>
    <p>${tr('No vitals recorded', '?? ???? ?????? ?????? ??????')}</p>
    <p style="font-size:12px;margin-top:8px">${tr('Vitals are recorded by nursing staff', '?????? ?????????? ???
???? ?? ???? ?????????')}</p>
  </div>`;

  const latest = {};
  vitals.forEach(v => { if (!latest[v.score_type]) latest[v.score_type] = v; });

  const vitalDefs = [
    { key: 'blood_pressure', icon: '?', label: { en: 'Blood Pressure', ar: '??? ????', unit: 'mmHg', abnormal

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l: v => false },
  { key: 'temperature', icon: '??', label: { en: 'Temperature', ar: '???????' }, unit: '°C', abnormal: v =>
parseFloat(v) > 38.5 || parseFloat(v) < 36 },
  { key: 'pulse', icon: '?', label: { en: 'Heart Rate', ar: '?????' }, unit: 'bpm', abnormal: v => parseFloa
t(v) < 60 || parseFloat(v) > 100 },
  { key: 'spo2', icon: '?', label: { en: 'SpO2', ar: '???? ????????' }, unit: '%', abnormal: v => parseFloat
(v) < 95 },
  { key: 'weight', icon: '??', label: { en: 'Weight', ar: '?????' }, unit: 'kg', abnormal: () => false },
  { key: 'height', icon: '?', label: { en: 'Height', ar: '?????' }, unit: 'cm', abnormal: () => false },
  { key: 'glucose', icon: '?', label: { en: 'Glucose', ar: '?????????' }, unit: 'mg/dL', abnormal: v => parse
Float(v) > 200 || parseFloat(v) < 70 },
  { key: 'pain_score', icon: '?', label: { en: 'Pain Score', ar: '???? ??????' }, unit: '/10', abnormal: v =>
parseFloat(v) >= 7 },
];

return `
<div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
  ${vitalDefs.filter(d => latest[d.key]).map(d => {
    const v = latest[d.key];
    const val = v.score_value || '?';
    const abn = d.abnormal(val);
    return `<div class="ds-vital-card ${abn ? 'abnormal' : ''}">
      <div style="font-size:22px">${d.icon}</div>
      <div class="ds-vital-value">${escapeHTML(String(val))}</div>
      <div class="ds-vital-unit">${d.unit}</div>
      <div class="ds-vital-label">${tr(d.label.en, d.label.ar)}</div>
      <div style="font-size:10px;color:var(--text-dim);margin-top:4px">${escapeHTML(v.recorded_at?.split('
T')[0] || '')}</div>
    </div>`;
  }).join('')}
</div>

${vitals.length ? `<div style="font-size:12px;color:var(--text-dim);margin-top:12px;text-align:center">
  ${tr('Showing latest reading per vital sign', '????? ??? ????? ??? ?????')} · ${vitals.length} ${tr('total
readings', '????? ???????')}
</div>` : ''}
`;
};

/* ===== */
/* TAB: NOTES (SOAP) */
/* ===== */
window.dsTabNotes = function({ chart = {} }) {
  const records = chart.records || [];
  const last = records[0] || {};
  return `
    <div style="margin-bottom:16px">
      <div style="font-weight:700;font-size:13px;margin-bottom:12px">? ${tr('SOAP Encounter Notes', '??????? ?
????? (SOAP)')}</div>
      <div class="ds-soap-grid">
        <div class="ds-soap-box s">
          <div class="ds-soap-label">S ? ${tr('Subjective', '???? (?????)')}</div>
          <textarea class="ds-soap-input" id="soapS" placeholder="${tr('Chief complaint, history...', '??????
????????? ???????...')}">${escapeHTML(last.symptoms || '')}</textarea>
        </div>

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<div class="ds-soap-box o">
  <div class="ds-soap-label">O ? ${tr('Objective', '????? (???)')}</div>
  <textarea class="ds-soap-input" id="soapO" placeholder="${tr('Exam findings, vitals...', '????? ???
?? ????????...')}"></textarea>
</div>
<div class="ds-soap-box a">
  <div class="ds-soap-label">A ? ${tr('Assessment', '????? (????)')}</div>
  <textarea class="ds-soap-input" id="soapA" placeholder="${tr('Diagnosis, ICD-10...', '??????? ??? I
CD-10...')}">${escapeHTML(last.diagnosis || '')}</textarea>
</div>
<div class="ds-soap-box p">
  <div class="ds-soap-label">P ? ${tr('Plan', '????? ????????')}</div>
  <textarea class="ds-soap-input" id="soapP" placeholder="${tr('Treatment plan, referrals...', '?????
?????????...')}">${escapeHTML(last.treatment || '')}</textarea>
</div>
</div>
<div style="display:flex;gap:10px;margin-top:12px">
  <button class="btn btn-primary" onclick="window.dsSaveSoapNotes()" style="flex:1">
    ? ${tr('Save Notes', '??? ?????????')}
  </button>
  <button class="btn" onclick="window.dsShowMedReportMenu()" style="flex:1;background:#fff3e0;border:1px
solid #ff9800;color:#e65100">
    ?? ${tr('Print Report', '????? ?????')}
  </button>
</div>
</div>
${records.length > 0 ? `
<div style="margin-top:20px">
  <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:upperc
ase">${tr('Previous Notes', '??????? ?????')}</div>
  ${records.slice(0, 5).map(r => `
    <div class="ds-timeline-card" style="margin-bottom:8px">
      <div class="ds-timeline-date">${escapeHTML(r.created_at?.split('T')[0] || '')}</div>
      <div class="ds-timeline-title">${escapeHTML(r.diagnosis || r.treatment || '-')}</div>
      ${r.notes ? `<div class="ds-timeline-detail">${escapeHTML(r.notes)}</div>` : ''}
      ${r.is_signed ? `<div style="margin-top:6px;font-size:11px;color:#16a34a">? ${tr('Signed by:', '????
? ??:')}$ ${escapeHTML(r.signed_by || '?')}</div>` : ''}
    </div>
  `).join('')}
</div>` : ''}
`;
};

window.dsInitSoapNotes = function() {
  // Auto-expand textareas
  document.querySelectorAll('.ds-soap-input').forEach(ta => {
    ta.style.height = 'auto';
    ta.style.height = (ta.scrollHeight + 10) + 'px';
    ta.addEventListener('input', () => {
      ta.style.height = 'auto';
      ta.style.height = (ta.scrollHeight + 10) + 'px';
    });
  });
};

```

```

window.dsSaveSoapNotes = async function() {
  const pid = window._DS.selectedPatientId;
  if (!pid) return showToast(tr('No patient selected', '?? ????' ), 'error');
  const S = document.getElementById('soapS')?.value || '';
  const O = document.getElementById('soapO')?.value || '';
  const A = document.getElementById('soapA')?.value || '';
  const P = document.getElementById('soapP')?.value || '';
  if (!A.trim()) return showToast(tr('Please enter a diagnosis/assessment', '???? ???????/????????'), 'error');
  try {
    await API.post('/api/medical/records', {
      patient_id: pid,
      symptoms: S,
      diagnosis: A,
      treatment: P,
      notes: O,
    });
    showToast(tr('Notes saved!', '?? ??? ?????????!'));
    // Reload chart data
    setTimeout(() => window.dsSelectPatient(pid), 500);
  } catch (e) { showToast(e?.message || tr('Save failed', '??? ?????'), 'error'); }
};

window.dsShowMedReportMenu = function() {
  if (!window._DS.selectedPatientId) return showToast(tr('No patient selected', '?? ????' ), 'error')
  ;
  window._selectedPatientId = window._DS.selectedPatientId;
  window._selectedPatientName = window._DS.selectedPatientData?.patient?.name_ar || window._DS.selectedPatient
Data?.patient?.name_en || '';
  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
  modal.innerHTML = `<div style="background:var(--bg-card);border-radius:16px;padding:24px;width:360px;text-al
ign:center">
    <h3 style="margin:0 0 16px;color:var(--primary)">?? ${tr('Print Report', '????? ?????')}</h3>
    <div style="display:flex;flex-direction:column;gap:10px">
      <button class="btn btn-primary" onclick="showMedicalReportForm('sick_leave');this.closest('.modal-overla
y,.fixed').remove()">? ${tr('Sick Leave', '????? ?????')}</button>
      <button class="btn btn-secondary" onclick="showMedicalReportForm('medical_report');this.closest('.modal-
overlay,.fixed').remove()">? ${tr('Medical Report', '????? ???')}</button>
      <button class="btn" onclick="showMedicalReportForm('fitness');this.closest('.modal-overlay,.fixed').remo
ve()" style="background:#e8f5e9;border:1px solid #2e7d32;color:#2e7d32">? ${tr('Fitness Certificate', '????? ?
????')}</button>
      <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '?????')}</but
ton>
    </div>
  </div>`;
  modal.classList.add('fixed');
  document.body.appendChild(modal);
  modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

/* ===== */
/* RIGHT PANEL: CPOE Orders Panel */

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/* ===== */
window.dsRenderOrdersPanel = function(patient, records = []) {
    const ordersPanel = document.getElementById('dsOrdersContent');
    if (!ordersPanel) return;
    const pid = safeId(patient.id);
    const patName = escapeHTML(isArabic ? (patient.name_ar || patient.name_en || '-') : (patient.name_en || patient.name_ar || '-'));
    const activeSection = window._DS.activeOrderSection || 'diagnosis';
    const sectionClass = id => `ds-order-section ${activeSection === id ? 'open' : 'collapsed'}`;

    ordersPanel.innerHTML = `
        <!-- Quick Diagnosis -->
        <div class="${sectionClass('diagnosis')}}" data-section="diagnosis">
            <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
                ? ${tr('Diagnosis', '???????')} <span style="margin-inline-start:auto">?</span>
            </div>
            <div class="ds-order-section-body">
                <div class="form-group mb-8">
                    <label style="font-size:11px;font-weight:700">${tr('Diagnosis (ICD-10)', '??????? (ICD-10)')}</label>
                    <input class="form-input" id="dsIcd" placeholder="${tr('Type or search diagnosis...', '???? ?? ????'
                    ?? ???????...')}"
                        oninput="window.handleDiagAutocomplete && handleDiagAutocomplete(this)" autocomplete="off">
                    <div id="drDiagSuggestions" style="position:absolute;top:100%;left:0;right:0;background:var(--bg-card,#fff);border:1px solid var(--border);border-radius:8px;z-index:100;max-height:150px;overflow-y:auto;display:none;box-shadow:0 4px 12px rgba(0,0,0,0.1)"></div>
                </div>
                <div class="form-group mb-8">
                    <label style="font-size:11px;font-weight:700">${tr('Symptoms', '???????')}</label>
                    <input class="form-input" id="dsSymp" placeholder="${tr('Chief complaint...', '????? ?????????...')}"
                >
                </div>
                <div class="form-group mb-8">
                    <label style="font-size:11px;font-weight:700">${tr('Notes', '???????')}</label>
                    <textarea class="form-input" id="dsNotes" rows="2" placeholder="${tr('Clinical notes...', '??????? ?
                    ??????...')}"></textarea>
                </div>
                <button class="btn btn-primary w-full" onclick="window.dsSaveRecord()" style="height:36px;font-size:12
                px">
                    ? ${tr('Save Diagnosis', '??? ????????')}
                </button>
            </div>
        </div>

        <!-- E-Prescription -->
        <div class="${sectionClass('rx')}}" data-section="rx">
            <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
                ? ${tr('E-Prescription', '?????? ?????????????')} <span style="margin-inline-start:auto">?</span>
            </div>
            <div class="ds-order-section-body">
                <div id="dsRxItems"></div>
                <div class="form-group mb-8" style="position:relative">
                    <label style="font-size:11px;font-weight:700">${tr('Add Drug', '????? ?????')}</label>
                    <input class="form-input" id="dsRxSearch" placeholder="${tr('Search drug...', '???? ?? ?????...')}"
    
```



```

        oninput="window.dsFilterDrugs(this.value)" autocomplete="off">
        <div id="dsRxDropdown" style="position:absolute;top:100%;left:0;right:0;background:var(--bg-card);border:1px solid var(--border);border-radius:8px;z-index:100;max-height:160px;overflow-y:auto;display:none;box-shadow:0 4px 12px rgba(0,0,0,0.12)"></div>
    </div>
    <div id="dsRxCdsAlerts"></div>
    <button class="btn btn-success w-full" id="dsSendRxBtn"
        onclick="window.dsSendPrescription(${pid})"
        style="height:36px;font-size:12px;display:${window._DS.rxItems.length ? 'flex' : 'none'};align-items:center;justify-content:center;gap:8px">
        ? ${tr('Send to Pharmacy', '???? ?????')} <span id="dsRxCount" class="ds-tab-badge">${window._DS.rxItems.length}</span>
    </button>
</div>
</div>

<!-- Lab Order -->
<div class="${sectionClass('lab')}" data-section="lab">
    <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
        ? ${tr('Lab Order', '??? ?????')} <span style="margin-inline-start:auto">?</span>
    </div>
    <div class="ds-order-section-body">
        <div class="form-group mb-8">
            <label style="font-size:11px;font-weight:700">${tr('Select Tests', '???? ?????????')}</label>
            <select class="form-input" id="dsLabTest" multiple size="4" style="height:auto">
                <optgroup label="${tr('Common', '????')}">
                    <option value="CBC">CBC ? ${tr('Complete Blood Count', '???? ?? ?????')}</option>
                    <option value="CMP">CMP ? ${tr('Comprehensive Metabolic', '????????? ?????')}</option>
                    <option value="HbA1c">HbA1c ? ${tr('Glycated Hemoglobin', '????? ?????????')}</option>
                    <option value="TSH">TSH ? ${tr('Thyroid Function', '????? ????? ?????')}</option>
                    <option value="Lipid Profile">Lipid ? ${tr('Cholesterol Profile', '???? ?????')}</option>
                    <option value="Urinalysis">UA ? ${tr('Urine Analysis', '????? ?????')}</option>
                    <option value="CRP">CRP ? ${tr('C-Reactive Protein', '?????? C ?????????')}</option>
                    <option value="Troponin">Troponin ? ${tr('Cardiac Markers', '???????? ?????')}</option>
                </optgroup>
            </select>
        </div>
        <div class="form-group mb-8">
            <label style="font-size:11px;font-weight:700">${tr('Urgency', '???????')}</label>
            <select class="form-input" id="dsLabUrgency">
                <option value="Routine">? ${tr('Routine', '????')}</option>
                <option value="Urgent">? ${tr('Urgent', '???????')}</option>
                <option value="STAT">? STAT ? ${tr('Immediate', '????')}</option>
            </select>
        </div>
        <div class="form-group mb-8">
            <label style="font-size:11px;font-weight:700">${tr('Clinical Note', '?????? ?????')}</label>
            <input class="form-input" id="dsLabNote" placeholder="${tr('Reason for order...', '??? ?????...')}">
        </div>
        <button class="btn w-full" onclick="window.dsOrderLab(${pid})"
            style="height:36px;font-size:12px;background:#0ea5e9;color:#fff;border:none">
            ? ${tr('Order Lab', '??? ?????')}
        </button>
    </div>
</div>

```

```

</div>

<!-- Radiology Order -->
<div class="{sectionClass('radiology')}}" data-section="radiology">
  <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
    ? ${tr('Radiology Order', '??? ???')} <span style="margin-inline-start:auto">?</span>
  </div>
  <div class="ds-order-section-body">
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Study Type', '??? ?????')}</label>
      <select class="form-input" id="dsRadType">
        <optgroup label="${tr('X-Ray', '???? ?????')}">
          <option>CXR ? Chest X-Ray</option>
          <option>X-Ray Abdomen</option>
          <option>X-Ray Spine (Cervical/Lumbar)</option>
          <option>X-Ray Extremity</option>
        </optgroup>
        <optgroup label="${tr('Ultrasound', '??????? ??? ?????')}">
          <option>US Abdomen & Pelvis</option>
          <option>US Thyroid</option>
          <option>US Cardiac (Echo)</option>
          <option>US Obstetric</option>
        </optgroup>
        <optgroup label="CT Scan">
          <option>CT Brain</option>
          <option>CT Chest</option>
          <option>CT Abdomen & Pelvis</option>
          <option>CT Spine</option>
        </optgroup>
        <optgroup label="MRI">
          <option>MRI Brain</option>
          <option>MRI Spine</option>
          <option>MRI Knee / Joint</option>
        </optgroup>
      </select>
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Clinical Indication', '????? ?????')}</label>
      <input class="form-input" id="dsRadNote" placeholder="${tr('Clinical reason...', '????? ???????.')}"
    >
  </div>
  <button class="btn w-full" onclick="window.dsOrderRadiology(${pid})"
    style="height:36px;font-size:12px;background:#7c3aed;color:#fff;border:none">
    ? ${tr('Order Radiology', '??? ???')}
  </button>
</div>
</div>

<!-- Referral -->
<div class="{sectionClass('referral')}}" data-section="referral">
  <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
    ? ${tr('Referral', '????')} <span style="margin-inline-start:auto">?</span>
  </div>
  <div class="ds-order-section-body">

```

```

<div class="form-group mb-8">
  <label style="font-size:11px;font-weight:700">${tr('Referred to', '????? ???')}</label>
  <select class="form-input" id="dsRefTo">
    <option value="cardiology">${tr('Cardiology', '????? ?????')}</option>
    <option value="neurology">${tr('Neurology', '????????')}</option>
    <option value="orthopedics">${tr('Orthopedics', '???????')}</option>
    <option value="gastroenterology">${tr('Gastroenterology', '??????? ?????')}</option>
    <option value="endocrinology">${tr('Endocrinology', '????? ?????')}</option>
    <option value="nephrology">${tr('Nephrology', '?????')}</option>
    <option value="pulmonology">${tr('Pulmonology', '?????')}</option>
    <option value="rheumatology">${tr('Rheumatology', '???????????')}</option>
    <option value="dermatology">${tr('Dermatology', '????????')}</option>
    <option value="psychiatry">${tr('Psychiatry', '???? ?????')}</option>
    <option value="ophthalmology">${tr('Ophthalmology', '???????')}</option>
    <option value="ent">${tr('ENT', '????? ?????? ????????')}</option>
    <option value="surgery">${tr('Surgery', '????????? ?????')}</option>
    <option value="urology">${tr('Urology', '????????? ?????')}</option>
  </select>
</div>
<div class="form-group mb-8">
  <label style="font-size:11px;font-weight:700">${tr('Reason', '????')}</label>
  <textarea class="form-input" id="dsRefNote" rows="2" placeholder="${tr('Clinical reason for referral
...', '??? ???????.')}></textarea>
</div>
<button class="btn w-full" onclick="window.dsOrderReferral(${pid})"
  style="height:36px;font-size:12px;background:#16a34a;color:#fff;border:none">
  ? ${tr('Send Referral', '????? ?????')}
</button>
</div>
</div>

<!-- Diet Order -->
<div class="${sectionClass('diet')}" data-section="diet">
  <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
    ? ${tr('Diet Order', '??? ?????')} <span style="margin-inline-start:auto">?</span>
  </div>
  <div class="ds-order-section-body">
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Diet Type', '??? ?????')}</label>
      <select class="form-input" id="dsDietType">
        <option value="Normal Diet">?? ${tr('Normal Diet', '???? ?????')}</option>
        <option value="Diabetic Diet">? ${tr('Diabetic Diet', '???? ?????')}</option>
        <option value="Renal Diet">? ${tr('Renal Diet', '???? ?????')}</option>
        <option value="Low Sodium Diet">? ${tr('Low Sodium', '????? ?????')}</option>
        <option value="Cardiac Diet">?? ${tr('Cardiac Diet', '???? ?????')}</option>
        <option value="High Protein Diet">? ${tr('High Protein', '???? ?????')}</option>
        <option value="NPO (Nothing by Mouth)">? NPO ? ${tr('Nothing by Mouth', '????? ?????')}</
option>
        <option value="Clear Liquid Diet">? ${tr('Clear Liquids Only', '????? ?????')}</option>
      </select>
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Duration', '????')}</label>
      <select class="form-input" id="dsDietDuration">

```

```

        <option>24 ???</option>
        <option>48 ???</option>
        <option>??? ???</option>
    </select>
</div>
<button class="btn w-full" onclick="window.dsOrderDiet({pid})"
    style="height:36px;font-size:12px;background:#16a34a;color:#fff;border:none">
    ? ${tr('Order Diet', '???')}</button>
</div>
</div>

<!-- IV Fluids Order -->
<div class="${sectionClass('iv')}" data-section="iv">
    <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
        ? ${tr('IV Fluids', '???')} <span style="margin-inline-start:auto"></span>
    </div>
    <div class="ds-order-section-body">
        <div class="form-group mb-8">
            <label style="font-size:11px;font-weight:700">${tr('Fluid Type', '???')}</label>
            <select class="form-input" id="dsIvType">
                <option value="Normal Saline 0.9% (NS)">Normal Saline 0.9% (NS)</option>
                <option value="Dextrose 5% in Water (D5W)">Dextrose 5% in Water (D5W)</option>
                <option value="Lactated Ringer's (LR)">Lactated Ringer's (LR)</option>
                <option value="Half-Normal Saline (0.45% NS)">Half-Normal Saline (0.45% NS)</option>
                <option value="D5 in 0.45% NS">D5 in 0.45% NS</option>
                <option value="D5 in Lactated Ringer's">D5 in Lactated Ringer's</option>
            </select>
        </div>
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:6px" class="mb-8">
            <div>
                <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Rate (mL/hr)', '??? m
L/???')}</label>
                <input class="form-input" id="dsIvRate" type="number" min="10" max="500" value="100" style="height
:32px;font-size:12px;padding:4px 8px">
            </div>
            <div>
                <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Duration (hr)', '???? (
????')}</label>
                <input class="form-input" id="dsIvDuration" type="number" min="1" max="72" value="8" style="height
:32px;font-size:12px;padding:4px 8px">
            </div>
        </div>
        <div class="form-group mb-8">
            <label style="font-size:11px;font-weight:700">${tr('Additive', '???')}</label>
            <input class="form-input" id="dsIvAdditive" placeholder="${tr('e.g. KCl 20mEq, MgSO4...', '??? : KCl
20mEq')}">
        </div>
        <button class="btn w-full" onclick="window.dsOrderIV({pid})"
            style="height:36px;font-size:12px;background:#0369a1;color:#fff;border:none">
            ? ${tr('Order IV Fluids', '???')}</button>
    </div>
</div>

```

```

<!-- Nursing Order -->
<div class="{sectionClass('nursing')}}" data-section="nursing">
  <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
    ? ${tr('Nursing Orders', '????? ?????')} <span style="margin-inline-start:auto">?</span>
  </div>
  <div class="ds-order-section-body">
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Order Type', '??? ?????')}</label>
      <select class="form-input" id="dsNursingType">
        <option value="Monitor BP every 4h">? ${tr('Monitor BP q4h', '???? ????? ?? 4 ?????')}</option>
        <option value="Monitor BP every 2h">? ${tr('Monitor BP q2h', '???? ????? ?? ?????')}</option>
        <option value="Vital Signs every 4h">?? ${tr('Vital Signs q4h', '????? ????? ?? 4 ?????')}</option>
n>
        <option value="Strict I&O monitoring">? ${tr('Strict I&O Monitoring', '????? ?????? ?????')}</
option>
        <option value="Daily weight">?? ${tr('Daily Weight', '??? ?????')}</option>
        <option value="Fall precautions">?? ${tr('Fall Precautions', '???????? ?????')}</option>
        <option value="Bed rest">?? ${tr('Bed Rest', '???? ?????')}</option>
        <option value="Wound care daily">? ${tr('Wound Care Daily', '???? ????? ?????')}</option>
        <option value="Foley catheter care">? ${tr('Foley Catheter Care', '????? ?????')}</option>
        <option value="Oxygen 2-4L/min via nasal cannula">? ${tr('O2 2-4L via NC', '????? 2-4 ??/?')}</option>
      </select>
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Additional Notes', '??????? ?????')}</label>
      <input class="form-input" id="dsNursingNote" placeholder="${tr('Specify if needed...', '??? ?? ???..
.')}"/>
    </div>
    <button class="btn w-full" onclick="window.dsOrderNursing(${pid})"
      style="height:36px;font-size:12px;background:#7c3aed;color:#fff;border:none">
      ? ${tr('Issue Nursing Order', '????? ??? ?????')}
    </button>
  </div>
</div>

<!-- Medical Consents -->
<div class="{sectionClass('consent')}}" data-section="consent">
  <div class="ds-order-section-header" onclick="window.dsToggleSection(this)">
    ? ${tr('Medical Consents', '????????? ?????')} <span style="margin-inline-start:auto">?</span>
  </div>
  <div class="ds-order-section-body">
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700">${tr('Consent Type', '??? ?????')}</label>
      <select class="form-input" id="dsConsentType">
        <option value="general_medical">${tr('General Medical Procedure', '????? ??? ???')}</option>
        <option value="surgical">${tr('Surgical Consent', '????? ????? ?????')}</option>
        <option value="anesthesia">${tr('Anesthesia Consent', '????? ?????')}</option>
        <option value="blood_transfusion">${tr('Blood Transfusion', '??? ??')}</option>
        <option value="treatment_refusal">${tr('Treatment Refusal', '??? ?????')}</option>
        <option value="privacy">${tr('Privacy Consent', '????? ?????')}</option>
      </select>
    </div>
  </div>

```

```

<div class="form-group mb-8">
  <label style="font-size:11px;font-weight:700">${tr('Procedure / Notes', '?????? / ??????')}</label
>

  <textarea class="form-input" id="dsConsentNote" rows="2" placeholder="${tr('Procedure, risks explain
ed, or consent notes...', '?????? ?????? ??????? ?? ?????? ???????...')}"></textarea>
</div>
<div class="ds-order-actions-grid">
  <button class="btn btn-primary" onclick="window.dsCreateConsentFromStation(${pid})" style="height:36
px;font-size:12px">
    ? ${tr('Create', '????')}
  </button>
  <button class="btn" onclick="window.dsSwitchTab('consents')" style="height:36px;font-size:12px">
    ? ${tr('Registry', '????')}
  </button>
</div>
</div>
</div>

<!-- Sign & Close -->
<div style="padding:12px 0">
  <button class="btn w-full" onclick="window.dsSignEncounter()"
    style="background:linear-gradient(135deg,#7c3aed,#0369a1);color:#fff;border:none;height:44px;font-size
:13px;font-weight:700;border-radius:12px">
    ?? ${tr('Sign & Close Encounter', '???? ?????? ??????')}
  </button>
</div>
`;

window.dsApplyOrdersAccordion(ordersPanel, activeSection);
window.dsRenderRxItems();
};

/* ===== */
/* Collapsible Sections */
/* ===== */
window.dsApplyOrdersAccordion = function(panel, activeSection) {
  const root = panel || document.getElementById('dsOrdersContent');
  if (!root) return;
  const sections = Array.from(root.querySelectorAll('.ds-order-section'));
  const target = activeSection || sections[0]?.dataset?.section || '';
  sections.forEach(section => {
    const isOpen = section.dataset.section === target;
    section.classList.toggle('open', isOpen);
    section.classList.toggle('collapsed', !isOpen);
    const arrow = section.querySelector('.ds-order-section-header span:last-child');
    if (arrow) arrow.textContent = isOpen ? '?' : '?';
  });
  window._DS.activeOrderSection = target || 'diagnosis';
};

window.dsToggleSection = function(header) {
  const section = header.closest('.ds-order-section');
  const root = header.closest('#dsOrdersContent');

```

```

if (!section || !root) return;
document.querySelectorAll('#dsOrdersContent .ds-order-section').forEach(item => {
    item.classList.add('collapsed');
    item.classList.remove('open');
    const arrow = item.querySelector('.ds-order-section-header span:last-child');
    if (arrow) arrow.textContent = '?';
});
section.classList.remove('collapsed');
section.classList.add('open');
const arrow = section.querySelector('.ds-order-section-header span:last-child');
if (arrow) arrow.textContent = '?';
window._DS.activeOrderSection = section.dataset.section || 'diagnosis';
};

/* ===== */
/* E-PRESCRIPTION */
/* ===== */
window.dsFilterDrugs = function(query) {
    const dropdown = document.getElementById('dsRxDropdown');
    if (!dropdown) return;
    const drugs = window._DS.drugs || [];
    const q = (query || '').trim().toLowerCase();
    if (!q) { dropdown.style.display = 'none'; return; }
    const matches = drugs.filter(d => {
        const n = ((d.name_ar || '') + ' ' + (d.name_en || '') + ' ' + (d.generic_name || '')).toLowerCase();
        return n.includes(q);
    }).slice(0, 12);
    if (!matches.length) { dropdown.style.display = 'none'; return; }
    dropdown.style.display = 'block';
    dropdown.innerHTML = matches.map(d => `
        <div style="padding:8px 12px;cursor:pointer;font-size:12px;border-bottom:1px solid var(--border)"
            class="autocomplete-item"
            onmousedown="event.preventDefault();window.dsAddRxItem(${safeId(d.id)},'${jsStr(d.name_ar || d.name_en |
| ')}','${jsStr(d.name_en || ')}','${jsStr(d.generic_name || ')}','${parseFloat(d.price || 0)}')">
                <strong>${escapeHTML(d.name_ar || d.name_en || '-')}</strong>
                ${d.name_en ? `<span style="color:var(--text-dim);font-size:11px"> ? ${escapeHTML(d.name_en)}</span>` :
                ''}
                ${d.generic_name ? `<span style="color:var(--text-dim);font-size:10px"> (${escapeHTML(d.generic_name)})<
/strong>` : ''}
            </div>
        `).join('');
};

window.dsAddRxItem = function(id, nameAr, nameEn, generic, price) {
    const dropdown = document.getElementById('dsRxDropdown');
    const input = document.getElementById('dsRxSearch');
    if (dropdown) dropdown.style.display = 'none';
    if (input) input.value = '';

    // Check for duplicate
    if (window._DS.rxItems.find(x => x.id === id)) {
        showToast(tr('Drug already added', '????? ???? ?????'), 'error'); return;
    }
}

```

```

// CDS: Check allergies
const allergies = window._DS.selectedPatientData?.allergies || [];
const allergyMatch = allergies.find(a => {
  const allergen = (a.allergen || '').toLowerCase();
  return nameEn.toLowerCase().includes(allergen) || nameAr.includes(allergen) || (generic || '').toLowerCase().includes(allergen);
});
if (allergyMatch) {
  const alertsDiv = document.getElementById('dsRxCdsAlerts');
  if (alertsDiv) {
    alertsDiv.innerHTML = `<div class="ds-cds-alert critical">
      <div class="ds-cds-alert-icon">?</div>
      <div class="ds-cds-alert-body">
        <div class="ds-cds-alert-title">${tr('ALLERGY ALERT', '????? ?????????')}: ${escapeHTML(nameAr || nameEn)}</div>
        <div class="ds-cds-alert-desc">${tr('Patient has recorded allergy to:', '?????? ???? ?????? ?????? ??:')} ${escapeHTML(allergyMatch.allergen)}</div>
      </div>
      <button class="ds-cds-alert-dismiss" onclick="this.parentElement.remove()">?</button>
    </div>`;
  }
  // Still allow override but warn
  if (!confirm(tr('?? ALLERGY ALERT! This drug may conflict with recorded allergies. Add anyway?', '?? ????? ???????! ?? ?????? ?? ?????? ?? ??????? ????????. ?? ????? ??????? ??????')))) return;
}

window._DS.rxItems.push({
  id, nameAr, nameEn, generic,
  dose: '', frequency: '??? ??????', duration: '7 ?????', route: '???? (PO)', qty: 1, notes: '', price
});
window.dsRenderRxItems();
};

window.dsRenderRxItems = function() {
  const container = document.getElementById('dsRxItems');
  const btn = document.getElementById('dsSendRxBtn');
  const countEl = document.getElementById('dsRxCount');
  if (countEl) countEl.textContent = window._DS.rxItems.length;
  if (btn) btn.style.display = window._DS.rxItems.length ? 'flex' : 'none';
  if (!container) return;

  if (!window._DS.rxItems.length) {
    container.innerHTML = `<div style="font-size:12px;color:var(--text-dim);padding:8px 0;margin-bottom:8px">${tr('No medications added yet', '?? ??? ?????? ?????? ???')}</div>`;
    return;
  }

  container.innerHTML = window._DS.rxItems.map((item, idx) => {
    // Auto-calculate Qty Total based on duration days × frequency multiplier
    const durDays = parseInt(item.duration) || 7;
    const freqMult = item.frequency?.includes('4 ?????') ? 4 : item.frequency?.includes('3 ?????') ? 3 : item.frequency?.includes('?????') ? 2 : item.frequency?.includes('6') ? 4 : item.frequency?.includes('8') ? 3 : 1;
    const qtyTotal = (item.qty || 1) * freqMult * durDays;

```



```

return `
<div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:10px;
padding:10px;margin-bottom:8px">
  <div style="display:flex;align-items:center;justify-content:space-between;margin-bottom:8px">
    <strong style="font-size:12px">? ${escapeHTML(item.nameAr || item.nameEn)}</strong>
    <button class="ds-rx-remove" onclick="window.dsRemoveRxItem(${idx})">?</button>
  </div>
  <div style="display:grid;grid-template-columns:1fr 1fr;gap:6px">
    <div>
      <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Dose', '?????')}</label>
      <input class="form-input" style="height:30px;font-size:12px;padding:4px 8px"
        value="${escapeHTML(item.dose)}" placeholder="${tr('e.g. 500mg', '????: 500mg')}"
        oninput="window._DS.rxItems[${idx}].dose=this.value">
    </div>
    <div>
      <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Route', '?????')}</label>
      <select class="form-input" style="height:30px;font-size:12px;padding:2px 6px"
        onchange="window._DS.rxItems[${idx}].route=this.value">
        <option ${item.route === '???? (PO)' ? 'selected' : ''}>???? (PO)</option>
        <option ${item.route === '???? (IV)' ? 'selected' : ''}>???? (IV)</option>
        <option ${item.route === '???? (IM)' ? 'selected' : ''}>???? (IM)</option>
        <option ${item.route === '???? (Topical)' ? 'selected' : ''}>???? (Topical)</option>
        <option ${item.route === '??? ????? (SC)' ? 'selected' : ''}>??? ????? (SC)</option>
        <option ${item.route === '??????? (INH)' ? 'selected' : ''}>??????? (INH)</option>
      </select>
    </div>
    <div>
      <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Frequency', '???????')}</label>
      <select class="form-input" style="height:30px;font-size:12px;padding:2px 6px"
        onchange="window._DS.rxItems[${idx}].frequency=this.value;window.dsRenderRxItems()">
        <option ${item.frequency === '??? ??????' ? 'selected' : ''}>??? ??????</option>
        <option ${item.frequency === '????? ??????' ? 'selected' : ''}>????? ??????</option>
        <option ${item.frequency === '3 ???? ??????' ? 'selected' : ''}>3 ???? ??????</option>
        <option ${item.frequency === '4 ???? ??????' ? 'selected' : ''}>4 ???? ??????</option>
        <option ${item.frequency === '?? 6 ??????' ? 'selected' : ''}>?? 6 ??????</option>
        <option ${item.frequency === '?? 8 ??????' ? 'selected' : ''}>?? 8 ??????</option>
        <option ${item.frequency === '??? ?????? (PRN)' ? 'selected' : ''}>??? ?????? (PRN)</option>
      </select>
    </div>
    <div>
      <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Duration', '????')}</label>
      <select class="form-input" style="height:30px;font-size:12px;padding:2px 6px"
        onchange="window._DS.rxItems[${idx}].duration=this.value;window.dsRenderRxItems()">
        <option>3 ????</option>
        <option ${item.duration === '5 ????' ? 'selected' : ''}>5 ????</option>
        <option ${item.duration === '7 ????' ? 'selected' : ''}>7 ????</option>
        <option>10 ????</option>
        <option>14 ????</option>
        <option>30 ????</option>
        <option>???????</option>
      </select>
    </div>
  </div>

```

```

<div>
  <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Refill (0-3)', '?????? ?
????')}</label>
  <select class="form-input" style="height:30px;font-size:12px;padding:2px 6px"
    onchange="window._DS.rxItems[${idx}].refill=parseInt(this.value)">
    <option value="0" ${!item.refill ? 'selected' : ''}>0 ? ${tr('No Refill', '???? ?????')}</option>
    <option value="1" ${item.refill===1 ? 'selected' : ''}>1x Refill</option>
    <option value="2" ${item.refill===2 ? 'selected' : ''}>2x Refill</option>
    <option value="3" ${item.refill===3 ? 'selected' : ''}>3x Refill</option>
  </select>
</div>
<div>
  <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Qty Total', '????? ?????
???'')}</label>
  <div style="height:30px;display:flex;align-items:center;padding:0 8px;background:var(--primary-glow,
#eff6ff);border-radius:8px;border:1px solid var(--primary);font-size:12px;font-weight:700;color:var(--primary)
">
    ${isNaN(qtyTotal) ? '?' : qtyTotal} ${tr('units', '????')}
  </div>
</div>
</div>`;
}).join('');
};

```

```

window.dsRemoveRxItem = function(idx) {

```

```

  window._DS.rxItems.splice(idx, 1);
  window.dsRenderRxItems();
};

```

```

window.dsSendPrescription = async function(patientId) {
  if (!window._DS.rxItems.length) return showToast(tr('Add at least one drug', '??? ????? ??? ?????'), 'error'
);
  if (!patientId) return showToast(tr('No patient selected', '?? ???? ???? ?????'), 'error');

  // Check drug interactions
  if (window._DS.rxItems.length >= 2) {
    const drugNames = window._DS.rxItems.map(x => x.nameEn || x.nameAr);
    const interactionResult = await window.checkDrugInteractions(drugNames).catch(() => ({ hasCritical: false,
failed: false }));
    if (interactionResult.hasCritical) {
      showToast(tr('Prescription blocked due to CRITICAL interaction', '????? ?????? ??? ???? ????'), 'error
');
    }
    return;
  }
}

```

```

try {
  const btn = document.getElementById('dsSendRxBtn');
  if (btn) btn.disabled = true;
  await Promise.all(window._DS.rxItems.map(item => API.post('/api/orders', {
    patient_id: patientId,
    type: 'medication',

```

```

        description: `${item.nameAr || item.nameEn} ${item.dose} ? ${item.frequency} × ${item.duration} (${item.
route})`,
        quantity: item.qty || 1,
        status: 'Pending',
        notes: item.notes || '',
    }));
    showToast(tr(`? ${window._DS.rxItems.length} drug(s) sent to pharmacy!`, `? ? ???? ${window._DS.rxItems.
length} ???? ??????`));
    window._DS.rxItems = [];
    window.dsRenderRxItems();
    if (btn) btn.disabled = false;
} catch (e) {
    showToast(e?.message || tr('Failed to send prescription', '??? ????? ??????'), 'error');
    const btn = document.getElementById('dsSendRxBtn');
    if (btn) btn.disabled = false;
}
};

/* ===== */
/* MEDICAL CONSENTS */
/* ===== */
window.dsCreateConsentFromStation = async function(patientId, selectId = 'dsConsentType') {
    if (!patientId) return showToast(tr('No patient selected', '?? ???? ???? ?????'), 'error');
    const patient = window._DS.selectedPatientData?.patient || {};
    const templateType = document.getElementById(selectId)?.value || document.getElementById('dsConsentType')?.v
alue || 'general_medical';
    const note = document.getElementById('dsConsentNote')?.value || '';
    const doctorName = window._DS.currentUser?.display_name || window._DS.currentUser?.name || window._DS.curren
tUser?.username || 'Doctor';
    let templates = [];
    try { templates = await API.get('/api/consent-forms/templates/list'); } catch { templates = []; }
    const template = templates.find(t => t.type === templateType) || templates.find(t => t.type === 'general_med
ical') || {
        type: 'general_medical',
        title: 'General Medical Procedure Consent',
        title_ar: '????? ???? ????',
        content: ''
    };
    const patientName = isArabic ? (patient.name_ar || patient.name_en || '') : (patient.name_en || patient.name
_ar || '');
    const content = [
        template.content || '',
        note.trim() ? `${tr('Doctor notes', '?????? ??????')}: ${note.trim()}` : ''
    ].filter(Boolean).join('\n\n');

    try {
        await API.post('/api/consent-forms', {
            patient_id: patientId,
            patient_name: patientName,
            form_type: template.type,
            form_title: template.title || template.title_ar || tr('Medical Consent', '????? ???'),
            form_title_ar: template.title_ar || template.title || tr('Medical Consent', '????? ???'),
            content,
            doctor_name: doctorName,

```

```

        notes: note
    });
    showToast(tr('Consent form created', '?? ????? ?????? ?????'));
    const noteEl = document.getElementById('dsConsentNote');
    if (noteEl) noteEl.value = '';
    window.dsSwitchTab('consents');
} catch (e) {
    showToast(e?.message || tr('Failed to create consent', '??? ????? ???????'), 'error');
}
};

window.dsTabConsents = async function(container) {
    const pid = window._DS.selectedPatientId;
    if (!pid) return;
    container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading consents...', '???? ????? ??????????..')}</p></div>`;
    let forms = [], templates = [];
    try {
        [forms, templates] = await Promise.all([
            API.get(`/api/consent-forms?patient_id=${encodeURIComponent(pid)}`).catch(() => []),
            API.get('/api/consent-forms/templates/list').catch(() => []),
        ]);
    } catch {
        forms = [];
        templates = [];
    }
    const signedCount = forms.filter(f => String(f.status || '').toLowerCase() === 'signed' || f.signed_at).length;
    const pendingCount = Math.max(forms.length - signedCount, 0);
    container.innerHTML = `
        <div class="ds-consent-toolbar">
            <div>
                <strong style="font-size:13px">? ${tr('Medical Consents', '????????? ??????')}</strong>
                <div style="font-size:11px;color:var(--text-dim);margin-top:3px">
                    ${tr('Signed', '????')}: ${signedCount} · ${tr('Pending', '???????? ??????')}: ${pendingCount}
                </div>
            </div>
            <div class="ds-consent-create">
                <select class="form-input" id="dsConsentTemplate" style="height:34px;font-size:12px">
                    ${templates.map(t => `<option value="${escapeHTML(t.type || '')}">${escapeHTML(isArabic ? (t.title_ar || t.title || '') : (t.title || t.title_ar || ''))}</option>`).join('')}
                </select>
                <button class="btn btn-primary" onclick="window.dsCreateConsentFromStation(${safeId(pid)}, 'dsConsentTemplate')" style="height:34px;font-size:12px">
                    ? ${tr('Create Consent', '????? ?????')}
                </button>
            </div>
        </div>
        ${forms.length ? `
            <div class="ds-consent-list">
                ${forms.map(f => {
                    const title = isArabic ? (f.form_title_ar || f.form_title || '-') : (f.form_title || f.form_title_ar || '-');
                    const isSigned = String(f.status || '').toLowerCase() === 'signed' || f.signed_at;

```

```

const created = f.created_at ? new Date(f.created_at).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US') : '-';
const signed = f.signed_at ? new Date(f.signed_at).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US') : '-';
return `
  <div class="ds-consent-card">
    <div class="ds-consent-card-main">
      <div style="font-weight:700;font-size:12px">${escapeHTML(title)}</div>
      <div style="font-size:11px;color:var(--text-dim);margin-top:3px">
        ${tr('Created', '????')}: ${escapeHTML(created)} · ${tr('Signed', '???????')}: ${escapeHTML(signed)}
      </div>
    </div>
    <span class="ds-consent-status ${isSigned ? 'signed' : 'pending'}">${isSigned ? tr('Signed', '??') : tr('Pending', '?????? ?????')}</span>
    <div class="ds-consent-actions">
      ${!isSigned ? `<button class="btn btn-sm btn-primary" onclick="window.dsOpenConsentSignModal(${safeId(f.id)})">?? ${tr('Sign', '????')}</button>` : ''}
      <button class="btn btn-sm" onclick="window.printConsentForm ? window.printConsentForm(${safeId(f.id)}) : window.dsOpenConsentSignModal(${safeId(f.id)})">?? ${tr('Print', '????')}</button>
    </div>
  </div>`;
  }).join('')
</div>` : `
<div style="text-align:center;padding:32px;color:var(--text-dim)">
  <div style="font-size:48px;opacity:.3"></div>
  <p>${tr('No consent forms for this patient', '?? ??? ?????? ??? ?????')}</p>
</div>`
`;
};

```

```

window.dsOpenConsentSignModal = async function(formId) {
  let form = null;
  try { form = await API.get(`/api/consent-forms/${formId}`); }
  catch (e) { return showToast(e.message || tr('Consent not found', '?? ??? ?????? ??? ?????'), 'error'); }
  const modal = document.createElement('div');
  modal.className = 'ds-consent-modal';
  modal.innerHTML = `
    <div class="ds-consent-modal-card">
      <div style="display:flex;justify-content:space-between;align-items:center;gap:12px;margin-bottom:12px">
        <strong style="font-size:15px">?? ${escapeHTML(isArabic ? (form.form_title_ar || form.form_title || '')) : (form.form_title || form.form_title_ar || '')}</strong>
        <button class="btn btn-sm" onclick="this.closest('.ds-consent-modal').remove()">?</button>
      </div>
      <div class="ds-consent-preview">
        <p style="white-space:pre-wrap;margin:0">${escapeHTML(form.content || '')}</p>
        <div style="margin-top:12px;font-size:12px;color:var(--text-dim)">
          <strong>${tr('Patient', '????')}</strong> ${escapeHTML(form.patient_name || '')}
          <br><strong>${tr('Doctor', '????')}</strong> ${escapeHTML(form.doctor_name || '')}
        </div>
      </div>
      <label style="font-size:12px;font-weight:700;margin-top:12px;display:block">${tr('Patient Signature', '???? ?????')}</label>
      <canvas id="dsConsentSigCanvas" width="560" height="180" class="ds-consent-canvas"></canvas>
    </div>
  `;

```

```

<div style="display:grid;grid-template-columns:1fr auto;gap:8px;margin-top:10px">
  <input class="form-input" id="dsConsentWitness" placeholder="{tr('Witness name', '??? ?????')}">
  <button class="btn" onclick="window.dsClearConsentSignature()" style="height:38px">{tr('Clear', '???')}</button>
</div>
<div style="display:flex;justify-content:flex-end;gap:8px;margin-top:14px">
  <button class="btn" onclick="this.closest('.ds-consent-modal').remove()">{tr('Cancel', '????')}</button>
  <button class="btn btn-primary" onclick="window.dsSignConsentFromStation({safeId(formId)})">?? {tr('Sign Consent', '???? ?????')}</button>
</div>
</div>
`;
document.body.appendChild(modal);
setTimeout(window.dsInitConsentSignatureCanvas, 0);
};

window.dsInitConsentSignatureCanvas = function() {
  const canvas = document.getElementById('dsConsentSigCanvas');
  if (!canvas) return;
  const ctx = canvas.getContext('2d');
  ctx.lineWidth = 2;
  ctx.lineCap = 'round';
  ctx.strokeStyle = '#111827';
  let drawing = false;
  const point = e => {
    const rect = canvas.getBoundingClientRect();
    return { x: (e.clientX - rect.left) * (canvas.width / rect.width), y: (e.clientY - rect.top) * (canvas.height / rect.height) };
  };
  canvas.onpointerdown = e => { drawing = true; canvas.setPointerCapture(e.pointerId); const p = point(e); ctx.beginPath(); ctx.moveTo(p.x, p.y); };
  canvas.onpointermove = e => { if (!drawing) return; const p = point(e); ctx.lineTo(p.x, p.y); ctx.stroke(); };
  canvas.onpointerup = () => { drawing = false; };
  canvas.onpointercancel = () => { drawing = false; };
};

window.dsClearConsentSignature = function() {
  const canvas = document.getElementById('dsConsentSigCanvas');
  if (canvas) canvas.getContext('2d').clearRect(0, 0, canvas.width, canvas.height);
};

window.dsSignConsentFromStation = async function(formId) {
  const canvas = document.getElementById('dsConsentSigCanvas');
  const signature = canvas ? canvas.toDataURL('image/png') : '';
  const witnessName = document.getElementById('dsConsentWitness')?.value || '';
  try {
    await API.put(`/api/consent-forms/${formId}/sign`, {
      patient_signature: signature,
      witness_name: witnessName
    });
    showToast(tr('Consent signed', '?? ????? ?????'));
    document.querySelector('.ds-consent-modal')?.remove();
  }
};

```

```

    window.dsSwitchTab('consents');
  } catch (e) {
    showToast(e?.message || tr('Failed to sign consent', '??? ????? ??????'), 'error');
  }
};

/* ===== */
/* CLINICAL ORDERS: Lab, Radiology, Referral */
/* ===== */

window.dsOrderLab = async function(patientId) {
  if (!patientId) return showToast(tr('No patient selected', '?? ????? ?????'), 'error');
  const sel = document.getElementById('dsLabTest');
  const tests = sel ? [...sel.selectedOptions].map(o => o.value) : [];
  if (!tests.length) return showToast(tr('Select at least one test', '???? ?????? ?????? ?? ????'), 'error');
  const urgency = document.getElementById('dsLabUrgency')?.value || 'Routine';
  const note = document.getElementById('dsLabNote')?.value || '';
  try {
    await API.post('/api/orders', {
      patient_id: patientId,
      type: 'lab',
      description: tests.join(', '),
      status: urgency === 'STAT' ? 'STAT' : 'Pending',
      notes: note || '',
    });
    showToast(tr(`? Lab order sent: ${tests.join(', ')}`, `? ? ??: ${tests.join(', ')}`));
    if (sel) { [...sel.options].forEach(o => o.selected = false); }
    document.getElementById('dsLabNote') && (document.getElementById('dsLabNote').value = '');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.dsOrderRadiology = async function(patientId) {
  if (!patientId) return showToast(tr('No patient selected', '?? ????? ?????'), 'error');
  const type = document.getElementById('dsRadType')?.value || '';
  if (!type) return showToast(tr('Select a study type', '???? ?? ????'), 'error');
  const note = document.getElementById('dsRadNote')?.value || '';
  try {
    await API.post('/api/orders', {
      patient_id: patientId,
      type: 'radiology',
      description: type,
      status: 'Pending',
      notes: note,
    });
    showToast(tr(`? Radiology ordered: ${type}`, `? ? ? ??: ${type}`));
    document.getElementById('dsRadNote') && (document.getElementById('dsRadNote').value = '');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.dsOrderReferral = async function(patientId) {
  if (!patientId) return showToast(tr('No patient selected', '?? ????? ?????'), 'error');
  const dept = document.getElementById('dsRefTo')?.value || '';
  const note = document.getElementById('dsRefNote')?.value || '';
  if (!note.trim()) return showToast(tr('Enter referral reason', '???? ?? ????'), 'error');
  try {

```

```

    await API.post('/api/orders', {
      patient_id: patientId,
      type: 'referral',
      description: `Referral to ${dept}`,
      status: 'Pending',
      notes: note,
    });
    showToast(tr(`? Referral sent to ${dept}`, `? ? ???? ?????? ??? ${dept}`));
    document.getElementById('dsRefNote') && (document.getElementById('dsRefNote').value = '');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* CLINICAL ORDERS: Diet, IV Fluids, Nursing */
/* ===== */
window.dsOrderDiet = async function(patientId) {
  if (!patientId) return showToast(tr('No patient selected', '?? ???? ???? ????'), 'error');
  const dietType = document.getElementById('dsDietType')?.value || 'Normal Diet';
  const duration = document.getElementById('dsDietDuration')?.value || '24 ????';
  try {
    await API.post('/api/orders', {
      patient_id: patientId,
      type: 'diet',
      description: `[DIET] ${dietType}`,
      status: 'Pending',
      notes: `${tr('Duration', '????')}: ${duration}`,
    });
    showToast(tr(`? Diet ordered: ${dietType}`, `? ? ? ? ? ? ? ? : ${dietType}`));
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.dsOrderIV = async function(patientId) {
  if (!patientId) return showToast(tr('No patient selected', '?? ???? ???? ????'), 'error');
  const fluidType = document.getElementById('dsIvType')?.value || 'Normal Saline 0.9% (NS)';
  const rate = document.getElementById('dsIvRate')?.value || '100';
  const duration = document.getElementById('dsIvDuration')?.value || '8';
  const additive = document.getElementById('dsIvAdditive')?.value || '';
  const totalVol = Math.round(parseFloat(rate) * parseFloat(duration));
  const desc = `[IV] ${fluidType} @ ${rate}mL/hr × ${duration}hr${additive ? ' ' + ' ' + additive : ''}`;
  try {
    await API.post('/api/orders', {
      patient_id: patientId,
      type: 'iv',
      description: desc,
      status: 'Pending',
      notes: `${tr('Total Volume', '???? ????')}: ${totalVol}mL${additive ? ' ' | ' ' + tr('Additive', '????')
+ ': ' + additive : ''}`,
    });
    showToast(tr(`? IV ordered: ${fluidType}`, `? ? ? ? ? ? ? ? : ${fluidType}`));
    document.getElementById('dsIvAdditive') && (document.getElementById('dsIvAdditive').value = '');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.dsOrderNursing = async function(patientId) {

```



```

if (!patientId) return showToast(tr('No patient selected', '?? ????' 'error'));
const orderType = document.getElementById('dsNursingType')?.value || '';
const note = document.getElementById('dsNursingNote')?.value || '';
try {
  await API.post('/api/orders', {
    patient_id: patientId,
    type: 'nursing',
    description: `[NURSING] ${orderType}`,
    status: 'Pending',
    notes: note,
  });
  showToast(tr(`? Nursing order issued`, `? ? ???? ? ? ????`));
  document.getElementById('dsNursingNote') && (document.getElementById('dsNursingNote').value = '');
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

```

```

/* ===== */
/* SAVE DIAGNOSIS RECORD */
/* ===== */
window.dsSaveRecord = async function() {
  const pid = window._DS.selectedPatientId;
  if (!pid) return showToast(tr('No patient selected', '?? ????' 'error'));
  const diag = document.getElementById('dsIcd')?.value || '';
  const symp = document.getElementById('dsSymp')?.value || '';
  const notes = document.getElementById('dsNotes')?.value || '';
  if (!diag.trim()) return showToast(tr('Enter diagnosis', '???? ?????'), 'error');
  try {
    const rec = await API.post('/api/medical/records', {
      patient_id: pid,
      diagnosis: diag,
      symptoms: symp,
      notes: notes,
      treatment: '',
    });
    window._DS.activeEncounterId = rec.id;
    showToast(tr('Diagnosis saved!', '?? ? ? ? ? ? ? !'));
    document.getElementById('dsIcd') && (document.getElementById('dsIcd').value = '');
    document.getElementById('dsSymp') && (document.getElementById('dsSymp').value = '');
    document.getElementById('dsNotes') && (document.getElementById('dsNotes').value = '');
    // Refresh chart
    setTimeout(() => window.dsSelectPatient(pid), 600);
  } catch (e) { showToast(e?.message || tr('Save failed', '??? ?????'), 'error'); }
};

```

```

/* ===== */
/* MARK WITH DOCTOR */
/* ===== */
window.dsMarkWithDoctor = async function(patientId) {
  if (!patientId) return;
  try {
    await API.put('/api/patients/' + patientId, { status: 'With Doctor' });
    showToast(tr('Patient moved to "With Doctor"', '?? ? ? ? ? ? ? ? ? "?? ??????"));
  }
};

```

```

window.dsRefreshWaitQueue();
// Update the banner button
const btn = document.getElementById('btnMarkDoctor');
if (btn) { btn.disabled = true; btn.innerHTML = '? ' + tr('With Doctor', '?? ?????'); }
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* SIGN & CLOSE ENCOUNTER */
/* ===== */
window.dsSignEncounter = function() {
  const encId = window._DS.activeEncounterId;
  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.6);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = `
    <div class="ds-sign-modal">
      <div class="ds-sign-icon">??</div>
      <h3 style="margin:0 0 8px;font-size:18px;font-weight:800">${tr('Sign Encounter', '???? ?????')}</h3>
      <p style="font-size:13px;color:var(--text-dim);margin-bottom:16px">
        ${tr('Enter your 4-6 digit PIN to sign and lock the encounter', '???? ?? PIN ?????? ?? 4-6 ?????? ??
        ????? ?????? ??????')}
      </p>
      <input type="password" class="ds-pin-input" id="dsPinInput" maxlength="6" inputmode="numeric"
        placeholder="? ? ? ?" oninput="this.value=this.value.replace(/\\D/g, '')">
      <div style="display:flex;gap:10px;margin-top:8px">
        <button class="btn btn-primary" onclick="window.dsConfirmSign(${encId})" style="flex:1;height:44px;font-size:14px;font-weight:700">
          ? ${tr('Sign Now', '???? ?????')}
        </button>
        <button class="btn" onclick="this.closest('.fixed-modal').remove()" style="flex:1;height:44px">
          ${tr('Cancel', '????')}
        </button>
      </div>
      <div id="dsSignError" style="color:#dc2626;font-size:12px;margin-top:8px"></div>
    </div>
  `;
  modal.classList.add('fixed-modal');
  document.body.appendChild(modal);
  modal.onclick = e => { if (e.target === modal) modal.remove(); };
  setTimeout(() => document.getElementById('dsPinInput')?.focus(), 100);
};

window.dsConfirmSign = async function(encId) {
  const pin = document.getElementById('dsPinInput')?.value || '';
  const errEl = document.getElementById('dsSignError');
  if (!/^\\d{4,6}$/.test(pin)) {
    if (errEl) errEl.textContent = tr('PIN must be 4-6 digits', '??? ?? ??? PIN ?? 4-6 ?????');
    return;
  }
  const doctorName = window._DS.currentUser?.name || window._DS.currentUser?.username || 'Doctor';
  const targetId = encId || (window._DS.activeEncounterId) || 0;
  try {
    const r = await API.post(`/api/encounters/${targetId || 0}/sign`, {

```

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        pin,
        doctor_name: doctorName,
        signature_note: `Signed electronically by ${doctorName} on ${new Date().toISOString()}`,
    });
    document.querySelector('.fixed-modal')?.remove();
    if (r.degraded) {
        showToast(tr('Encounter signed (with notice)', '?? ?????? ?? ??????'));
    } else {
        showToast(tr('? Encounter signed and closed!', '? ?? ?????? ?????? ??????!'));
    }
    // Refresh patient to With Doctor / Completed
    if (window._DS.selectedPatientId) {
        await API.put('/api/patients/' + window._DS.selectedPatientId, { status: 'Done' }).catch(() => {});
        window.dsRefreshWaitQueue();
    }
} catch (e) {
    if (errEl) errEl.textContent = e?.message || tr('Signing failed', '??? ??????');
}
};

/* ===== */
/* ADD PROBLEM (stub) */
/* ===== */
window.dsAddProblem = function() {
    const pid = window._DS.selectedPatientId;
    if (!pid) return showToast(tr('No patient selected', '?? ??? ???? ????'), 'error');
    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
    modal.innerHTML = `<div style="background:var(--bg-card);border-radius:16px;padding:24px;width:400px">
        <h3 style="margin:0 0 16px;color:var(--primary)">?? ${tr('Add Problem', '???? ????')}</h3>
        <div class="form-group mb-8"><label>${tr('Problem Name', '??? ?????')}</label><input class="form-input"
        id="dpName" placeholder="Hypertension, DM Type 2..."></div>
        <div class="form-group mb-8"><label>ICD-10</label><input class="form-input" id="dpIcd" placeholder="I10, E
        11..."></div>
        <div class="form-group mb-8"><label>${tr('Status', '????')}</label>
        <select class="form-input" id="dpStatus">
            <option value="active">${tr('Active', '???')}</option>
            <option value="controlled">${tr('Controlled', '??? ?????')}</option>
            <option value="resolved">${tr('Resolved', '????')}</option>
        </select></div>
        <div class="form-group mb-12"><label>${tr('Onset Date', '???? ?????')}</label><input type="date" class=
        "form-input" id="dpOnset"></div>
        <div style="display:flex;gap:10px">
            <button class="btn btn-primary" style="flex:1" onclick="window.dsSubmitProblem(${pid})">? ${tr('Save', '
            ???')}</button>
            <button class="btn btn-secondary" style="flex:1" onclick="this.closest('.fixed-p').remove()">${tr('Cancel', '
            ?????')}</button>
        </div>
    </div>`;
    modal.classList.add('fixed-p');
    document.body.appendChild(modal);
    modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

```

```

window.dsSubmitProblem = async function(pid) {
  const name = document.getElementById('dpName')?.value?.trim() || '';
  if (!name) return showToast(tr('Enter problem name', '???? ?? ??????'), 'error');
  try {
    await API.post('/api/patients/' + pid + '/problems', {
      problem_name: name,
      icd_code: document.getElementById('dpIcd')?.value || '',
      status: document.getElementById('dpStatus')?.value || 'active',
      onset_date: document.getElementById('dpOnset')?.value || null,
    }).catch(() =>
      // Fallback: save via medical records notes
      API.post('/api/medical/records', {
        patient_id: pid,
        diagnosis: name + (document.getElementById('dpIcd')?.value ? ' [' + document.getElementById('dpIcd').v
alue + ']' : ''),
        treatment: '',
        notes: 'Problem: ' + name,
      })
    );
    showToast(tr('Problem added!', '??? ????? ??????!'));
    document.querySelector('.fixed-p')?.remove();
    setTimeout(() => window.dsSelectPatient(pid), 500);
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* REGISTRATION ? makes app.js delegation stub work */
/* ===== */
window.renderDoctorStation = renderDoctor;

// Also expose utility for drug interaction check (stub ? extend later with real DB)
window.checkDrugInteractions = async function(drugNames) {
  const knownInteractions = [
    { drugs: ['warfarin', 'aspirin'], severity: 'critical', message: tr('Major bleeding risk', '??? ????'
) },
    { drugs: ['metformin', 'contrast'], severity: 'critical', message: tr('Risk of lactic acidosis', '??? ????'
????????') },
    { drugs: ['ssri', 'maoi'], severity: 'critical', message: tr('Serotonin syndrome risk', '??? ?????? ?????
????????') },
    { drugs: ['warfarin', 'nsaid'], severity: 'critical', message: tr('Increased bleeding risk', '??? ????'
??') },
  ];
  const lowerNames = drugNames.map(n => n.toLowerCase());
  let hasCritical = false;
  let alerts = [];
  knownInteractions.forEach(ix => {
    const matched = ix.drugs.filter(d => lowerNames.some(n => n.includes(d)));
    if (matched.length >= 2) {
      if (ix.severity === 'critical') hasCritical = true;
      alerts.push({ ...ix, matchedDrugs: matched });
    }
  });
  return { hasCritical, alerts };
};

```

```

};

console.log('[DoctorStation] Module v3 loaded ? ???? ?????? ?????? Epic/Cerner/Oracle ??????');

/* ===== */
/* TAB: ORDERS BOARD (???? ?????? ??????) */
/* ===== */
window.dsTabOrders = async function(container) {
  const pid = window._DS.selectedPatientId;
  if (!pid) return;
  container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading orders...', '???? ?????? ???????...')}</p></div>`;
  let data = { orders: [], prescriptions: [] };
  try { data = await API.get(`/api/patients/${pid}/active-orders`); } catch { data = { orders: [], prescriptions: [] }; }
  const { orders = [], prescriptions = [] } = data;
  const allItems = [
    ...orders.map(o => ({ ...o, _cat: o.order_category || (o.is_radiology==1?'radiology':'lab'), _icon: o.is_radiology==1?'':'?'})),
    ...prescriptions.map(p => ({ ...p, _cat: 'medication', _icon: '?', description: p.drug_name || p.description })),
  ].sort((a, b) => new Date(b.created_at) - new Date(a.created_at));

  if (!allItems.length) {
    container.innerHTML = `<div style="text-align:center;padding:32px;color:var(--text-dim)"><div style="font-size:48px;opacity:.3">?</div><p>${tr('No active orders', '?? ????? ??????')}</p></div>`;
    return;
  }
  const statusColor = { Pending: '#ca8a04', STAT: '#dc2626', 'In Progress': '#0ea5e9', Completed: '#16a34a', Dispensed: '#16a34a', Cancelled: '#6b7280' };
  const catLabel = { lab: tr('Lab', '?????'), radiology: tr('Radiology', '?????'), medication: tr('Medication', '?????'), referral: tr('Referral', '?????'), nursing: tr('Nursing', '?????'), diet: tr('Diet', '?????'), iv: tr('IV Fluids', '?????') };
  container.innerHTML = `
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
      <strong style="font-size:13px">? ${tr('Active Orders Board', '???? ???????')} (${allItems.length})</strong>
    </div>
    ${allItems.map(o => `
      <div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:10px;padding:10px 12px;margin-bottom:8px;display:flex;align-items:center;gap:10px">
        <div style="font-size:22px">${escapeHTML(o._icon||'')}</div>
        <div style="flex:1">
          <div style="font-weight:700;font-size:12px">${escapeHTML(o.description || '-')}</div>
          <div style="font-size:11px;color:var(--text-dim)">${escapeHTML(catLabel[o._cat]||o._cat||'')} · ${escapeHTML(o.created_at?.split('T')[0]||'')}</div>
        </div>
        <span style="background:${statusColor[o.status]||'#6b7280'}22;color:${statusColor[o.status]||'#6b7280'};padding:2px 10px;border-radius:20px;font-size:11px;font-weight:700;white-space:nowrap">${escapeHTML(o.status||'')}</span>
      </div>
    `).join('')}
  `;
};

```

```

/* ===== */
/* TAB: RESULTS (???? ???? ???? ) */
/* ===== */
window.dsTabResults = async function(container) {
  const pid = window._DS.selectedPatientId;
  if (!pid) return;
  container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading results...', '???? ???? ??????...')}</p></div>`;
  let data = { labOrders: [], radOrders: [], labResults: [] };
  try { data = await API.get(`/api/patients/${pid}/lab-results`); } catch { data = { labOrders: [], radOrders: [], labResults: [] }; }
  const { labOrders = [], radOrders = [], labResults = [] } = data;

  if (!labOrders.length && !radOrders.length && !labResults.length) {
    container.innerHTML = `<div style="text-align:center;padding:32px;color:var(--text-dim)"><div style="font-size:48px;opacity:.3">?</div><p>${tr('No lab results found', '?? ???? ????? ????? ?????')}</p></div>`;
    return;
  }

  // Group labResults by test_name for latest value
  const byTest = {};
  labResults.forEach(r => { if (!byTest[r.test_name]) byTest[r.test_name] = []; byTest[r.test_name].push(r); });

  container.innerHTML = `
    ${labResults.length ? `
      <div style="margin-bottom:16px">
        <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">? ${tr('Lab Results', '???? ????')} (${labResults.length})</div>
        <table style="width:100%;border-collapse:collapse;font-size:12px">
          <thead>
            <tr style="background:var(--surface-container,#f1f5f9);">
              <th style="padding:6px 8px;text-align:start;font-weight:700">${tr('Test', '????')}</th>
              <th style="padding:6px 8px;text-align:start;font-weight:700">${tr('Result', '????')}</th>
              <th style="padding:6px 8px;text-align:start;font-weight:700">${tr('Unit', '????')}</th>
              <th style="padding:6px 8px;text-align:start;font-weight:700">${tr('Reference', '????')}</th>
              <th style="padding:6px 8px;text-align:start;font-weight:700">${tr('Date', '????')}</th>
            </tr>
          </thead>
          <tbody>
            ${labResults.map(r => {
              const isCrit = r.is_critical;
              const isAbn = r.is_abnormal || isCrit;
              const rowColor = isCrit ? '#fee2e2' : isAbn ? '#fef9c3' : '';
              const textColor = isCrit ? '#dc2626' : isAbn ? '#ca8a04' : 'inherit';
              const refRange = (r.reference_low!=null && r.reference_high!=null) ? `${r.reference_low} - ${r.reference_high}` : (r.reference_range || '-');
              return `<tr style="background:${rowColor};border-bottom:1px solid var(--border)">
                <td style="padding:6px 8px;font-weight:600;color:${textColor}">
                  ${isCrit ? '?' : isAbn ? '??' : ''}${escapeHTML(r.test_name||'-')}
                </td>
                <td style="padding:6px 8px;font-weight:700;color:${textColor}">${escapeHTML(String(r.result_value??r.result_text??'-'))}</td>
            `;
            })}
          </tbody>
        </table>
      `
    } : ''}
  `;

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        <td style="padding:6px 8px;color:var(--text-dim)">${escapeHTML(r.unit||'')}</td>
        <td style="padding:6px 8px;color:var(--text-dim)">${escapeHTML(refRange)}</td>
        <td style="padding:6px 8px;color:var(--text-dim)">${escapeHTML(r.order_date?.split('T')[0]||r.result_date?.split('T')[0]||'')}</td>
    </tr>`;
    }).join('')}
</tbody>
</table>
</div>` : ''}
${radOrders.length ? `
<div style="margin-top:16px">
    <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">? ${tr('Radiology Orders', '????? ?????')} (${radOrders.length})</div>
    ${radOrders.map(o => `
        <div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:8px;padding:10px 12px;margin-bottom:6px">
            <div style="font-weight:700;font-size:12px">${escapeHTML(o.description||o.order_type||'-')}</div>
            <div style="font-size:11px;color:var(--text-dim);margin-top:4px">${escapeHTML(o.created_at?.split('T')[0]||'')} · ???? : <strong>${escapeHTML(o.status||'-')}</strong></div>
            ${o.results ? `<div style="font-size:12px;margin-top:6px;padding:6px;background:#f0f9ff;border-radius:6px;border-inline-start:3px solid #0ea5e9">${escapeHTML(o.results)}</div>` : ''}
        </div>
    `).join('')}
</div>` : ''}
`;
};

/* ===== */
/* TAB: HISTORY+ (????? ???? : ??????? ?????? ??????) */
/* ===== */
window.dsTabHistoryExt = async function(container) {
    const pid = window._DS.selectedPatientId;
    if (!pid) return;
    container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading extended history...', '???? ????? ???????...')}</p></div>`;
    let data = { social: {}, family: [], surgical: [], immunizations: [] };
    try { data = await API.get(`/api/patients/${pid}/history-extended`); } catch { /* graceful */ }
    const { social = {}, family = [], surgical = [], immunizations = [] } = data;
    const smokingMap = { never: tr('Never Smoked', '?? ???? ??'), former: tr('Former Smoker', '??? ??????'), current: tr('Current Smoker', '???? ???????') };

    container.innerHTML = `
    <!-- Social History -->
    <div style="margin-bottom:16px">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:8px">
            <strong style="font-size:12px;color:var(--text-dim);text-transform:uppercase">? ${tr('Social History', '??????? ?????????')}</strong>
            <button class="btn btn-sm" style="font-size:10px;padding:4px 8px" onclick="window.dsEditSocialHistory(${pid})">?? ${tr('Edit', '????')}</button>
        </div>
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px">
            ${[['smoking_status', tr('Smoking', '???????'), smokingMap[social.smoking_status] || social.smoking_status || '-'],
            ['occupation', tr('Occupation', '???????'), social.occupation || '-'],

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        ['marital_status', tr('Marital Status', '????? ?????????'), social.marital_status || '-'],
        ['exercise_frequency', tr('Exercise', '???????'), social.exercise_frequency || '-'],
    ].map(([k, label, val]) => `
        <div style="background:var(--surface-container,#f8fafc);border-radius:8px;padding:8px 10px">
            <div style="font-size:10px;color:var(--text-dim);font-weight:700">${escapeHTML(label)}</div>
            <div style="font-size:13px;font-weight:600">${escapeHTML(val)}</div>
        </div>
    `).join(''))
</div>
</div>

<!-- Family History -->
<div style="margin-bottom:16px">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:8px">
        <strong style="font-size:12px;color:var(--text-dim);text-transform:uppercase">????? ${tr('Family History', '??????? ????????')} (${family.length})</strong>
        <button class="btn btn-sm" style="font-size:10px;padding:4px 8px" onclick="window.dsAddFamilyHistory(${pid})">+ ${tr('Add', '?????')}</button>
    </div>
    ${family.length ? family.map(f => `
        <div style="display:flex;gap:8px;align-items:center;padding:6px 0;border-bottom:1px solid var(--border)">
            <div style="font-size:18px">?????</div>
            <div><div style="font-weight:700;font-size:12px">${escapeHTML(f.condition||'-')}</div>
            <div style="font-size:11px;color:var(--text-dim)">${escapeHTML(f.relation||'')} ${f.icd_code?'· '+f.icd_code:''}</div></div>
        </div>
    `).join('') : `<div style="color:var(--text-dim);font-size:12px;padding:8px">? ${tr('None recorded', '????')}</div>`}
</div>

<!-- Surgical History -->
<div style="margin-bottom:16px">
    <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">? ${tr('Surgical History', '??????? ????????')} (${surgical.length})</div>
    ${surgical.length ? surgical.map(s => `
        <div style="background:var(--surface-container,#f8fafc);border-radius:8px;padding:8px 10px;margin-bottom:6px">
            <div style="font-weight:700;font-size:12px">${escapeHTML(s.procedure_name||'-')}</div>
            <div style="font-size:11px;color:var(--text-dim)">${escapeHTML(s.procedure_date||'')} ${s.hospital?'· '+s.hospital:''}</div>
        </div>
    `).join('') : `<div style="color:var(--text-dim);font-size:12px;padding:8px">? ${tr('None recorded', '????')}</div>`}
</div>

<!-- Immunizations -->
<div>
    <div style="font-weight:700;font-size:12px;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">? ${tr('Immunizations', '????????')} (${immunizations.length})</div>
    ${immunizations.length ? immunizations.map(i => `
        <div style="display:flex;gap:8px;align-items:center;padding:6px 0;border-bottom:1px solid var(--border)">
            <div style="font-size:18px">?</div>

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        <div><div style="font-weight:700;font-size:12px">${escapeHTML(i.vaccine_name||i.vaccine||'-')}</div>
        <div style="font-size:11px;color:var(--text-dim)">${escapeHTML(i.given_date?.split('T')[0]||i.vaccin
e_date?.split('T')[0]||'')} ${i.dose_number?. ' ' + i.dose_number:' '}</div></div>
    </div>
    `).join('') : `<div style="color:var(--text-dim);font-size:12px;padding:8px">? ${tr('None recorded', '??
????')}</div>`
</div>
`;
};

/* ===== */
/* EDIT SOCIAL HISTORY MODAL */
/* ===== */
window.dsEditSocialHistory = function(pid) {
    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.55);z-index:9999;display:flex;align-it
ems:center;justify-content:center';
    modal.innerHTML = `<div style="background:var(--bg-card);border-radius:16px;padding:24px;width:480px;max-hei
ght:90vh;overflow-y:auto">
    <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('Social History', '?????? ?????????')}</h3>
    <div class="form-group mb-8"><label>${tr('Smoking Status', '???? ?????????')}</label>
    <select class="form-input" id="shSmoke">
        <option value="never">${tr('Never', '?? ?????')}</option>
        <option value="former">${tr('Former Smoker', '??? ?????')}</option>
        <option value="current">${tr('Current Smoker', '???? ?????')}</option>
    </select></div>
    <div class="form-group mb-8"><label>${tr('Marital Status', '?????? ?????????')}</label>
    <select class="form-input" id="shMarital">
        <option value="Single">${tr('Single', '????')}</option>
        <option value="Married">${tr('Married', '????')}</option>
        <option value="Divorced">${tr('Divorced', '????')}</option>
        <option value="Widowed">${tr('Widowed', '????')}</option>
    </select></div>
    <div class="form-group mb-12"><label>${tr('Occupation', '??????')}</label><input class="form-input" id="sh
Occupation" placeholder="${tr('Job title', '?????? ?????????')}"></div>
    <div class="form-group mb-12"><label>${tr('Exercise Frequency', '????? ?????????')}</label>
    <select class="form-input" id="shExercise">
        <option value="None">${tr('None', '?? ?????')}</option>
        <option value="Occasional">${tr('Occasional', '????????')}</option>
        <option value="3x per week">${tr('3x per week', '3 ?????/????')}</option>
        <option value="Daily">${tr('Daily', '????????')}</option>
    </select></div>
    <div style="display:flex;gap:10px">
        <button class="btn btn-primary" style="flex:1" onclick="window.dsSaveSocialHistory(${pid})">? ${tr('Save
', '????')}</button>
        <button class="btn btn-secondary" style="flex:1" onclick="this.closest('[style*=fixed]').remove()">${tr(
'Cancel', '????')}</button>
    </div>
</div>`;
    document.body.appendChild(modal);
    modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

window.dsSaveSocialHistory = async function(pid) {

```

```

try {
  await API.post(`/api/patients/${pid}/social-history`, {
    smoking_status: document.getElementById('shSmoke')?.value || '',
    marital_status: document.getElementById('shMarital')?.value || '',
    occupation: document.getElementById('shOccupation')?.value || '',
    exercise_frequency: document.getElementById('shExercise')?.value || '',
  });
  showToast(tr('Social history saved!', '?? ??? ?????? ?????????!'));
  document.querySelector('[style*="position:fixed"][style*="inset:0"]')?.remove();
  window.dsSwitchTab('history_ext');
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.dsAddFamilyHistory = function(pid) {
  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.55);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = `<div style="background:var(--bg-card);border-radius:16px;padding:24px;width:400px">
    <h3 style="margin:0 0 16px;color:var(--primary)">????? ${tr('Add Family History', '????? ?????? ??????')}</h3>
    <div class="form-group mb-8"><label>${tr('Relation', '??? ??????')}</label>
      <select class="form-input" id="fhRelation">
        <option>${tr('Father', '????')}</option><option>${tr('Mother', '????')}</option>
        <option>${tr('Sibling', '??/????')}</option><option>${tr('Grandparent', '??/????')}</option>
      </select></div>
      <div class="form-group mb-8"><label>${tr('Condition', '????????')}</label><input class="form-input" id="fhCondition" placeholder="${tr('e.g. DM, HTN, Cancer', '????: ????, ???')}"></div>
      <div class="form-group mb-12"><label>ICD-10</label><input class="form-input" id="fhIcd" placeholder="E11, I10..."></div>
      <div style="display:flex;gap:10px">
        <button class="btn btn-primary" style="flex:1" onclick="window.dsSaveFamilyHistory(${pid})">? ${tr('Save', '????')}</button>
        <button class="btn btn-secondary" style="flex:1" onclick="this.closest('[style*=fixed]').remove()">${tr('Cancel', '????')}</button>
      </div>
    </div>`;
  document.body.appendChild(modal);
  modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

window.dsSaveFamilyHistory = async function(pid) {
  const condition = document.getElementById('fhCondition')?.value?.trim() || '';
  if (!condition) return showToast(tr('Enter condition', '???? ?????????'), 'error');
  try {
    await API.post(`/api/patients/${pid}/family-history`, {
      relation: document.getElementById('fhRelation')?.value || '',
      condition,
      icd_code: document.getElementById('fhIcd')?.value || '',
    });
    showToast(tr('Family history added!', '?? ?????? ????????? ?????????!'));
    document.querySelector('[style*="position:fixed"][style*="inset:0"]')?.remove();
    window.dsSwitchTab('history_ext');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

```

```

/* ===== */
/* TAB: PULMONOLOGY (S-MODE Gold Standard) */
/* ===== */
window.dsTabPulmonology = async function({ patient = {}, chart = {} }) {
  const records = chart.records || [];
  const pulmRecords = records.filter(r => r.department === 'pulmonology');
  const lastPulm = pulmRecords[0] || {};

  return `
    <div style="display:flex;flex-direction:column;gap:16px">
      <!-- PFT Analysis Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Pulmonary Function Test (PFT)', '????? ?????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenPftModal()">+ ${tr('New PFT', '????? ?
          ???')}</button>
        </div>
        <div id="pulmPftChart" style="height:200px;background:#f9fafb;border-radius:8px;margin-bottom:12px;dis
        play:flex;align-items:center;justify-content:center;color:var(--text-dim);border:1px dashed var(--border)">
          <div style="text-align:center">
            <div style="font-size:24px">?</div>
            <p style="font-size:11px">${tr('Flow-Volume Loop Visualization', '????? ?????? ??????')}</p>
          </div>
        </div>
        <div class="ds-vital-grid" style="grid-template-columns:repeat(3,1fr)">
          <div class="ds-vital-card">
            <div class="ds-vital-label">FEV1/FVC</div>
            <div class="ds-vital-value">${escapeHTML(lastPulm.fev1_fvc_ratio || '?')}</div>
          </div>
          <div class="ds-vital-card">
            <div class="ds-vital-label">GOLD Stage</div>
            <div class="ds-vital-value" style="color:var(--primary);font-weight:700">${escapeHTML(lastPulm.gol
            d_stage || '?')}</div>
          </div>
          <div class="ds-vital-card">
            <div class="ds-vital-label">DLCO</div>
            <div class="ds-vital-value">${escapeHTML(lastPulm.dlco_actual || '?')}</div>
          </div>
        </div>
      </div>

      <!-- Sleep Study Section -->
      <div class="card mb-12" style="border-radius:12px;padding:16px;border:1px solid var(--border)">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
          <strong style="font-size:13px">? ${tr('Sleep Study (Polysomnography)', '????? ?????')}</strong>
          <button class="btn btn-sm btn-primary" onclick="window.dsOpenSleepModal()">+ ${tr('New Study', '????
          ? ?????')}</button>
        </div>
        <div class="ds-problem-row" style="margin-bottom:8px">
          <div class="ds-med-icon">?</div>
          <div class="ds-med-info">
            <div class="ds-med-name">${tr('AHI Index:', '???? AHI:')} ${escapeHTML(lastPulm.ahi_index || '?')}

```

```

</div>
    <div class="ds-med-detail">${tr('Severity:', '?????:')} ${escapeHTML(lastPulm.sleep_severity || '?'
    ')}</div>
    </div>
    <div class="ds-med-status active">${escapeHTML(lastPulm.sleep_diagnosis || '?')}</div>
</div>

<!-- Cor Pulmonale Bridge (S-MODE) -->
<div class="ds-cds-alert warning" style="background:#fff7ed;border-left:4px solid #ea580c">
    <div class="ds-cds-alert-icon">?</div>
    <div class="ds-cds-alert-body">
        <div class="ds-cds-alert-title">${tr('Cardio-Pulmonary Bridge', '??? ????? ?????')}</div>
        <div class="ds-cds-alert-desc">${tr('Check for Cor Pulmonale risk based on current FEV1 and Right Ve
ntricle Pressure', '???? ?? ????? ?? ????? ?????? ?????? ??? FEV1 ??? ?????? ??????')}</div>
        <button class="btn btn-sm" onclick="window.dsCheckCorPulmonale()" style="margin-top:8px;background:#
ea580c;color:#fff;border:none;padding:4px 8px;font-size:10px">
            ? ${tr('Run Risk Analysis', '???? ?????')}
        </button>
    </div>
</div>
</div>
`;
};

```

```

window.dsOpenPftModal = async function() {
    const pid = window._DS.selectedPatientId;
    const encounterId = window._DS.activeEncounterId;
    if (!pid) return;

    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
    modal.innerHTML = `
        <div style="background:var(--bg-card);border-radius:16px;padding:24px;width:450px;box-shadow:0 10px 25px r
gba(0,0,0,0.2)">
            <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('PFT Results', '???? ?????? ????? ?????')}</h3>
            <div class="ds-soap-grid" style="grid-template-columns:1fr 1fr;gap:12px">
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FEV1 Actual</label>
                    <input class="form-input" id="pft_fev1_act" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FEV1 Predicted</label>
                    <input class="form-input" id="pft_fev1_pred" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FVC Actual</label>
                    <input class="form-input" id="pft_fvc_act" type="number" step="0.01">
                </div>
                <div class="form-group">
                    <label style="font-size:11px;font-weight:700">FVC Predicted</label>
                    <input class="form-input" id="pft_fvc_pred" type="number" step="0.01">
                </div>
            </div>
        </div>
    `;
}

```

```

    </div>
    <div class="form-group mt-12">
      <label style="font-size:11px;font-weight:700">${tr('Interpretation', '??????')}</label>
      <textarea class="form-input" id="pft_interp" rows="2"></textarea>
    </div>
    <div style="display:flex;gap:10px;margin-top:20px">
      <button class="btn btn-danger" onclick="this.closest('.fixed').remove()">? ${tr('Cancel', '????')}</b
utton>
      <button class="btn btn-primary" style="flex:1" onclick="window.dsSavePft()">? ${tr('Save & Analyze', '
??? ?????')}</button>
    </div>
  </div>
  `;
  modal.classList.add('fixed');
  document.body.appendChild(modal);
};

window.dsSavePft = async function() {
  const pid = window._DS.selectedPatientId;
  const data = {
    encounter_id: window._DS.activeEncounterId,
    fev1_actual: document.getElementById('pft_fev1_act').value,
    fev1_predicted: document.getElementById('pft_fev1_pred').value,
    fvc_actual: document.getElementById('pft_fvc_act').value,
    fvc_predicted: document.getElementById('pft_fvc_pred').value,
    interpretation: document.getElementById('pft_interp').value,
  };
  try {
    const res = await API.post('/api/pulmonology/pft', data);
    showToast(tr('PFT Saved & Analyzed!', '?? ??? ?????? PFT!'));
    window.dsSelectPatient(pid);
    document.querySelector('.fixed').remove();
  } catch (e) { showToast(e.message, 'error'); }
};

window.dsCheckCorPulmonale = async function() {
  const pid = window._DS.selectedPatientId;
  // In a real scenario, we'd fetch current FEV1 and Right Ventricle Pressure from DB
  const mockData = {
    patient_id: pid,
    fev1: 25, // Severe
    oxygenSat: 88,
    rightVentriclePressure: 30, // High
    ecgFindings: 'Right Axis Deviation'
  };
  try {
    const res = await API.post('/api/pulmonology/cor-pulmonale-check', mockData);
    const risk = res.riskAssessment;
    const color = risk.risk === 'HIGH' ? '#dc2626' : '#16a34a';
    showToast(`Risk: ${risk.risk} - ${risk.finding}`, risk.risk === 'HIGH' ? 'error' : 'success');
  } catch (e) { showToast(e.message, 'error'); }
};

```

=====

FILE: public\js\encounter-workflow.js

=====

/**

* public/js/encounter-workflow.js

* Safe Encounter Workflow Module for Multi-Facility Platform.

* Contains encounter types, journey steps, and access validation rules.

*/

window.ENCOUNTER_TYPES = {

OPD: { id: 'opd', name_en: 'Outpatient Department', name_ar: '????????? ?????????' },

INPATIENT: { id: 'inpatient', name_en: 'Inpatient Admission', name_ar: '????????? ?????????' },

ER_OBSERVATION: { id: 'er_obs', name_en: 'ER Observation', name_ar: '????????? ?????????' },

DAY_CASE: { id: 'day_case', name_en: 'Day Case Surgery', name_ar: '?????? ?????? ???????' }

};

window.getEncounterTypesForFacility = function(facilityType) {

if (facilityType === 'health_unit') {

return [window.ENCOUNTER_TYPES.OPD];

}

if (facilityType === 'phc' || facilityType === 'polyclinic') {

return [window.ENCOUNTER_TYPES.OPD];

}

// General Hospital and Medical City allow all

return Object.values(window.ENCOUNTER_TYPES);

};

window.canOpenEncounterForDepartment = function(facilityType, deptPage, encounterTypeId) {

// Check if department itself is allowed in the facility

const allowedDepts = window.getDepartmentsByFacilityType(facilityType);

if (allowedDepts !== null && !allowedDepts.includes(deptPage)) {

return false;

}

// Hospital-only inpatient rule

if (encounterTypeId === 'inpatient') {

if (['health_unit', 'phc', 'polyclinic'].includes(facilityType)) {

return false;

}

}

return true;

};

window.getEncounterJourneySteps = function(encounterTypeId) {

if (encounterTypeId === 'inpatient') {

return [

{ step: 1, name_en: 'Admission Request', name_ar: '??? ???????' },

{ step: 2, name_en: 'Bed Assignment', name_ar: '????? ???????' },

{ step: 3, name_en: 'Nursing Assessment', name_ar: '????????? ?????????' },

{ step: 4, name_en: 'Physician Rounds', name_ar: '????????? ?????? ???????' },

{ step: 5, name_en: 'Discharge Planning', name_ar: '??? ???????' }

];

```

    }
    // Default OPD Journey
    return [
      { step: 1, name_en: 'Registration / Triage', name_ar: '?????? ?????' },
      { step: 2, name_en: 'Nursing Vitals', name_ar: '?????? ?????' },
      { step: 3, name_en: 'Doctor Consultation', name_ar: '????? ??????' },
      { step: 4, name_en: 'Prescription / Referrals', name_ar: '????? ?????? ????????' }
    ];
  };

window.getEncounterWorkspaceCards = function(encounterTypeId) {
  return [
    { id: 'overview', name_en: 'Encounter Overview', name_ar: '???? ???? ???? ??????' },
    { id: 'triage', name_en: 'Nursing Assessment & Vitals', name_ar: '?????? ???? ?????? ????????' },
    { id: 'assessment', name_en: 'Clinical Assessment & SOAP', name_ar: '?????? ???? ???? ?????? ????????' },
    { id: 'orders', name_en: 'Orders (Labs/Radiology/Meds)', name_ar: '?????? (??????/??????/???????)' },
    { id: 'discharge', name_en: 'Discharge Summary / Follow-up', name_ar: '???? ?????? ????????' }
  ];
};

=====
FILE: public\js\endocrine-station.js
=====

/**
 * endocrine-station.js
 * Endocrinology & Diabetes Specialist Station - Stitch Google Design
 */

const EndocrineStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Endocrine Command Center', '???? ????
????? ??????')}</h1>
          <div class="flex gap-2">
            <button onclick="EndocrineStation.openGlucoseLog()" class="stitch-btn-primary px-4 py-
2 bg-blue-500 text-white rounded-lg shadow-sm hover:bg-blue-600 transition-all">
              ${tr('Log Glucose', '???? ???????')}
            </button>
            <button onclick="EndocrineStation.openThyroidMetrics()" class="stitch-btn-secondary px
-4 py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all
">
              ${tr('Thyroid Metrics', '????? ???? ???????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <!-- Left: Endocrine Context -->
    
```

```
<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
    <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Metabolic Vitals', '??????? ??????')}</h3>
    <div id="endocrine-vitals-grid" class="grid grid-cols-2 gap-4">
      <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
        <span class="block text-xs text-slate-500">${tr('HbA1c', '???? ?????????')}
      </span>

        <span class="text-lg font-bold text-slate-800">7.2%</span>
      </div>
      <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
        <span class="block text-xs text-slate-500">${tr('TSH', '???? TSH')}</span>

        <span class="text-lg font-bold text-slate-800">4.5 mIU/L</span>
      </div>
    </div>
  </div>

  <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
    <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('TIR (Time In Range)', '???? ?? ??????')}</h3>
    <div class="flex flex-col items-center">
      <div class="stitch-radial-gauge w-32 h-32 relative">
        <svg class="w-full h-full" viewBox="0 0 100 100">
          <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

          <circle cx="50" cy="50" r="40" stroke="#3b82f6" stroke-width="8" fill=
"none"

          stroke-dasharray="251.2" stroke-dashoffset="100" stroke-linecap="r
ound" />

        </svg>
        <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">65%</span>
      </div>
      <p class="mt-4 text-xs text-blue-600 font-medium">${tr('Stable - Target 70%',
'???? - ????? 70%')}</p>
    </div>
  </div>
</div>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-blue-500 text-blue-500 font-medium"
>${tr('Glucose Trends', '??????? ?????????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Thyroid Pa
nel', '???? ????? ?????????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('AI Predict
ion', '???? ?????? ?????????')}</button>
    </div>
    <div id="endocrine-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
```



```
-200 rounded-xl">
        ${tr('No active trend. Please log glucose data to begin.', '?? ??? ????
???. ??? ???? ???? ???? ???? ????.'))}
    </div>
</div>
</div>
</div>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-blue-50 rounded-xl shadow-sm border border-blue-100">
        <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">?</span>
            <h3 class="text-sm font-bold text-blue-900 uppercase tracking-wider">${tr('AI
Insights', '??? ????? ????')}</h3>
        </div>
        <div id="ai-insights-panel" class="space-y-4">
            <div class="p-3 bg-white rounded-lg border border-blue-200 shadow-sm">
                <p class="text-xs text-blue-700 font-semibold mb-1">${tr('Insulin Adjustme
nt', '???? ?????? ????')}</p>
                <p class="text-sm text-slate-600 italic">"Pattern suggests dawn phenomenon
. Suggest reducing basal insulin by 10%."</p>
                <button class="mt-2 text-xs text-blue-600 underline font-medium">${tr('Vie
w Evidence', '??? ?????')}</button>
            </div>
        </div>
    </div>
</div>
</div>
</div>
    `;
},

    openGlucoseLog: () => { alert('Glucose Log Modal Triggered'); },
    openThyroidMetrics: () => { alert('Thyroid Metrics Modal Triggered'); }
};

=====
FILE: public\js\enterprise-contracts.js
=====

/**
 * public/js/enterprise-contracts.js
 * Safe Enterprise Module Contracts and DTO / Schema Definitions.
 * Prepares the platform for API integration by defining read-only contracts and DTO shapes.
 */

window.CONTRACT_VERSIONS = {
    v1: '1.0.0-draft'
};

window.RESOURCE_TYPES = {
```

```
FACILITY: 'FACILITY',
DEPARTMENT: 'DEPARTMENT',
APPOINTMENT: 'APPOINTMENT',
QUEUE: 'QUEUE',
ENCOUNTER: 'ENCOUNTER',
ORDER: 'ORDER',
RESULT: 'RESULT',
DOCUMENTATION: 'DOCUMENTATION',
PHARMACY: 'PHARMACY',
BILLING: 'BILLING',
AUDIT: 'AUDIT'
};
```

```
window.ENDPOINT_DRAFTS = {
  GET_FACILITIES: '/api/v1/facilities',
  GET_DEPARTMENTS: '/api/v1/departments',
  GET_QUEUE: '/api/v1/queue',
  GET_ENCOUNTERS: '/api/v1/encounters',
  GET_ORDERS: '/api/v1/clinical-orders',
  GET_RESULTS: '/api/v1/clinical-results',
  GET_PHARMACY_REVIEW: '/api/v1/clinical-pharmacy',
  GET_BILLING_PREVIEW: '/api/v1/clinical-billing',
  POST_DISABLED_FINAL_ACTION: '/api/v1/actions/finalize'
};
```

```
window.DTO_SCHEMAS = {
  FacilityDTO: {
    id: 'string',
    name_en: 'string',
    name_ar: 'string',
    type: 'string'
  },
  DepartmentDTO: {
    id: 'number',
    name_en: 'string',
    name_ar: 'string',
    facilityType: 'string'
  },
  AppointmentDTO: {
    id: 'string',
    type: 'string',
    preferredDate: 'string'
  },
  QueueItemDTO: {
    queueNo: 'string',
    type: 'string',
    status: 'string',
    priority: 'string',
    waitTimeMin: 'number',
    deptPage: 'number'
  },
  EncounterDTO: {
    id: 'string',
    patientId: 'string',
```

```

        encounterTypeId: 'string'
    },
    ClinicalOrderDTO: {
        id: 'string',
        type: 'string',
        itemCode: 'string',
        status: 'string'
    },
    ClinicalResultDTO: {
        id: 'string',
        code: 'string',
        value: 'string',
        abnormal: 'boolean'
    },
    PharmacyReviewDTO: {
        id: 'string',
        medicationCode: 'string',
        status: 'string'
    },
    BillingPreviewDTO: {
        encounterId: 'string',
        totalAmount: 'number',
        copayPercent: 'number'
    },
    AuditPreviewDTO: {
        timestamp: 'string',
        role: 'string',
        action: 'string',
        status: 'string'
    }
};

window.getContractVersion = function() {
    return window.CONTRACT_VERSIONS.v1;
};

window.getResourceContract = function(resourceType) {
    return window.DTO_SCHEMAS[`${resourceType.charAt(0) + resourceType.slice(1).toLowerCase()}DTO`] || null;
};

window.getEndpointDraft = function(actionKey) {
    return window.ENDPOINT_DRAFTS[actionKey] || null;
};

window.validateDtoShapePreview = function(dtoName, data) {
    const schema = window.DTO_SCHEMAS[dtoName];
    if (!schema) return false;

    // Basic structural validation
    for (const key in schema) {
        if (!data.hasOwnProperty(key)) {
            return false;
        }
    }
}

```

```
        return true;
    };

    window.isLiveEndpointEnabled = function() {
        return false;
    };

    window.isWriteOperationEnabled = function() {
        return false;
    };
};
```

```
=====
FILE: public\js\enterprise-security.js
=====
```

```
/**
 * public/js/enterprise-security.js
 * Safe Enterprise RBAC/ABAC & Audit Hardening Module.
 * Defines roles, actions, risk levels, scope validations, and simulated audit logs.
 */
```

```
window.ROLES = {
    ADMIN: 'ADMIN',
    PHYSICIAN: 'PHYSICIAN',
    NURSE: 'NURSE',
    PHARMACIST: 'PHARMACIST',
    BILLING_OFFICER: 'BILLING_OFFICER',
    LAB_TECH: 'LAB_TECH',
    RADIOLOGY_TECH: 'RADIOLOGY_TECH',
    RECEPTIONIST: 'RECEPTIONIST',
    QUALITY_OFFICER: 'QUALITY_OFFICER',
    READ_ONLY_AUDITOR: 'READ_ONLY_AUDITOR'
};
```

```
window.SECURITY_ACTIONS = {
    VIEW_FACILITY: 'VIEW_FACILITY',
    VIEW_DEPARTMENT: 'VIEW_DEPARTMENT',
    PREVIEW_APPOINTMENT: 'PREVIEW_APPOINTMENT',
    PREVIEW_QUEUE: 'PREVIEW_QUEUE',
    PREVIEW_ENCOUNTER: 'PREVIEW_ENCOUNTER',
    PREVIEW_ORDER: 'PREVIEW_ORDER',
    PREVIEW_RESULT: 'PREVIEW_RESULT',
    PREVIEW_DOCUMENTATION: 'PREVIEW_DOCUMENTATION',
    PREVIEW_PHARMACY: 'PREVIEW_PHARMACY',
    PREVIEW BILLING: 'PREVIEW BILLING',

    // Final Actions (Always blocked in this phase)
    FINAL_BOOKING: 'FINAL_BOOKING',
    FINAL_CHECKIN: 'FINAL_CHECKIN',
    FINAL_CLINICAL_ORDER: 'FINAL_CLINICAL_ORDER',
    FINAL_SIGNATURE: 'FINAL_SIGNATURE',
    FINAL_DISPENSE: 'FINAL_DISPENSE',
```

```

    FINAL_INVOICE: 'FINAL_INVOICE',
    SUBMIT_CLAIM: 'SUBMIT_CLAIM',
    FINANCIAL_POSTING: 'FINANCIAL_POSTING'
  };

  window.canPreviewAction = function(role, action) {
    // Auditors and Admins can preview everything
    if (role === window.ROLES.ADMIN || role === window.ROLES.READ_ONLY_AUDITOR) {
      return true;
    }

    // Role-specific preview allowances
    if (action === window.SECURITY_ACTIONS.PREVIEW_BILLING && role !== window.ROLES.BILLING_OFFICER) {
      return false;
    }
    if (action === window.SECURITY_ACTIONS.PREVIEW_PHARMACY && role !== window.ROLES.PHARMACIST) {
      return false;
    }

    return true;
  };

  window.canPerformFinalAction = function(role, action) {
    // All final actions are strictly blocked in this simulation phase
    return false;
  };

  window.getActionRiskLevel = function(action) {
    const highRiskActions = [
      window.SECURITY_ACTIONS.FINAL_SIGNATURE,
      window.SECURITY_ACTIONS.FINAL_DISPENSE,
      window.SECURITY_ACTIONS.FINAL_INVOICE,
      window.SECURITY_ACTIONS.SUBMIT_CLAIM,
      window.SECURITY_ACTIONS.FINANCIAL_POSTING
    ];
    return highRiskActions.includes(action) ? 'high' : 'normal';
  };

  window.getRequiredRoleForAction = function(action) {
    switch (action) {
      case window.SECURITY_ACTIONS.PREVIEW_PHARMACY:
      case window.SECURITY_ACTIONS.FINAL_DISPENSE:
        return window.ROLES.PHARMACIST;
      case window.SECURITY_ACTIONS.PREVIEW_BILLING:
      case window.SECURITY_ACTIONS.FINAL_INVOICE:
      case window.SECURITY_ACTIONS.SUBMIT_CLAIM:
      case window.SECURITY_ACTIONS.FINANCIAL_POSTING:
        return window.ROLES.BILLING_OFFICER;
      case window.SECURITY_ACTIONS.FINAL_SIGNATURE:
        return window.ROLES.PHYSICIAN;
      default:
        return window.ROLES.READ_ONLY_AUDITOR;
    }
  };
};

```

```

window.getHumanReviewRequirement = function(action) {
    return window.getActionRiskLevel(action) === 'high';
};

window.getDisabledFinalActionReason = function(action) {
    return 'Simulated read-only environment. Final actions require live environment authorization.';
};

window.buildAuditPreviewEvent = function(role, action, context = {}) {
    return {
        timestamp: new Date().toISOString(),
        event: 'simulated_audit_log',
        role: role,
        action: action,
        riskLevel: window.getActionRiskLevel(action),
        facility: context.facility || 'unspecified',
        department: context.department || 'unspecified',
        status: 'BLOCKED_BY_GUARD'
    };
};

```

```

=====
FILE: public\js\facility-catalog.js
=====

```

```

/**
 * public/js/facility-catalog.js
 * Static Facility Catalog and Helper Functions for Multi-Facility Platform.
 */

window.FACILITY_CATALOG = {
    medical_cities: [
        { id: 1, name_en: 'King Fahad Medical City', name_ar: '????? ????? ??? ??????', region: 'Riyadh' },
        { id: 2, name_en: 'King Abdullah Medical City', name_ar: '????? ????? ??????? ??????', region: 'Makkah' }
    ],
    facility_types: [
        { code: 'medical_city', name_en: 'Medical City', name_ar: '????? ????' },
        { code: 'general_hospital', name_en: 'General Hospital', name_ar: '??????? ???' },
        { code: 'specialized_hospital', name_en: 'Specialized Hospital', name_ar: '??????? ?????' },
        { code: 'tertiary_hospital', name_en: 'Tertiary Hospital', name_ar: '??????? ?????' },
        { code: 'polyclinic', name_en: 'Polyclinic / Medical Complex', name_ar: '???? ??? / ??????' },
        { code: 'phc', name_en: 'Primary Healthcare Center', name_ar: '???? ?????? ????' },
        { code: 'specialty_center', name_en: 'Specialty Center', name_ar: '???? ??????' },
        { code: 'diagnostic_center', name_en: 'Diagnostic Center', name_ar: '???? ??????' },
        { code: 'rehabilitation_center', name_en: 'Rehabilitation Center', name_ar: '???? ?????? ???' },
        { code: 'dialysis_center', name_en: 'Dialysis Center', name_ar: '???? ????? ??????' },
        { code: 'dental_center', name_en: 'Dental Center', name_ar: '???? ??????' },
        { code: 'mental_health_center', name_en: 'Mental Health Center', name_ar: '???? ??? ??????' },
        { code: 'home_healthcare_unit', name_en: 'Home Healthcare Unit', name_ar: '???? ?????? ????? ??????' },
        { code: 'mobile_clinic', name_en: 'Mobile Clinic', name_ar: '????? ??????' },
        { code: 'virtual_clinic', name_en: 'Virtual Clinic', name_ar: '????? ?????????' },
    ]
}

```

```

    { code: 'health_unit', name_en: 'Health Unit', name_ar: '???? ????' },
    { code: 'emergency_center', name_en: 'Emergency Center', name_ar: '???? ????? ??????' }
  ],
  facilities: [
    { id: 101, city_id: 1, type: 'medical_city', name_en: 'KFMC Main Campus', name_ar: '??? ????? ????? ??? ??
    ????? ??????', status: 'Active', deptsCount: 45, desc: 'Central medical city with advanced tertiary care.' },
    { id: 102, city_id: 1, type: 'general_hospital', name_en: 'KFMC General Hospital', name_ar: '?????? ?????
    ????? ??? ??????', status: 'Active', deptsCount: 28, desc: 'General hospital serving emergency and surgery.' },
    { id: 103, city_id: 1, type: 'polyclinic', name_en: 'Sulaimaniyah Complex', name_ar: '???? ??????????? ?????
    ?', status: 'Active', deptsCount: 15, desc: 'Outpatient polyclinic with primary care.' },
    { id: 104, city_id: 1, type: 'health_unit', name_en: 'Olaya Primary Care Unit', name_ar: '???? ????????? ???
    ??? ??????', status: 'Active', deptsCount: 6, desc: 'Basic primary care and vaccinations.' },
    { id: 105, city_id: 1, type: 'virtual_clinic', name_en: 'Riyadh Virtual Clinic', name_ar: '????? ?????? ??
    ?????????', status: 'Active', deptsCount: 4, desc: 'Telemedicine consultations.' },
    { id: 201, city_id: 2, type: 'specialized_hospital', name_en: 'KAMC Oncology Center', name_ar: '???? ?????
    ?? ?????? ????? ??????', status: 'Active', deptsCount: 20, desc: 'Cancer treatment and research center.' },
    { id: 202, city_id: 2, type: 'phc', name_en: 'Makkah Primary Care Center', name_ar: '???? ????????? ??????
    ?????', status: 'Active', deptsCount: 8, desc: 'Family medicine and vaccinations.' }
  ]
};

window.getFacilitiesByMedicalCity = function(cityId) {
  return window.FACILITY_CATALOG.facilities.filter(f => f.city_id === cityId);
};

window.getDepartmentsByFacilityType = function(type) {
  // Returns list of departments enabled for a given facility type
  const map = {
    medical_city: null, // all enabled
    general_hospital: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 2
    4, 30, 33, 34, 42],
    specialized_hospital: [0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 13, 14, 15, 17, 18, 20, 34, 42],
    tertiary_hospital: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23,
    24, 30, 33, 34, 42],
    polyclinic: [0, 1, 2, 3, 4, 6, 8, 9, 13, 14, 15, 20, 34, 42],
    phc: [0, 1, 2, 3, 4, 6, 14, 15, 33, 34],
    specialty_center: [0, 1, 2, 3, 4, 6, 8, 14, 15, 20, 34],
    diagnostic_center: [3, 4, 14, 15],
    rehabilitation_center: [0, 1, 2, 14, 15, 24],
    dialysis_center: [0, 1, 2, 14, 15, 20],
    dental_center: [0, 1, 2, 14, 15, 21],
    mental_health_center: [0, 1, 2, 14, 15, 22],
    home_healthcare_unit: [0, 1, 2, 14, 15, 33],
    mobile_clinic: [0, 1, 2, 14, 15, 33],
    virtual_clinic: [0, 1, 2, 14, 15, 33],
    health_unit: [0, 1, 2, 3, 14, 15, 33],
    emergency_center: [0, 1, 2, 3, 14, 15, 19]
  };
  return map[type] !== undefined ? map[type] : map.general_hospital;
};

```

=====

FILE: public\js\gastro-station.js

=====

```
/**
 * gastro-station.js
 * Gastroenterology Specialist Station - Stitch Google Design
 */

const GastroStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Gastroenterology Command Center', '???
? ????? ?????? ?????? ??????')}</h1>
          <div class="flex gap-2">
            <button onclick="GastroStation.openEndoscopyReport()" class="stitch-btn-primary px-4 p
y-2 bg-orange-600 text-white rounded-lg shadow-sm hover:bg-orange-700 transition-all">
              ${tr('New Endoscopy Report', '????? ?????? ???')}
            </button>
            <button onclick="GastroStation.openHepatologyMetrics()" class="stitch-btn-secondary px
-4 py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all
">
              ${tr('Liver Metrics', '?????? ?????')}
            </button>
          </div_header>

          <div class="grid grid-cols-12 gap-6">
            <!-- Left: GI Context -->
            <div class="col-span-3 space-y-6">
              <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
                <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Liver Function', '????? ?????')}</h3>
                <div id="gastro-vitals-grid" class="grid grid-cols-2 gap-4">
                  <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('Bilirubin', '????????')}</span>

                    <span class="text-lg font-bold text-slate-800">2.4 mg/dL</span>
                  </div>
                  <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('Albumin', '????????')}</span>

                    <span class="text-lg font-bold text-slate-800">3.1 g/dL</span>
                  </div>
                </div_la>
              </div_la>

              <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
                <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('MELD Score', '???? MELD')}</h3>
                <div class="flex flex-col items-center">
                  <div class="stitch-radial-gauge w-32 h-32 relative">
                    <svg class="w-full h-full" viewBox="0 0 100 100">
```



```

        <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

        <circle cx="50" cy="50" r="40" stroke="#f59e0b" stroke-width="8" fill=
"none"

        stroke-dasharray="251.2" stroke-dashoffset="120" stroke-linecap="r
ound" />

        </svg>
        <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">22</span>
    </div>
    <p class="mt-4 text-xs text-amber-600 font-medium">${tr('Moderate Risk - Monit
or Closely', '?? ???? - ?????? ?????')}</p>
    </div_ la>
</div_ la>
</div_ la>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
    <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">

        <div class="flex border-b border-slate-200 mb-6">
            <button class="px-4 py-2 border-b-2 border-orange-600 text-orange-600 font-med
ium">${tr('Endoscopy', '????????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Hepatology
', '????? ?????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Motility',
'????? ?????? ?????')}</button>
        </div_ la>
        <div id="gastro-workspace-content" class="space-y-4">
            <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">

                ${tr('No active report. Please start a new endoscopy session.', '?? ???? ?
???? ????. ??? ???? ???? ?????? ??????.')}
            </div_ la>
        </div_ la>
    </div_ la>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-orange-50 rounded-xl shadow-sm border border-orange-100
">

        <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">?</span>
            <h3 class="text-sm font-bold text-orange-900 uppercase tracking-wider">${tr('A
I Insights', '?? ?????? ?????????')}</h3>
        </div_ la>
        <div id="ai-insights-panel" class="space-y-4">
            <div class="p-3 bg-white rounded-lg border border-orange-200 shadow- la>
                <p class="text-xs text-orange-700 font-semibold mb-1">${tr('Suggested Path
ology', '??????????? ?????????')}</p>
                <p class="text-sm text-slate-600 italic">"Findings are highly suggestive o
f Ulcerative Colitis. Similar to Case #1102."</p>
                <button class="mt-2 text-xs text-orange-600 underline font-medium">${tr('V
```

```

iew Evidence', '??? ?????')}</button>
        </div_ la
      </div_ la
    </div_ la
  </div_ la
</div_ la
</div_ la
  `;
},

openEndoscopyReport: () => {
  alert('Endoscopy Report Modal Triggered');
},

openHepatologyMetrics: () => {
  alert('Hepatology Metrics Modal Triggered');
}
};

```

```

=====
FILE: public\js\infectious-station.js
=====

```

```

/**
 * infectious-station.js
 * Infectious Disease Specialist Station - Stitch Google Design
 */

```

```

const InfectiousStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Infectious Disease Command Center', '?
          ??? ????? ??????? ???????')}</h1>
          <div class="flex gap-2">
            <button onclick="InfectiousStation.openCultureLog()" class="stitch-btn-primary px-4 py
            -2 bg-red-600 text-white rounded-lg shadow-sm hover:bg-red-700 transition-all">
              ${tr('Log Culture Result', '????? ????? ???????')}
            </button>
            <button onclick="InfectiousStation.openASPReview()" class="stitch-btn-secondary px-4 p
            y-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('ASP Review', '?????? ????????? ???????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t

```

```
r('Isolation Status', '???? ?????')}</h3>
    <div class="p-3 bg-red-100 text-red-700 rounded-lg text-center font-bold">
        ${tr('Contact Precautions', '???????? ???????')}
    </div>
</div>
<div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
    <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Sepsis Risk', '??? ??????')}</h3>
    <div class="flex flex-col items-center">
        <div class="stitch-radial-gauge w-32 h-32 relative">
            <svg class="w-full h-full" viewBox="0 0 100 100">
                <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
                <circle cx="50" cy="50" r="40" stroke="#ef4444" stroke-width="8" fill=
"none"
                    stroke-dasharray="251.2" stroke-dashoffset="50" stroke-linecap="ro
und" />
            </svg>
            <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">High</span>
        </div>
        <p class="mt-4 text-xs text-red-600 font-medium">${tr('Sepsis Bundle Pending',
'???? ??????? ???? ??????')}</p>
    </div>
</div>

<div class="col-span-6 space-y-6">
    <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
        <div class="flex border-b border-slate-200 mb-6">
            <button class="px-4 py-2 border-b-2 border-red-600 text-red-600 font-medium">${
tr('Culture Results', '????? ??????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('ASP Review
', '????? ??????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('AI Analysi
s', '????? ?????? ??????')}</button>
        </div>
        <div id="infectious-workspace-content" class="space-y-4">
            <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
                ${tr('No active culture. Please log a new result to begin.', '?? ???? ????
? ?????. ??? ???? ???? ???? ????')}</div>
            </div>
        </div>
    </div>

<div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-red-50 rounded-xl shadow-sm border border-red-100">
        <div class="flex items-center gap-2 mb-4">
            <span class="text-xl"?</span>
            <h3 class="text-sm font-bold text-red-900 uppercase tracking-wider">${tr('AI I
nsights', '??? ?????? ??????')}</h3>
```

```

        </div>
        <div id="ai-insights-panel" class="space-y-4">
          <div class="p-3 bg-white rounded-lg border border-red-200 shadow-sm">
            <p class="text-xs text-red-700 font-semibold mb-1">${tr('Antibiotic Suggest
tion', '????? ?????? ??????')}</p>
            <p class="text-sm text-slate-600 italic">"Pathogen shows resistance to Cef
triaxone. Suggest switching to Meropenem based on local antibiogram."</p>
            <button class="mt-2 text-xs text-red-600 underline font-medium">${tr('View
Evidence', '??? ??????')}</button>
          </div>
        </div>
      </div>
    </div>
  </div>
  `;
},

openCultureLog: () => { alert('Culture Log Modal Triggered'); },
openASPReview: () => { alert('ASP Review Modal Triggered'); }
};

```

```

/**
 * public/js/local-api-preview-ui.js
 * =====
 * Premium Read-Only Local Mock API Preview Panel.
 * Renders a glassmorphic sidebar/widget showing in-memory API & Contract simulation.
 * 100% client-side, zero fetch, zero database, zero external connections.
 * =====
 */

(function() {
  if (typeof window === 'undefined') return;

  window.addEventListener('DOMContentLoaded', () => {
    if (window.location.hostname === 'jumanasoft.com' || window.location.hostname === 'www.jumanasoft.com') {
      return;
    }

    // Create the toggle button
    const toggleBtn = document.createElement('button');
    toggleBtn.id = 'mockApiPreviewToggle';
    toggleBtn.innerHTML = '?? ?????? ?????? ???????? (Mock API)';
    toggleBtn.style.cssText = `
      position: fixed;
      bottom: 20px;
      left: 20px;
      z-index: 9999;
      background: linear-gradient(135deg, #0f172a, #1e293b);
    `;
  });
})();

```

```

color: #38bdf8;
border: 1px solid #38bdf8;
padding: 10px 16px;
border-radius: 30px;
font-family: 'IBM Plex Sans Arabic', sans-serif;
font-size: 13px;
font-weight: 600;
cursor: pointer;
box-shadow: 0 4px 20px rgba(0, 0, 0, 0.4);
transition: all 0.3s ease;
`;

// Create the main preview panel
const panel = document.createElement('div');
panel.id = 'mockApiPreviewPanel';
panel.style.cssText = `
position: fixed;
bottom: 80px;
left: 20px;
width: 420px;
max-height: 80vh;
z-index: 9999;
background: rgba(15, 23, 42, 0.85);
backdrop-filter: blur(16px);
-webkit-backdrop-filter: blur(16px);
border: 1px solid rgba(255, 255, 255, 0.1);
border-radius: 16px;
font-family: 'IBM Plex Sans Arabic', sans-serif;
color: #f1f5f9;
box-shadow: 0 10px 30px rgba(0, 0, 0, 0.5);
display: none;
flex-direction: column;
overflow: hidden;
transition: all 0.3s ease;
`;

// Panel Header
const header = document.createElement('div');
header.style.cssText = `
padding: 14px 16px;
background: rgba(30, 41, 59, 0.5);
border-bottom: 1px solid rgba(255, 255, 255, 0.05);
display: flex;
justify-content: space-between;
align-items: center;
`;
header.innerHTML = `
<div style="display: flex; align-items: center; gap: 8px;">
  <span style="font-size: 18px;">??</span>
  <span style="font-weight: 700; font-size: 14px; color: #38bdf8;">????? ???????? ???????? ??????????</spa
n>

</div>
<button id="closeMockApiPanel" style="background:none; border:none; color:#94a3b8; cursor:pointer; font-
size: 16px;">?</button>

```

```

`;

// Panel Body
const body = document.createElement('div');
body.style.cssText = `
padding: 16px;
overflow-y: auto;
flex: 1;
display: flex;
flex-direction: column;
gap: 12px;
font-size: 12px;
`;

body.innerHTML = `
  <!-- Warning Alert -->
  <div style="background: rgba(239, 68, 68, 0.15); border: 1px solid rgba(239, 68, 68, 0.3); border-radius
: 8px; padding: 10px; color: #fca5a5;">
    <strong>???? ????:</strong> ??? ???? ???? ???? ???? ??. ??? ???? ? ???? ???? ? ? ?
    ???? ???????.
  </div>

  <!-- Resource Selector -->
  <div>
    <label style="display:block; margin-bottom: 6px; font-weight:600; color: #94a3b8;">??? ?????? ?????/?
    ?????:</label>
    <select id="mockResourceSelector" style="width:100%; padding: 8px; background: #1e293b; border: 1px so
lid #475569; border-radius: 6px; color:#fff; font-family: inherit;">
      ${window.supportedResources || []}.map(r => `<option value="${r}">${r}</option>`).join('')}
    </select>
  </div>

  <!-- Action Selector (GET vs POST Simulation) -->
  <div>
    <label style="display:block; margin-bottom: 6px; font-weight:600; color: #94a3b8;">??? ?????? ??????
    ?:</label>
    <select id="mockActionSelector" style="width:100%; padding: 8px; background: #1e293b; border: 1px soli
d #475569; border-radius: 6px; color:#fff; font-family: inherit;">
      <option value="GET">????? ???? (GET)</option>
      <option value="POST">???? ???? (POST)</option>
    </select>
  </div>

  <!-- Metadata Panel -->
  <div style="background: rgba(30, 41, 59, 0.3); border: 1px solid rgba(255, 255, 255, 0.05); border-radiu
s: 8px; padding: 12px; display: flex; flex-direction: column; gap: 8px;">
    <div style="display:flex; justify-content:space-between;"><span style="color:#94a3b8;">????? ??????
    :</span> <code id="mockEndpoint" style="color:#60a5fa;">-</code></div>
    <div style="display:flex; justify-content:space-between;"><span style="color:#94a3b8;">???? ???????:
    </span> <span id="mockResponseStatus" style="font-weight:700;">-</span></div>
    <div style="display:flex; justify-content:space-between;"><span style="color:#94a3b8;">??????? ????
    ?? (RBAC):</span> <span id="mockRbac" style="color:#e2e8f0;">-</span></div>
    <div style="display:flex; justify-content:space-between;"><span style="color:#94a3b8;">???? ??????:</
    span> <span style="color:#f87171; font-weight:600;">???? ?</span></div>

```

```

        <div style="display:flex; justify-content:space-between;"><span style="color:#94a3b8;">??????? </span> <span style="color:#f87171; font-weight:600;">???? </span></div>
    </div>

    <!-- Audit Log Preview -->
    <div>
        <span style="font-weight:600; color:#94a3b8; display:block; margin-bottom: 6px;">????? (Audit Event Preview):</span>
        <pre id="mockAuditPreview" style="margin:0; padding:10px; background:#0f172a; border: 1px solid #334155; border-radius:6px; color:#34d399; font-size:11px; overflow-x:auto; max-height: 80px;"></pre>
    </div>

    <!-- Mock Response JSON -->
    <div style="flex: 1; display: flex; flex-direction: column; min-height: 150px;">
        <span style="font-weight:600; color:#94a3b8; display:block; margin-bottom: 6px;">??????? (Mock JSON Response):</span>
        <pre id="mockJsonResponse" style="margin:0; padding:10px; background:#0f172a; border: 1px solid #334155; border-radius:6px; color:#38bdf8; font-size:11px; overflow:auto; flex: 1; max-height: 200px;"></pre>
    </div>
`;

panel.appendChild(header);
panel.appendChild(body);
document.body.appendChild(toggleBtn);
document.body.appendChild(panel);

// Toggle logic
toggleBtn.addEventListener('click', () => {
    if (panel.style.display === 'none') {
        panel.style.display = 'flex';
        toggleBtn.style.background = '#38bdf8';
        toggleBtn.style.color = '#0f172a';
        updatePreview();
    } else {
        panel.style.display = 'none';
        toggleBtn.style.background = 'linear-gradient(135deg, #0f172a, #1e293b)';
        toggleBtn.style.color = '#38bdf8';
    }
});

document.getElementById('closeMockApiPanel').addEventListener('click', () => {
    panel.style.display = 'none';
    toggleBtn.style.background = 'linear-gradient(135deg, #0f172a, #1e293b)';
    toggleBtn.style.color = '#38bdf8';
});

// Event Listeners for Selector changes
const resourceSelect = document.getElementById('mockResourceSelector');
const actionSelect = document.getElementById('mockActionSelector');

resourceSelect.addEventListener('change', updatePreview);
actionSelect.addEventListener('change', updatePreview);

function updatePreview() {

```

```

const resource = resourceSelect.value;
const action = actionSelect.value;

const endpoint = window.getEndpointDraft ? window.getEndpointDraft(`GET_${resource.toUpperCase().replace(
('-', '_'))}`) || `/api/v1/${resource}` : `/api/v1/${resource}`;
document.getElementById('mockEndpoint').innerText = action === 'GET' ? endpoint : `/api/v1/actions/final
ize`;

// Get RBAC requirement from enterprise-security.js if loaded
let requiredRole = 'READ_ONLY_AUDITOR';
if (window.SECURITY_ACTIONS && window.getRequiredRoleForAction) {
  const actionKey = `PREVIEW_${resource.toUpperCase().replace('-', '_')}`;
  const secAction = window.SECURITY_ACTIONS[actionKey];
  if (secAction) {
    requiredRole = window.getRequiredRoleForAction(secAction);
  }
}
document.getElementById('mockRbac').innerText = requiredRole;

// Call mock api runtime
if (window.getMockApiResponse) {
  const response = window.getMockApiResponse(resource, {
    action: action,
    endpoint: action === 'GET' ? endpoint : '/api/v1/actions/finalize'
  });

  // Update UI
  const statusEl = document.getElementById('mockResponseStatus');
  statusEl.innerText = response.status;
  if (response.status === 'SUCCESS') {
    statusEl.style.color = '#34d399';
  } else if (response.status === 'BLOCKED') {
    statusEl.style.color = '#f87171';
  } else {
    statusEl.style.color = '#fbbf24';
  }

  document.getElementById('mockJsonResponse').innerText = JSON.stringify(response, null, 2);

  // Update Audit Log
  if (window.getMockAuditPreview) {
    const auditEvent = window.getMockAuditPreview(resource, action, requiredRole);
    document.getElementById('mockAuditPreview').innerText = JSON.stringify(auditEvent, null, 2);
  }
} else {
  document.getElementById('mockJsonResponse').innerText = '// Mock API Runtime not loaded.';
}
});
})();

```

=====

FILE: public\js\login-ui.js

=====

```
// Login page UI interactions (externalized from inline <script> for CSP readiness).
// Behavior is identical to the prior inline block; loaded at end of <body> so DOM elements exist.
```

```
// Modal toggle interactions
```

```
const showBtn = document.getElementById('showLoginModalBtn');
const closeBtn = document.getElementById('closeLoginModalBtn');
const modal = document.getElementById('loginModal');
const modalContainer = modal.querySelector('div');
```

```
function openModal() {
  modal.classList.remove('hidden');
  // Force reflow
  modal.offsetHeight;
  modal.classList.add('opacity-100');
  modalContainer.classList.remove('scale-95');
  modalContainer.classList.add('scale-100');
  document.getElementById('username').focus();
}
```

```
function closeModal() {
  modal.classList.remove('opacity-100');
  modalContainer.classList.remove('scale-100');
  modalContainer.classList.add('scale-95');
  setTimeout(() => {
    modal.classList.add('hidden');
  }, 300);
}
```

```
showBtn.addEventListener('click', openModal);
closeBtn.addEventListener('click', closeModal);
modal.addEventListener('click', (e) => {
  if (e.target === modal) closeModal();
});
```

```
// Also trigger modal from any call-to-action button
document.querySelectorAll('.trigger-login-modal').forEach(btn => {
  btn.addEventListener('click', (e) => {
    e.preventDefault();
    openModal();
  });
});
```

```
// Simple scroll reveal interaction
```

```
document.addEventListener('DOMContentLoaded', () => {
  const observerOptions = {
    threshold: 0.1
  };

  const observer = new IntersectionObserver((entries) => {
    entries.forEach(entry => {
      if (entry.isIntersecting) {
```

```

        entry.target.classList.add('opacity-100', 'translate-y-0');
        entry.target.classList.remove('opacity-0', 'translate-y-10');
    }
});
}, observerOptions);

document.querySelectorAll('section > div').forEach(el => {
    el.classList.add('transition-all', 'duration-700', 'opacity-0', 'translate-y-10');
    observer.observe(el);
});
});

```

```

=====
FILE: public\js\login.js
=====

```

```

document.getElementById('loginForm').addEventListener('submit', async (e) => {
    e.preventDefault();
    const btn = document.getElementById('loginBtn');
    const errEl = document.getElementById('errorMsg');

    // Premium loading state
    btn.disabled = true;
    btn.classList.add('loading', 'opacity-80', 'cursor-not-allowed');
    const originalText = btn.innerHTML;
    btn.innerHTML = `
        <svg class="animate-spin h-5 w-5 text-white" xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0
0 24 24" style="margin-inline-end: 8px; display: inline-block;">
            <circle class="opacity-25" cx="12" cy="12" r="10" stroke="currentColor" stroke-width="4"></circle>
            <path class="opacity-75" fill="currentColor" d="M4 12a8 8 0 018-8V0C5.373 0 0 5.373 0 12h4zm2 5.29
1A7.962 7.962 0 014 12H0c0 3.042 1.135 5.824 3 7.938l3-2.647z"></path>
        </svg>
        ??? ?????.
    `;
    errEl.classList.remove('show');
    errEl.classList.add('hidden');

    try {
        const res = await fetch('/api/auth/login', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({
                username: document.getElementById('username').value.trim(),
                password: document.getElementById('password').value.trim()
            })
        });
    };
    const data = await res.json();
    if (data.success) {
        window.location.href = '/';
    } else {
        errEl.textContent = '?? ???? ?? ???? ???? ????!';
        errEl.classList.add('show');
    }
}

```

```

        errEl.classList.remove('hidden');
        btn.disabled = false;
        btn.classList.remove('loading', 'opacity-80', 'cursor-not-allowed');
        btn.innerHTML = originalText;
    }
} catch (err) {
    errEl.textContent = '??? ?? ??????? ????????? ????? ????????? ??????';
    errEl.classList.add('show');
    errEl.classList.remove('hidden');
    btn.disabled = false;
    btn.classList.remove('loading', 'opacity-80', 'cursor-not-allowed');
    btn.innerHTML = originalText;
}
});

```

```

// Check if already logged in
fetch('/api/auth/me').then(r => r.json()).then(d => {
    if (d.user) window.location.href = '/';
}).catch(() => { });

// Load Theme
fetch('/api/settings').then(r => r.json()).then(s => {
    if (s.theme) document.documentElement.setAttribute('data-theme', s.theme);
}).catch(() => { });

```

```

=====
FILE: public\js\mock-api-runtime.js
=====

```

```

/**
 * public/js/mock-api-runtime.js
 * =====
 * Local Mock API Runtime Prototype for NamaMedical.
 * Provides a 100% read-only, client-side mock runtime for enterprise modules.
 * Strictly decoupled from any real database, external network, or staging/production servers.
 * =====
 */

```

```

(function() {
    const root = typeof window !== 'undefined' ? window : (global.window || {});

    // 1. Core Flags & Config
    root.mockApiRuntimeEnabled = true;
    root.liveApiRuntimeEnabled = false;
    root.writeRuntimeEnabled = false;

    root.supportedResources = [
        'facilities',
        'departments',
        'appointments',
        'queue',
        'encounters',
    ]

```

```

'clinical-orders',
'clinical-results',
'clinical-pharmacy',
'clinical-billing',
'audit-events'
];

// 2. Mock Data Store (Purely synthetic, 100% compliant with enterprise DTO contracts, zero PHI)
root.MOCK_DATA_STORE = {
  facilities: [
    { id: 'FAC-001', name_en: 'Al-Faisal Medical City', name_ar: '????? ?????? ??????', type: 'HOSPITAL' },
    { id: 'FAC-002', name_en: 'Nama Outpatient Clinic', name_ar: '?????? ??? ?????????', type: 'CLINIC' }
  ],
  departments: [
    { id: 101, name_en: 'Cardiology Clinic', name_ar: '????? ?????? ?????????? ?????????', facilityType: 'HOSPITAL' },
    { id: 102, name_en: 'Radiology Imaging Unit', name_ar: '????? ?????? ?????????? ??????', facilityType: 'HOSPITAL' },
    { id: 103, name_en: 'Main Pharmacy', name_ar: '????????? ??????????', facilityType: 'HOSPITAL' }
  ],
  appointments: [
    { id: 'APT-9901', type: 'CARDIOLOGY', preferredDate: '2026-07-01T10:00:00Z' },
    { id: 'APT-9902', type: 'RADIOLOGY', preferredDate: '2026-07-01T11:30:00Z' }
  ],
  queue: [
    { queueNo: 'C-01', type: 'CLINIC', status: 'WAITING', priority: 'NORMAL', waitTimeMin: 15, deptPage: 101 },
    { queueNo: 'R-02', type: 'RADIOLOGY', status: 'ACTIVE', priority: 'URGENT', waitTimeMin: 5, deptPage: 102 }
  ],
  encounters: [
    { id: 'ENC-2026-001', patientId: 'PAT-MOCK-8801', encounterTypeId: 'OPD_CONSULTATION' },
    { id: 'ENC-2026-002', patientId: 'PAT-MOCK-8802', encounterTypeId: 'IPD_ADMISSION' }
  ],
  'clinical-orders': [
    { id: 'ORD-5501', type: 'LABORATORY', itemCode: 'CBC_TEST', status: 'PENDING' },
    { id: 'ORD-5502', type: 'RADIOLOGY', itemCode: 'CHEST_XRAY', status: 'COMPLETED' }
  ],
  'clinical-results': [
    { id: 'RES-3301', code: 'WBC_COUNT', value: '7.5 x10^9/L', abnormal: false },
    { id: 'RES-3302', code: 'HEMOGLOBIN', value: '11.2 g/dL', abnormal: true }
  ],
  'clinical-pharmacy': [
    { id: 'RX-7701', medicationCode: 'AMOXICILLIN_500MG', status: 'REVIEWED' },
    { id: 'RX-7702', medicationCode: 'WARFARIN_5MG', status: 'HOLD_INTERACTION_RISK' }
  ],
  'clinical-billing': [
    { encounterId: 'ENC-2026-001', totalAmount: 150.00, copayPercent: 20 },
    { encounterId: 'ENC-2026-002', totalAmount: 2500.00, copayPercent: 10 }
  ],
  'audit-events': [
    { timestamp: '2026-06-30T04:00:00Z', role: 'PHYSICIAN', action: 'READ_CLINICAL_RECORD', status: 'SUCCESS' },
    { timestamp: '2026-06-30T04:15:00Z', role: 'PHARMACIST', action: 'REVIEW_MEDICATION', status: 'SUCCESS' }
  ]
};

```

```

}
]
};

// 3. Safety Assertion Helpers
root.assertNoWriteOperation = function(action) {
  const upperAction = (action || '').toUpperCase();
  if (['POST', 'PUT', 'DELETE', 'PATCH', 'WRITE', 'FINALIZE', 'DISPENSE', 'SUBMIT'].includes(upperAction)) {
    throw new Error('WRITE_OPERATION_DISABLED: Cannot perform write/final actions in local mock runtime.');
```

```
  }
};
```

```

root.assertNoLiveEndpoint = function(endpoint) {
  if (root.isLiveEndpointEnabled && root.isLiveEndpointEnabled()) {
    throw new Error('SECURITY_VIOLATION: Live endpoints must remain disabled.');
```

```
  }
};
```

```

root.assertNoPhiPayload = function(payload) {
  if (!payload) return;
  const phiKeys = ['ssn', 'nationalid', 'iqama', 'phone', 'patientname', 'realname', 'creditcard'];
  const jsonStr = JSON.stringify(payload).toLowerCase();
  for (const key of phiKeys) {
    if (jsonStr.includes(`"${key}"`)) {
      throw new Error('SECURITY_VIOLATION: PHI payload or sensitive personal data is strictly forbidden in this mock runtime.');
```

```
    }
  }
};
```

```

// 4. Core Query & Router Simulation
root.validateMockRequestScope = function(resource, action) {
  if (!root.supportedResources.includes(resource)) {
    return { valid: false, reason: 'RESOURCE_NOT_SUPPORTED' };
  }
  try {
    root.assertNoWriteOperation(action);
    return { valid: true };
  } catch (err) {
    return { valid: false, reason: 'WRITE_OPERATION_DISABLED', message: err.message };
  }
};
```

```

root.buildReadOnlyResponse = function(resource, data) {
  return {
    status: 'SUCCESS',
    source: 'LOCAL MOCK API RUNTIME',
    timestamp: new Date().toISOString(),
    resource: resource,
  };
};
```

```

        count: Array.isArray(data) ? data.length : 1,
        data: data
    };
};

root.buildErrorResponse = function(status, reason, message) {
    return {
        status: status || 'ERROR',
        source: 'LOCAL MOCK API RUNTIME',
        timestamp: new Date().toISOString(),
        reason: reason || 'UNKNOWN_ERROR',
        message: message || 'An error occurred.'
    };
};

root.getMockAuditPreview = function(resource, action, userRole) {
    return {
        timestamp: new Date().toISOString(),
        role: userRole || 'GUEST',
        action: `${action || 'READ'}_${(resource || '').toUpperCase()}`,
        status: 'MOCK_AUDITED'
    };
};

root.getMockApiResponse = function(resource, params = {}) {
    const action = params.action || 'GET';
    const endpoint = params.endpoint || `/api/v1/${resource}`;

    // Safety boundaries check
    root.assertNoLiveEndpoint(endpoint);
    if (params.payload) {
        root.assertNoPhiPayload(params.payload);
    }

    const scope = root.validateMockRequestScope(resource, action);
    if (!scope.valid) {
        return root.buildErrorResponse('BLOCKED', scope.reason, scope.message);
    }

    const data = root.MOCK_DATA_STORE[resource];
    if (!data) {
        return root.buildErrorResponse('NOT_FOUND', 'RESOURCE_NOT_FOUND', `Resource ${resource} has no mock data
    `);
    }

    // Attempt DTO shape validation if enterprise-contracts.js is loaded
    if (root.validateDtoShapePreview && root.getResourceContract) {
        // Find DTO name based on resource mapping
        const dtoMap = {
            facilities: 'FacilityDTO',
            departments: 'DepartmentDTO',
            appointments: 'AppointmentDTO',
            queue: 'QueueItemDTO',
            encounters: 'EncounterDTO',

```

```

    'clinical-orders': 'ClinicalOrderDTO',
    'clinical-results': 'ClinicalResultDTO',
    'clinical-pharmacy': 'PharmacyReviewDTO',
    'clinical-billing': 'BillingPreviewDTO',
    'audit-events': 'AuditPreviewDTO'
  };

  const dtoName = dtoMap[resource];
  if (dtoName) {
    const items = Array.isArray(data) ? data : [data];
    for (const item of items) {
      const isValid = root.validateDtoShapePreview(dtoName, item);
      if (!isValid) {
        return root.buildErrorResponse(
          'CONTRACT_VIOLATION',
          'DTO_SHAPE_MISMATCH',
          `Mock data item in ${resource} does not match the contract DTO: ${dtoName}`
        );
      }
    }
  }

  return root.buildReadOnlyResponse(resource, data);
};

// Expose to Node.js if applicable
if (typeof module !== 'undefined' && module.exports) {
  module.exports = root;
}
})();

=====
FILE: public\js\nephrology-station.js
=====

/**
 * nephrology-station.js
 * Nephrology Specialist Station - Stitch Google Design
 */

const NephrologyStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Nephrology Command Center', '???? ????
? ????? ????? ????? ?????')}</h1>
          <div class="flex gap-2">
            <button onclick="NephrologyStation.openDialysisSession()" class="stitch-btn-primary px
-4 py-2 bg-blue-700 text-white rounded-lg shadow-sm hover:bg-blue-800 transition-all">

```

```
        ${tr('New Dialysis Session', '???? ???? ?????')}
    </button>
    <button onclick="NephrologyStation.openTransplantFollowup()" class="stitch-btn-secondary px-4 py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">

        ${tr('Transplant Follow-up', '?????? ???????')}
    </button>
</div_header>

<div class="grid grid-cols-12 gap-6">
    <!-- Left: Renal Context -->
    <div class="col-span-3 space-y-6">
        <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
            <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${tr('Renal Vitals', '???????? ???????')}</h3>
            <div id="nephro-vitals-grid" class="grid grid-cols-2 gap-4">
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('Creatinine', '????????????')}
                </span>

                    <span class="text-lg font-bold text-slate-800">2.1 mg/dL</span>
                </div>
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('Potassium', '????????????')}
                </span>

                    <span class="text-lg font-bold text-slate-800">5.2 mEq/L</span>
                </div>
            </div_la>
        </div_la>

        <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
            <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${tr('GFR Stage', '????? GFR')}</h3>
            <div class="flex flex-col items-center">
                <div class="stitch-radial-gauge w-32 h-32 relative">
                    <svg class="w-full h-full" viewBox="0 0 100 100">
                        <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

                        <circle cx="50" cy="50" r="40" stroke="#3b82f6" stroke-width="8" fill=
"none"

                        stroke-dasharray="251.2" stroke-dashoffset="150" stroke-linecap="r
ound" />

                    </svg>
                    <span class="absolute inset-0 flex items-center justify-center text-xl font-
t-bold text-slate-800">28 ml/min</span>
                </div>
                <p class="mt-4 text-xs text-blue-600 font-medium">${tr('Stage 4 CKD - Severe',
'???????? ??????? - ????? ?????')}</p>
            </div_la>
        </div_la>
    </div_la>

    <!-- Center: Active Workspace -->
    <div class="col-span-6 space-y-6">
        <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
```



```
h-[600px]">
```

```
        <div class="flex border-b border-slate-200 mb-6">
            <button class="px-4 py-2 border-b-2 border-blue-700 text-blue-700 font-medium"
>${tr('Dialysis Log', '??? ?????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Transplant
', '????????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Biopsy AI'
, '???? ?????????')}</button>
        </div_ la
        <div id="nephro-workspace-content" class="space-y-4">
            <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
                ${tr('No active session. Please start a new dialysis log to begin.', '?? ?
??? ???? ?????. ??? ???? ???? ???? ???? ?????.')}
            </div_ la
        </div_ la
    </div_ la
</div_ la

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-blue-50 rounded-xl shadow-sm border border-blue-100">
        <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">?</span>
            <h3 class="text-sm font-bold text-blue-900 uppercase tracking-wider">${tr('AI
Insights', '??? ?????? ?????????')}</h3>
        </div_ la
        <div id="ai-insights-panel" class="space-y-4">
            <div class="p-3 bg-white rounded-lg border border-blue-200 shadow-sm">
                <p class="text-xs text-blue-700 font-semibold mb-1">${tr('Biopsy Suggestio
n', '?????? ??????')}</p>
                <p class="text-sm text-slate-600 italic">"Pattern strongly suggests FSGS.
Similar to Case #2201. Recommend ACEi titration."</p>
                <button class="mt-2 text-xs text-blue-600 underline font-medium">${tr('Vie
w Evidence', '??? ??????')}</button>
            </div_ la
        </div_ la
    </div_ la
</div_ la
</div_ la
</div_ la
    `;
},

openDialysisSession: () => {
    alert('Dialysis Session Modal Triggered');
},

openTransplantFollowup: () => {
    alert('Transplant Follow-up Modal Triggered');
}
};
```

```

=====
FILE: public\js\nursing-station.js
=====

/**
 * =====
 * NURSING STATION v1 ? World-Class Clinical Workstation
 * ??? ???? ? ???? (Epic / Cerner / Oracle Health)
 * =====
 * Three-Panel Layout:
 *   LEFT  ? Patient Worklist ?? EWS Colors ???? ????
 *   CENTER ? 10 ???? ????
 *   RIGHT ? Quick Actions: Vitals / Alerts / Escalate / SBAR
 * =====
 * v1 Features:
 *   1. Vitals + EWS/NEWS2 Auto-Score + Trend
 *   2. eMAR ? MAR Sheet ???? + Five Rights
 *   3. Head-to-Toe Assessment (Neuro/Resp/CV/GI/GU/Skin)
 *   4. Nursing Care Plan (NANDA/NIC/NOC)
 *   5. Intake & Output Tracker + Daily Balance
 *   6. SBAR Handover Report + Print
 *   7. Triage (ESI 1-5) + Room Assignment
 *   8. Risk Assessments (Morse + Braden + DVT + qSOFA)
 *   9. Nursing Orders Board (Acknowledge + Execute)
 *  10. Nursing Notes (SOAP + Progress + Templates)
 * =====
 */

'use strict';

/* ---- Global State ---- */
window._NS = window._NS || {
  selectedPatientId: null,
  selectedPatientData: null,
  activeTab: 'vitals',
  ioEntries: [],          // Intake/Output entries for current session
  worklist: [],           // Patient worklist
  refreshTimer: null,
};

/* ===== */
/* NEWS2 SCORING ENGINE */
/* ===== */
function calcNEWS2({ rr, spo2, temp, sbp, hr, avpu, onO2 }) {
  let score = 0;
  const details = {};

  // Respiratory Rate
  if (rr <= 8) { score += 3; details.rr = { val: rr, pts: 3, flag: 'critical' }; }
  else if (rr <= 11) { score += 1; details.rr = { val: rr, pts: 1, flag: 'low' }; }
  else if (rr <= 20) { score += 0; details.rr = { val: rr, pts: 0, flag: 'normal' }; }
  else if (rr <= 24) { score += 1; details.rr = { val: rr, pts: 1, flag: 'low' }; }
  else { score += 3; details.rr = { val: rr, pts: 3, flag: 'critical' }; }
}

```

```

// SpO2 (Scale 1)
if      (spo2 <= 91) { score += 3; details.spo2 = { val: spo2, pts: 3, flag: 'critical' }; }
else if (spo2 <= 93) { score += 2; details.spo2 = { val: spo2, pts: 2, flag: 'high' }; }
else if (spo2 <= 95) { score += 1; details.spo2 = { val: spo2, pts: 1, flag: 'low' }; }
else      { score += 0; details.spo2 = { val: spo2, pts: 0, flag: 'normal' }; }

// O2 supplemental
if (onO2) { score += 2; }

// Temperature
if      (temp <= 35.0) { score += 3; details.temp = { val: temp, pts: 3, flag: 'critical' }; }
else if (temp <= 36.0) { score += 1; details.temp = { val: temp, pts: 1, flag: 'low' }; }
else if (temp <= 38.0) { score += 0; details.temp = { val: temp, pts: 0, flag: 'normal' }; }
else if (temp <= 39.0) { score += 1; details.temp = { val: temp, pts: 1, flag: 'low' }; }
else      { score += 2; details.temp = { val: temp, pts: 2, flag: 'high' }; }

// Systolic BP
if      (sbp <= 90) { score += 3; details.sbp = { val: sbp, pts: 3, flag: 'critical' }; }
else if (sbp <= 100) { score += 2; details.sbp = { val: sbp, pts: 2, flag: 'high' }; }
else if (sbp <= 110) { score += 1; details.sbp = { val: sbp, pts: 1, flag: 'low' }; }
else if (sbp <= 219) { score += 0; details.sbp = { val: sbp, pts: 0, flag: 'normal' }; }
else      { score += 3; details.sbp = { val: sbp, pts: 3, flag: 'critical' }; }

// Heart Rate
if      (hr <= 40) { score += 3; details.hr = { val: hr, pts: 3, flag: 'critical' }; }
else if (hr <= 50) { score += 1; details.hr = { val: hr, pts: 1, flag: 'low' }; }
else if (hr <= 90) { score += 0; details.hr = { val: hr, pts: 0, flag: 'normal' }; }
else if (hr <= 110) { score += 1; details.hr = { val: hr, pts: 1, flag: 'low' }; }
else if (hr <= 130) { score += 2; details.hr = { val: hr, pts: 2, flag: 'high' }; }
else      { score += 3; details.hr = { val: hr, pts: 3, flag: 'critical' }; }

// Consciousness (AVPU)
if (avpu && avpu !== 'A') { score += 3; details.avpu = { val: avpu, pts: 3, flag: 'critical' }; }
else { details.avpu = { val: avpu || 'A', pts: 0, flag: 'normal' }; }

const risk = score >= 7 ? 'high' : score >= 5 ? 'medium' : 'low';
const riskColor = score >= 7 ? '#dc2626' : score >= 5 ? '#ca8a04' : '#16a34a';
const riskLabel = score >= 7
  ? tr('HIGH ? Escalate Urgently', '???? ? ???? ????')
  : score >= 5 ? tr('MEDIUM ? Monitor Closely', '????? ? ?????? ??????')
  : tr('LOW ? Routine Monitoring', '????? ? ?????? ??????');
return { score, risk, riskColor, riskLabel, details };
}

/* ===== */
/* MORSE FALL RISK CALCULATOR */
/* ===== */
function calcMorse({ fallHistory, secDx, aidType, iv, gait, mentalStatus }) {
  const pts = {
    fallHistory: fallHistory ? 25 : 0,
    secDx:      secDx ? 15 : 0,
    aid:        aidType === 'furniture' ? 30 : aidType === 'crutch' ? 15 : 0,
    iv:         iv ? 20 : 0,
  };

```

```

    gait:      gait === 'impaired' ? 20 : gait === 'weak' ? 10 : 0,
    mental:    mentalStatus === 'forgets' ? 15 : 0,
  };
  const total = Object.values(pts).reduce((a, b) => a + b, 0);
  const risk = total >= 45 ? 'high' : total >= 25 ? 'low' : 'no';
  const label = risk === 'high' ? tr('High Risk', '????? ?????') : risk === 'low' ? tr('Low Risk', '????? ?????') : tr('No Risk', '?? ?????');
  const color = risk === 'high' ? '#dc2626' : risk === 'low' ? '#ca8a04' : '#16a34a';
  return { total, risk, label, color, pts };
}

/* ===== */
/*  BRADEN PRESSURE INJURY CALCULATOR  */
/* ===== */
function calcBraden({ sensory, moisture, activity, mobility, nutrition, friction }) {
  const total = (sensory || 4) + (moisture || 4) + (activity || 4) + (mobility || 4) + (nutrition || 4) + (friction || 3);
  const risk = total <= 9 ? 'very-high' : total <= 12 ? 'high' : total <= 14 ? 'moderate' : total <= 18 ? 'mild' : 'no';
  const label = {
    'very-high': tr('Very High Risk', '????? ????? ?????'),
    'high':      tr('High Risk', '????? ?????'),
    'moderate':  tr('Moderate Risk', '????? ??????'),
    'mild':      tr('Mild Risk', '????? ?????'),
    'no':        tr('No Risk', '?? ?????'),
  }[risk];
  const color = total <= 12 ? '#dc2626' : total <= 18 ? '#ca8a04' : '#16a34a';
  return { total, risk, label, color };
}

/* ===== */
/*  MAIN ENTRY POINT  */
/* ===== */
async function renderNursingStation(el) {
  if (window._NS.refreshTimer) { clearInterval(window._NS.refreshTimer); window._NS.refreshTimer = null; }
  window._NS.selectedPatientId = null;
  window._NS.ioEntries = [];

  // Skeleton
  el.innerHTML = `
    <div class="page-title">???? ${tr('Nursing Station', '???? ???????')}
      <span style="font-size:12px;font-weight:400;color:var(--text-dim);margin-inline-start:12px">
        ${tr('Epic-class Clinical Workstation', '???? ?????? ?????? ?????')}
      </span>
    </div>
    <div class="ns-layout" id="nsLayout">
      <div class="ns-left-panel"><div class="ns-panel-header"><span>? ${tr('Loading...', '???? ???????...')}</span></div>
      <div class="ns-center-panel" style="align-items:center;justify-content:center">
        <div style="text-align:center;padding:40px;color:var(--text-dim)"><div style="font-size:64px;opacity:0.3">????</div><p>${tr('Loading...', '???? ???????...')}</p></div>
      </div>
      <div class="ns-right-panel"><div class="ns-panel-header"><span>? ${tr('Actions', '??????')}</span></div>
    </div>
  `

```

```

    </div>
`;

let worklist = [], emarOrders = [], currentUser = {};
try {
    [worklist, emarOrders, currentUser] = await Promise.all([
        API.get('/api/queue/patients').catch(() => API.get('/api/patients').then(p => p.filter(x => x.status ===
'Waiting' || x.status === 'With Nurse' || x.status === 'With Doctor'))),
        API.get('/api/emar/orders').catch(() => []),
        API.get('/api/auth/me').catch(() => ({})),
    ]);
} catch (e) {
    el.innerHTML = `<div class="page-title">???? ${tr('Nursing Station', '???? ???????')}</div>
<div class="error-card-premium"><div class="error-card-icon">??</div>
<h3>${tr('Failed to load', '??? ???????')}</h3><p>${escapeHTML(e.message)}</p>
<button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ???????')}</
button>
</div>`;
    return;
}

if (!Array.isArray(worklist)) worklist = [];
if (!Array.isArray(emarOrders)) emarOrders = [];
window._NS.worklist = worklist;
window._NS.emarOrders = emarOrders;
window._NS.currentUser = currentUser.user || currentUser;

el.innerHTML = `
<div class="page-title" style="margin-bottom:12px">
    ??? ${tr('Nursing Station', '???? ???????')}
    <span style="font-size:12px;font-weight:500;color:var(--text-dim);margin-inline-start:12px">
        ${tr('Nurse:', '?????/?')} <strong>${escapeHTML(window._NS.currentUser?.name || window._NS.currentUs
er?.username || 'Nurse')}</strong>
    </span>
    <span style="font-size:12px;font-weight:500;color:var(--text-dim);margin-inline-start:12px">
        ? ${new Date().toLocaleDateString(isArabic ? 'ar-SA' : 'en-US', { weekday:'long', year:'numeric', mont
h:'long', day:'numeric' })}
    </span>
    <span style="font-size:11px;font-weight:600;color:var(--text-dim);margin-inline-start:auto;background:va
r(--surface-container,#f1f5f9);padding:4px 12px;border-radius:20px">
        ? ${tr('Shift', '??????')}: ${getShiftName()}
    </span>
</div>
<div class="ns-layout" id="nsLayout">

    <!-- LEFT PANEL: Patient Worklist -->
    <div class="ns-left-panel" id="nsWorklist">
        <div class="ns-panel-header">
            <span>? ${tr('Patient Worklist', '????? ???????')}</span>
            <div class="ns-badge-count" id="nsWorklistCount">${worklist.length}</div>
        </div>
        <div id="nsWorklistBody" style="flex:1;overflow-y:auto"></div>
        <div style="padding:10px;border-top:1px solid var(--border)">
            <button class="btn btn-sm w-full" onclick="window.nsRefreshWorklist()"

```

```

        style="background:var(--primary-glow);color:var(--primary);border:1px solid var(--primary);font-size:11px">
            ? ${tr('Refresh', '????')}
        </button>
    </div>
</div>

<!-- CENTER PANEL -->
<div class="ns-center-panel" id="nsCenterPanel">
    <div id="nsCenterContent" style="flex:1;display:flex;align-items:center;justify-content:center;flex-direction:column;padding:40px;color:var(--text-dim);text-align:center">
        <div style="font-size:72px;margin-bottom:20px;opacity:0.3">?</div>
        <h3 style="font-size:18px;font-weight:700;margin-bottom:8px;color:var(--text-dim)">${tr('Select a patient', '???? ?????')}</h3>
        <p style="font-size:13px;opacity:0.7">${tr('Patient chart will appear here', '???? ????? ?????? ?')}</p>
    </div>
</div>

<!-- RIGHT PANEL: Quick Actions -->
<div class="ns-right-panel" id="nsActionsPanel">
    <div class="ns-panel-header"><span>? ${tr('Quick Actions', '????? ????')}</span></div>
    <div id="nsActionsContent" style="padding:16px;flex:1;overflow-y:auto">
        <div style="text-align:center;padding:32px;color:var(--text-dim)">
            <div style="font-size:36px;opacity:0.25">?</div>
            <p style="font-size:12px;margin-top:8px">${tr('Select a patient first', '???? ????? ????')}</p>
        </div>
    </div>
</div>

</div>
`;

window.nsRenderWorklist(worklist);
window._NS.refreshTimer = setInterval(window.nsRefreshWorklist, 90000);
}

/* ===== */
/*  HELPERS                                     */
/* ===== */
function getShiftName() {
    const h = new Date().getHours();
    if (h >= 7 && h < 15) return tr('Morning (07:00-15:00)', '????? (07-15)');
    if (h >= 15 && h < 23) return tr('Evening (15:00-23:00)', '????? (15-23)');
    return tr('Night (23:00-07:00)', '????? (23-07)');
}

function getEWSColor(score) {
    if (score >= 7) return '#dc2626';
    if (score >= 5) return '#ca8a04';
    if (score >= 1) return '#0ea5e9';
    return '#16a34a';
}

```

```

/* ===== */
/*  LEFT PANEL: WORKLIST                                     */
/* ===== */
window.nsRenderWorklist = function(patients) {
    const body = document.getElementById('nsWorklistBody');
    const count = document.getElementById('nsWorklistCount');
    if (!body) return;
    // Queue rows carry the queue-row id in `id` and the real patient id in `patient_id`.
    // Resolve to the patient id and drop duplicate queue rows for the same patient.
    if (!Array.isArray(patients)) patients = [];
    const seenPids = new Set();
    patients = patients.filter(p => {
        const pid = p.patient_id || p.id;
        if (!pid || seenPids.has(pid)) return false;
        seenPids.add(pid);
        return true;
    });
    if (count) count.textContent = patients.length;

    if (!patients.length) {
        body.innerHTML = `
```

```

        <span style="background:${statusColor}22;color:${statusColor};font-size:9px;padding:1px 6px;border
-radius:10px;font-weight:700">${escapeHTML(statusLabel)}</span>
        ${ewsScore > 0 ? `<span style="background:${ewsColor}22;color:${ewsColor};font-size:9px;padding:1p
x 6px;border-radius:10px;font-weight:700">NEWS2: ${ewsScore}</span>` : ''}
        ${p.triage_level ? `<span style="background:${tColor}22;color:${tColor};font-size:9px;padding:1px
6px;border-radius:10px;font-weight:700">T${p.triage_level}</span>` : ''}
    </div>
</div>
    <div style="font-size:11px;color:var(--text-dim);white-space:nowrap">${p.exam_room_name ? '? ' + escap
eHTML(p.exam_room_name) : ''}</div>
</div>`;
    }).join('');
};

window.nsRefreshWorklist = async function() {
    try {
        const q = await API.get('/api/queue/patients').catch(() =>
            API.get('/api/patients').then(p => p.filter(x => x.status === 'Waiting' || x.status === 'With Nurse' ||
x.status === 'With Doctor'))
        );
        window._NS.worklist = Array.isArray(q) ? q : [];
        window.nsRenderWorklist(window._NS.worklist);
    } catch (e) { /* silent */ }
};

/* ===== */
/*  SELECT PATIENT ? Load Chart                                */
/* ===== */
window.nsSelectPatient = async function(patientId) {
    if (!patientId) return;
    window._NS.selectedPatientId = patientId;
    window._NS.activeTab = 'vitals';
    window._NS.ioEntries = [];

    document.querySelectorAll('.ns-worklist-item').forEach(el => {
        el.classList.toggle('active', parseInt(el.dataset.pid) === patientId);
    });

    const center = document.getElementById('nsCenterContent') || document.getElementById('nsCenterPanel');
    if (center) center.innerHTML = `<div style="padding:40px;text-align:center;color:var(--text-dim)"><div style
="font-size:32px;margin-bottom:12px">?</div><p>${tr('Loading chart...', '???? ????? ??????...')}</p></div>`;

    let chart = {}, vitals = [], problems = [], allergies = [], medications = [], emarOrders = [];
    try {
        [chart, vitals, problems, allergies, medications, emarOrders] = await Promise.all([
            API.get(`/api/patients/${patientId}/chart`).catch(() => ({})),
            API.get(`/api/nursing/vitals/${patientId}`).catch(() => []),
            API.get(`/api/patients/${patientId}/problems`).catch(() => []),
            API.get(`/api/patients/${patientId}/allergies`).catch(() => []),
            API.get(`/api/patients/${patientId}/medications`).catch(() => []),
            API.get('/api/emar/orders').catch(() => []),
        ]);
    } catch (e) { console.error('Chart load error:', e); }
}

```



```

// API.get resolves with the error JSON (not a throw) on 4xx/5xx ? normalize shapes
const asArray = v => (Array.isArray(v) ? v : []);
vitals = asArray(vitals);
problems = asArray(problems);
allergies = asArray(allergies);
medications = asArray(medications);
emarOrders = asArray(emarOrders);
if (!chart || typeof chart !== 'object' || chart.error) chart = {};

const patient = chart.patient || {};
window._NS.selectedPatientData = { patient, chart, vitals, problems, allergies, medications, emarOrders };

// Compute NEWS2 from latest vitals
const latestV = {};
vitals.forEach(v => { if (!latestV[v.score_type]) latestV[v.score_type] = v.score_value; });
const bpSys = parseInt((latestV['blood_pressure'] || '0/0').split('/')[0]) || 0;
const news2 = calcNEWS2({
  rr: parseFloat(latestV['respiratory_rate'] || 0),
  spo2: parseFloat(latestV['spo2'] || 99),
  temp: parseFloat(latestV['temperature'] || 37),
  sbp: bpSys,
  hr: parseFloat(latestV['pulse'] || 75),
  avpu: latestV['consciousness'] || 'A',
  onO2: false,
});
window._NS.selectedPatientData.news2 = news2;

window.nsRenderPatientChart(patient, chart, vitals, problems, allergies, medications);
window.nsRenderActionsPanel(patient, news2);
};

/* ===== */
/*  CENTER PANEL: Patient Chart                                */
/* ===== */
window.nsRenderPatientChart = function(patient, chart, vitals, problems, allergies, medications) {
  const centerPanel = document.getElementById('nsCenterPanel');
  if (!centerPanel) return;

  const name = isArabic ? (patient.name_ar || patient.name_en || '-') : (patient.name_en || patient.name_ar || '-');
  const initial = (name || '?').charAt(0).toUpperCase();
  const news2 = window._NS.selectedPatientData?.news2 || { score: 0, riskColor: '#16a34a', riskLabel: '' };

  // Allergy chips
  const allergyBanner = allergies.length
    ? `
```

```

<div class="ns-patient-avatar">${escapeHTML(initial)}</div>
<div style="flex:1">
  <div style="display:flex;align-items:center;gap:8px;flex-wrap:wrap">
    <div class="ns-patient-name">${escapeHTML(name)}</div>
    <span style="background:${news2.riskColor}22;color:${news2.riskColor};padding:2px 10px;border-radius
:20px;font-size:11px;font-weight:800">
      NEWS2: ${news2.score} ? ${news2.riskLabel}
    </span>
  </div>
  <div style="display:flex;flex-wrap:wrap;gap:6px;margin-top:4px">
    ${patient.file_number ? `<span class="ns-meta-chip">?? ${escapeHTML(String(patient.file_number))}</s
pan>` : ''}
    ${patient.age ? `<span class="ns-meta-chip">? ${escapeHTML(String(patient.age))} ${tr('yrs', '???')}<
span>` : ''}
    ${patient.gender ? `<span class="ns-meta-chip">${patient.gender === '???'||patient.gender==='Male'?
?':'?'} ${escapeHTML(patient.gender)}</span>` : ''}
    ${patient.blood_type ? `<span class="ns-meta-chip">? ${escapeHTML(patient.blood_type)}</span>` : ''}
    ${patient.insurance_company ? `<span class="ns-meta-chip">? ${escapeHTML(patient.insurance_company)}
</span>` : ''}
  </div>
  ${allergyBanner}
</div>
<div style="display:flex;gap:8px;flex-shrink:0">
  <button class="btn btn-sm" onclick="window.nsMarkWithNurse(${safeId(patient.id)})"
    style="font-size:11px;padding:6px 10px;background:#0ea5e9;color:#fff;border:none">
    ??? ${tr('With Nurse', '?? ?????')}
  </button>
</div>
</div>

<!-- Tabs -->
<div class="ns-chart-tabs" id="nsChartTabs">
  ${[
    { id: 'vitals', icon: '??', en: 'Vitals + EWS', ar: '????????' },
    { id: 'emar', icon: '?', en: 'eMAR', ar: '????? ??????' },
    { id: 'h2t', icon: '?', en: 'Assessment', ar: '????????' },
    { id: 'careplan', icon: '?', en: 'Care Plan', ar: '??? ??????' },
    { id: 'io', icon: '?', en: 'I & O', ar: '????????' },
    { id: 'handover', icon: '?', en: 'Handover', ar: '????? ???????' },
    { id: 'triage', icon: '??', en: 'Triage', ar: '?????' },
    { id: 'risks', icon: '?', en: 'Risk Assess.', ar: '????? ????????' },
    { id: 'orders', icon: '?', en: 'Nsg Orders', ar: '????? ????????' },
    { id: 'notes', icon: '?', en: 'Notes', ar: '?????????' },
  ].map(t => `
    <div class="ns-tab ${t.id === window._NS.activeTab ? 'active' : ''}"
      onclick="window.nsSwitchTab('${t.id}')" data-tab="${t.id}">
      ${t.icon} ${tr(t.en, t.ar)}
    </div>
  `).join('')}
</div>

<!-- Tab Content -->
<div class="ns-tab-content" id="nsTabContent"></div>
';

```

```

    window.nsSwitchTab(window._NS.activeTab);
};

/* ===== */
/*  TAB SWITCHING */
/* ===== */
window.nsSwitchTab = function(tabId) {
    window._NS.activeTab = tabId;
    document.querySelectorAll('.ns-tab').forEach(el => el.classList.toggle('active', el.dataset.tab === tabId));
    const content = document.getElementById('nsTabContent');
    if (!content) return;
    const d = window._NS.selectedPatientData || {};
    const pid = window._NS.selectedPatientId;

    switch (tabId) {
        case 'vitals':    window.nsTabVitals(content, d, pid); break;
        case 'emar':      window.nsTabEMAR(content, d, pid); break;
        case 'h2t':       content.innerHTML = window.nsTabH2T(d); break;
        case 'careplan':  window.nsTabCarePlan(content, d, pid); break;
        case 'io':        window.nsTabIO(content, pid); break;
        case 'handover':  content.innerHTML = window.nsTabHandover(d, pid); window.nsLoadHandovers(pid); break;
        case 'triage':    content.innerHTML = window.nsTabTriage(d, pid); break;
        case 'risks':     content.innerHTML = window.nsTabRisks(d, pid); break;
        case 'orders':    window.nsTabOrders(content, pid); break;
        case 'notes':     content.innerHTML = window.nsTabNotes(d, pid); break;
        default: content.innerHTML = `<p>${tr('Coming soon', '?????')}</p>`;
    }
};

/* ===== */
/*  TAB 1: VITALS + EWS/NEWS2 */
/* ===== */
window.nsTabVitals = function(container, { patient = {}, vitals = [] }, pid) {
    const latestV = {};
    vitals.forEach(v => { if (!latestV[v.score_type]) latestV[v.score_type] = v.score_value; });
    const news2 = window._NS.selectedPatientData?.news2 || calcNEWS2({});

    container.innerHTML = `
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px">
            <!-- Vitals Entry Form -->
            <div>
                <div style="font-weight:700;font-size:13px;margin-bottom:12px;color:var(--primary)">
                    ?? ${tr('Record New Vitals', '????? ?????? ?????? ?????')}
                </div>
                <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px;margin-bottom:12px">
                    <div>
                        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('BP (mmHg)', '?????')}</l
abel>
                        <input class="form-input" id="nsVBp" placeholder="120/80" style="height:34px;font-size:13px">
                    </div>
                    <div>
                        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Heart Rate (bpm)', '????
?')}</label>

```

```

        <input class="form-input" id="nsVHr" type="number" placeholder="75" style="height:34px;font-size:1
3px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Temp (°C)', '??????')}<
/label>
        <input class="form-input" id="nsVTemp" type="number" step="0.1" placeholder="37.0" style="height:3
4px;font-size:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('RR (/min)', '???? ?????
')}</label>
        <input class="form-input" id="nsVRr" type="number" placeholder="18" style="height:34px;font-size:1
3px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">SpO2 (%)</label>
        <input class="form-input" id="nsVSpo2" type="number" placeholder="98" style="height:34px;font-size
:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Pain (0-10)', '???? 0-1
0')}</label>
        <input class="form-input" id="nsVPain" type="number" min="0" max="10" placeholder="0" style="heigh
t:34px;font-size:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Weight (kg)', '????')}<
/label>
        <input class="form-input" id="nsVWeight" type="number" step="0.1" placeholder="70" style="height:3
4px;font-size:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">GCS (3-15)</label>
        <input class="form-input" id="nsVGcs" type="number" min="3" max="15" placeholder="15" style="heigh
t:34px;font-size:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Blood Glucose', '????')
} (mg/dL)</label>
        <input class="form-input" id="nsVGlucose" type="number" placeholder="100" style="height:34px;font-
size:13px">
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">AVPU</label>
        <select class="form-input" id="nsVAvpu" style="height:34px;font-size:13px">
            <option value="A">A ? ${tr('Alert', '????')}</option>
            <option value="V">V ? ${tr('Voice', '????? ?????')}</option>
            <option value="P">P ? ${tr('Pain', '????? ?????')}</option>
            <option value="U">U ? ${tr('Unresponsive', '?? ?????')}</option>
        </select>
    </div>
    <div style="display:flex;align-items:flex-end;gap:6px">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim);white-space:nowrap">${tr('On 02

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?', '??? ?????????')}</label>
    <input type="checkbox" id="nsVOn02" style="width:18px;height:18px;margin-bottom:8px">
  </div>
</div>
<div class="form-group mb-8">
  <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Nursing Notes', '??????? ?
?????')}</label>
  <textarea class="form-input" id="nsVNotes" rows="2" placeholder="${tr('Any observations...', '?? ??
????...')}"></textarea>
</div>
<div style="display:flex;gap:8px">
  <button class="btn btn-primary" onclick="window.nsSaveVitals(${pid})" style="flex:1;height:40px;font
-size:13px">
    ? ${tr('Save & Calculate EWS', '??? ????? EWS')}
  </button>
</div>
</div>

<!-- NEWS2 Score + History -->
<div>
  <!-- Current NEWS2 Score -->
  <div style="background:${news2.riskColor}11;border:2px solid ${news2.riskColor};border-radius:16px;padding:16px;margin-bottom:16px;text-align:center">
    <div style="font-size:11px;font-weight:700;color:var(--text-dim);text-transform:uppercase;margin-bottom:4px">
      ? NEWS2 Score
    </div>
    <div style="font-size:52px;font-weight:900;color:${news2.riskColor};line-height:1">${news2.score}</div>
    <div style="font-size:12px;font-weight:700;color:${news2.riskColor};margin-top:4px">${news2.riskLabel}</div>
    ${news2.score >= 5 ? `
      <button class="btn" onclick="window.nsEscalate(${pid})"
        style="margin-top:12px;background:#dc2626;color:#fff;border:none;font-size:12px;font-weight:700;
width:100%">
        ? ${tr('ESCALATE TO DOCTOR', '????? ??? ????')}
      </button>
    ` : ''}
  </div>

  <!-- Vitals History -->
  <div style="font-weight:700;font-size:12px;margin-bottom:8px;color:var(--text-dim)">
    ? ${tr('Vitals History', '??? ???????? ?????')} (${vitals.length})
  </div>
  ${vitals.length ? `
    <div style="overflow-x:auto;font-size:11px">
      <table style="width:100%;border-collapse:collapse">
        <thead>
          <tr style="background:var(--surface-container,#f1f5f9)">
            <th style="padding:5px 8px;text-align:start">${tr('Time', '????')}</th>
            <th style="padding:5px 8px">BP</th>
            <th style="padding:5px 8px">HR</th>
            <th style="padding:5px 8px">Temp</th>
            <th style="padding:5px 8px">SpO2</th>

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        </tr>
    </thead>
    <tbody>
        ${vitals.slice(0, 8).map(v => `
            <tr style="border-bottom:1px solid var(--border)">
                <td style="padding:5px 8px;color:var(--text-dim)">${escapeHTML(v.recorded_at?.split('T')[1]
]??.slice(0,5) || v.created_at?.split('T')[1]??.slice(0,5) || '-')}</td>
                <td style="padding:5px 8px;text-align:center;font-weight:700">${escapeHTML(v.score_type ==
= 'blood_pressure' ? v.score_value : '-')}</td>
                <td style="padding:5px 8px;text-align:center">${escapeHTML(v.score_type === 'pulse' ? v.sc
ore_value : '-')}</td>
                <td style="padding:5px 8px;text-align:center">${escapeHTML(v.score_type === 'temperature'
? v.score_value : '-')}</td>
                <td style="padding:5px 8px;text-align:center">${escapeHTML(v.score_type === 'spo2' ? v.sco
re_value + '%' : '-')}</td>
            </tr>`}).join('')}
        </tbody>
    </table>
</div>
` : `<div style="text-align:center;padding:20px;color:var(--text-dim);font-size:12px">${tr('No previou
s vitals', '?? ???? ?????? ????? ?????')}</div>`
</div>
</div>
`;
};

```

```

window.nsSaveVitals = async function(pid) {
    if (!pid) return showToast(tr('No patient selected', '?? ???? ???? ?????'), 'error');
    const bp    = document.getElementById('nsVBp')?.value || '';
    const hr    = parseInt(document.getElementById('nsVHr')?.value) || 0;
    const temp  = parseFloat(document.getElementById('nsVTemp')?.value) || 0;
    const rr    = parseInt(document.getElementById('nsVRr')?.value) || 0;
    const spo2  = parseInt(document.getElementById('nsVSpo2')?.value) || 0;
    const pain  = parseInt(document.getElementById('nsVPain')?.value) || 0;
    const weight= parseFloat(document.getElementById('nsVWeight')?.value) || 0;
    const gcs   = parseInt(document.getElementById('nsVGcs')?.value) || 15;
    const glucose=parseInt(document.getElementById('nsVGlucose')?.value)||0;
    const avpu  = document.getElementById('nsVAvpu')?.value || 'A';
    const onO2  = document.getElementById('nsVOnO2')?.checked || false;
    const notes = document.getElementById('nsVNotes')?.value || '';

    const sbp = parseInt(bp.split('/')[0]) || 0;
    const news2 = calcNEWS2({ rr, spo2, temp, sbp, hr, avpu, onO2 });

    try {
        // Post vitals bundle
        const vitalsData = [
            { type: 'blood_pressure', value: bp },
            { type: 'pulse',          value: hr },
            { type: 'temperature',    value: temp },
            { type: 'respiratory_rate', value: rr },
            { type: 'spo2',           value: spo2 },
            { type: 'pain_score',     value: pain },
            { type: 'weight',         value: weight },

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    { type: 'gcs_score',      value: gcs },
    { type: 'glucose',       value: glucose },
    { type: 'consciousness', value: avpu },
  ].filter(x => x.value && x.value !== 0 && x.value !== '0');

  await API.post('/api/nursing/vitals', {
    patient_id: pid,
    bp, temp, weight,
    pulse: hr, o2_sat: spo2,
    respiratory_rate: rr,
    blood_sugar: glucose,
    notes: `NEWS2=${news2.score} | GCS=${gcs} | AVPU=${avpu} | Pain=${pain}/10 | O2=${onO2?'Yes':'No'} | ${notes}`,
  });

  showToast(`? ${tr('Vitals saved!', '?? ?? ???? ???? !')} NEWS2 = ${news2.score} ? ${news2.riskLabel}`);

  if (news2.score >= 7) {
    showToast(`? ${tr('HIGH NEWS2 SCORE! Consider escalating to doctor.', '??? NEWS2 ???? ! ?? ???? ???? ? ????')}`, 'error');
  }

  setTimeout(() => window.nsSelectPatient(pid), 800);
} catch (e) { showToast(e?.message || tr('Save failed', '???'), 'error'); }
};

window.nsEscalate = async function(pid) {
  if (!pid) return;
  const news2 = window._NS.selectedPatientData?.news2;
  const patient = window._NS.selectedPatientData?.patient || {};
  const name = isArabic ? (patient.name_ar || patient.name_en) : (patient.name_en || patient.name_ar);
  try {
    await API.post('/api/orders', {
      patient_id: pid,
      type: 'nursing',
      description: `[ESCALATE] NEWS2 Score = ${news2?.score || '?'} ? URGENT: ${name}`,
      status: 'STAT',
      notes: `Nurse escalation ? ${news2?.riskLabel || ''} ? ${new Date().toLocaleTimeString()}`,
    });
    showToast(`? ${tr('Escalated to doctor!', '?? ???? ???? !')}`, 'error');
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* TAB 2: eMAR ? Medication Administration Record */
/* ===== */

window.nsTabEMAR = function(container, { patient = {}, emarOrders = [], allergies = [] }, pid) {
  const patOrders = emarOrders.filter(o => String(o.patient_id) === String(pid));
  const shiftMap = { morning: '?? 07:00-15:00', evening: '? 15:00-23:00', night: '? 23:00-07:00' };

  container.innerHTML = `
    <div style="font-weight:700;font-size:13px;margin-bottom:12px;color:var(--primary)">
      ? ${tr('Electronic MAR ? Medication Administration Record', '??? ???? ????')}
  `

```

```

</div>
${allergies.length ? `
<div class="ns-cds-alert critical" style="margin-bottom:12px">
  ? ${tr('ALLERGY ALERT', '???? ?????')}: ${allergies.map(a => escapeHTML(a.allergen)).join(', ')}
</div>` : ''}
${!patOrders.length ? `
<div style="text-align:center;padding:32px;color:var(--text-dim)">
  <div style="font-size:48px;opacity:0.3;margin-bottom:12px">?</div>
  <p>${tr('No medication orders for this patient', '?? ???? ????? ?????? ????? ??????')}</p>
</div>
` : `
<div style="overflow-x:auto">
  <table style="width:100%;border-collapse:collapse;font-size:12px">
    <thead>
      <tr style="background:var(--surface-container,#f1f5f9)">
        <th style="padding:8px;text-align:start;font-weight:700">${tr('Medication', '?????')}</th>
        <th style="padding:8px;text-align:center">${tr('Dose', '?????')}</th>
        <th style="padding:8px;text-align:center">${tr('Route', '?????')}</th>
        <th style="padding:8px;text-align:center">${tr('Frequency', '???????')}</th>
        <th style="padding:8px;text-align:center">${tr('Due', '?????')}</th>
        <th style="padding:8px;text-align:center">${tr('Status', '?????')}</th>
        <th style="padding:8px;text-align:center">${tr('Action', '?????')}</th>
      </tr>
    </thead>
    <tbody>
      ${patOrders.map(o => {
        const isHighAlert = ['insulin','heparin','warfarin','morphine','fentanyl','potassium','digoxin','epinephrine','norepinephrine'].some(h => (o.medication || o.drug_name || '').toLowerCase().includes(h));
        const statusColor = o.status === 'Completed' || o.status === 'Dispensed' ? '#16a34a' : o.status === 'Pending' ? '#ca8a04' : '#6b7280';
        return `
          <tr style="border-bottom:1px solid var(--border);${isHighAlert ? 'background:#fef9c3;' : ''}">
            <td style="padding:8px;font-weight:700">
              ${isHighAlert ? '<span style="background:#dc2626;color:#fff;font-size:9px;padding:1px 5px;border-radius:4px;margin-left:4px">?? HIGH ALERT</span>' : ''}
              ${escapeHTML(o.medication || o.drug_name || '-')}
            </td>
            <td style="padding:8px;text-align:center">${escapeHTML(o.dose || '-')}</td>
            <td style="padding:8px;text-align:center">${escapeHTML(o.route || '-')}</td>
            <td style="padding:8px;text-align:center">${escapeHTML(o.frequency || '-')}</td>
            <td style="padding:8px;text-align:center;color:var(--text-dim)">${escapeHTML(o.scheduled_at?.split('T')[1]?[0,5] || 'PRN')}</td>
            <td style="padding:8px;text-align:center">
              <span style="background:${statusColor}22;color:${statusColor};padding:2px 8px;border-radius:10px;font-size:10px;font-weight:700">
                ${escapeHTML(o.status || 'Pending')}
              </span>
            </td>
            <td style="padding:8px;text-align:center">
              <button class="btn btn-sm" onclick="window.nsGiveMed(${safeId(o.id)},{safeId(pid),'${jsStr(o.medication||o.drug_name||'')}','${jsStr(o.dose||'')}','${jsStr(o.route||'')}','${isHighAlert}')"
                style="background:#16a34a;color:#fff;border:none;font-size:11px;padding:4px 10px">
                ? ${tr('Give', '????')}
              </button>
          `
      })}
    </tbody>
  </table>
`

```



```

        </td>
      </tr>`;
    }).join('')}
  </tbody>
</table>
</div>
`}
<!-- eMAR Legend -->
<div style="display:flex;gap:12px;margin-top:16px;flex-wrap:wrap;font-size:11px">
  <span style="display:flex;align-items:center;gap:4px"><span style="background:#f0fdf4;border:1px solid #
16a34a;width:16px;height:16px;border-radius:3px"></span> ${tr('Administered', '????')}</span>
  <span style="display:flex;align-items:center;gap:4px"><span style="background:#fef9c3;border:1px solid #
ca8a04;width:16px;height:16px;border-radius:3px"></span> ${tr('Pending', '????')}</span>
  <span style="display:flex;align-items:center;gap:4px"><span style="background:#fef9c3;width:16px;height:
16px;border-radius:3px;border:1px solid #eab308"></span> ?? ${tr('High Alert Drug', '???? ???? ?????')}</spa
n>
</div>
`;
};

```

```

window.nsGiveMed = async function(orderId, patientId, med, dose, route, isHighAlert) {
  if (isHighAlert) {
    const w = prompt(tr('HIGH-ALERT MEDICATION ? Enter witness nurse ID:', '???? ???? ?????? ? ???? ????? ????
???? ??????'));
    if (!w) { showToast(tr('Witness required', '???? ????'), 'error'); return; }
    try {
      await API.post('/api/mar/administer', {
        emar_order_id: orderId, patient_id: patientId,
        scanned_patient_id: patientId, scanned_drug: med,
        scanned_dose: dose, scanned_route: route,
        scheduled_at: new Date().toISOString(),
        witness_user_id: parseInt(w),
      });
      showToast(tr('? High-alert medication administered with witness', '? ?? ????? ?????? ????? ????'));
      setTimeout(() => window.nsSelectPatient(patientId), 800);
    } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
  } else {
    if (!confirm(tr(`Give ${med} ? ${dose} ? ${route}?`, `????? ${med} ? ${dose} ? ${route}?`))) return;
    try {
      await API.post('/api/mar/administer', {
        emar_order_id: orderId, patient_id: patientId,
        scanned_patient_id: patientId, scanned_drug: med,
        scanned_dose: dose, scanned_route: route,
        scheduled_at: new Date().toISOString(),
      });
      showToast(tr(`? ${med} administered`, `? ?? ?????? ${med}`));
      setTimeout(() => window.nsSelectPatient(patientId), 800);
    } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
  }
};

```

```

/* ===== */
/* TAB 3: HEAD-TO-TOE ASSESSMENT */
/* ===== */

```

```

window.nsTabH2T = function({ patient = {}, vitals = [] }) {
  return `
    <div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
      ? ${tr('Head-to-Toe Assessment', '?????? ?????? ?????? (?? ????? ??????)')}
    </div>
    <form id="nsH2TForm" style="display:grid;grid-template-columns:1fr 1fr;gap:12px">

      <!-- Neurological -->
      <div style="border:1px solid var(--border);border-radius:12px;padding:12px;grid-column:span 2">
        <div style="font-weight:700;font-size:12px;color:#7c3aed;margin-bottom:10px">? ${tr('Neurological', '?
        ??????')}</div>
        <div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:8px">
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">GCS</label>
            <input class="form-input" name="gcs" type="number" min="3" max="15" placeholder="15" style="height
            :32px;font-size:12px">
          </div>
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">AVPU</label>
            <select class="form-input" name="avpu" style="height:32px;font-size:12px">
              <option value="A">A ? Alert</option>
              <option value="V">V ? Voice</option>
              <option value="P">P ? Pain</option>
              <option value="U">U ? Unresponsive</option>
            </select>
          </div>
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Pupils', '????????')}</l
            abel>
            <select class="form-input" name="pupils" style="height:32px;font-size:12px">
              <option>PERRL (?????)</option>
              <option>Unequal (?????)</option>
              <option>Fixed/Dilated (????????)</option>
              <option>Constricted (????????)</option>
            </select>
          </div>
          <div style="grid-column:span 3">
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Orientation', '?????')}
          </label>
            <div style="display:flex;gap:8px;margin-top:4px">
              ${[ 'Person', 'Place', 'Time', 'Event' ].map(o => `<label style="font-size:11px;display:flex;align
              -items:center;gap:4px"><input type="checkbox" name="orient_${o.toLowerCase()}" checked> ${tr(o, { Person:'???'
              , Place:'????', Time:'????', Event:'???' }[o])}</label>`).join('')}
            </div>
          </div>
        </div>
      </div>

      <!-- Respiratory -->
      <div style="border:1px solid var(--border);border-radius:12px;padding:12px">
        <div style="font-weight:700;font-size:12px;color:#0ea5e9;margin-bottom:10px">? ${tr('Respiratory', '?
        ??????')}</div>
        <div style="display:grid;gap:8px">
          <div>

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        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Breath Sounds', '???? ?
????')}</label>
        <select class="form-input" name="breath_sounds" style="height:32px;font-size:12px">
            <option>Clear Bilaterally (??? ?????)</option>
            <option>Wheezing (???)</option>
            <option>Crackles (????)</option>
            <option>Diminished (????)</option>
            <option>Absent (????)</option>
        </select>
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('O2 Therapy', '????? ??
????')}</label>
        <select class="form-input" name="o2_therapy" style="height:32px;font-size:12px">
            <option value="Room Air">Room Air (??? ?????)</option>
            <option value="Nasal Cannula">Nasal Cannula (?????)</option>
            <option value="Face Mask">Face Mask (??? ?)</option>
            <option value="Non-rebreather">Non-rebreather Mask</option>
            <option value="Mechanical Ventilation">Mechanical Ventilation</option>
        </select>
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Cough', '????')}</label>
        <select class="form-input" name="cough" style="height:32px;font-size:12px">
            <option>None (?? ?)</option>
            <option>Productive (?? ?)</option>
            <option>Nonproductive (???)</option>
        </select>
    </div>
</div>
</div>
</div>

<!-- Cardiovascular -->
<div style="border:1px solid var(--border);border-radius:12px;padding:12px">
    <div style="font-weight:700;font-size:12px;color:#dc2626;margin-bottom:10px"> ${tr('Cardiovascular',
'????? ?????')}</div>
    <div style="display:grid;gap:8px">
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Heart Rhythm', '?? ?
?')}</label>
            <select class="form-input" name="heart_rhythm" style="height:32px;font-size:12px">
                <option>Regular (????)</option>
                <option>Irregular (?? ?)</option>
                <option>Atrial Fibrillation</option>
                <option>Bradycardia</option>
                <option>Tachycardia</option>
            </select>
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Capillary Refill', '???
?? ?????')}</label>
            <select class="form-input" name="cap_refill" style="height:32px;font-size:12px">
                <option>< 2 sec (????)</option>
            </select>
        </div>
    </div>

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        <option>2-3 sec (???? ????)</option>
        <option>> 3 sec (????)</option>
    </select>
</div>
<div>
    <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Peripheral Edema', '???
?????')}</label>
    <select class="form-input" name="edema" style="height:32px;font-size:12px">
        <option>None (?? ????)</option>
        <option>1+ Mild</option><option>2+ Moderate</option>
        <option>3+ Severe</option><option>4+ Pitting</option>
    </select>
</div>
</div>
</div>
</div>

<!-- GI -->
<div style="border:1px solid var(--border);border-radius:12px;padding:12px">
    <div style="font-weight:700;font-size:12px;color:#ca8a04;margin-bottom:10px">? ${tr('Gastrointestinal'
, '????? ????')}</div>
    <div style="display:grid;gap:8px">
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Bowel Sounds', '???? ?
????')}</label>
            <select class="form-input" name="bowel_sounds" style="height:32px;font-size:12px">
                <option>Active (????)</option><option>Hypoactive (?????)</option>
                <option>Hyperactive (????)</option><option>Absent (????)</option>
            </select>
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Last BM', '??? ????')}</
label>
            <input class="form-input" name="last_bm" type="date" style="height:32px;font-size:12px">
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Oral Intake %', '???? ?
???')}</label>
            <select class="form-input" name="oral_intake" style="height:32px;font-size:12px">
                <option>100% (????)</option><option>75%</option><option>50%</option>
                <option>25%</option><option>0% / NPO</option>
            </select>
        </div>
    </div>
</div>
</div>

<!-- GU -->
<div style="border:1px solid var(--border);border-radius:12px;padding:12px">
    <div style="font-weight:700;font-size:12px;color:#0369a1;margin-bottom:10px">? ${tr('Genitourinary', '
????? ????')}</div>
    <div style="display:grid;gap:8px">
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Urine Output', '???? ?
???')}</label>
            <select class="form-input" name="urine_output" style="height:32px;font-size:12px">

```

```

        <option>Normal (?????)</option><option>Reduced (?????)</option>
        <option>Oliguria < 0.5mL/kg/hr</option><option>Anuria (????)</option>
    </select>
</div>
<div>
    <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Catheter', '????')}</la
bel>

    <select class="form-input" name="catheter" style="height:32px;font-size:12px">
        <option>None (?? ????)</option><option>Foley Catheter</option>
        <option>Suprapubic</option><option>Condom Catheter</option>
    </select>
</div>
</div>
</div>

<!-- Skin -->
<div style="border:1px solid var(--border);border-radius:12px;padding:12px;grid-column:span 2">
    <div style="font-weight:700;font-size:12px;color:#16a34a;margin-bottom:10px">? ${tr('Skin & Wound', '?
???? ???????')}</div>
    <div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:8px">
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Skin Color', '??? ?????'
)}}</label>

            <select class="form-input" name="skin_color" style="height:32px;font-size:12px">
                <option>Normal (?????)</option><option>Pale (????)</option>
                <option>Jaundiced (????)</option><option>Flushed (????)</option>
                <option>Cyanotic (????)</option>
            </select>
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Skin Turgor', '????? ??
??')}</label>

            <select class="form-input" name="skin_turgor" style="height:32px;font-size:12px">
                <option>Good (???)</option><option>Fair (?????)</option>
                <option>Poor / Tenting (????)</option>
            </select>
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Wound/IV Site', '?????/?
??? ??????')}</label>

            <select class="form-input" name="wound_status" style="height:32px;font-size:12px">
                <option>Intact (????)</option><option>Redness (?????)</option>
                <option>Pressure Injury Stage I</option>
                <option>Pressure Injury Stage II</option>
                <option>Pressure Injury Stage III-IV</option>
                <option>Surgical Wound</option>
            </select>
        </div>
    </div>
    <div style="grid-column:span 3">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Wound Description', '???
????')}</label>

        <textarea class="form-input" name="wound_notes" rows="2" placeholder="${tr('Size, depth, exudate,
dressing...', '????? ?????? ??????? ???????...')}"></textarea>
    </div>

```

```

    </div>
</div>

<!-- Pain Assessment -->
<div style="border:1px solid var(--border);border-radius:12px;padding:12px;grid-column:span 2">
    <div style="font-weight:700;font-size:12px;color:#dc2626;margin-bottom:10px">? ${tr('Pain Assessment',
'????? ?????')}</div>
    <div style="display:grid;grid-template-columns:repeat(5,1fr);gap:8px;margin-bottom:10px">
        ${[0,1,2,3,4,5,6,7,8,9,10].map(n => `
            <label style="text-align:center;cursor:pointer">
                <input type="radio" name="pain_nrs" value="${n}" ${n===0?'checked':''} style="display:block;margin:0 auto 2px">
                <span style="font-size:16px">${n<=2?'':n<=4?'':n<=6?'':n<=8?'':'?'}</span>
                <div style="font-size:11px;font-weight:700">${n}</div>
            </label>
        `).join('')}</div>
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px">
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Pain Location', '???? ??
????')}</label>
            <input class="form-input" name="pain_location" placeholder="${tr('Where?', '????')}" style="height
:32px;font-size:12px">
        </div>
        <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Pain Character', '?????
?????')}</label>
            <select class="form-input" name="pain_character" style="height:32px;font-size:12px">
                <option>Sharp (???)</option><option>Dull (????)</option>
                <option>Burning (????)</option><option>Throbbing (????)</option>
                <option>Cramping (????)</option><option>Pressure (???)</option>
            </select>
        </div>
    </div>
</div>

<!-- Submit -->
<div style="grid-column:span 2;display:flex;gap:10px">
    <button type="button" class="btn btn-primary" onclick="window.nsSaveH2T(${window._NS.selectedPatientId
})" style="flex:1;height:44px;font-size:13px">
        ? ${tr('Save Assessment', '??? ??????')}
    </button>
    <button type="button" class="btn" onclick="window.nsPrintH2T()" style="height:44px;font-size:13px;back
ground:#fff3e0;border:1px solid #ff9800;color:#e65100">
        ?? ${tr('Print', '????')}
    </button>
</div>
</form>
`;
};

```

```

window.nsSaveH2T = async function(pid) {
    if (!pid) return showToast(tr('No patient selected', '?? ???? ???? ????'), 'error');
    const form = document.getElementById('nSH2TForm');

```

```

if (!form) return;
const data = Object.fromEntries(new FormData(form).entries());
data.pain_nrs = document.querySelector('input[name="pain_nrs"]:checked')?.value || '0';
try {
  await API.post('/api/nursing/assessments', {
    patient_id: pid,
    assessment_type: 'Head-to-Toe',
    pain_score: parseInt(data.pain_nrs),
    gcs_score: parseInt(data.gcs) || 15,
    assessment_data: JSON.stringify(data),
    nurse: window._NS.currentUser?.name || '',
    shift: getShiftName(),
  }).catch(() =>
    API.post('/api/medical/records', {
      patient_id: pid,
      diagnosis: 'Nursing Head-to-Toe Assessment',
      notes: JSON.stringify(data),
      treatment: '',
    })
  );
  showToast(tr('? Assessment saved!', '? ?? ??? ???????!'));
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* TAB 4: NURSING CARE PLAN */
/* ===== */
window.nsTabCarePlan = async function(container, { patient = {} }, pid) {
  container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading...', '???? ???????...')}</p></div>`;
  let plans = [];
  try { plans = await API.get('/api/nursing/care-plans'); } catch {}
  const patPlans = plans.filter(p => String(p.patient_id) === String(pid));

  const nandaDiagnoses = [
    { code: '00001', label: 'Imbalanced Nutrition: Less Than Body Requirements | ??? ???????' },
    { code: '00002', label: 'Deficient Fluid Volume | ??? ??? ???????' },
    { code: '00004', label: 'Risk for Infection | ??? ???????' },
    { code: '00007', label: 'Hyperthermia | ?????? ???????' },
    { code: '00030', label: 'Impaired Gas Exchange | ?????? ?????? ???????' },
    { code: '00046', label: 'Impaired Skin Integrity | ??? ?????? ??????' },
    { code: '00061', label: 'Caregiver Role Strain | ?????? ??? ?????? ???????' },
    { code: '00078', label: 'Ineffective Health Management | ?????? ?????? ??? ???????' },
    { code: '00085', label: 'Impaired Physical Mobility | ?????? ??????? ????????' },
    { code: '00108', label: 'Bathing Self-Care Deficit | ??? ??????? ????????' },
    { code: '00118', label: 'Disturbed Body Image | ?????? ?????? ??????' },
    { code: '00128', label: 'Acute Confusion | ????????' },
    { code: '00132', label: 'Acute Pain | ??? ???' },
    { code: '00133', label: 'Chronic Pain | ??? ?????' },
    { code: '00146', label: 'Anxiety | ??????' },
    { code: '00148', label: 'Fear | ??????' },
    { code: '00155', label: 'Risk for Falls | ??? ??????' },
    { code: '00179', label: 'Risk for Unstable Blood Glucose | ??? ??? ??????? ????????' },
    { code: '00204', label: 'Ineffective Peripheral Tissue Perfusion | ?????? ??????? ????????' },
  ]

```

```

    { code: '00215', label: 'Decreased Cardiac Output | ?????? ?????? ??????' },
  ];

container.innerHTML = `
  <div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
    ? ${tr('Nursing Care Plan (NANDA/NIC/NOC)', '??? ?????? ?????????')}
  </div>

  <!-- Current Plans -->
  ${patPlans.length ? `
    <div style="margin-bottom:20px">
      <div style="font-size:12px;font-weight:700;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">${tr('Active Care Plans', '??? ?????? ??????')} (${patPlans.length})</div>
      ${patPlans.map(p => `
        <div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:10px;padding:12px;margin-bottom:8px">
          <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:6px">
            <strong style="font-size:13px">${escapeHTML(p.diagnosis || '-')}</strong>
            <span class="badge badge-${p.priority==='High'? 'danger':p.priority==='Low'? 'success': 'warning'}">${escapeHTML(p.priority || 'Medium')}</span>
          </div>
          ${p.goals ? `<div style="font-size:12px;color:var(--text-dim)">? ${tr('Goals', '???????')}: ${escapeHTML(p.goals)}</div>` : ''}
          ${p.interventions ? `<div style="font-size:12px;color:var(--text-dim);margin-top:4px">? ${tr('Interventions', '????????')}: ${escapeHTML(p.interventions)}</div>` : ''}
        </div>
      `).join('')}
    </div>
  ` : ''}

  <!-- New Care Plan Form -->
  <div style="border:1px solid var(--border);border-radius:12px;padding:16px">
    <div style="font-size:12px;font-weight:700;margin-bottom:12px">? ${tr('Add New Care Plan', '????? ??? ???? ?????')}</div>
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:10px">
      <div style="grid-column:span 2">
        <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('NANDA Nursing Diagnosis', '????? ?????? NANDA')}</label>
        <select class="form-input" id="cpNanda">
          <option value="">${tr('-- Select NANDA Diagnosis --', '-- ???? ?????? ????????? --')}</option>
          ${nandaDiagnoses.map(d => `<option value="${d.code}": ${d.label}">${d.code}: ${escapeHTML(d.label)}</option>`).join('')}
        </select>
      </div>
      <div>
        <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Priority', '????????')}</label>
        <select class="form-input" id="cpPriority">
          <option value="Low">${tr('Low', '?????')}</option>
          <option value="Medium" selected>${tr('Medium', '?????')}</option>
          <option value="High">${tr('High', '?????')}</option>
        </select>
      </div>
    </div>
  </div>
`

```



```

        <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Target Date', '?????? ????
?????')}</label>
        <input class="form-input" id="cpTarget" type="date" value="${new Date().toISOString().slice(0,10)}">
    </div>
    <div style="grid-column:span 2">
        <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('NOC Outcomes / Goals', '??
???? ???? ???? ???? ????')}</label>
        <textarea class="form-input" id="cpGoals" rows="2" placeholder="${tr('Measurable goals...', '???? ?
???? ????...')}></textarea>
    </div>
    <div style="grid-column:span 2">
        <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('NIC Interventions', '????
??? ????????? NIC')}</label>
        <textarea class="form-input" id="cpInterventions" rows="3" placeholder="${tr('Nursing interventions.
..', '??????? ?????????...')}></textarea>
    </div>
</div>
<button class="btn btn-primary" onclick="window.nsSaveCarePlan(${pid})" style="margin-top:12px;height:40
px;font-size:13px">
    ? ${tr('Save Care Plan', '??? ??? ???????')}
</button>
</div>
`;
};

```

```

window.nsSaveCarePlan = async function(pid) {
    const nanda = document.getElementById('cpNanda')?.value || '';
    const priority= document.getElementById('cpPriority')?.value || 'Medium';
    const goals = document.getElementById('cpGoals')?.value || '';
    const intv = document.getElementById('cpInterventions')?.value || '';
    if (!nanda) return showToast(tr('Select a NANDA diagnosis', '???? ????????? ?????????'), 'error');
    try {
        const patient = window._NS.selectedPatientData?.patient || {};
        await API.post('/api/nursing/care-plans', {
            patient_id: pid,
            patient_name: isArabic ? (patient.name_ar || patient.name_en) : (patient.name_en || patient.name_ar),
            diagnosis: nanda, priority, goals, interventions: intv,
            status: 'Active',
        });
        showToast(tr('? Care plan saved!', '? ?? ??? ??? ?????????!'));
        window.nsSwitchTab('careplan');
    } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

```

```

/* ===== */
/* TAB 5: INTAKE & OUTPUT */
/* ===== */
window.nsTabIO = async function(container, pid) {
    // Load persisted entries from server; fall back to local cache offline
    try {
        const data = await API.get(`/api/nursing/io/${pid}`);
        window._NS.ioEntries = (data.entries || []).map(e => ({
            type: e.entry_type,
            source: e.source,

```

```

        volume: e.volume_ml || 0,
        time: e.entry_time || String(e.created_at || '').slice(11, 16),
    }));
} catch (e) { /* offline/mock mode ? keep local cache */ }
const entries = window._NS.ioEntries;
const intake = entries.filter(e => e.type === 'intake').reduce((s, e) => s + (e.volume || 0), 0);
const output = entries.filter(e => e.type === 'output').reduce((s, e) => s + (e.volume || 0), 0);
const balance = intake - output;
const balColor= balance > 500 ? '#ca8a04' : balance > 1000 ? '#dc2626' : '#16a34a';

container.innerHTML = `
<div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
    ? ${tr('Intake & Output Tracking', '????? ?????? ?????? ??????')}
</div>

<!-- Balance Summary -->
<div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:12px;margin-bottom:20px">
    <div style="background:#e0f2fe;border-radius:12px;padding:14px;text-align:center">
        <div style="font-size:11px;font-weight:700;color:#0369a1;margin-bottom:4px">? ${tr('Total Intake', '??
        ???? ??????')}</div>
        <div style="font-size:28px;font-weight:900;color:#0369a1">${intake}</div>
        <div style="font-size:11px;color:#0369a1">mL</div>
    </div>
    <div style="background:#fef9c3;border-radius:12px;padding:14px;text-align:center">
        <div style="font-size:11px;font-weight:700;color:#ca8a04;margin-bottom:4px">? ${tr('Total Output', '??
        ???? ??????')}</div>
        <div style="font-size:28px;font-weight:900;color:#ca8a04">${output}</div>
        <div style="font-size:11px;color:#ca8a04">mL</div>
    </div>
    <div style="background:${balance}>=0?'#f0fdf4':'#fee2e2';border-radius:12px;padding:14px;text-align:center;border:2px solid ${balColor}">
        <div style="font-size:11px;font-weight:700;color:${balColor};margin-bottom:4px">?? ${tr('Balance', '??
        ?????')}</div>
        <div style="font-size:28px;font-weight:900;color:${balColor}">${balance >= 0 ? '+' : ''}${balance}</div>
    </div>
    <div style="font-size:11px;color:${balColor}">mL</div>
</div>

<!-- Add Entry -->
<div style="display:grid;grid-template-columns:1fr 1fr;gap:16px;margin-bottom:16px">
    <!-- Intake -->
    <div style="border:1px solid #0ea5e922;background:#e0f2fe11;border-radius:12px;padding:14px">
        <div style="font-weight:700;font-size:12px;color:#0369a1;margin-bottom:10px">? ${tr('Add Intake', '??
        ?? ???? ?????')}</div>
        <div class="form-group mb-8">
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Source', '??????')}</label>
            <select class="form-input" id="ioIntakeSource" style="height:32px;font-size:12px">
                <option value="IV Fluid">IV Fluids (???? ?????)</option>
                <option value="PO Oral">PO Oral (????)</option>
                <option value="Tube Feed">Tube Feeding (???? ?????)</option>
                <option value="Blood Product">Blood Product (???? ?????)</option>
                <option value="Medication Flush">Med Flush</option>
            </select>
        </div>
    </div>
</div>

```

```

        </select>
    </div>
    <div class="form-group mb-8">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Volume (mL)', '???? (??)'
    )}</label>
        <input class="form-input" id="ioIntakeVol" type="number" min="1" placeholder="250" style="height:32p
x;font-size:12px">
    </div>
    <div class="form-group mb-8">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Time', '????')}</label>
        <input class="form-input" id="ioIntakeTime" type="time" value="${new Date().toTimeString().slice(0,5
    )}" style="height:32px;font-size:12px">
    </div>
    <button class="btn w-full" onclick="window.nsAddIO('intake')"
        style="background:#0369a1;color:#fff;border:none;height:34px;font-size:12px">
        ? ${tr('Add Intake', '???? ???')}
    </button>
</div>

<!-- Output -->
<div style="border:1px solid #ca8a0422;background:#fef9c311;border-radius:12px;padding:14px">
    <div style="font-weight:700;font-size:12px;color:#ca8a04;margin-bottom:10px">? ${tr('Add Output', '???
?? ???? ???')}</div>
    <div class="form-group mb-8">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Type', '????')}</label>
        <select class="form-input" id="ioOutputType" style="height:32px;font-size:12px">
            <option value="Urine">Urine (???)</option>
            <option value="Vomit">Vomit (???)</option>
            <option value="Drain">Drain (???)</option>
            <option value="Nasogastric">Nasogastric (???? ????)</option>
            <option value="Stool">Stool (???)</option>
            <option value="Blood Loss">Blood Loss (???)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Volume (mL)', '???? (??)'
    )}</label>
        <input class="form-input" id="ioOutputVol" type="number" min="1" placeholder="300" style="height:32p
x;font-size:12px">
    </div>
    <div class="form-group mb-8">
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Time', '????')}</label>
        <input class="form-input" id="ioOutputTime" type="time" value="${new Date().toTimeString().slice(0,5
    )}" style="height:32px;font-size:12px">
    </div>
    <button class="btn w-full" onclick="window.nsAddIO('output')"
        style="background:#ca8a04;color:#fff;border:none;height:34px;font-size:12px">
        ? ${tr('Add Output', '???? ???')}
    </button>
</div>
</div>

<!-- I&O Log Table -->
${entries.length ? `

```

```

<div>
  <div style="font-size:12px;font-weight:700;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">${tr('Shift Log', '??? ?????')}</div>
  <table style="width:100%;border-collapse:collapse;font-size:12px">
    <thead>
      <tr style="background:var(--surface-container,#f1f5f9)">
        <th style="padding:6px 8px;text-align:start">${tr('Time', '????')}</th>
        <th style="padding:6px 8px;text-align:start">${tr('Type', '????')}</th>
        <th style="padding:6px 8px;text-align:start">${tr('Source', '????')}</th>
        <th style="padding:6px 8px;text-align:end">${tr('Volume', '????')} (mL)</th>
      </tr>
    </thead>
    <tbody>
      ${entries.map(e => `
        <tr style="border-bottom:1px solid var(--border);background:${e.type==='intake'?'#e0f2fe11':'#fef9c311'}">
          <td style="padding:6px 8px;color:var(--text-dim)">${escapeHTML(e.time)}</td>
          <td style="padding:6px 8px">
            <span style="color:${e.type==='intake'?'#0369a1':'#ca8a04'};font-weight:700">
              ${e.type === 'intake' ? '? IN' : '? OUT'}
            </span>
          </td>
          <td style="padding:6px 8px">${escapeHTML(e.source)}</td>
          <td style="padding:6px 8px;text-align:end;font-weight:700">${e.volume}</td>
        </tr>`).join('')}
      <tr style="background:var(--surface-container,#f1f5f9);font-weight:700">
        <td colspan="3" style="padding:6px 8px">${tr('Balance', '????')}</td>
        <td style="padding:6px 8px;text-align:end;color:${balColor}">${balance >= 0 ? '+' : ''}${balance}<
    </td>
  </tr>
</tbody>
</table>
</div>
` : `<div style="text-align:center;padding:20px;color:var(--text-dim);font-size:12px">${tr('No entries yet
? add intake/output above', '?? ???? ??????? ? ??? ??????? ?????')}</div>`
`;
};

```

```

window.nsAddIO = async function(type) {
  let source, vol, time;
  if (type === 'intake') {
    source = document.getElementById('ioIntakeSource')?.value || 'IV Fluid';
    vol = parseInt(document.getElementById('ioIntakeVol')?.value) || 0;
    time = document.getElementById('ioIntakeTime')?.value || new Date().toString().slice(0,5);
  } else {
    source = document.getElementById('ioOutputType')?.value || 'Urine';
    vol = parseInt(document.getElementById('ioOutputVol')?.value) || 0;
    time = document.getElementById('ioOutputTime')?.value || new Date().toString().slice(0,5);
  }
  if (!vol) return showToast(tr('Enter volume', '???? ?????'), 'error');
  try {
    await API.post('/api/nursing/io', {
      patient_id: window._NS.selectedPatientId,
      entry_type: type,

```

```

        source,
        volume_ml: vol,
        entry_time: time,
        shift: getShiftName(),
    });
} catch (e) {
    // offline/mock mode ? keep entry in local cache so the shift log still works
    window._NS.ioEntries.push({ type, source, volume: vol, time });
}
showToast(tr('Entry added', '?? ??????'));
window.nsSwitchTab('io');
};

/* ===== */
/* TAB 6: HANDOVER / SBAR */
/* ===== */
window.nsTabHandover = function({ patient = {}, chart = {}, vitals = [] }, pid) {
    const name = isArabic ? (patient.name_ar || patient.name_en || '-') : (patient.name_en || patient.name_ar || '-');
    const news2 = window._NS.selectedPatientData?.news2 || { score: 0 };
    const lastRec = (chart.records || [])[0] || {};

    return `
        <div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
            ? ${tr('Shift Handover ? SBAR Format', '????? ? ?????? SBAR')}
        </div>
        <div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:12px;
padding:14px;margin-bottom:16px;font-size:12px">
            ? <strong>${escapeHTML(name)}</strong> · ?? ${escapeHTML(String(patient.file_number || '-'))} ·
            NEWS2: <strong style="color:${news2.riskColor || '#16a34a'}">${news2.score}</strong> ·
            ? ${new Date().toLocaleDateString(isArabic ? 'ar-SA' : 'en-US')}
        </div>

        <div style="display:grid;gap:12px">
            <!-- S: Situation -->
            <div style="border:2px solid #0369a1;border-radius:12px;overflow:hidden">
                <div style="background:#0369a1;color:#fff;padding:10px 14px;font-weight:700;font-size:12px">
                    ? S ? ${tr('SITUATION (Current patient status)', '????? ?????? ??????')}
                </div>
                <div style="padding:12px">
                    <textarea class="form-input" id="sbarS" rows="3"
                        placeholder="${tr('Patient name, age, diagnosis, reason for hospitalization, current vital status.
..', '?? ?????? ?????? ?????? ???? ?????? ?????? ?????? ??????..')}"
                        style="font-size:12px">${escapeHTML(lastRec.diagnosis ? `????? ${name} ??? ???? : ${lastRec.diagn
osis}. NEWS2 = ${news2.score}` : '')}</textarea>
                    </div>
                </div>

            <!-- B: Background -->
            <div style="border:2px solid #7c3aed;border-radius:12px;overflow:hidden">
                <div style="background:#7c3aed;color:#fff;padding:10px 14px;font-weight:700;font-size:12px">
                    ? B ? ${tr('BACKGROUND (Medical history & context)', '????? ?????? ?????? ??????')}
                </div>
                <div style="padding:12px">

```

```

        <textarea class="form-input" id="sbarB" rows="3"
            placeholder="${tr('PMH, allergies, current medications, procedures done...', '?????? ?????? ???
????? ?????? ??????? ????????? ???????...')}"
            style="font-size:12px"></textarea>
    </div>
</div>

<!-- A: Assessment -->
<div style="border:2px solid #ca8a04;border-radius:12px;overflow:hidden">
    <div style="background:#ca8a04;color:#fff;padding:10px 14px;font-weight:700;font-size:12px">
        ? A ? ${tr('ASSESSMENT (Nursing assessment & concerns)', '?????? ??????? ???????')}
    </div>
    <div style="padding:12px">
        <textarea class="form-input" id="sbarA" rows="3"
            placeholder="${tr('Clinical assessment, trends, concerns, pain score, any changes...', '?????? ??
?????? ?????????? ????????? ???? ?????? ?? ???????...')}"
            style="font-size:12px"></textarea>
    </div>
</div>

<!-- R: Recommendation -->
<div style="border:2px solid #16a34a;border-radius:12px;overflow:hidden">
    <div style="background:#16a34a;color:#fff;padding:10px 14px;font-weight:700;font-size:12px">
        ? R ? ${tr('RECOMMENDATION (Actions needed)', '?????? ??????? ???????')}
    </div>
    <div style="padding:12px">
        <textarea class="form-input" id="sbarR" rows="3"
            placeholder="${tr('Pending tasks, follow-up orders, medications due, things to watch...', '?????
???????? ????????? ????????? ???? ???? ???????...')}"
            style="font-size:12px"></textarea>
    </div>
</div>

<div style="display:flex;gap:10px">
    <button class="btn btn-primary" onclick="window.nsSaveSBAR(${pid})" style="flex:1;height:44px;font-siz
e:13px">
        ? ${tr('Save Handover Report', '??? ????? ???????')}
    </button>
    <button class="btn" onclick="window.nsPrintSBAR()" style="height:44px;font-size:13px;background:#fff3e
0;border:1px solid #ff9800;color:#e65100">
        ?? ${tr('Print SBAR', '????? SBAR')}
    </button>
</div>

<div id="nsHandoverHistory"></div>
</div>
`;
};

/* Load previous handover reports for this patient */
window.nsLoadHandovers = async function(pid) {
    const box = document.getElementById('nsHandoverHistory');
    if (!box) return;
    let rows = [];

```

```

try { rows = await API.get(`/api/nursing/handover/${pid}`); } catch (e) { return; }
if (!Array.isArray(rows) || !rows.length) return;
box.innerHTML = `
  <div style="font-size:12px;font-weight:700;color:var(--text-dim);margin:16px 0 8px;text-transform:uppercase">
    ?? ${tr('Previous Handovers', '????? ??????? ???????')}
  </div>
  ${rows.map(h => `
    <div style="border:1px solid var(--border);border-radius:10px;padding:10px 12px;margin-bottom:8px;font-size:12px;background:var(--surface-container,#f8fafc)">
      <div style="display:flex;gap:8px;flex-wrap:wrap;color:var(--text-dim);font-size:11px;margin-bottom:4px">
        <span>???? ${escapeHTML(h.nurse_name || '-')}</span>
        <span>? ${escapeHTML(h.shift || '-')}</span>
        <span>? ${escapeHTML(String(h.created_at || '').slice(0, 16).replace('T', ' '))}</span>
        ${h.news2_score ? `<span style="font-weight:700">NEWS2: ${parseInt(h.news2_score, 10) || 0}</span>` : ''}
      </div>
      <div style="white-space:pre-wrap">${escapeHTML(h.sbar_s || '')}</div>
      ${h.sbar_r ? `<div style="white-space:pre-wrap;margin-top:4px;color:#16a34a"><strong>R:</strong> ${escapeHTML(h.sbar_r)}</div>` : ''}
    </div>`).join('')}
  `;
};

```

```

window.nsSaveSBAR = async function(pid) {
  const S = document.getElementById('sbarS')?.value || '';
  const B = document.getElementById('sbarB')?.value || '';
  const A = document.getElementById('sbarA')?.value || '';
  const R = document.getElementById('sbarR')?.value || '';
  if (!S.trim()) return showToast(tr('Fill in Situation section', '???? ??? ??????'), 'error');
  try {
    await API.post('/api/nursing/handover', {
      patient_id: pid,
      sbar_s: S, sbar_b: B, sbar_a: A, sbar_r: R,
      shift: getShiftName(),
      news2_score: window._NS.selectedPatientData?.news2?.score || 0,
    });
    // best-effort copy into the medical record so it shows in the patient chart
    API.post('/api/medical/records', {
      patient_id: pid,
      diagnosis: 'Nursing SBAR Handover Report',
      symptoms: S,
      notes: `BACKGROUND: ${B}\n\nASSESSMENT: ${A}`,
      treatment: `RECOMMENDATION: ${R}`,
    }).catch(() => {});
    showToast(tr('? SBAR Handover saved!', '? ?? ??? ????? ??????!'));
    window.nsLoadHandovers(pid);
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

```

```

window.nsPrintSBAR = function() {
  const patient = window._NS.selectedPatientData?.patient || {};
  const patientName = isArabic ? (patient.name_ar || patient.name_en || '-') : (patient.name_en || patient.name_ar || '-');
}

```

```

e_ar || '-');
const S = document.getElementById('sbarS')?.value || '';
const B = document.getElementById('sbarB')?.value || '';
const A = document.getElementById('sbarA')?.value || '';
const R = document.getElementById('sbarR')?.value || '';
const w = window.open('', '_blank');
w.document.write(`
<html><head><title>SBAR Handover</title>
<style>body{font-family:Arial,sans-serif;padding:24px;direction:rtl}h1{font-size:18px;color:#0369a1}.section{margin:16px 0;padding:14px;border-radius:8px}.s{background:#e0f2fe;border-left:5px solid #0369a1}.b{background:#f3e8ff;border-left:5px solid #7c3aed}.a{background:#fef9c3;border-left:5px solid #ca8a04}.r{background:#f0fdf4;border-left:5px solid #16a34a}h3{margin:0 0 8px;font-size:14px}p{margin:0;white-space:pre-wrap;font-size:13px}</style>
</head><body>
<h1>? ????? ?????? ? SBAR</h1>
<p><strong>?????:</strong> ${escapeHTML(patientName)} | <strong>?????:</strong> ${new Date().toLocaleString('ar-SA')} | <strong>?????:</strong> ${escapeHTML(window._NS.currentUser?.name || '')}</p>
<div class="section s"><h3>S ? ?????</h3><p>${escapeHTML(S)}</p></div>
<div class="section b"><h3>B ? ?????</h3><p>${escapeHTML(B)}</p></div>
<div class="section a"><h3>A ? ?????</h3><p>${escapeHTML(A)}</p></div>
<div class="section r"><h3>R ? ?????</h3><p>${escapeHTML(R)}</p></div>
</body></html>
`);
w.document.close();
w.print();
};

/* ===== */
/* TAB 7: TRIAGE (ESI 1-5) */
/* ===== */
window.nsTabTriage = function({ patient = {} }, pid) {
  const esiData = [
    { level: 1, color: '#dc2626', bg: '#fee2e2', label: tr('Resuscitation', '????'), desc: tr('Life/limb-threatening ? immediate', '???? ???? ? ???'), icon: '?' },
    { level: 2, color: '#ea580c', bg: '#fff7ed', label: tr('Emergent', '????'), desc: tr('High risk ? immediate evaluation', '?? ???? ? ??? ????'), icon: '?' },
    { level: 3, color: '#ca8a04', bg: '#fef9c3', label: tr('Urgent', '????'), desc: tr('Multiple resources required', '???? ???? ????'), icon: '?' },
    { level: 4, color: '#3b82f6', bg: '#eff6ff', label: tr('Less Urgent', '?? ?????'), desc: tr('One resource needed', '???? ???? ????'), icon: '?' },
    { level: 5, color: '#16a34a', bg: '#f0fdf4', label: tr('Non-Urgent', '?? ????'), desc: tr('No resources needed', '?? ???? ????'), icon: '?' },
  ];

  return `
<div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
  ?? ${tr('Triage Assessment ? ESI (Emergency Severity Index)', '???? ???? ? ??? ????')}
</div>
<div class="form-group mb-12">
  <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Chief Complaint', '???? ????')}</label>
  <input class="form-input" id="tgComplaint" placeholder="${tr('What brings the patient in today?', '?? ?? ?? ?? ????')}">
</div>

```



```

<div style="display:grid;gap:10px;margin-bottom:16px">
  ${esiData.map(e => `
    <label style="display:flex;align-items:center;gap:12px;padding:12px;border:2px solid ${e.color}33;border-radius:12px;cursor:pointer;background:${e.bg};transition:all 0.15s"
      onmouseover="this.style.borderColor='${e.color}'" onmouseout="this.style.borderColor='${e.color}33'"
    >
      <input type="radio" name="esiLevel" value="${e.level}" style="width:18px;height:18px;accent-color:${e.color}">
      <span style="font-size:24px">${e.icon}</span>
      <div>
        <div style="font-weight:800;font-size:14px;color:${e.color}">ESI-${e.level}: ${e.label}</div>
        <div style="font-size:11px;color:var(--text-dim)">${e.desc}</div>
      </div>
    </label>
  `).join('')}
</div>
<div style="display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-bottom:16px">
  <div>
    <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Initial BP', '????? ?????')}
  </label>
    <input class="form-input" id="tgBp" placeholder="120/80">
  </div>
  <div>
    <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Initial HR', '????? ?????')}
  </label>
    <input class="form-input" id="tgHr" type="number" placeholder="75">
  </div>
  <div>
    <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('SpO2', '???? ?????????')}</label>
    <input class="form-input" id="tgSpo2" type="number" placeholder="98">
  </div>
  <div>
    <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Temp', '????????')}</label>
    <input class="form-input" id="tgTemp" type="number" step="0.1" placeholder="37.0">
  </div>
</div>
<div class="form-group mb-12">
  <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Triage Notes', '??????? ?????')}
</label>
  <textarea class="form-input" id="tgNotes" rows="2" placeholder="${tr('Additional observations...', '????
??? ??????...')}"></textarea>
</div>
  <button class="btn btn-primary w-full" onclick="window.nsSaveTriage(${pid})" style="height:44px;font-size:
14px;font-weight:700">
    ?? ${tr('Save Triage & Route to Doctor', '??? ????? ????????? ?????')}
  </button>
`;
};

```

```

window.nsSaveTriage = async function(pid) {
  const level = parseInt(document.querySelector('input[name="esiLevel"]:checked')?.value || '3');
  const complaint = document.getElementById('tgComplaint')?.value || '';
  const notes = document.getElementById('tgNotes')?.value || '';

```

```

if (!complaint.trim()) return showToast(tr('Enter chief complaint', '???? ?????? ????????'), 'error');
try {
  await API.post('/api/nursing/triage', {
    patient_id: pid, triage_level: level, acuity_notes: notes, chief_complaint: complaint,
  }).catch(() =>
    API.post('/api/orders', {
      patient_id: pid, type: 'nursing',
      description: `[TRIAGE] ESI-${level} | ${complaint}`,
      status: level <= 2 ? 'STAT' : 'Pending', notes,
    })
  );
  showToast(tr(`? Triage ESI-${level} saved!`, `? ?? ????? ESI-${level}!`));
  if (level <= 2) showToast(`? ${tr('CRITICAL ? Immediate attention required!', '??? ? ??? ?????? ??????!')}`
    `, 'error');
  setTimeout(() => window.nsSelectPatient(pid), 800);
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* TAB 8: RISK ASSESSMENTS (Morse + Braden + DVT + qSOFA) */
/* ===== */
window.nsTabRisks = function({ patient = {} }, pid) {
  return `
    <div style="font-weight:700;font-size:13px;margin-bottom:16px;color:var(--primary)">
      ? ${tr('Clinical Risk Assessments', '??????? ???????? ????????')}
    </div>
    <div style="display:grid;gap:16px">

      <!-- Morse Fall Risk -->
      <div style="border:1px solid var(--border);border-radius:12px;padding:14px">
        <div style="font-weight:700;font-size:13px;color:#ca8a04;margin-bottom:12px">? ${tr('Morse Fall Risk S
cale', '????? ??? ???? ?????')}</div>
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px">
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('History of Falls', '????
? ?????')}</label>
            <select class="form-input" id="morseFall" style="height:32px;font-size:12px">
              <option value="0">No ? 0 pts</option><option value="25">Yes ? 25 pts</option>
            </select>
          </div>
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Secondary Diagnosis', '?
???? ?????')}</label>
            <select class="form-input" id="morseSecDx" style="height:32px;font-size:12px">
              <option value="0">No ? 0 pts</option><option value="15">Yes ? 15 pts</option>
            </select>
          </div>
          <div>
            <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Ambulatory Aid', '?????
????')}</label>
            <select class="form-input" id="morseAid" style="height:32px;font-size:12px">
              <option value="none">None/Bed rest ? 0 pts</option>
              <option value="crutch">Crutches/Cane/Walker ? 15 pts</option>
              <option value="furniture">Furniture ? 30 pts</option>
            </select>
          </div>
        </div>
      </div>
    </div>
  `
}

```

```

        </select>
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">IV Therapy</label>
        <select class="form-input" id="morseIV" style="height:32px;font-size:12px">
            <option value="0">No ? 0 pts</option><option value="20">Yes ? 20 pts</option>
        </select>
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Gait', '????? ?????')}</
label>
        <select class="form-input" id="morseGait" style="height:32px;font-size:12px">
            <option value="normal">Normal ? 0 pts</option>
            <option value="weak">Weak ? 10 pts</option>
            <option value="impaired">Impaired ? 20 pts</option>
        </select>
    </div>
    <div>
        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${tr('Mental Status', '??????
???????')}</label>
        <select class="form-input" id="morseMental" style="height:32px;font-size:12px">
            <option value="oriented">Oriented ? 0 pts</option>
            <option value="forgets">Forgets Limitations ? 15 pts</option>
        </select>
    </div>
</div>
<div id="morseResult" style="margin-top:10px;padding:10px;border-radius:8px;text-align:center;font-we
ght:700;background:#f1f5f9"></div>
<div style="display:flex;gap:8px;margin-top:8px">
    <button class="btn" onclick="window.nsCalcMorse()" style="flex:1;background:#ca8a04;color:#fff;borde
r:none;font-size:12px;height:36px">
        ? ${tr('Calculate', '?????')}
    </button>
    <button class="btn" onclick="window.nsSaveMorse(${pid})" style="flex:1;background:#16a34a;color:#fff
;border:none;font-size:12px;height:36px">
        ? ${tr('Save', '???')}
    </button>
</div>
</div>

<!-- Braden Scale -->
<div style="border:1px solid var(--border);border-radius:12px;padding:14px">
    <div style="font-weight:700;font-size:13px;color:#7c3aed;margin-bottom:12px">?? ${tr('Braden Pressure
Injury Scale', '????? ?????? ?????? ??????')}</div>
    <div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:8px">
        ${[
            { id: 'brSensory', label: 'Sensory Perception', max: 4 },
            { id: 'brMoisture', label: 'Moisture', max: 4 },
            { id: 'brActivity', label: 'Activity', max: 4 },
            { id: 'brMobility', label: 'Mobility', max: 4 },
            { id: 'brNutrition', label: 'Nutrition', max: 4 },
            { id: 'brFriction', label: 'Friction/Shear', max: 3 },
        ]}.map(f => `
            <div>

```

```

        <label style="font-size:10px;font-weight:700;color:var(--text-dim)">${f.label}</label>
        <select class="form-input" id="${f.id}" style="height:32px;font-size:12px">
            ${Array.from({length:f.max}, (_, i) => i+1).map(n => `<option value="${n}">${n}</option>`).join('')}
        </select>
    </div>
    `).join('')}
</div>
<div id="bradenResult" style="margin-top:10px;padding:10px;border-radius:8px;text-align:center;font-weight:700;background:#f1f5f9"></div>
<div style="display:flex;gap:8px;margin-top:8px">
    <button class="btn" onclick="window.nsCalcBraden()" style="flex:1;background:#7c3aed;color:#fff;border:none;font-size:12px;height:36px">
        ? ${tr('Calculate', '????')}
    </button>
    <button class="btn" onclick="window.nsSaveBraden(${pid})" style="flex:1;background:#16a34a;color:#fff;border:none;font-size:12px;height:36px">
        ? ${tr('Save', '???')}
    </button>
</div>
</div>

<!-- qSOFA Sepsis Screening -->
<div style="border:2px solid #dc262633;border-radius:12px;padding:14px">
    <div style="font-weight:700;font-size:13px;color:#dc2626;margin-bottom:12px">? ${tr('qSOFA Sepsis Screening', '??? ?????? qSOFA')}</div>
    <div style="display:grid;gap:8px">
        ${[
            { id: 'qsRr', label: tr('Respiratory Rate ? 22/min', '???? ?????? ? 22/????') },
            { id: 'qsAlt', label: tr('Altered Mentation (GCS < 15)', '???? ?????? (GCS < 15)') },
            { id: 'qsSbp', label: tr('Systolic BP ? 100 mmHg', '??? ?????? ? 100') },
        ].map(c => `
            <label style="display:flex;align-items:center;gap:10px;padding:8px 12px;background:var(--surface-container,#f8fafc);border-radius:8px;cursor:pointer">
                <input type="checkbox" id="${c.id}" style="width:18px;height:18px">
                <span style="font-size:12px">${c.label}</span>
            </label>
            `).join('')}
    </div>
    <div id="qsofaResult" style="margin-top:10px;padding:10px;border-radius:8px;text-align:center;font-weight:700;background:#f1f5f9"></div>
    <button class="btn w-full" onclick="window.nsCalcQSOFA()" style="margin-top:8px;background:#dc2626;color:#fff;border:none;font-size:12px;height:36px">
        ? ${tr('Screen for Sepsis', '??? ??????')}
    </button>
</div>
</div>
`;
};

```

```

window.nsCalcMorse = function() {
    const result = calcMorse({
        fallHistory: parseInt(document.getElementById('morseFall')?.value) > 0,
        secDx:      parseInt(document.getElementById('morseSecDx')?.value) > 0,
    })
}

```

```

    aidType:      document.getElementById('morseAid')?.value,
    iv:           parseInt(document.getElementById('morseIV')?.value) > 0,
    gait:         document.getElementById('morseGait')?.value,
    mentalStatus: document.getElementById('morseMental')?.value,
  });
  const el = document.getElementById('morseResult');
  if (el) {
    el.style.background = result.color + '22';
    el.style.color = result.color;
    el.innerHTML = `${tr('Morse Score', '???? ????')}: <strong>${result.total}</strong> ? ${result.label}`;
  }
  window._NS._lastMorse = result;
};

window.nsSaveMorse = async function(pid) {
  window.nsCalcMorse();
  const r = window._NS._lastMorse;
  if (!r) return;
  try {
    await API.post('/api/nursing/risk-assessment', {
      patient_id: pid, assessment_type: 'Morse Fall Risk',
      total_score: r.total, risk_level: r.label,
      assessed_by: window._NS.currentUser?.name || '',
    });
    showToast(tr('? Morse saved!', '? ?? ??? ????? ????!'));
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.nsCalcBraden = function() {
  const result = calcBraden({
    sensory:   parseInt(document.getElementById('brSensory')?.value) || 4,
    moisture:  parseInt(document.getElementById('brMoisture')?.value) || 4,
    activity:  parseInt(document.getElementById('brActivity')?.value) || 4,
    mobility:  parseInt(document.getElementById('brMobility')?.value) || 4,
    nutrition: parseInt(document.getElementById('brNutrition')?.value) || 4,
    friction:  parseInt(document.getElementById('brFriction')?.value) || 3,
  });
  const el = document.getElementById('bradenResult');
  if (el) {
    el.style.background = result.color + '22';
    el.style.color = result.color;
    el.innerHTML = `${tr('Braden Score', '???? ?????')}: <strong>${result.total}/23</strong> ? ${result.label}`;
  };
  }
  window._NS._lastBraden = result;
};

window.nsSaveBraden = async function(pid) {
  window.nsCalcBraden();
  const r = window._NS._lastBraden;
  if (!r) return;
  try {
    await API.post('/api/nursing/risk-assessment', {
      patient_id: pid, assessment_type: 'Braden Scale',

```

```

        total_score: r.total, risk_level: r.label,
        assessed_by: window._NS.currentUser?.name || '',
    });
    showToast(tr('? Braden saved!', '? ?? ??? ????? ?????!'));
} catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

window.nsCalcQSofa = function() {
    const rr = document.getElementById('qsRr')?.checked ? 1 : 0;
    const alt = document.getElementById('qsAlt')?.checked ? 1 : 0;
    const sbp = document.getElementById('qsSbp')?.checked ? 1 : 0;
    const total = rr + alt + sbp;
    const el = document.getElementById('qsofaResult');
    const positive = total >= 2;
    if (el) {
        el.style.background = positive ? '#fee2e2' : '#f0fdf4';
        el.style.color = positive ? '#dc2626' : '#16a34a';
        el.innerHTML = `qSOFA: <strong>${total}/3</strong> ? ${positive
            ? `? ${tr('POSITIVE ? Suspected Sepsis! Escalate!', '????? ? ?????? ?????! ????? ?????!')}`
            : `? ${tr('Negative', '????')} `}`;
    }
};

/* ===== */
/* TAB 9: NURSING ORDERS BOARD */
/* ===== */

window.nsTabOrders = async function(container, pid) {
    container.innerHTML = `<div style="padding:24px;text-align:center;color:var(--text-dim)"><div style="font-size:32px">?</div><p>${tr('Loading orders...', '???? ???????...')}</p></div>`;
    let data = { orders: [], prescriptions: [] };
    try { data = await API.get(`/api/patients/${pid}/active-orders`); } catch {}
    const nursingOrders = (data.orders || []).filter(o => o.order_category === 'nursing' || (o.description || '').includes(['NURSING']) || (o.description || '').includes(['DIET']) || (o.description || '').includes(['IV']) || (o.description || '').includes(['ESCALATE']));
    const allOrders = data.orders || [];

    container.innerHTML = `
        <div style="font-weight:700;font-size:13px;margin-bottom:12px;color:var(--primary)">
            ? ${tr('Nursing Orders Board', '???? ?????? ???????')} (${allOrders.length})
        </div>
        ${!allOrders.length ? `<div style="text-align:center;padding:32px;color:var(--text-dim)"><div style="font-size:48px;opacity:0.3">?</div><p>${tr('No orders for this patient', '?? ????? ??????')}</p></div>` : `
        <div style="overflow-x:auto">
            <table style="width:100%;border-collapse:collapse;font-size:12px">
                <thead>
                    <tr style="background:var(--surface-container,#f1f5f9)">
                        <th style="padding:8px;text-align:start">${tr('Type', '????')}</th>
                        <th style="padding:8px;text-align:start">${tr('Order', '????')}</th>
                        <th style="padding:8px;text-align:center">${tr('Status', '?????')}</th>
                        <th style="padding:8px;text-align:center">${tr('Time', '????')}</th>
                        <th style="padding:8px;text-align:center">${tr('Action', '????')}</th>
                    </tr>
                </thead>
                <tbody>

```

```

    ${allOrders.map(o => {
        const isNursing = (o.description || '').includes('[NURSING]') || (o.description || '').includes('[
DIET]') || (o.description || '').includes('[IV]') || o.order_category === 'nursing';
        const isStat = o.status === 'STAT';
        const statusColor = { Pending: '#ca8a04', STAT: '#dc2626', 'In Progress': '#0ea5e9', Completed: '#
16a34a' }[o.status] || '#6b7280';
        const icon = (o.description || '').includes('[DIET]') ? '?' : (o.description || '').includes('[IV
]') ? '?' : (o.description || '').includes('[NURSING]') ? '?' : (o.description || '').includes('[ESCALATE]') ?
'?' : o.order_category === 'lab' ? '?' : '?';
        return `
            <tr style="border-bottom:1px solid var(--border);${isStat ? 'background:#fee2e2;' : isNursing ? 'b
ackground:#e0f2fe11;' : ''}">
                <td style="padding:8px;font-size:16px">${icon}</td>
                <td style="padding:8px;font-weight:600;max-width:200px;word-break:break-word">${escapeHTML(o.des
cription || '-')}</td>
                <td style="padding:8px;text-align:center">
                    <span style="background:${statusColor}22;color:${statusColor};padding:2px 8px;border-radius:10
px;font-size:10px;font-weight:700">
                        ${isStat ? '?' : ''}${escapeHTML(o.status || 'Pending')}
                    </span>
                </td>
                <td style="padding:8px;text-align:center;color:var(--text-dim)">${escapeHTML(o.created_at?.split
('T')[1]?.slice(0,5) || '-')}</td>
                <td style="padding:8px;text-align:center">
                    ${o.status !== 'Completed' ? `<button class="btn btn-sm" onclick="window.nsAckOrder(${safeId(o
.id)},${safeId(pid)})"
                        style="background:#16a34a;color:#fff;border:none;font-size:10px;padding:3px 8px">
                            ? ${tr('Done', '????')}
                        </button>` : `<span style="color:#16a34a;font-size:11px">?</span>`}
                </td>
            </tr>`;
        }).join('')}
    </tbody>
</table>
</div>
`
`
};

```

```

window.nsAckOrder = async function(orderId, pid) {
    try {
        await API.put(`/api/orders/${orderId}`, { status: 'Completed' }).catch(() =>
            API.post('/api/medical/records', {
                patient_id: pid, diagnosis: `Order completed by nurse`,
                notes: `Order ID ${orderId} marked done`, treatment: '',
            })
        );
        showToast(tr(`? Order marked as done`, `? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?`));
        window.nsTabOrders(document.getElementById('nsTabContent'), pid);
    } catch (e) { showToast(e?.message || tr('Error', `???`), 'error'); }
};

```

```

/* ===== */
/* TAB 10: NURSING NOTES */

```

```

/* ===== */
window.nsTabNotes = function({ chart = {} }, pid) {
  const records = chart.records || [];
  const nursingNotes = records.filter(r => r.diagnosis?.includes('Nursing') || r.treatment?.includes('Nursing') || r.notes?.includes('SBAR'));

  const templates = [
    { label: tr('Pre-op Check', '??? ?? ??? ??????'), text: 'Pre-operative check completed. Patient verified ID, allergies reviewed, consent signed, NPO confirmed, IV access patent.' },
    { label: tr('Fall Prevention', '??? ??????'), text: 'Fall prevention measures implemented: bed in low position, call bell within reach, non-slip footwear, fall risk sign on door.' },
    { label: tr('Pain Management', '????? ??????'), text: 'Pain assessed using NRS. Pain score: ____/10. Medication given as ordered. Reassessment in 30 minutes.' },
    { label: tr('IV Site Check', '??? ???? ??????'), text: 'IV site assessed: no redness, swelling, or infiltration noted. IV patent, infusing at prescribed rate.' },
    { label: tr('Discharge Teaching', '????? ??????'), text: 'Discharge teaching provided: medication compliance, diet, wound care, when to seek emergency care. Patient verbalized understanding.' },
  ];

  return `
    <div style="font-weight:700;font-size:13px;margin-bottom:12px;color:var(--primary)">
      ? ${tr('Nursing Notes', '????????? ??????????')}
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Quick Templates', '????? ??????')}</label>
      <div style="display:flex;flex-wrap:wrap;gap:6px">
        ${templates.map(t => `
          <button class="btn btn-sm" onclick="document.getElementById('nsNoteText').value='${t.text.replace(/'/g, '\\\\\'')}'"
            style="font-size:10px;padding:4px 8px;background:var(--surface-container,#f1f5f9);border:1px solid var(--border)">
              ${t.label}
            </button>
          `).join('')}
      </div>
    </div>
    <div class="form-group mb-8">
      <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Note Type', '??? ?????????')}</label>
      <select class="form-input" id="nsNoteType">
        <option value="Progress Note">Progress Note (????? ????)</option>
        <option value="Focus Charting">Focus Charting (????? ????)</option>
        <option value="Shift Note">Shift Note (????? ????)</option>
        <option value="Incident Note">Incident Note (????? ????)</option>
        <option value="Wound Assessment">Wound Assessment (????? ????)</option>
      </select>
    </div>
    <div class="form-group mb-12">
      <label style="font-size:11px;font-weight:700;color:var(--text-dim)">${tr('Note', '????????')}</label>
      <textarea class="form-input" id="nsNoteText" rows="6"
        placeholder="${tr('Enter nursing note...', '???? ????????? ??????????...')}"></textarea>
    </div>
    <button class="btn btn-primary w-full" onclick="window.nsSaveNote(${pid})" style="height:44px;font-size:13

```



```

px;margin-bottom:20px">
    ? ${tr('Save & Sign Note', '??? ?????? ?????????')}
</button>

<!-- Previous Notes -->
${nursingNotes.length ? `
<div>
    <div style="font-size:12px;font-weight:700;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">${tr('Previous Nursing Notes', '????????? ?????????? ?????????')} (${nursingNotes.length})</div>
    ${nursingNotes.slice(0, 5).map(r => `
        <div style="background:var(--surface-container,#f8fafc);border:1px solid var(--border);border-radius:10px;padding:10px;margin-bottom:8px">
            <div style="font-size:11px;color:var(--text-dim);margin-bottom:4px">${escapeHTML(r.created_at?.split('T')[0] || '-')} · ${escapeHTML(r.diagnosis || '-')}</div>
            <div style="font-size:12px">${escapeHTML((r.notes || r.treatment || '').slice(0, 200))}${(r.notes || '').length > 200 ? '...' : ''}</div>
            ${r.is_signed ? `<div style="margin-top:4px;font-size:10px;color:#16a34a">? ${tr('Signed', '????')}
? ${escapeHTML(r.signed_by || '')}</div>` : ''}
        </div>
    `).join('')}
</div>
` : ''}
`;
};

```

```

window.nsSaveNote = async function(pid) {
    const text = document.getElementById('nsNoteText')?.value?.trim() || '';
    const type = document.getElementById('nsNoteType')?.value || 'Progress Note';
    if (!text) return showToast(tr('Enter note text', '???? ?? ?????????'), 'error');
    try {
        await API.post('/api/medical/records', {
            patient_id: pid,
            diagnosis: `Nursing ${type}`,
            notes: text,
            treatment: `Nurse: ${window._NS.currentUser?.name || ''} | Shift: ${getShiftName()}`,
            symptoms: '',
        });
        showToast(tr('? Note saved!', '? ?? ??? ?????????!'));
        document.getElementById('nsNoteText') && (document.getElementById('nsNoteText').value = '');
        setTimeout(() => window.nsSwitchTab('notes'), 500);
    } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

```

```

/* ===== */
/* RIGHT PANEL: Quick Actions */
/* ===== */

```

```

window.nsRenderActionsPanel = function(patient, news2) {
    const panel = document.getElementById('nsActionsContent');
    if (!panel) return;
    const pid = safeId(patient.id);

    panel.innerHTML = `
        <!-- NEWS2 Quick Badge -->
        <div style="background:${news2.riskColor}11;border:2px solid ${news2.riskColor};border-radius:14px;padding
    `

```

```

:14px;text-align:center;margin-bottom:16px">
    <div style="font-size:10px;font-weight:700;color:${news2.riskColor};text-transform:uppercase">NEWS2 Score</div>
    <div style="font-size:44px;font-weight:900;color:${news2.riskColor};line-height:1.1">${news2.score}</div>
>
    <div style="font-size:10px;font-weight:700;color:${news2.riskColor};margin-top:2px">${news2.riskLabel}</div>
</div>

<!-- Quick Action Buttons -->
<div style="display:grid;gap:8px">
    <button class="btn" onclick="window.nsSwitchTab('vitals')"
        style="background:#0ea5e9;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ?? ${tr('Quick Vitals', '???? ???')}
    </button>
    <button class="btn" onclick="window.nsSwitchTab('emar')"
        style="background:#16a34a;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ? ${tr('Give Medication', '????? ???')}
    </button>
    <button class="btn" onclick="window.nsSwitchTab('io')"
        style="background:#7c3aed;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ? ${tr('I & O Entry', '????? ?????')}
    </button>
    ${news2.score >= 5 ? `
    <button class="btn" onclick="window.nsEscalate(${pid})"
        style="background:#dc2626;color:#fff;border:none;font-size:12px;height:40px;font-weight:800;border-radius:10px;animation:pulse 1s infinite">
        ? ${tr('ESCALATE TO DOCTOR', '????? ??????')}
    </button>` : ''}
    <button class="btn" onclick="window.nsSwitchTab('handover')"
        style="background:#ca8a04;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ? ${tr('Write SBAR', '????? SBAR')}
    </button>
    <button class="btn" onclick="window.nsSwitchTab('triage')"
        style="background:#ea580c;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ?? ${tr('Triage', '??? ??????')}
    </button>
    <button class="btn" onclick="window.nsMarkWithNurse(${pid})"
        style="background:#0369a1;color:#fff;border:none;font-size:12px;height:40px;font-weight:700;border-radius:10px">
        ??? ${tr('Mark: With Nurse', '??? ??????: ?? ??????')}
    </button>
</div>

<!-- Patient Info Summary -->
<div style="margin-top:16px;border:1px solid var(--border);border-radius:12px;padding:12px">
    <div style="font-size:11px;font-weight:700;color:var(--text-dim);margin-bottom:8px;text-transform:uppercase">? ${tr('Patient', '?????')}</div>
    <div style="font-size:12px;font-weight:700">${escapeHTML(isArabic ? (patient.name_ar || patient.name_en

```

```

|| '-' ) : (patient.name_en || patient.name_ar || '-'))}</div>
    ${patient.age ? `<div style="font-size:11px;color:var(--text-dim);margin-top:2px">? ${escapeHTML(String(
patient.age))} ${tr('yrs', '???')} . ${escapeHTML(patient.gender || '-')}</div>` : ''}
    ${patient.blood_type ? `<div style="font-size:11px;color:var(--text-dim)">? ${escapeHTML(patient.blood_t
ype)}</div>` : ''}
    ${patient.file_number ? `<div style="font-size:11px;color:var(--text-dim)">?? ${escapeHTML(String(patien
t.file_number))}</div>` : ''}
    </div>
`;
};

/* ===== */
/* MARK WITH NURSE */
/* ===== */
window.nsMarkWithNurse = async function(patientId) {
  if (!patientId) return;
  try {
    await API.put('/api/patients/' + patientId, { status: 'With Nurse' });
    showToast(tr('? Marked as With Nurse', '? ??? ?????: ?? ??????'));
    window.nsRefreshWorklist();
  } catch (e) { showToast(e?.message || tr('Error', '???'), 'error'); }
};

/* ===== */
/* REGISTRATION ? Override old renderNursing */
/* ===== */
window.renderNursingStation = renderNursingStation;

// Override the old renderNursing to use the new module
if (typeof window.renderNursingOld === 'undefined') {
  window.renderNursingOld = window.renderNursing || function() {};
}
window.renderNursing = async function(el) {
  return renderNursingStation(el);
};

console.log('[NursingStation] Module v1 loaded ? ??? ???? ???? Epic/Cerner ???? ?');

=====
FILE: public\js\obgyn-peds-station.js
=====

/**
 * obgyn-peds-station.js
 * OBGYN & Pediatrics Specialist Station - Stitch Google Design
 */

const ObgynPedsStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">

```

```

<header class="flex justify-between items-center mb-6">
  <h1 class="text-2xl font-bold text-slate-800">${tr('Maternal & Child Command Center', '???
? ????? ???? ??????')}</h1>
  <div class="flex gap-2">
    <button onclick="ObgynPedsStation.openMFMScan()" class="stitch-btn-primary px-4 py-2 b
g-rose-500 text-white rounded-lg shadow-sm hover:bg-rose-600 transition-all">
      ${tr('New Fetal Scan', '??? ????? ?????')}
    </button>
    <button onclick="ObgynPedsStation.openIVFCycle()" class="stitch-btn-secondary px-4 py-
2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
      ${tr('IVF Cycle Log', '??? ????? IVF')}
    </button>
  </div>
</header>

<div class="grid grid-cols-12 gap-6">
  <div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
      <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Fetal Vitals', '???????? ?????????')}</h3>
      <div class="grid grid-cols-2 gap-4">
        <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
          <span class="block text-xs text-slate-500">${tr('FHR', '??? ??????')}</spa
n>

          <span class="text-lg font-bold text-slate-800">145 bpm</span>
        </div>
        <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
          <span class="block text-xs text-slate-500">${tr('Gest. Age', '??? ??????')}

</span>

          <span class="text-lg font-bold text-slate-800">24w 3d</span>
        </div>
      </div>
    </div>

    <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
      <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Growth Percentile', '????? ??????')}</h3>
      <div class="flex flex-col items-center">
        <div class="stitch-radial-gauge w-32 h-32 relative">
          <svg class="w-full h-full viewBox="0 0 100 100">
            <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

            <circle cx="50" cy="50" r="40" stroke="#f43f5e" stroke-width="8" fill=
"none"

            stroke-dasharray="251.2" stroke-dashoffset="100" stroke-linecap="r
ound" />
          </svg>
          <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">15th</span>
        </div>
        <p class="mt-4 text-xs text-rose-600 font-medium">${tr('SGA - Small for Gestat
ional Age', '???? ????????? ????? ??????')}</p>
      </div>
    </div>
  </div>

```

```

    </div>

    <div class="col-span-6 space-y-6">
      <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-h-[600px]">
        <div class="flex border-b border-slate-200 mb-6">
          <button class="px-4 py-2 border-b-2 border-rose-500 text-rose-500 font-medium">
            <tr('MFM Scans', '????? MFM')></button>
          <button class="px-4 py-2 text-slate-500 hover:text-slate-800">
            <tr('IVF Cycle', '???? IVF')></button>
          <button class="px-4 py-2 text-slate-500 hover:text-slate-800">
            <tr('NICU Monitor', '????? NICU')></button>
        </div>
        <div id="obgyn-peds-workspace-content" class="space-y-4">
          <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate-200 rounded-xl">
            <tr('No active scan. Please upload a 4D ultrasound to begin.', '?? ??? ? ?? ??? . ??? ???? 4D ??????')></div>
          </div>
        </div>
      </div>

      <div class="col-span-3 space-y-6">
        <div class="stitch-card p-4 bg-rose-50 rounded-xl shadow-sm border border-rose-100">
          <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">?</span>
            <h3 class="text-sm font-bold text-rose-900 uppercase tracking-wider">
              <tr('AI Insights', '??? ?????? ?????????')></h3>
            </div>
            <div id="ai-insights-panel" class="space-y-4">
              <div class="p-3 bg-white rounded-lg border border-rose-200 shadow-sm">
                <p class="text-xs text-rose-700 font-semibold mb-1">
                  <tr('Fetal Anomaly', '???? ?????? ?????')></p>
                <p class="text-sm text-slate-600 italic">
                  Pattern suggests mild Ventricular Septal Defect. Similar to Case # la- la- la.</p>
                <button class="mt-2 text-xs text-rose-600 underline font-medium">
                  <tr('View Evidence', '??? ?????')></button>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  `;
},

openMFMScan: () => { alert('MFM Scan Modal Triggered'); },
openIVFCycle: () => { alert('IVF Cycle Modal Triggered'); }
};

```

FILE: public\js\onboarding-wizard.js

=====

```
/* =====
 * onboarding-wizard.js ? E0 Facility Onboarding Wizard (client)
 * Self-contained multi-step wizard launched from Settings (Admin only). POSTs to
 *   POST /api/admin/facilities/provision.
 *
 * Security: ALL dynamic output goes through escapeHTML; ids via safeId; bilingual via tr() (ar/en, RTL).
 * No secrets handled client-side beyond the one-time admin password the server returns (shown once).
 * Does not modify app.js globals; exposes window>NamaOnboardingWizard.open().
 * Depends on globals already present in app.js: tr, escapeHTML, safeId, showToast, isArabic, API.
 * ===== */
(function () {
  'use strict';

  // Mirror of server-side onboarding.js mapping (kept in sync). NAV index space 0..42.
  var ALL = []; for (var i = 0; i <= 42; i++) ALL.push(i);
  var HEALTH_CENTER = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 20, 21, 30, 33, 34, 35, 41, 42];
  var POLYCLINIC = [0, 1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 15, 20, 30, 34, 42];
  var GENERAL_HOSPITAL = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23
, 24, 30, 33, 34, 42];

  var ARCHETYPE_MODULES = {
    medical_city: ALL,
    large_hospital: ALL,
    general_hospital: GENERAL_HOSPITAL,
    polyclinic: POLYCLINIC,
    health_center: HEALTH_CENTER
  };

  var BED_ARCHETYPES = { medical_city: 1, large_hospital: 1, general_hospital: 1 };
  var ALLOWED_INTEGRATIONS = ['MOH', 'CBAHI', 'NPHIES', 'ZATCA', 'SCFHS', 'PACS'];

  // Module labels (mirror of NAV_ITEMS in app.js, for the module-matrix step).
  var MODULE_LABELS = [
    ['Dashboard', '???? ?????'], ['Reception', '????????'], ['Appointments', '????????'],
    ['Doctor Station', '???? ?????'], ['Laboratory', '????????'], ['Radiology', '?????'],
    ['Pharmacy', '????????'], ['HR', '???????? ?????'], ['Finance', '????????'], ['Insurance', '????????'],
    ['Inventory', '????????'], ['Nursing', '????????'], ['Waiting Queue', '????? ?????????'],
    ['Patient Accounts', '?????? ?????'], ['Reports', '????????'], ['Messaging', '????????'],
    ['Catalog', '????????'], ['Dept Requests', '????? ?????'], ['Surgery & Pre-Op', '????????? ?? ????'],
    ['Blood Bank', '??? ?????'], ['Consent Forms', '????????'], ['Emergency', '????????'],
    ['Inpatient ADT', '????????'], ['ICU', '???????? ?????'], ['CSSD', '???????? ?????'],
    ['Dietary', '????????'], ['Infection Control', '?????? ?????'], ['Quality', '?????'],
    ['Maintenance', '????????'], ['Transport', '??? ?????'], ['Medical Records', '???????? ?????'],
    ['Clinical Pharmacy', '???????? ?????'], ['Rehabilitation', '????? ?????'],
    ['Patient Portal', '????? ?????'], ['ZATCA E-Invoice', '????? ?????????'], ['Telemedicine', '???? ?? ???']
  ],

  ['Pathology', '??? ?????'], ['Social Work', '?????? ?????????'], ['Mortuary', '???? ?????'],
  ['CME', '????????'], ['Cosmetic Surgery', '????? ?????'], ['OB/GYN', '?????? ?????'],
  ['Settings', '????????']
];

  var ARCHETYPE_CARDS = [
```

```

    { key: 'medical_city', en: 'Medical City', ar: '????? ???', icon: '?', descEn: 'Multi-facility tenant, a
ll modules', descAr: '?????? ????? ????????? ?? ???????' },
    { key: 'large_hospital', en: 'Large Hospital', ar: '?????? ???', icon: '?', descEn: '>300 beds, ER+ICU+OR
', descAr: '???? ?? 300 ????? ????? ????????? ???????' },
    { key: 'general_hospital', en: 'General Hospital', ar: '?????? ???', icon: '?', descEn: 'Secondary care +
inpatient', descAr: '???? ?????? + ??????' },
    { key: 'polyclinic', en: 'Polyclinic', ar: '??????', icon: '?', descEn: 'Multi-specialty outpatient, no be
ds', descAr: '?????? ??????? ????????? ??? ??????' },
    { key: 'health_center', en: 'Health Center', ar: '???? ???', icon: '?', descEn: 'Primary care, no beds', d
escAr: '????? ??????? ??? ??????' }
];

var STEPS = [
    ['Archetype', '?????'], ['Identity & License', '?????? ?????????'], ['Structure', '??????'],
    ['Modules', '????????'], ['Admin User', '????? ??????'], ['Integrations', '?????????'],
    ['Branding & Language', '?????? ????????? ??????'], ['Confirm', '?????']
];

// wizard state
var state = null;
var step = 0;

function T(en, ar) { return (typeof tr === 'function') ? tr(en, ar) : en; }
function esc(v) { return (typeof escapeHTML === 'function') ? escapeHTML(v) : String(v == null ? '' : v); }
function toast(m, t) { if (typeof showToast === 'function') showToast(m, t); }

function resetState() {
    state = {
        archetype: '', tenant_name: '', subdomain: '', moh_license: '', cr_no: '', vat_no: '',
        facility_name: '', beds: 0, currency: 'SAR', timezone: 'Asia/Riyadh',
        modules: null, admin_username: '', admin_display_name: '', admin_password: '',
        integrations: {}, language: (typeof isArabic !== 'undefined' && isArabic) ? 'ar' : 'en', brand_color: ''
    };
    step = 0;
}

function rootEl() {
    var r = document.getElementById('nama-onboarding-root');
    if (!r) {
        r = document.createElement('div');
        r.id = 'nama-onboarding-root';
        document.body.appendChild(r);
    }
    return r;
}

function close() {
    var r = document.getElementById('nama-onboarding-root');
    if (r) r.innerHTML = '';
}

function defaultModulesFor(arch) {
    var list = ARCHETYPE_MODULES[arch] || [];
    var set = {};

```

```

    list.forEach(function (i) { set[i] = true; });
    return set;
}

function stepper() {
    var html = '<div style="display:flex;flex-wrap:wrap;gap:6px;margin-bottom:16px">';
    for (var i = 0; i < STEPS.length; i++) {
        var active = i === step;
        var done = i < step;
        var bg = active ? '#0d6efd' : (done ? '#198754' : '#e9ecef');
        var col = (active || done) ? '#fff' : '#555';
        html += '<span style="padding:4px 10px;border-radius:12px;font-size:12px;background:' + bg + ';color:' +
col + '">'
            + (i + 1) + '. ' + esc(T(STEPS[i][0], STEPS[i][1])) + '</span>';
    }
    return html + '</div>';
}

function stepArchetype() {
    var html = '<h3 style="margin:0 0 12px">' + esc(T('Choose facility type', '???? ??? ??????')) + '</h3>';
    html += '<div style="display:grid;grid-template-columns:repeat(auto-fill,minmax(180px,1fr));gap:10px">';
    ARCHETYPE_CARDS.forEach(function (c) {
        var sel = state.archetype === c.key;
        html += '<button type="button" data-arch="' + esc(c.key) + '" class="nama-arch-card" '
            + 'style="text-align:start;padding:14px;border-radius:10px;cursor:pointer;border:2px solid '
            + (sel ? '#0d6efd' : '#ddd') + ';background:' + (sel ? '#eaf2ff' : '#fff') + '">'
            + '<div style="font-size:26px">' + esc(c.icon) + '</div>'
            + '<div style="font-weight:700;margin-top:6px">' + esc(T(c.en, c.ar)) + '</div>'
            + '<div style="font-size:12px;color:#666;margin-top:4px">' + esc(T(c.descEn, c.descAr)) + '</div>'
            + '</button>';
    });
    html += '</div>';
    return html;
}

function field(label, key, type, placeholder) {
    type = type || 'text';
    return '<label style="display:block;margin-bottom:10px">'
        + '<span style="display:block;font-size:13px;font-weight:600;margin-bottom:3px">' + esc(label) + '</span>'
        + '<input data-key="' + esc(key) + '" type="' + esc(type) + '" value="' + esc(state[key]) + '" '
        + 'placeholder="' + esc(placeholder || '') + '" '
        + 'style="width:100%;padding:8px;border:1px solid #ccc;border-radius:6px;box-sizing:border-box"></label>'
        + '</div>';
}

function stepIdentity() {
    var html = '<h3 style="margin:0 0 12px">' + esc(T('Identity & License', '????? ??????')) + '</h3>';
    html += field(T('Tenant / Organization name', '??? ?????? / ?????'), 'tenant_name', 'text', 'jumanaMedic
al City');
    html += field(T('Subdomain (a-z, 0-9, hyphen)', '????? ?????? (a-z ?0-9 ?????)'), 'subdomain', 'text', 'n
ama-city');
    html += field(T('MOH License (optional)', '????? ?????? ????? (?????)'), 'moh_license', 'text', '');
    html += field(T('Commercial Registration (optional)', '????? ?????? (?????)'), 'cr_no', 'text', '');
}

```



```

    html += field(T('VAT Number (optional)', '????? ??????? (??????)'), 'vat_no', 'text', '');
    return html;
}

function stepStructure() {
    var hasBeds = !!BED_ARCHETYPES[state.archetype];
    var html = '<h3 style="margin:0 0 12px">' + esc(T('Structure', '????? ???????')) + '</h3>';
    html += field(T('Primary facility name', '??? ?????? ???????'), 'facility_name', 'text', '');
    if (hasBeds) {
        html += field(T('Inpatient beds', '??? ??????'), 'beds', 'number', '0');
    } else {
        html += '<p style="font-size:13px;color:#666">' + esc(T('This archetype is bed-less (outpatient only).',
'??? ?????? ??? ?????? (?????? ?????? ???).')) + '</p>';
    }
    html += field(T('Currency (ISO 3-letter)', '????? (??? ?????)'), 'currency', 'text', 'SAR');
    html += field(T('Timezone', '?????? ???????'), 'timezone', 'text', 'Asia/Riyadh');
    return html;
}

function stepModules() {
    if (!state.modules) state.modules = defaultModulesFor(state.archetype);
    var allowed = {};
    (ARCHETYPE_MODULES[state.archetype] || []).forEach(function (i) { allowed[i] = true; });
    var html = '<h3 style="margin:0 0 8px">' + esc(T('Enabled modules', '?????? ???????')) + '</h3>';
    html += '<p style="font-size:12px;color:#666;margin:0 0 10px">' + esc(T('Toggle within the archetype set.
Dashboard & Settings are always on.', '???/??? ??? ?????? ??????. ??? ?????? ?????????? ?????????? ??????'))
+ '</p>';
    html += '<div style="display:grid;grid-template-columns:repeat(auto-fill,minmax(190px,1fr));gap:4px;max-height:280px;overflow:auto;border:1px solid #eee;padding:8px;border-radius:8px">';
    for (var i = 0; i < MODULE_LABELS.length; i++) {
        var inSet = !!allowed[i];
        var locked = (i === 0 || i === 42);
        var checked = inSet && (state.modules[i] || locked);
        var dis = (!inSet || locked) ? ' disabled' : '';
        html += '<label style="display:flex;align-items:center;gap:6px;font-size:13px;opacity:' + (inSet ? '1' :
'0.4') + '>'
            + '<input type="checkbox" data-mod="' + (typeof safeId === 'function' ? safeId(i) : i) + '" + (checked ? ' checked' : '') + dis + '> '
            + esc(T(MODULE_LABELS[i][0], MODULE_LABELS[i][1])) + (inSet ? '' : '?') + '</label>';
    }
    html += '</div>';
    return html;
}

function stepAdmin() {
    var html = '<h3 style="margin:0 0 12px">' + esc(T('Administrator account', '???? ???? ??????')) + '</h3>';
    html += field(T('Admin username', '??? ?????? ??????'), 'admin_username', 'text', 'admin');
    html += field(T('Admin display name', '????? ??????'), 'admin_display_name', 'text', '');
    html += field(T('Password (leave blank to auto-generate a strong one)', '???? ?????? (????? ?????? ??????
???? ?????????)'), 'admin_password', 'password', '');
    html += '<p style="font-size:12px;color:#a00">' + esc(T('No default passwords. If blank, a strong random password is generated and shown once.', '?? ?????? ????? ??????????. ?? ?????? ?????? ?????????? ?????? ?????? ?????? ??? ??
???.')) + '</p>';
    return html;
}

```

```

}

function stepIntegrations() {
  var html = '<h3 style="margin:0 0 8px">' + esc(T('Integrations (gated)', '???????? (??????)')) + '</h3>'
;
  html += '<p style="font-size:12px;color:#666;margin:0 0 10px">' + esc(T('Mark integrations to enable later
. No keys, secrets, or certificates are entered here.', '???? ????????? ????????? ??????. ?? ????? ?????? ?? ??
??? ?? ?????? ???.')) + '</p>';
  ALLOWED_INTEGRATIONS.forEach(function (nm) {
    var on = !!state.integrations[nm];
    html += '<label style="display:flex;align-items:center;gap:8px;margin-bottom:6px;font-size:14px">'
      + '<input type="checkbox" data-int="' + esc(nm) + '"' + (on ? ' checked' : '') + '>' + esc(nm) + '</l
abel>';
    });
  return html;
}

function stepBranding() {
  var html = '<h3 style="margin:0 0 12px">' + esc(T('Branding & Language', '????? ?????? ??????')) + '</h3
>';
  html += '<label style="display:block;margin-bottom:10px"><span style="display:block;font-size:13px;font-we
ight:600;margin-bottom:3px">' + esc(T('Default language', '???? ?????????')) + '</span>'
    + '<select data-key="language" style="width:100%;padding:8px;border:1px solid #ccc;border-radius:6px">'
    + '<option value="ar"' + (state.language === 'ar' ? ' selected' : '') + '>' + esc(T('Arabic', '????????'))
) + '</option>'
    + '<option value="en"' + (state.language === 'en' ? ' selected' : '') + '>' + esc(T('English', '????????
??')) + '</option>'
    + '</select></label>';
  html += field(T('Brand color (#RRGGBB, optional)', '??? ?????? (#RRGGBB ??????)'), 'brand_color', 'text',
'#0d6efd');
  return html;
}

function stepConfirm() {
  var mods = collectModules();
  var ints = Object.keys(state.integrations).filter(function (k) { return state.integrations[k]; });
  var rows = [
    [T('Archetype', '????'), state.archetype],
    [T('Tenant', '????????'), state.tenant_name],
    [T('Subdomain', '????? ??????'), state.subdomain],
    [T('Facility', '????????'), state.facility_name],
    [T('Beds', '????????'), String(state.beds || 0)],
    [T('Currency', '?????'), state.currency],
    [T('Timezone', '??????? ????????'), state.timezone],
    [T('Enabled modules', '??????? ?????????'), String(mods.length)],
    [T('Admin username', '????? ??????'), state.admin_username],
    [T('Password', '???? ??????'), state.admin_password ? T('(provided)', '(?????))' : T('(auto-generate)',
'(??????? ?????????)')],
    [T('Integrations', '????????'), ints.length ? ints.join(', ') : T('none', '?? ??')],
    [T('Language', '?????'), state.language]
  ];
  var html = '<h3 style="margin:0 0 12px">' + esc(T('Review & confirm', '???????? ?????????')) + '</h3>';
  html += '<table style="width:100%;border-collapse:collapse;font-size:14px">';
  rows.forEach(function (r) {

```

```

    html += '<tr><td style="padding:6px 8px;border-bottom:1px solid #eee;font-weight:600;width:45%">' + esc(
r[0]) + '</td>'
    + '<td style="padding:6px 8px;border-bottom:1px solid #eee">' + esc(r[1]) + '</td></tr>';
  });
  html += '</table>';
  return html;
}

function collectModules() {
  // returns array of enabled indices intersected with archetype set
  var allowed = ARCHETYPE_MODULES[state.archetype] || [];
  if (!state.modules) return allowed.slice();
  var out = [];
  allowed.forEach(function (i) {
    if (i === 0 || i === 42 || state.modules[i]) out.push(i);
  });
  return out;
}

function renderBody() {
  switch (step) {
    case 0: return stepArchetype();
    case 1: return stepIdentity();
    case 2: return stepStructure();
    case 3: return stepModules();
    case 4: return stepAdmin();
    case 5: return stepIntegrations();
    case 6: return stepBranding();
    case 7: return stepConfirm();
    default: return '';
  }
}

function validateStep() {
  if (step === 0 && !state.archetype) return T('Select an archetype', '???? ?????');
  if (step === 1) {
    if (!state.tenant_name || state.tenant_name.trim().length < 2) return T('Tenant name is required', '???
???????? ?????');
    if (!/^[a-z0-9]([a-z0-9-]{0,61}[a-z0-9])?$/i.test(state.subdomain)) return T('Invalid subdomain', '???? ?
??? ??? ?????');
  }
  if (step === 2) {
    if (!state.facility_name || state.facility_name.trim().length < 2) return T('Facility name is required',
'??? ????????? ?????');
    if (!/^[A-Za-z]{3}$/i.test(state.currency || '')) return T('Currency must be 3 letters', '?????? 3 ?????')
;
  }
  if (step === 4) {
    if (!/^[A-Za-z0-9._-]{3,50}$/i.test(state.admin_username || '')) return T('Invalid admin username', '???
?????? ??? ?????');
    if (state.admin_password && state.admin_password.length < 8) return T('Password must be >= 8 chars', '??
?? ?????? 8 ????? ??? ?????');
  }
  return null;
}

```

```

}

function render() {
  var dir = (typeof isArabic !== 'undefined' && isArabic) ? 'rtl' : 'ltr';
  var r = rootEl();
  var isLast = step === STEPS.length - 1;
  var html = ''
    + '<div style="position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:9999;display:flex;align-items:flex-start;justify-content:center;overflow:auto" dir="' + dir + '">'
    + '<div style="background:#fff;border-radius:12px;max-width:680px;width:94%;margin:32px auto;padding:20px;box-shadow:0 10px 40px rgba(0,0,0,.3)">'
    + '<div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">'
    + '<h2 style="margin:0;font-size:18px">' + esc(T('Facility Onboarding Wizard', '????? ????? ???????')) + '</h2>'
    + '<button id="nama-onb-close" type="button" style="border:none;background:#eee;border-radius:6px;padding:4px 10px;cursor:pointer"></button>'
    + '</div>'
    + stepper()
    + '<div id="nama-onb-body" style="min-height:240px">' + renderBody() + '</div>'
    + '<div style="display:flex;justify-content:space-between;margin-top:18px">'
    + '<button id="nama-onb-back" type="button" ' + (step === 0 ? 'disabled' : '') + ' style="padding:8px 16px;border-radius:6px;border:1px solid #ccc;background:#f8f9fa;cursor:pointer">' + esc(T('Back', '?????')) + '</button>'
    + '<button id="nama-onb-next" type="button" style="padding:8px 18px;border-radius:6px;border:none;background:#0d6efd;color:#fff;cursor:pointer">'
    + esc(isLast ? T('Provision facility', '????? ???????') : T('Next', '?????')) + '</button>'
    + '</div></div></div>';
  r.innerHTML = html;
  bind();
}

function bind() {
  var r = rootEl();
  var closeBtn = r.querySelector('#nama-onb-close');
  if (closeBtn) closeBtn.onclick = close;
  var backBtn = r.querySelector('#nama-onb-back');
  if (backBtn) backBtn.onclick = function () { if (step > 0) { step--; render(); } };
  var nextBtn = r.querySelector('#nama-onb-next');
  if (nextBtn) nextBtn.onclick = onNext;

  // archetype cards
  Array.prototype.forEach.call(r.querySelectorAll('.nama-arch-card'), function (btn) {
    btn.onclick = function () {
      var k = btn.getAttribute('data-arch');
      if (ARCHETYPE_MODULES[k]) {
        state.archetype = k;
        state.modules = defaultModulesFor(k);
        if (!BED_ARCHETYPES[k]) state.beds = 0;
        render();
      }
    };
  });
  // text/select inputs
  Array.prototype.forEach.call(r.querySelectorAll('[data-key]'), function (inp) {

```

```

inp.oninput = inp.onChange = function () {
  var key = inp.getAttribute('data-key');
  var val = inp.value;
  if (key === 'beds') val = parseInt(val, 10) || 0;
  if (key === 'currency') val = String(val).toUpperCase();
  state[key] = val;
};
});
// module toggles
Array.prototype.forEach.call(r.querySelectorAll('[data-mod]'), function (cb) {
  cb.onChange = function () {
    var idx = parseInt(cb.getAttribute('data-mod'), 10);
    if (!state.modules) state.modules = defaultModulesFor(state.archetype);
    state.modules[idx] = cb.checked;
  };
});
// integration toggles
Array.prototype.forEach.call(r.querySelectorAll('[data-int]'), function (cb) {
  cb.onChange = function () { state.integrations[cb.getAttribute('data-int')] = cb.checked; };
});
}

function onNext() {
  var err = validateStep();
  if (err) { toast(err, 'error'); return; }
  if (step < STEPS.length - 1) { step++; render(); return; }
  submit();
}

function submit() {
  var integrations = Object.keys(state.integrations)
    .filter(function (k) { return state.integrations[k]; })
    .map(function (k) { return { name: k, enabled: true }; });
  var payload = {
    archetype: state.archetype,
    tenant_name: state.tenant_name,
    subdomain: state.subdomain,
    moh_license: state.moh_license,
    cr_no: state.cr_no,
    vat_no: state.vat_no,
    facility_name: state.facility_name,
    beds: state.beds,
    currency: state.currency,
    timezone: state.timezone,
    modules: collectModules(),
    admin_username: state.admin_username,
    admin_display_name: state.admin_display_name,
    admin_password: state.admin_password || undefined,
    integrations: integrations,
    language: state.language,
    brand_color: state.brand_color || undefined
  };
  if (!window.API || typeof window.API.post !== 'function') { toast(T('API unavailable', '?????? ????'), 'error'); return; }
}

```

```

var endpoint = window.location.pathname.includes('/super-admin')
? '/api/super-admin/tenants/provision'
: '/api/admin/facilities/provision';
window.API.post(endpoint, payload).then(function (res) {
    showResult(res);
}).catch(function (e) {
    toast((e && e.message) ? e.message : T('Provisioning failed', '??? ?????'), 'error');
});
}

function showResult(res) {
    var r = rootEl();
    var dir = (typeof isArabic !== 'undefined' && isArabic) ? 'rtl' : 'ltr';
    var pwLine = res.admin_password
    ? '<div style="margin-top:10px;padding:10px;background:#fff3cd;border:1px solid #ffe69c;border-radius:8px;font-size:14px">'
    + '<b>' + esc(T('Admin password (shown once):', '???? ???? ?????? (???? ???? ????):')) + '</b><br>'
    + '<code style="font-size:15px">' + esc(res.admin_password) + '</code><br>'
    + '<span style="font-size:12px;color:#a00">' + esc(T('Copy now and deliver securely. It will not be shown again.', '????? ???? ?????? ?????? ?? ????? ??????.')) + '</span></div>'
    : '';
    r.innerHTML = '<div style="position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:9999;display:flex;align-items:flex-start;justify-content:center" dir="' + dir + '">'
    + '<div style="background:#fff;border-radius:12px;max-width:520px;width:92%;margin:48px auto;padding:24px;text-align:center">'
    + '<div style="font-size:40px"></div>'
    + '<h2 style="margin:8px 0">' + esc(T('Facility provisioned', '?? ????? ??????')) + '</h2>'
    + '<p style="margin:4px 0">' + esc(T('Tenant ID', '????? ??????')) + ': <b>' + esc(res.tenant_id) + '</b></p>'
    + '<p style="margin:4px 0;font-size:14px;color:#555">' + esc(T('Enabled modules', '??????? ??????'))
    + ': ' + esc((res.enabled_modules || []).length) + '</p>'
    + pwLine
    + '<button id="nama-onb-done" type="button" style="margin-top:16px;padding:8px 22px;border:none;border-radius:6px;background:#198754;color:#fff;cursor:pointer">' + esc(T('Done', '??')) + '</button>'
    + '</div></div>';
    var done = r.querySelector('#nama-onb-done');
    if (done) done.onclick = close;
    toast(T('Facility provisioned successfully', '?? ????? ?????? ?????'), 'success');
}

function open() {
    resetState();
    render();
}

window>NamaOnboardingWizard = { open: open, close: close };
})();

```

```

=====
FILE: public\js\oncology-station.js
=====

```

```

/**
 * oncology-station.js
 * Oncology & Hematology Specialist Station - Stitch Google Design
 */

const OncologyStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Oncology Command Center', '???? ?????
??????? ?????? ?????')}</h1>
          <div class="flex gap-2">
            <button onclick="OncologyStation.openChemoCycle()" class="stitch-btn-primary px-4 py-2
bg-purple-700 text-white rounded-lg shadow-sm hover:bg-purple-800 transition-all">
              ${tr('Log Chemo Cycle', '????? ??? ????')}
            </button>
            <button onclick="OncologyStation.openBMTMonitoring()" class="stitch-btn-secondary px-4
py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('BMT Monitoring', '????? ????? ?????')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <!-- Left: Oncology Context -->
          <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Toxicity Status', '???? ?????')}</h3>
              <div id="onco-vitals-grid" class="grid grid-cols-2 gap-4">
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                  <span class="block text-xs text-slate-500">${tr('ANC Count', '??? ???????
??')}</span>
                  <span class="text-lg font-bold text-red-600">450 /uL</span>
                </div>
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                  <span class="block text-xs text-slate-500">${tr('Platelets', '??????')}</
span>
                  <span class="text-lg font-bold text-slate-800">110k</span>
                </div>
              </div>
            </div>

            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Genomic Profile', '????? ?????')}</h3>
              <div class="flex flex-col items-center">
                <div class="stitch-radial-gauge w-32 h-32 relative">
                  <svg class="w-full h-full viewBox="0 0 100 100">
                    <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
                    <circle cx="50" cy="50" r="40" stroke="#7e22ce" stroke-width="8" fill=

```

```
"none"

stroke-dasharray="251.2" stroke-dashoffset="80" stroke-linecap="ro
und" />

</svg>
<span class="absolute inset-0 flex items-center justify-center text-xl font-b
t-bold text-slate-800">Match</span>
</div>
<p class="mt-4 text-xs text-purple-600 font-medium">${tr('Targeted Therapy Can
didate', '???? ?????? ??????')}</p>
</div>
</div>
</div>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-purple-700 text-purple-700 font-med
ium">${tr('Chemo Logs', '????? ?????????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('BMT Tracki
ng', '???? ?????????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Genomics A
I', '???? ?????????')}</button>
    </div>
    <div id="onco-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
        ${tr('No active cycle. Please log a new chemotherapy session.', '?? ???? ?
??? ?????. ???? ?????? ???? ?????? ??????')}
      </div>
    </div>
  </div>
</div>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-purple-50 rounded-xl shadow-sm border border-purple-100
">
    <div class="flex items-center gap-2 mb-4">
      <span class="text-xl">?</span>
      <h3 class="text-sm font-bold text-purple-900 uppercase tracking-wider">${tr('A
I Insights', '??? ?????? ?????????')}</h3>
    </div>
    <div id="ai-insights-panel" class="space-y-4">
      <div class="p-3 bg-white rounded-lg border border-purple-200 shadow-sm">
        <p class="text-xs text-purple-700 font-semibold mb-1">${tr('Targeted Thera
py', '?????? ?????? ?????????')}</p>
        <p class="text-sm text-slate-600 italic">"Genomic profile suggests high re
sponse to Pembrolizumab. Similar to Case # la-992."</p>
        <button class="mt-2 text-xs text-purple-600 underline font-medium">${tr('V
iew Evidence', '??? ??????')}</button>
      </div>
    </div>
  </div>
</div>
```



```

        </div>
      </div>
    </div>
  </div>
  `;
},

openChemoCycle: () => { alert('Chemo Cycle Modal Triggered'); },
openBMTMonitoring: () => { alert('BMT Monitoring Modal Triggered'); }
};

=====
FILE: public\js\pulmonology-station.js
=====

/**
 * pulmonology-station.js
 * Pulmonology Specialist Station - Stitch Google Design
 */

const PulmonologyStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Pulmonology Command Center', '???? ??
          ?? ????? ?????? ??????')}</h1>
          <div class="flex gap-2">
            <button onclick="PulmonologyStation.openPFTUpload()" class="stitch-btn-primary px-4 py
            -2 bg-emerald-600 text-white rounded-lg shadow-sm hover:bg-emerald-700 transition-all">
              ${tr('Upload PFT', '??? ????? ?????')}
            </button>
            <button onclick="PulmonologyStation.openSleepStudy()" class="stitch-btn-secondary px-4
            py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('Sleep Study', '????? ?????')}
            </button>
          </div_header>

          <div class="grid grid-cols-12 gap-6">
            <!-- Left: Respiratory Context -->
            <div class="col-span-3 space-y-6">
              <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
                <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
                r('Respiratory Vitals', '???????? ?????????')}</h3>
                <div id="pulmo-vitals-grid" class="grid grid-cols-2 gap-4">
                  <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('SpO2', '???? ?????????')}<
            /span>

                    <span class="text-lg font-bold text-slate-800">94%</span>
                  </div>
                  <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">

```

```
<span class="block text-xs text-slate-500">${tr('RR', '???? ?????')}</spa
n>

<span class="text-lg font-bold text-slate-800">18 bpm</span>
</div>
</div_la>
</div_la>

<div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
  <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Lung Function Index', '???? ????? ?????')}</h3>
  <div class="flex flex-col items-center">
    <div class="stitch-radial-gauge w-32 h-32 relative">
      <svg class="w-full h-full" viewBox="0 0 100 100">
        <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
        <circle cx="50" cy="50" r="40" stroke="#10b981" stroke-width="8" fill=
"none"
        stroke-dasharray="251.2" stroke-dashoffset="100" stroke-linecap="r
ound" />
      </svg>
      <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">60%</span>
    </div>
    <p class="mt-4 text-xs text-emerald-600 font-medium">${tr('Moderate Obstructio
n', '?????? ?????')}</p>
  </div_la>
</div_la>
</div_la>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-emerald-600 text-emerald-600 font-m
edium">${tr('Spirometry', '???? ?????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Sleep Stud
y', '????? ?????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Bronchosco
py', '????? ?????')}</button>
    </div_la>
    <div id="pulmo-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
        ${tr('No active analysis. Please upload PFT data to begin.', '?? ???? ????
? ????. ??? ???? ???? ???? ???? ???? ????')}
      </div>
    </div_la>
  </div_la>
</div_la>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-emerald-50 rounded-xl shadow-sm border border-emerald-1
```

```

00">
    <div class="flex items-center gap-2 mb-4">
      <span class="text-xl">?</span>
      <h3 class="text-sm font-bold text-emerald-900 uppercase tracking-wider">${tr('
AI Insights', '??? ?????? ?????????')}</h3>
    </div_ la>
    <div id="ai-insights-panel" class="space-y-4">
      <div class="p-3 bg-white rounded-lg border border-emerald-200 shadow-sm">
        <p class="text-xs text-emerald-700 font-semibold mb-1">${tr('Suggested Dia
gnosis', '?????? ???????')}</p>
        <p class="text-sm text-slate-600 italic">"Pattern suggests COPD with mild
exacerbation. Similar to Case #4412."</p>
        <button class="mt-2 text-xs text-emerald-600 underline font-medium">${tr('
View Evidence', '??? ??????')}</button>
      </div>
    </div_ la>
  </div_ la>
</div_ la>
</div_ la>
</div_ la>
</div_ la>
  `;
},

openPFTUpload: () => {
  alert('PFT Upload Modal Triggered');
},

openSleepStudy: () => {
  alert('Sleep Study Modal Triggered');
}
};

```

```

=====
FILE: public\js\rheuma-station.js
=====

```

```

/**
 * rheuma-station.js
 * Rheumatology & Immunology Specialist Station - Stitch Google Design
 */

const RheumaStation = {
  render: async (patientId) => {
    const container = document.getElementById('app-content');
    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr('Rheumatology Command Center', '???? ??
          ??? ?????????? ?????????')}</h1>
          <div class="flex gap-2">
            <button onclick="RheumaStation.openScoreLog()" class="stitch-btn-primary px-4 py-2 bg-
            rose-600 text-white rounded-lg shadow-sm hover:bg-rose-700 transition-all">

```

```

        ${tr('Log DAS28/SLEDAI', '???? DAS28/SLEDAI')}}
    </button>
    <button onclick="RheumaStation.openSerologyLog()" class="stitch-btn-secondary px-4 py-
2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
        ${tr('Serology Markers', '????? ????')}
    </button>
</div>
</header>

<div class="grid grid-cols-12 gap-6">
    <!-- Left: Rheuma Context -->
    <div class="col-span-3 space-y-6">
        <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
            <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Disease Activity', '???? ?????')}</h3>
            <div id="rheuma-vitals-grid" class="grid grid-cols-2 gap-4">
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('DAS28', '???? DAS28')}</
span>

                    <span class="text-lg font-bold text-slate-800">5.4</span>
                </div>
                <div class="stitch-gauge-item p-3 bg-slate-100 rounded-lg text-center">
                    <span class="block text-xs text-slate-500">${tr('SLEDAI', '???? SLEDAI')}
</span>

                    <span class="text-lg font-bold text-slate-800">8.0</span>
                </div>
            </div>
        </div>

        <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
            <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${t
r('Autoimmune Status', '????? ????????')}</h3>
            <div class="flex flex-col items-center">
                <div class="stitch-radial-gauge w-32 h-32 relative">
                    <svg class="w-full h-full" viewBox="0 0 100 100">
                        <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />

                        <circle cx="50" cy="50" r="40" stroke="#e11d48" stroke-width="8" fill=
"none"

                        stroke-dasharray="251.2" stroke-dashoffset="120" stroke-linecap="r
ound" />
                    </svg>
                    <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">High</span>
                </div>
                <p class="mt-4 text-xs text-rose-600 font-medium">${tr('Active Inflammation',
'????? ?')}</p>
            </div>
        </div>
    </div>

    <!-- Center: Active Workspace -->
    <div class="col-span-6 space-y-6">
        <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-

```

```

        <div class="flex border-b border-slate-200 mb-6">
            <button class="px-4 py-2 border-b-2 border-rose-600 text-rose-600 font-medium"
>${tr('Symptom Cluster', '????? ??????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Serology T
rends', '??????? ?????')}</button>
            <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('AI Differe
ntial', '??????? ???????? AI')}</button>
        </div>
        <div id="rheuma-workspace-content" class="space-y-4">
            <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
                ${tr('No active cluster. Please log serology markers to begin.', '?? ???
????? ????. ??? ???? ?????? ?????? ??????.'))}
            </div>
        </div>
    </div>
</div>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
    <div class="stitch-card p-4 bg-rose-50 rounded-xl shadow-sm border border-rose-100">
        <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">?</span>
            <h3 class="text-sm font-bold text-rose-900 uppercase tracking-wider">${tr('AI
Insights', '??? ?????? ?????????')}</h3>
        </div>
        <div id="ai-insights-panel" class="space-y-4">
            <div class="p-3 bg-white rounded-lg border border-rose-200 shadow-sm">
                <p class="text-xs text-rose-700 font-semibold mb-1">${tr('Suggested Diagno
sis', '??????? ????????')}</p>
                <p class="text-sm text-slate-600 italic">"Pattern suggests Systemic Lupus
Erythematosus (SLE) with renal involvement. Similar to Case # la- la- la."</p>
                <button class="mt-2 text-xs text-rose-600 underline font-medium">${tr('Vie
w Evidence', '??? ??????')}</button>
            </div>
        </div>
    </div>
</div>
</div>
</div>
    `;
},

openScoreLog: () => { alert('Score Log Modal Triggered'); },
openSerologyLog: () => { alert('Serology Log Modal Triggered'); }
};

```

```
=====
FILE: public\js\surgery-station.js
=====
```

```

/**
 * surgery-station.js
 * Unified Surgical Specialist Station - Stitch Google Design
 * Adapts based on the surgeon's specialty (S-MODE)
 */

const SurgeryStation = {
  render: async (patientId, specialty = 'general') => {
    const container = document.getElementById('app-content');

    const specialtyConfig = {
      'general': { title: 'General Surgery', color: 'bg-slate-700', btn: 'Log Procedure', icon: '?' },
      'cardio': { title: 'Cardiothoracic Surgery', color: 'bg-red-700', btn: 'Log Heart Surgery', icon: '?' },
      'neuro': { title: 'Neurosurgery', color: 'bg-indigo-700', btn: 'Log Neuro Session', icon: '?' },
      'ortho': { title: 'Orthopedic Surgery', color: 'bg-blue-700', btn: 'Log Joint Replacement', icon: '?' },
      'eye': { title: 'Ophthalmology', color: 'bg-sky-600', btn: 'Log Eye Surgery', icon: '?' },
      'ent': { title: 'ENT Surgery', color: 'bg-orange-600', btn: 'Log ENT Session', icon: '?' },
      'urology': { title: 'Urology Surgery', color: 'bg-yellow-600', btn: 'Log Urology Procedure', icon: '?' },
      'plastic': { title: 'Plastic Surgery', color: 'bg-pink-600', btn: 'Log Aesthetic Procedure', icon: '?' }
    };

    const config = specialtyConfig[specialty] || specialtyConfig['general'];

    container.innerHTML = `
      <div class="stitch-station-container p-6 bg-slate-50 min-h-screen">
        <header class="flex justify-between items-center mb-6">
          <h1 class="text-2xl font-bold text-slate-800">${tr(config.title + ' Command Center', config.title + ' ??? ???? ')}</h1>
          <div class="flex gap-2">
            <button onclick="SurgeryStation.openChecklist()" class="stitch-btn-primary px-4 py-2 ${config.color} text-white rounded-lg shadow-sm hover:opacity-90 transition-all">
              ${tr(config.btn, '??? ???? ')}
            </button>
            <button onclick="SurgeryStation.openRecoveryPlan()" class="stitch-btn-secondary px-4 py-2 bg-white border border-slate-300 text-slate-700 rounded-lg shadow-sm hover:bg-slate-100 transition-all">
              ${tr('Recovery Plan', '??? ')}
            </button>
          </div>
        </header>

        <div class="grid grid-cols-12 gap-6">
          <!-- Left: Surgical Context -->
          <div class="col-span-3 space-y-6">
            <div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
              <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${tr('Pre-Op Status', '??? ? ????')}</h3>
              <div class="space-y-3">
                <div class="flex justify-between text-sm">
                  <span>${tr('Surgical Site Marked', '??? ')}</span>
                  <span class="text-emerald-600 font-bold">?</span>
                </div>
              </div>
            </div>
          </div>
        </div>
      </div>
    `;
  }
};

```

```

    </div>
    <div class="flex justify-between text-sm">
      <span>${tr('Consent Signed', '????? ??????')}</span>
      <span class="text-emerald-600 font-bold">?</span>
    </div>
    <div class="flex justify-between text-sm">
      <span>${tr('Anesthesia Clear', '????? ?????')}</span>
      <span class="text-amber-500 font-bold">?</span>
    </div>
  </div>
</div>

<div class="stitch-card p-4 bg-white rounded-xl shadow-sm border border-slate-200">
  <h3 class="text-sm font-semibold text-slate-500 mb-4 uppercase tracking-wider">${tr('Surgical Risk', '??????? ?????????')}</h3>
  <div class="flex flex-col items-center">
    <div class="stitch-radial-gauge w-32 h-32 relative">
      <svg class="w-full h-full" viewBox="0 0 100 100">
        <circle cx="50" cy="50" r="40" stroke="#e2e8f0" stroke-width="8" fill=
"none" />
        <circle cx="50" cy="50" r="40" stroke="#ef4444" stroke-width="8" fill=
"none"
        stroke-dasharray="251.2" stroke-dashoffset="120" stroke-linecap="r
ound" />
      </svg>
      <span class="absolute inset-0 flex items-center justify-center text-xl fon
t-bold text-slate-800">Med</span>
    </div>
    <p class="mt-4 text-xs text-slate-600 font-medium">${tr('ASA Class III - Moder
ate Risk', '????? ASA 3 - ??? ?????')}</p>
  </div>
</div>

<!-- Center: Active Workspace -->
<div class="col-span-6 space-y-6">
  <div class="stitch-card p-6 bg-white rounded-xl shadow-sm border border-slate-200 min-
h-[600px]">
    <div class="flex border-b border-slate-200 mb-6">
      <button class="px-4 py-2 border-b-2 border-slate-700 text-slate-700 font-medium">${tr('WHO Checklist', '????? ?????? WHO')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('Intra-op L
og', '??? ?????????')}</button>
      <button class="px-4 py-2 text-slate-500 hover:text-slate-800">${tr('AI Recover
y', '????? AI')}</button>
    </div>
    <div id="surgery-workspace-content" class="space-y-4">
      <div class="p-8 text-center text-slate-400 border-2 border-dashed border-slate
-200 rounded-xl">
        ${tr('No active session. Please start a new surgical procedure to begin.',
'?? ???? ???? ?????. ???? ??? ?????? ?????? ??????')}
      </div>
    </div>
  </div>
</div>
</div>
```

```
</div>

<!-- Right: AI Insights -->
<div class="col-span-3 space-y-6">
  <div class="stitch-card p-4 bg-slate-100 rounded-xl shadow-sm border border-slate-200">

    <div class="flex items-center gap-2 mb-4">
      <span class="text-xl">?</span>
      <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">${tr('Su
rgical AI', '???? ?????')}</h3>
    </div>
    <div id="ai-insights-panel" class="space-y-4">
      <div class="p-3 bg-white rounded-lg border border-slate-200 shadow-sm">
        <p class="text-xs text-slate-700 font-semibold mb-1">${tr('Recovery Predic
tion', '???? ?????')}</p>
        <p class="text-sm text-slate-600 italic">"Based on intra-op blood loss and
age, patient is likely to require 48h ICU monitoring."</p>
        <button class="mt-2 text-xs text-slate-600 underline font-medium">${tr('Vi
ew Evidence', '??? ?????')}</button>
      </div>
    </div>
  </div>
</div>
</div>
`;
},

openChecklist: () => { alert('WHO Checklist Modal Triggered'); },
openRecoveryPlan: () => { alert('Recovery Plan Modal Triggered'); }
};
```

```
/**
 * Cardiology Module UI - Google Material Design 3 + Stitch Integration
 * Handles the rendering and interaction for General Cardiology
 */
```



```

},

/**
 * Render the Main Cardiology Dashboard
 * @param {HTMLElement} container - The element to render into
 */
renderDashboard: async (container) => {
  container.innerHTML = `
    <div class="m3-cardiology-container" style="display: flex; height: 100vh; background: ${Cardiology
UI.colors.surface}; font-family: 'Roboto', sans-serif;">
      <!-- Navigation Rail (Material 3) -->
      <nav class="m3-nav-rail" style="width: 80px; background: #f3f4f9; display: flex; flex-directio
n: column; align-items: center; padding: 20px 0; border-right: 1px solid ${CardiologyUI.colors.outline};">
        <div class="m3-nav-item active" onclick="CardiologyUI.switchTab('overview')" style="margin
-bottom: 20px; cursor: pointer; text-align: center;">
          <div class="m3-icon-wrapper" style="width: 56px; height: 32px; background: ${Cardiolog
yUI.colors.primaryContainer}; color: ${CardiologyUI.colors.onPrimaryContainer}; border-radius: 16px; display:
flex; align-items: center; justify-content: center; font-weight: bold;">
            <i class="fas fa-heartbeat"></i>
          </div>
          <span style="font-size: 12px; display: block; margin-top: 4px;">Overview</span>
        </div>
        <div class="m3-nav-item" onclick="CardiologyUI.switchTab('ecg')" style="margin-bottom: 20p
x; cursor: pointer; text-align: center;">
          <div class="m3-icon-wrapper" style="width: 56px; height: 32px; border-radius: 16px; di
splay: flex; align-items: center; justify-content: center;">
            <i class="fas fa-wave-square"></i>
          </div>
          <span style="font-size: 12px; display: block; margin-top: 4px;">ECG</span>
        </div>
        <div class="m3-nav-item" onclick="CardiologyUI.switchTab('meds')" style="margin-bottom: 20
px; cursor: pointer; text-align: center;">
          <div class="m3-icon-wrapper" style="width: 56px; height: 32px; border-radius: 16px; di
splay: flex; align-items: center; justify-content: center;">
            <i class="fas fa-pills"></i>
          </div>
          <span style="font-size: 12px; display: block; margin-top: 4px;">Meds</span>
        </div>
      </nav>

      <!-- Main Content Area -->
      <main class="m3-main-content" style="flex: 1; padding: 24px; overflow-y: auto;">
        <header style="display: flex; justify-content: space-between; align-items: center; margin-
bottom: 32px;">
          <h1 style="font-size: 24px; color: #1c1b1f;">Cardiology Patient Overview</h1>
          <div id="patient-badge" style="background: ${CardiologyUI.colors.primaryContainer}; co
lor: ${CardiologyUI.colors.onPrimaryContainer}; padding: 8px 16px; border-radius: 20px; font-weight: 500;">
            Patient: #12345 - Ahmed Ali
          </div>
        </header>

        <!-- Vitals Grid (Material 3 Cards) -->
        <div class="m3-vitals-grid" style="display: grid; grid-template-columns: repeat(auto-fit,
minmax(250px, 1fr)); gap: 16px; margin-bottom: 32px;">

```

```

        <div class="m3-card" style="background: white; padding: 20px; border-radius: 12px; border: 1px solid ${CardiologyUI.colors.outline}; box-shadow: 0 2px 4px rgba(0,0,0,0.05);">
            <span style="color: ${CardiologyUI.colors.secondary}; font-size: 14px;">Blood Pressure</span>

            <div style="font-size: 32px; font-weight: bold; margin: 8px 0;">120/80 <span style="font-size: 16px; color: gray;">mmHg</span></div>
            <div style="color: green; font-size: 12px;">? Stable</div>
        </div>
        <div class="m3-card" style="background: white; padding: 20px; border-radius: 12px; border: 1px solid ${CardiologyUI.colors.outline}; box-shadow: 0 2px 4px rgba(0,0,0,0.05);">
            <span style="color: ${CardiologyUI.colors.secondary}; font-size: 14px;">Heart Rate</span>

            <div style="font-size: 32px; font-weight: bold; margin: 8px 0;">72 <span style="font-size: 16px; color: gray;">bpm</span></div>
            <div style="color: green; font-size: 12px;">? Normal</div>
        </div>
        <div class="m3-card" style="background: white; padding: 20px; border-radius: 12px; border: 1px solid ${CardiologyUI.colors.outline}; box-shadow: 0 2px 4px rgba(0,0,0,0.05);">
            <span style="color: ${CardiologyUI.colors.secondary}; font-size: 14px;">Ejection Fraction (EF%)</span>

            <div style="font-size: 32px; font-weight: bold; margin: 8px 0; color: ${CardiologyUI.colors.error}">35%</div>
            <div style="color: ${CardiologyUI.colors.error}; font-size: 12px;">? High Risk (HF rEF)</div>
        </div>
    </div>

    <!-- AI Insight Section (RAG Integrated) -->
    <div class="m3-ai-insight" style="background: ${CardiologyUI.colors.primaryContainer}; padding: 20px; border-radius: 16px; border-left: 6px solid ${CardiologyUI.colors.primary}; margin-bottom: 32px;">
        <h3 style="margin: 0 0 10px 0; display: flex; align-items: center; gap: 8px;">
            <i class="fas fa-robot"></i> AI Clinical Insight
        </h3>
        <p style="margin: 0; color: ${CardiologyUI.colors.onPrimaryContainer}; line-height: 1.5;">
            Based on EF% (35%) and current medications, the patient is a candidate for
            <strong>ARNI therapy (Sacubitril/Valsartan)</strong> according to 2023 ESC Guidelines.

            Consider optimizing Beta-blocker dosage.
        </p>
    </div>

    <!-- Action Area -->
    <div id="action-area">
        <button onclick="CardiologyUI.openVisitModal()" style="background: ${CardiologyUI.colors.primary}; color: white; border: none; padding: 12px 24px; border-radius: 100px; font-weight: 500; cursor: pointer; box-shadow: 0 4px 8px rgba(0,0,0,0.2);">
            + New Cardiology Visit
        </button>
    </div>
</main>

<!-- Floating Action Button (FAB) -->
<div class="m3-fab" style="position: absolute; bottom: 24px; right: 24px; width: 56px; height:

```

```

56px; background: ${CardiologyUI.colors.primaryContainer}; color: ${CardiologyUI.colors.onPrimaryContainer};
border-radius: 16px; display: flex; align-items: center; justify-content: center; cursor: pointer; box-shadow:
0 4px 12px rgba(0,0,0,0.3); font-size: 24px;">
    <i class="fas fa-plus"></i>
  </div>
</div>
`;
},

switchTab: (tab) => {
  console.log('Switching to tab:', tab);
  // Implementation for tab switching logic
},

openVisitModal: () => {
  alert('Opening New Visit Modal... (Stitch-Modal implementation)');
}
};

```

```

// Export to global window for app.js access
window.CardiologyUI = CardiologyUI;

```

```

=====
FILE: public\js\modules\interventional_cardiology_ui.js
=====

```

```

/**
 * Interventional Cardiology UI - Google Material Design 3 + Stitch Integration
 * Specialized for Cath Lab environments (High Contrast, High Precision)
 */

const ICUI = {
  colors: {
    primary: '#0061a4',
    onPrimary: '#ffffff',
    primaryContainer: '#d1e4ff',
    onPrimaryContainer: '#001d35',
    secondary: '#5b5e71',
    surface: '#0f1115', // Dark surface for Cath Lab to reduce glare
    onSurface: '#e2e2e6',
    error: '#ffb4ab',
    outline: '#44474e'
  },

  renderDashboard: async (container) => {
    container.innerHTML = `
      <div class="m3-ic-container" style="display: flex; height: 100vh; background: ${ICUI.colors.surface}; color: ${ICUI.colors.onSurface}; font-family: 'Roboto', sans-serif;">
        <!-- Navigation Rail (Material 3) -->
        <nav class="m3-nav-rail" style="width: 80px; background: #1c1b1f; display: flex; flex-direction: column; align-items: center; padding: 20px 0; border-right: 1px solid ${ICUI.colors.outline};">
          <div class="m3-nav-item active" onclick="ICUI.switchTab('session')" style="margin-bottom:

```

```
20px; cursor: pointer; text-align: center;">
        <div class="m3-icon-wrapper" style="width: 56px; height: 32px; background: ${ICUI.colors.primaryContainer}; color: ${ICUI.colors.onPrimaryContainer}; border-radius: 16px; display: flex; align-items: center; justify-content: center; font-weight: bold;">
            <i class="fas fa-syringe"></i>
        </div>
        <span style="font-size: 12px; display: block; margin-top: 4px;">Session</span>
    </div>
    <div class="m3-nav-item" onclick="ICUI.switchTab('stents')" style="margin-bottom: 20px; cursor: pointer; text-align: center;">
        <div class="m3-icon-wrapper" style="width: 56px; height: 32px; border-radius: 16px; display: flex; align-items: center; justify-content: center;">
            <i class="fas fa-circle-notch"></i>
        </div>
        <span style="font-size: 12px; display: block; margin-top: 4px;">Stents</span>
    </div>
    <div class="m3-nav-item" onclick="ICUI.switchTab('hemo')" style="margin-bottom: 20px; cursor: pointer; text-align: center;">
        <div class="m3-icon-wrapper" style="width: 56px; height: 32px; border-radius: 16px; display: flex; align-items: center; justify-content: center;">
            <i class="fas fa-chart-line"></i>
        </div>
        <span style="font-size: 12px; display: block; margin-top: 4px;">Hemo</span>
    </div>
</nav>

<!-- Main Content Area -->
<main class="m3-main-content" style="flex: 1; padding: 24px; overflow-y: auto;">
    <header style="display: flex; justify-content: space-between; align-items: center; margin-bottom: 32px;">
        <h1 style="font-size: 24px; color: ${ICUI.colors.onSurface}">Cath Lab Procedure Control</h1>
        <div id="session-badge" style="background: ${ICUI.colors.primaryContainer}; color: ${ICUI.colors.onPrimaryContainer}; padding: 8px 16px; border-radius: 20px; font-weight: 500;">
            Session: PCI-2026-001 | Patient: Ahmed Ali
        </div>
    </header>

    <!-- Critical Metrics Grid (Material 3 Cards) -->
    <div class="m3-metrics-grid" style="display: grid; grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); gap: 16px; margin-bottom: 32px;">
        <div class="m3-card" style="background: #2b2930; padding: 20px; border-radius: 12px; border: 1px solid ${ICUI.colors.outline}">
            <span style="color: #cac4d0; font-size: 14px;">Door-to-Balloon Time</span>
            <div style="font-size: 32px; font-weight: bold; margin: 8px 0; color: ${ICUI.colors.primaryContainer}">42 <span style="font-size: 16px; color: gray;">min</span></div>
            <div style="color: #4caf50; font-size: 12px;">? Optimal (Guideline < 90min)</div>
        </div>
        <div class="m3-card" style="background: #2b2930; padding: 20px; border-radius: 12px; border: 1px solid ${ICUI.colors.outline}">
            <span style="color: #cac4d0; font-size: 14px;">Contrast Volume</span>
            <div style="font-size: 32px; font-weight: bold; margin: 8px 0; color: ${ICUI.colors.primaryContainer}">85 <span style="font-size: 16px; color: gray;">ml</span></div>
            <div style="color: #ffb4ab; font-size: 12px;">? Approaching Limit (GFR: 40)</div>
    </div>
</main>
```

```

    </div>
    <div class="m3-card" style="background: #2b2930; padding: 20px; border-radius: 12px; border: 1px solid ${ICUI.colors.outline};">
        <span style="color: #cac4d0; font-size: 14px;">Post-Dilation Pressure</span>
        <div style="font-size: 32px; font-weight: bold; margin: 8px 0;">16 <span style="font-size: 16px; color: gray;">atm</span></div>
        <div style="color: #4caf50; font-size: 12px;">? Optimal Expansion</div>
    </div>
</div>

<!-- AI Procedural Insight (RAG Integrated) -->
<div class="m3-ai-insight" style="background: #211f26; padding: 20px; border-radius: 16px; border-left: 6px solid ${ICUI.colors.primary}; margin-bottom: 32px;">
    <h3 style="margin: 0 0 10px 0; display: flex; align-items: center; gap: 8px; color: ${ICUI.colors.onSurface};">
        <i class="fas fa-robot"></i> AI Procedural Guidance
    </h3>
    <p style="margin: 0; color: #cac4d0; line-height: 1.5;">
        Current vessel segment (LAD) shows high calcification.
        <strong style="color: ${ICUI.colors.primaryContainer};">Recommendation:</strong> Consider
        <strong style="color: ${ICUI.colors.primaryContainer};">Rotational Atherectomy</strong>
        before stent deployment to ensure optimal expansion.
    </p>
</div>

<!-- Action Area -->
<div id="action-area">
    <button onclick="ICUI.openStentModal()" style="background: ${ICUI.colors.primary}; color: white; border: none; padding: 12px 24px; border-radius: 100px; font-weight: 500; cursor: pointer; box-shadow: 0 4px 8px rgba(0,0,0,0.2);">
        + Record Stent Deployment
    </button>
</div>
</main>

<!-- Floating Action Button (FAB) -->
<div class="m3-fab" style="position: absolute; bottom: 24px; right: 24px; width: 56px; height: 56px; background: ${ICUI.colors.primaryContainer}; color: ${ICUI.colors.onPrimaryContainer}; border-radius: 16px; display: flex; align-items: center; justify-content: center; cursor: pointer; box-shadow: 0 4px 12px rgba(0,0,0,0.3); font-size: 24px;">
    <i class="fas fa-plus"></i>
</div>
</div>
</div>

switchTab: (tab) => {
    console.log('Switching to IC Tab:', tab);
},

openStentModal: () => {
    alert('Opening Stent Registry Modal... (Stitch-Material3 implementation)');
}

```

```

    }
};

window.ICUI = ICUI;

=====
FILE: public\super-admin\plans-admin.js
=====

/* Jumanasoft Super Admin ? Plans & Pricing client (Batch 3). CSP-friendly: no inline handlers, all output escaped. */
(function () {
    'use strict';
    var API = '/api/super-admin';
    function safeFetch(url, options) {
        options = options || {};
        options.credentials = 'same-origin';
        return fetch(url, options).then(function (r) {
            if (r.status === 401) {
                window.location.href = '/login.html';
                throw new Error('???? ????? ??????');
            }
            return r;
        });
    }
    var $ = function (id) { return document.getElementById(id); };
    function esc(s) {
        return String(s == null ? '' : s).replace(/[\<>"]/g, function (c) {
            return { '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' }[c];
        });
    }
    var CURRENCIES = ['SAR', 'USD', 'AED', 'EGP', 'KWD', 'BHD', 'QAR', 'OMR', 'JOD'];
    var SUPPORT = ['basic', 'standard', 'priority', 'enterprise'];
    var MODULES = ['dashboard', 'patients', 'appointments', 'doctor', 'nursing', 'lab', 'radiology', 'pharmacy',
        'inventory', 'invoices', 'accounts', 'finance', 'insurance', 'reports', 'messaging', 'settings', 'surgery',
        'icu', 'emergency', 'inpatient', 'bloodbank', 'obgyn', 'antenatal', 'cssd', 'quality', 'infection', 'him',
        'medical-records', 'pathology', 'hr', 'maintenance', 'api'];

    var loadedOnce = false;

    function pstate(msg, isError) {
        var el = $('pl-state'); if (!el) return;
        if (!msg) { el.hidden = true; el.textContent = ''; el.className = 'sa-state'; return; }
        el.hidden = false; el.textContent = msg; el.className = 'sa-state' + (isError ? ' sa-error' : '');
    }
    function money(v) { var n = Number(v); return (isFinite(n) ? n : 0).toFixed(2); }
    function activeBadge(a) {
        return a ? '<span class="sa-badge active">????</span>' : '<span class="sa-badge cancelled">?????</span>';
    }

    function rowHtml(p) {
        var act = '<button class="sa-btn sa-ghost" data-action="pl-edit" data-key="' + esc(p.plan_key) + '">?????<

```

```

/button>';
    if (p.active) act += '<button class="sa-btn sa-danger" data-action="pl-disable" data-key="' + esc(p.plan_key) + '">????</button>';
    else act += '<button class="sa-btn" data-action="pl-enable" data-key="' + esc(p.plan_key) + '">????</button>';
    return '<tr>' +
        '<td><code>' + esc(p.plan_key) + '</code></td>' +
        '<td>' + esc(p.name_ar) + '<div class="sa-muted">' + esc(p.name_en) + '</div></td>' +
        '<td>' + esc(p.currency) + '</td>' +
        '<td>' + money(p.monthly_price) + '</td>' +
        '<td>' + money(p.yearly_price) + '</td>' +
        '<td>' + (Number(p.trial_days) || 0) + '</td>' +
        '<td>' + activeBadge(p.active) + '</td>' +
        '<td><div class="sa-row-actions">' + act + '</div></td>' +
        '</tr>';
}

var cache = [];
function load() {
    pstate('???? ??????');
    $('pl-form').hidden = true;
    safeFetch(API + '/plans')
        .then(function (r) {
            if (r.status === 403) throw new Error('????? Super Admin. ');
            if (!r.ok) throw new Error('????? (' + r.status + '). ');
            return r.json();
        })
        .then(function (data) {
            cache = data.plans || [];
            var tb = $('pl-tbody'); tb.innerHTML = '';
            if (!cache.length) { pstate('?? ???? ??? ??? . ??? ????'); return; }
            pstate('');
            tb.innerHTML = cache.map(rowHtml).join('');
        })
        .catch(function (e) { pstate(e.message || '???.', true); $('pl-tbody').innerHTML = ''; });
}

function field(label, id, value, type, attrs) {
    return '<label class="pf-field"><span>' + esc(label) + '</span>' +
        '<input id="' + id + '" class="sa-input" type="' + (type || 'text') + '" value="' + esc(value == null ? '' : value) + '" ' + (attrs || '') + ' /></label>';
}

function selectField(label, id, options, value) {
    var opts = options.map(function (o) { return '<option value="' + esc(o) + '" ' + (o === value ? 'selected' : '') + '>' + esc(o) + '</option>'; }).join('');
    return '<label class="pf-field"><span>' + esc(label) + '</span><select id="' + id + '" class="sa-input">' +
        opts + '</select></label>';
}

function checkField(label, id, checked) {
    return '<label class="pf-check"><input id="' + id + '" type="checkbox" ' + (checked ? 'checked' : '') + ' /> <span>' + esc(label) + '</span></label>';
}

function showForm(plan) {

```

```

var f = $('#pl-form'); f.hidden = false;
var ent = (plan && plan.entitlements) || {};
var mods = ent.modules_enabled || [];
var isEdit = !!plan;
var modBoxes = MODULES.map(function (m) {
    return '<label class="pf-mod"><input type="checkbox" data-mod="' + esc(m) + '" + (mods.indexOf(m) > -1
? ' checked' : '') + ' /> ' + esc(m) + '</label>';
}).join('');
f.innerHTML =
    '<h3>' + (isEdit ? '????? ???? ? ' + esc(plan.plan_key) : '??? ?????') + '</h3>' +
    '<div class="pf-grid">' +
        field('??????? (plan_key)', 'pf-key', plan ? plan.plan_key : '', 'text', isEdit ? 'disabled' : 'placeholder="??? pro_2026"') +
        selectField('??????', 'pf-currency', CURRENCIES, plan ? plan.currency : 'SAR') +
        field('????? (????)', 'pf-name_ar', plan ? plan.name_ar : '') +
        field('????? (???????)', 'pf-name_en', plan ? plan.name_en : '') +
        field('????? ??????', 'pf-monthly', plan ? plan.monthly_price : '0', 'number', 'min="0" step="0.01"')
    +
        field('????? ??????', 'pf-yearly', plan ? plan.yearly_price : '0', 'number', 'min="0" step="0.01"') +
        field('????? ??????', 'pf-trial', plan ? plan.trial_days : '0', 'number', 'min="0" max="365"') +
        field('??????', 'pf-sort', plan ? plan.sort_order : '0', 'number') +
        field('?? ?????????? (???=??? ?????)', 'pf-max_users', ent.max_users, 'number', 'min="0"') +
        field('?? ?????? (???=??? ?????)', 'pf-max_branches', ent.max_branches, 'number', 'min="0"') +
        field('?? ?????????/??? (???=??? ?????)', 'pf-max_invoices', ent.max_invoices_per_month, 'number', 'min="0"') +
        selectField('????? ?????', 'pf-support', SUPPORT, ent.support_level || 'standard') +
    '</div>' +
    '<div class="pf-desc">' +
        '<label class="pf-field"><span>????? (????)</span><textarea id="pf-desc_ar" class="sa-input">' + esc(plan ? plan.description_ar : '') + '</textarea></label>' +
        '<label class="pf-field"><span>????? (???????)</span><textarea id="pf-desc_en" class="sa-input">' + esc(plan ? plan.description_en : '') + '</textarea></label>' +
    '</div>' +
    '<div class="pf-checks">' +
        checkField('???? API', 'pf-api', ent.api_access) +
        checkField('???? ?????', 'pf-domain', ent.custom_domain) +
    '</div>' +
    '<div class="pf-mods"><div class="sa-muted">??????? ??????????:</div>' + modBoxes + '</div>' +
    '<div id="pf-errors" class="sa-state sa-error" hidden></div>' +
    '<div class="pf-actions">' +
        '<button class="sa-btn" data-action="pl-save" data-key="' + esc(plan ? plan.plan_key : '') + '">???</button>' +
        '<button class="sa-btn sa-ghost" data-action="pl-cancel">?????</button>' +
    '</div>';
f.scrollToView({ behavior: 'smooth', block: 'nearest' });
}

function numOrEmpty(id) { var v = $(id).value.trim(); return v === '' ? '' : v; }
function collect() {
    var mods = [];
    Array.prototype.forEach.call(document.querySelectorAll('[data-mod]'), function (cb) { if (cb.checked) mods.push(cb.getAttribute('data-mod')); });
    return {
        plan_key: $('pf-key').value.trim(),

```



```

name_ar: $('pf-name_ar').value.trim(), name_en: $('pf-name_en').value.trim(),
description_ar: $('pf-desc_ar').value, description_en: $('pf-desc_en').value,
currency: $('pf-currency').value,
monthly_price: $('pf-monthly').value, yearly_price: $('pf-yearly').value,
trial_days: $('pf-trial').value, sort_order: $('pf-sort').value,
max_users: numOrEmpty('pf-max_users'), max_branches: numOrEmpty('pf-max_branches'),
max_invoices_per_month: numOrEmpty('pf-max_invoices'),
support_level: $('pf-support').value,
api_access: $('pf-api').checked, custom_domain: $('pf-domain').checked,
modules_enabled: mods
};
}
function showErrors(list) {
  var e = $('pf-errors'); if (!e) return;
  if (!list || !list.length) { e.hidden = true; e.textContent = ''; return; }
  e.hidden = false; e.textContent = '????? ?????: ' + list.join(' · ');
}

function save(key) {
  var body = collect();
  var isEdit = !!key;
  var url = isEdit ? (API + '/plans/' + encodeURIComponent(key)) : (API + '/plans');
  showErrors(null); pstate('???? ??????');
  safeFetch(url, { method: isEdit ? 'PUT' : 'POST', headers: { 'Content-Type': 'application/json' }, body: JSON.stringify(body) })
    .then(function (r) { return r.json().then(function (j) { return { status: r.status, j: j }; }); })
    .then(function (res) {
      if (res.status === 400) { showErrors(res.j.details || [res.j.error || '???? ??? ?????']); pstate(''); return; }
      if (res.status === 409) { showErrors([res.j.error || '??????? ??????']); pstate(''); return; }
      if (res.status >= 400) throw new Error('??? ????? (' + res.status + ').');
      $('pl-form').hidden = true; load();
    })
    .catch(function (e) { pstate(e.message || '???.', true); });
}

function setActive(key, enable) {
  if (!window.confirm((enable ? '?????' : '?????') + ' ????? ' + key + '?')) return;
  pstate('???? ??????????');
  safeFetch(API + '/plans/' + encodeURIComponent(key) + '/' + (enable ? 'enable' : 'disable'), { method: 'POST' })
    .then(function (r) { if (!r.ok) throw new Error('??? ???????? (' + r.status + ').'); return r.json(); })
    .then(function () { load(); })
    .catch(function (e) { pstate(e.message || '???.', true); });
}

// ----- tenant plan assignment (called by super-admin.js details view) -----
function mountTenantPlan(tenantId, el) {
  if (!el) return;
  el.innerHTML = '<div class="sa-muted">???? ?????? ??? ??????????</div>';
  Promise.all([
    safeFetch(API + '/tenants/' + encodeURIComponent(tenantId) + '/plan').then(function (r) { return r.ok ? r.json() : { current: null }; }),
    safeFetch(API + '/plans').then(function (r) { return r.ok ? r.json() : { plans: [] }; })
  ])

```

```

    }).then(function (out) {
        var current = out[0].current;
        var active = (out[1].plans || []).filter(function (p) { return p.active; });
        var opts = active.map(function (p) { return '<option value="' + esc(p.plan_key) + '">' + esc(p.plan_key)
+ ' ? ' + esc(p.name_ar) + '</option>'; }).join('');
        el.innerHTML =
            '<h4>????? ???????</h4>' +
            '<div class="sa-kv"><span class="k">?????</span><span>' + (current ? '<code>' + esc(current.plan_key)
+ '</code>' + '<span class="sa-muted">(' + esc(current.assignment_source) + ')</span>' : '? ? ? ? ?') + '</span><
/div>' +
            (active.length
                ? '<div class="pf-assign"><select id="sa-assign-plan" class="sa-input">' + opts + '</select>' +
                '<button class="sa-btn" data-action="pl-assign" data-id="' + esc(tenantId) + '">????? ??????</butto
n></div>'
                : '<div class="sa-muted">?? ???? ???? ???? ???????</div>') +
            '<div id="sa-ent-mount"></div>';
        mountEntitlements(tenantId, $('sa-ent-mount'));
    }).catch(function () { el.innerHTML = '<div class="sa-state sa-error">????? ?????? ??? ???????</div>'; })
;
}

```

// OBSERVE-only: show resolved entitlements when the resolver is enabled. Silently skipped otherwise (404/di
sabled).

```

function mountEntitlements(tenantId, el) {
    if (!el) return;
    safeFetch(API + '/tenants/' + encodeURIComponent(tenantId) + '/entitlements')
        .then(function (r) { return r.ok ? r.json() : null; })
        .then(function (d) {
            if (!d || !d.entitlements) return; // resolver disabled -> show nothing
            var e = d.entitlements;
            function lim(v) { return v == null ? '??? ?????' : esc(v); }
            el.innerHTML =
                '<h4>???????????? ??????? <span class="sa-badge ' + (d.source === 'plan' ? 'active' : 'trial') + '">
' + esc(d.source) + '</span> <span class="sa-muted">(' + esc(d.enforcement_mode) + ')</span></h4>' +
                '<div class="sa-kv">' +
                '<span class="k">?? ?????????</span><span>' + lim(e.max_users) + '</span>' +
                '<span class="k">?? ?????</span><span>' + lim(e.max_branches) + '</span>' +
                '<span class="k">?? ???????/?</span><span>' + lim(e.max_invoices_per_month) + '</span>' +
                '<span class="k">????? ?????</span><span>' + esc(e.support_level || 'standard') + '</span>' +
                '<span class="k">???? API</span><span>' + (e.api_access ? '???' : '??') + '</span>' +
                '<span class="k">???? ?????</span><span>' + (e.custom_domain ? '???' : '??') + '</span>' +
                '<span class="k">??????</span><span>' + ((e.modules_enabled && e.modules_enabled.length) ? e.module
s_enabled.map(esc).join(' ? ') : '?') + '</span>' +
                '</div>';
            })
        .catch(function () { /* observe-only: never disrupt the tenant view */ });
}

```

```

function assign(tenantId) {
    var sel = $('sa-assign-plan'); if (!sel) return;
    var key = sel.value;
    if (!window.confirm('????? ????? ' + key + ' ???????? #' + tenantId + '?')) return;
    safeFetch(API + '/tenants/' + encodeURIComponent(tenantId) + '/plan', {
        method: 'POST', headers: { 'Content-Type': 'application/json' },

```

```

        body: JSON.stringify({ plan_key: key, assignment_source: 'manual' })
    })
    .then(function (r) { return r.json().then(function (j) { return { status: r.status, j: j }; }); })
    .then(function (res) {
        if (res.status === 409) { window.alert(res.j.error || '?? ???? ????? ??? ??????'); return; }
        if (res.status >= 400) throw new Error(res.j.error || ('??? ?????? (' + res.status + ').'));
        mountTenantPlan(tenantId, $('sa-plan-mount'));
    })
    .catch(function (e) { window.alert(e.message || '???'); });
}

// delegation: act ONLY on pl-* actions (tenant actions handled by super-admin.js)
document.addEventListener('click', function (ev) {
    var b = ev.target.closest('[data-action]'); if (!b) return;
    var action = b.getAttribute('data-action');
    var key = b.getAttribute('data-key');
    if (action === 'pl-edit') { var p = cache.filter(function (x) { return x.plan_key === key; })[0]; if (p) s
howForm(p); }
    else if (action === 'pl-disable') setActive(key, false);
    else if (action === 'pl-enable') setActive(key, true);
    else if (action === 'pl-save') save(key);
    else if (action === 'pl-cancel') { $('pl-form').hidden = true; }
    else if (action === 'pl-assign') assign(b.getAttribute('data-id'));
});

var newBtn = $('pl-new'); if (newBtn) newBtn.addEventListener('click', function () { showForm(null); });
var refBtn = $('pl-refresh'); if (refBtn) refBtn.addEventListener('click', load);
document.addEventListener('sa-tab', function (ev) {
    if (ev.detail && ev.detail.tab === 'plans' && !loadedOnce) { loadedOnce = true; load(); }
});

window.SAPPlans = { mountTenantPlan: mountTenantPlan };
})();

```

```

=====
FILE: public\super-admin\super-admin.js
=====

```

```

/* Jumanasoft Super Admin ? Tenant Control Center client (CSP-friendly: no inline handlers). */
(function () {
    'use strict';
    var API = '/api/super-admin';
    function safeFetch(url, options) {
        options = options || {};
        options.credentials = 'same-origin';
        return fetch(url, options).then(function (r) {
            if (r.status === 401) {
                window.location.href = '/login.html';
                throw new Error('??? ???? ??????');
            }
            return r;
        });
    }
});

```

```

}
var $ = function (id) { return document.getElementById(id); };
function esc(s) {
    return String(s == null ? '' : s).replace(/[\<>"]/g, function (c) {
        return { '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' }[c];
    });
}
function state(msg, isError) {
    var el = $('sa-state');
    if (!msg) { el.hidden = true; el.textContent = ''; el.className = 'sa-state'; return; }
    el.hidden = false; el.textContent = msg; el.className = 'sa-state' + (isError ? ' sa-error' : '');
}

// Onboarding Wizard global dependencies mapping
window.tr = function(en, ar) { return ar; };
window.escapeHTML = function(s) { return esc(s); };
window.safeId = function(v) { const n = Number(v); return (v !== null && v !== undefined && String(v).trim() !== '' && Number.isFinite(n)) ? n : ''; };
window.showToast = function(msg, type) { state(msg, type === 'error'); };
window.isArabic = true;
function fmtDate(d) { if (!d) return '?'; try { return new Date(d).toLocaleDateString('ar'); } catch (e) { return esc(d); } }
function badge(status) {
    var label = { active: '??', trial: '?????', suspended: '????', cancelled: '????' }[status] || esc(status);
    return '<span class="sa-badge ' + esc(status) + '">' + label + '</span>';
}

function rowHtml(t) {
    var act = '';
    if (t.status === 'suspended') act += '<button class="sa-btn" data-action="reactivate" data-id="' + t.id + '">???? ????</button>';
    else if (t.status === 'active' || t.status === 'trial') act += '<button class="sa-btn sa-danger" data-action="suspend" data-id="' + t.id + '">????</button>';
    act += '<button class="sa-btn sa-ghost" data-action="details" data-id="' + t.id + '">????</button>';
    return '<tr>' +
        '<td>' + esc(t.id) + '</td>' +
        '<td>' + esc(t.name) + '</td>' +
        '<td>' + esc(t.subdomain) + '</td>' +
        '<td>' + badge(t.status) + '</td>' +
        '<td>' + esc(t.plan) + '</td>' +
        '<td>' + fmtDate(t.created_at) + '</td>' +
        '<td><div class="sa-row-actions">' + act + '</div></td>' +
        '</tr>';
}

function load() {
    state('??? ????');
    $('sa-detail').hidden = true;
    var qs = new URLSearchParams();
    if ($('#sa-status').value) qs.set('status', $('#sa-status').value);
    if ($('#sa-plan').value.trim()) qs.set('plan', $('#sa-plan').value.trim());
    if ($('#sa-search').value.trim()) qs.set('q', $('#sa-search').value.trim());
    safeFetch(API + '/tenants?' + qs.toString())

```

```

.then(function (r) {
  if (r.status === 403) throw new Error('????? ?????? Super Admin.');
```

```

  if (!r.ok) throw new Error('????? ??????? (' + r.status + ').');
```

```

  return r.json();
})
.then(function (data) {
  var tb = $('sa-tbody'); tb.innerHTML = '';
  if (!data.tenants || !data.tenants.length) { state('?? ???? ?????????? ????????.'); return; }
  state('');
  tb.innerHTML = data.tenants.map(rowHtml).join('');
})
.catch(function (e) { state(e.message || '???.', true); $('sa-tbody').innerHTML = ''; });
}

function details(id) {
  state('???? ?????? ??????????');
  safeFetch(API + '/tenants/' + encodeURIComponent(id))
    .then(function (r) { if (!r.ok) throw new Error('????? ?????? ?????????? (' + r.status + ').'); return r.json(); })
    .then(function (data) {
      state('');
      var t = data.tenant; var d = $('sa-detail'); d.hidden = false;
      d.innerHTML = '<h3>?????? ?????????? ? ' + esc(t.name) + ' ' + badge(t.status) + '</h3>' +
        '<div class="sa-kv">' +
        '<span class="k">??????</span><span>' + esc(t.id) + '</span>' +
        '<span class="k">?????? ?????</span><span>' + esc(t.subdomain) + '</span>' +
        '<span class="k">??????</span><span>' + esc(t.plan) + '</span>' +
        '<span class="k">??? ?????????</span><span>' + (t.users == null ? '?' : esc(t.users)) + '</span>' +
        '<span class="k">??? ?????????</span><span>' + (t.facilities == null ? '?' : esc(t.facilities)) + '</span>' +
        '<span class="k">??? ?????</span><span>' + fmtDate(t.last_activity) + '</span>' +
        '<span class="k">????</span><span>' + fmtDate(t.created_at) + '</span>' +
        '</div>' +
        '<div id="sa-plan-mount" class="sa-plan-mount"></div>';
      if (window.SAPPlans && window.SAPPlans.mountTenantPlan) {
        window.SAPPlans.mountTenantPlan(t.id, document.getElementById('sa-plan-mount'));
      }
    })
    .catch(function (e) { state(e.message || '???.', true); });
}

function changeStatus(id, action) {
  var verb = action === 'suspend' ? '?????' : '????? ??????';
  if (!window.confirm('????? ' + verb + ' ?????????? #' + id + '?')) return;
  state('????? ' + verb + '?');
  safeFetch(API + '/tenants/' + encodeURIComponent(id) + '/' + action, { method: 'POST' })
    .then(function (r) {
      if (r.status === 409) return r.json().then(function (j) { throw new Error(j.error || '?????? ???? ???? ??????'); });
      if (!r.ok) throw new Error('??? ?????????? (' + r.status + ').');
      return r.json();
    })
    .then(function () { load(); })
    .catch(function (e) { state(e.message || '???.', true); });
}

```

```

}

function loadStats() {
  safeFetch(API + '/stats')
    .then(function (r) { if (!r.ok) throw new Error('????? ????? ??????????'); return r.json(); })
    .then(function (data) {
      $('stat-revenue').textContent = parseFloat(data.total_revenue || 0).toLocaleString('ar-SA') + ' SAR';
      $('stat-tenants').textContent = parseInt(data.total_tenants || 0, 10).toLocaleString('ar-SA');
      $('stat-active-subs').textContent = parseInt(data.active_subscriptions || 0, 10).toLocaleString('ar-SA');
    });

  var chartContainer = $('sa-chart-bars');
  chartContainer.innerHTML = '';
  if (!data.plans_breakdown || !data.plans_breakdown.length) {
    chartContainer.innerHTML = '<div class="sa-muted">?? ???? ?????????? ???? ???.</div>';
    return;
  }

  var total = data.plans_breakdown.reduce(function (sum, row) { return sum + row.count; }, 0) || 1;
  var labels = {
    free_trial: '???? ???????? (Free Trial)',
    basic: '?????? ?????????? (Basic)',
    premium: '?????? ?????????? (Premium)',
    enterprise: '???? ?????????? (Enterprise)'
  };

  data.plans_breakdown.forEach(function (pb) {
    var label = labels[pb.plan_key] || pb.plan_key;
    var pct = Math.round((pb.count / total) * 100);
    var row = document.createElement('div');
    row.className = 'sa-chart-bar-row';
    row.innerHTML =
      '<div class="sa-chart-bar-lbl">' + esc(label) + '</div>' +
      '<div class="sa-chart-bar-outer"><div class="sa-chart-bar-inner" style="width: ' + pct + '%"></div>' +
      '<div class="sa-chart-bar-val">' + esc(pb.count) + ' (' + pct + '%)</div>';
    chartContainer.appendChild(row);
  });
});

.catch(function (e) { console.error(e); });
}

document.addEventListener('click', function (ev) {
  var b = ev.target.closest('[data-action]'); if (!b) return;
  var id = b.getAttribute('data-id'); var action = b.getAttribute('data-action');
  if (action === 'details') details(id);
  else if (action === 'suspend' || action === 'reactivate') changeStatus(id, action);
});

$('sa-refresh').addEventListener('click', load);
$('sa-search').addEventListener('keydown', function (e) { if (e.key === 'Enter') load(); });
$('sa-status').addEventListener('change', load);

// Onboarding Wizard open handler
var onboardBtn = $('sa-onboard-btn');
```

```

if (onboardBtn) {
  onboardBtn.addEventListener('click', function() {
    if (window>NamaOnboardingWizard) window>NamaOnboardingWizard.open();
  });
}

// Tab switching (CSP-friendly delegation). Emits a 'sa-tab' CustomEvent so plans-admin can lazy-load.
var tabs = document.getElementById('sa-tabs');
if (tabs) {
  tabs.addEventListener('click', function (ev) {
    var b = ev.target.closest('[data-tab]'); if (!b) return;
    var tab = b.getAttribute('data-tab');
    Array.prototype.forEach.call(tabs.querySelectorAll('[data-tab]'), function (el) {
      el.classList.toggle('active', el === b);
    });
    var pt = $('panel-tenants'), pp = $('panel-plans'), po = $('panel-overview');
    if (pt) pt.hidden = (tab !== 'tenants');
    if (pp) pp.hidden = (tab !== 'plans');
    if (po) po.hidden = (tab !== 'overview');
    if (tab === 'overview') loadStats();
    if (tab === 'tenants') load();
    document.dispatchEvent(new CustomEvent('sa-tab', { detail: { tab: tab } }));
  });
}
loadStats();
})();

```

```

=====
FILE: tests\verify_cardiology.js
=====

```

```

const { pool } = require('../db_postgres');
const assert = require('assert');

async function runCardiologyTests() {
  console.log('? Starting Cardiology Suite Verification...');

  try {
    // 1. Test Table Existence
    const tables = ['cardiology_exams', 'cardiology_echo', 'cath_lab_procedures', 'stent_registry', 'ep_ablation_logs', 'ep_device_registry'];
    for (const table of tables) {
      const res = await pool.query(`SELECT 1 FROM information_schema.tables WHERE table_name = $1`, [table]);
      if (res.rowCount === 0) throw new Error(`Table ${table} is missing!`);
      console.log(`? Table ${table} exists.`);
    }

    // 2. Test RLS / Tenant Isolation (Mocking tenant context)
    await pool.query('SET app.current_tenant = \'00000000-0000-0000-0000-000000000000\'');
    // This should not fail but return 0 rows if table is empty
    const rlsCheck = await pool.query('SELECT * FROM cardiology_exams');
  }
}

```

```
    console.log(`? RLS Policy verified (Returned ${rlsCheck.rowCount} rows).`);

    console.log(`\n? ALL CARDIOLOGY CORE COMPONENTS VERIFIED SUCCESSFULLY!`);
  } catch (err) {
    console.error(`\n? VERIFICATION FAILED:`, err.message);
    process.exit(1);
  } finally {
    await pool.end();
  }
}

runCardiologyTests();
```